Vadodara Municipal Corporation (VMC)

Request for Proposal (RFP)For
Selection of Agency for the Supply, Installation, Testing, Commissioning(SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city

Tender Ref No. :	262/26-27
Last Date Online Price Bid Submission:	24/08/2026 up to 16:00 Hrs. (https://tender.nprocure.com and www.vmc.gov.in)
Last Date of Receipt of Physical Bid:	27/08/2026 up to 16:00 Hrs.
Date of Opening of Physical Bid:	27/08/2026, 16:30 Hrs.
Bid Processing Fee:	Rs. 30,000 /- (Rupees Thirty Thousands only)
Bid Security EMD:	Rs. 1,00,00,000 /- (Rupees One Crore)

Invited by:

Director IT,
Information Technology Department
Vadodara Municipal Corporation, Khanderao Market Building, Palace Road,
Vadodara - 390001
E-mail: it-projects@vmc.gov.in

DISCLAIMER

The information contained in this Request for Proposal document ("**RFP**") whether subsequently provided to the bidders, ("**Bidder/s**") verbally or in documentary form by Vadodara Municipal Corporation (henceforth referred to as "**VMC**" in this document) or any of its employees or advisors, is provided to Bidders on the terms and conditions set out in this RFP document and any other terms and conditions subject to which such information is provided.

This RFP document is not an agreement and is not an offer or invitation to any party. The purpose of this RFP is to provide the Bidders or any other person with information to assist the formulation of their financial offers ("**Bid**"). This RFP includes statements, which reflect various assumptions and assessments arrived at by VMC in relation to this scope. This Tender document does not purport to contain all the information each Bidder may require. This Tender document may not be appropriate for all persons, and it is not possible for the Chief Executive Officer, VMC and their employees or advisors to consider the objectives, technical expertise and particular needs of each Bidder. The assumptions, assessments, statements and information contained in the Bid documents, may not be complete, accurate, adequate or correct. Each Bidder must therefore conduct its own analysis of the information contained in this RFP and to seek its own professional advice from appropriate sources.

Information provided in this Tender document to the Bidder is on a wide range of matters, some of which may depend upon interpretation of law. The information given is not intended to be an exhaustive account of statutory requirements and should not be regarded as a complete or authoritative statement of law. VMC accepts no responsibility for the accuracy or otherwise for any interpretation of opinion on law expressed herein.

VMC and their employees and advisors make no representation or warranty and shall incur no liability to any person, including the Bidder under law, statute, rules or regulations or tort, the principles of restitution or unjust enrichment or otherwise for any loss, cost, expense or damage which may arise from or be incurred or suffered on account of anything contained in this RFP or otherwise, including the accuracy, reliability or completeness of the RFP, and any assessment, assumption, statement or information contained therein or deemed to form part of this RFP or arising in any way in this Selection Process.

VMC also accepts no liability of any nature whether resulting from negligence or otherwise howsoever caused arising from reliance of any Bidder upon the statements contained in this RFP. VMC may in its absolute discretion, but without being under any obligation to do so, can amend or supplement the information in this RFP.

The issue of this RFP document does not imply that VMC is bound to select a Bidder or to appoint the Selected Bidder (as defined hereinafter), for implementation and VMC reserves the right to reject all or any of the Bidders or Bids without assigning any reason whatsoever.

The Bidder shall bear all its costs associated with or relating to the preparation and submission of its Bid including but not limited to preparation, copying, postage, delivery fees, expenses associated with any demonstrations or presentations which may be required by VMC or any other costs incurred in connection with or relating to its Bid. All such costs and expenses will remain with the Bidder and VMC shall not be liable in any manner whatsoever for the same or for any other costs or other expenses incurred by a Bidder in preparation for submission of the Bid, regardless of the conduct or outcome of the selection process.

Table of Contents

Table of Contents

SECTION: 1 PROJECT PROFILE *10*
1.1 Objective of this Project **11**
SECTION: 2 ELIGIBILITY CRITERIA *12*
2.1 Pre-Qualification (PQ) Criteria **14**
SECTION: 3 EVALUATION OF BID *19*
3.1 Evaluation Process **20**
SECTION: 4 ..... *22*
SCOPE OF WORK and *22*
TECHNICAL SPECIFICATIONS *22*
4.1 Detailed Scope of Work **23**
4.2 Supply & Delivery **25**
4.3 Resource Deployment and Support Requirements **27**
4.4 Installation, Integration, Testing & Commissioning **27**
4.5 Reporting, Database Administration and Trouble Shooting **28**
4.6 Operation & Maintenance **30**
4.7 Incident Management System **34**
4.8 Enterprise Management Systems (EMS) **35**
1.1 Inclusions *42*
1.2 Laying of Underground Optical Fiber across the city *42*
Optical Fiber Cable Execution Methodology *42*
Open Cut Method *43*
Optical Fiber Cable (OFC) Testing Methodology *44*
4.9 Functional Requirement **45**
4.10 Technical Specification Require Equipment **49**
SECTION: 5 ..... *83*
DELIVERABLE, TIMELINE & *83*
PAYMENT PROCESS *83*
Penalty = QuarterlyO&MCharges × ApplicablePenalty% **99**
SECTION: 6 ..... *105*
GENERAL TERMS & INSTRUCTIONS TO THE BIDDERS *105*
6.1. Instructions to the bidder **106**
6.2 Back-to-Back OEM Support **111**

6.3 Performance Bank Guarantee 111
6.4 Additional Quantity/Change Request 111
6.5 Termination of Contract 112
6.6 Confidentiality: 114
6.7 Insurance 115
6.8 Damages for Breach of Agreement: 116
6.9 Defect Liability Period (DLP) 116
6.10 Confidential Policy 116
6.11 Limitation of Liability 117
6.12 Data Security 117
6.13 Pole, Foundation & Structural Stability Certificate 118
6.14 Integration with Existing Systems 118
6.15 Critical Spare Inventory 118
6.16 Other Terms and Conditions 118
1. The Successful Bidder shall repaint all installed CCTV poles, mounting structures, and fabricated structures (where applicable) at least once after 2 years from the Go-Live date, so that repainting is carried out two (3) times during the entire Contract Period, including the initial Color painting at the time of installation. The repainting shall include surface preparation, rust removal (where required), primer, and weather-resistant paint/color, ensuring that the structures remain free from corrosion and in good condition throughout the Contract Period. 118
2. The Successful Bidder shall inspect all installed CCTV poles and fabricated structures at least once every year throughout the five (5) year Contract Period after Go-Live. For MS painted poles, repainting shall be carried out whenever required to maintain protection against corrosion and proper appearance. For Hot-Dip Galvanized (GI) poles, repainting shall not be mandatory, however, any damaged, scratched, corroded, or deteriorated galvanized surface shall be restored using an approved cold galvanizing compound or equivalent protective coating, or as per the OEM/manufacturer's recommendation, to ensure structural integrity and corrosion protection throughout the five (5) year Contract Period. The selected bidder should provide tape data cartridges for backup. The backup should be taken automatically. 119
3. The approx. quantity will be based on budget but it may be increase or decrease based on requirement. 119
4. The scope of warranty by the bidder also includes reinstallation and reconfiguration of the item in case of breakdown. 119
5. VMC will purchase CCTV Cameras' from the date of contract, with the same rate as purchase order within 2 years. 119
6. All the installation should be finished within 4 months from the work order. 119
7. A vendor is unconditionally bound to VMC's All Terns & General Conditions 119
8. Hand-over of the system at the end of contractual period along with all documentation required to operate and maintain the system in healthy condition. 120
9. No supplied hardware put on roadside to cause an accident and Damage of supplied parts. It is completely under vendor scope to prevent before final installation. 120
10. Bidder to ensure compliance to the entire cyber security standard required safeguarding the edge devices; connectivity & desktop to ensure cyber attach resilient operations 120
11. If required, the maintenance work & technical support will have to be carried out beyond VMC's & Bidder's normal working hours and on holidays too, without any extra charges. The maintenance engineers should be available & approachable round-the-clock (24X7) through telephone. 120
12. For Selected bidder supplied items are not fulfill the specification of RFP/Solution then bidder needs to supply higher specification items without any kind of additional cost. 120
13. Fraud and Corruption ..... Error! Bookmark not defined.
14. In case of Fraud and Corruption by bidder and/or manpower deputed for this service, VMC shall retain all rights to lodge Police Complaint/ FIR against bidder under various provisions of Indian Penal Code and any other penal law in force including Information Technology act and Prevention of Corruption act. Error! Bookmark not defined.
15. The Successful Bidder shall provide comprehensive Operations & Maintenance (O&M) support for a period of five (5) years from the date of Go-Live. The Bidder shall ensure 24x7x365 monitoring, operation, maintenance, fault rectification, and support of the entire CCTV Surveillance System, including cameras,

poles, junction boxes, OFC network, switches, servers, storage, video wall, VMS, and associated infrastructure. 120

16. The Bidder shall carry out Quarterly Preventive Maintenance (PM) of all field and central infrastructure and submit PM reports to VMC. Daily fault rectification, camera cleaning, focus adjustment, angle correction, OFC restoration, network troubleshooting, and equipment replacement shall be included within the scope of work. 120

17. The Bidder shall arrange Hydra/Crane services, vehicles, OFC splicing resources, and Field Response Teams (FRT) comprising technicians, splicers, supervisors, drivers, and necessary equipment for timely restoration of services. No separate payment shall be made for manpower, vehicles, tools, Hydra services, preventive maintenance activities, or fault rectification services, and the cost thereof shall be deemed included in the quoted O&M/CAMC charges. 120

18. The quoted prices shall be inclusive of supply, transportation, insurance, installation, testing, commissioning, integration, configuration, training, documentation, warranty, maintenance, and all other activities required for successful implementation of the project. 120

19. The Bidder shall clearly specify the OEM, Make, Model Number, and Part Number of all proposed equipment, software, and licenses. 120

20. Any hardware, software, accessory, component, fitting, mounting material, cable, connector, license, service, or item required for successful completion and operation of the project, whether explicitly mentioned in the BOQ or not, shall be deemed included in the Bidder's scope and quoted price without any additional cost to VMC. 120

21. Comprehensive Warranty and CAMC support obligations shall be clearly defined by the Bidder. CAMC rates shall be quoted separately for each year of the O&M period. 120

22. GST and other applicable taxes shall be indicated separately in the Financial Bid..... Error! Bookmark not defined.

23. All software licenses, Video Management Software (VMS) licenses, analytics licenses, operating system licenses, database licenses, and other application licenses shall be perpetual unless specifically stated otherwise in the tender. 120

24. The proposed solution shall be scalable and capable of supporting a minimum of 900 CCTV cameras, including future expansion without requiring replacement of the core infrastructure. 120

25. The quantities indicated in the BOQ are tentative and may increase or decrease based on site conditions, survey findings, operational requirements, and VMC directives. Payment shall be made strictly on the basis of actual quantities supplied, installed, tested, commissioned, and accepted by VMC at approved unit rates. 120

26. The proposed Storage System, Chassis, Servers, Video Management System, and associated central infrastructure shall support integration of approximately 250 existing CCTV cameras of the legacy project and shall be sized with a minimum 20% spare capacity for future expansion without requiring major hardware replacement. 120

27. The Bidder shall arrange and manage Hydra/Crane vehicles, tools, testing equipment, safety equipment, consumables, and all resources required for maintenance and restoration activities at no additional cost to VMC. 120

28. The Bidder shall deploy and maintain a dedicated Field Response Team (FRT) consisting of adequate skilled manpower, including technicians, OFC splicers, supervisors, drivers, vehicles, and required tools, for timely fault rectification and maintenance activities throughout the Contract Period. 121

29. The Bidder shall maintain sufficient inventory of spare equipment and consumables to ensure compliance with SLA requirements and minimize downtime of surveillance infrastructure. 121

Annexure 1: Technical Bid Format 122

Annexure 2: Commercial Bid Format &Instructions 137

Annexure 3: Agreement Format 143

Notice Inviting Tender (NIT)

Vadodara Municipal Corporation E tender - Request for Proposal (RFP) For Selection of Agency for the Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city	
Request for Proposal (RFP)For Selection of Agency for the Supply, Installation, Testing, Commissioning(SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city	
Last Date Online Price Bid Submission:	24/08/2026 up to 16:00 Hrs.
Last Date of Receipt of Physical Bid:	27/08/2026 up to 16:00 Hrs.
Detail & Tender available On	https://tender.nprocure.com & www.vmc.gov.in
Vadodara Municipal corporation (VMC) Office Address	Information Technology Department Vadodara Municipal Corporation, Khanderao Market, Vadodara 390001
Email	it-projects@vmc.gov.in

IMPORTANT DATES

Selection of Agency for the Supply, Installation, Testing, Commissioning (SITC)and Operation & maintenance of Surveillance Cameras in Vadodara city		
1.	**Contract Period**	5 Years after Go Live
2.	**Delivery**	Selection of Agency for the Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city
3.	**RFP No.**	262/26-27
4.	**Last Date Online Price Bid Submission:**	24/08/2026 up to 16:00 Hrs. (at https://tender.nprocure.com only)
5.	**Last Date of Receipt of Physical Bid:**	27/08/2026 up to 16:00 Hrs.
6.	**Date of Opening of Physical Bid:**	27/08/2026 16:30 Hrs.
7.	**Bid Processing Fee:** **(Non - refundable)**	INR 30,000 /- (Rupees Thirty thousand only) in form of DD only
8.	**Bid EMD:**	INR 1,00,00,000/- (Rupees One Crore only) shall be either in form of Demand Draft in favor of "Municipal Commissioner, Vadodara" from any nationalized/scheduled banks, payable at Vadodara for the **period of 6 years.** OR Bank Guarantee issued by the banks (specified in Finance Department's GR. No: FD/MSM/e-file/4/2024/ 2859/DMO dated 01/05/2025 only in favor of "Commissioner, Vadodara Municipal Corporation".
9.	**Pre-Bid Meeting Date & Time**	12/08/2026 upto 15:00 Hrs for Pre-bid of Queries through E-mail it-projects@vmc.gov.in only.
10.	**Date & Time of opening of Price Bid**	Will be intimated to the qualified bidders at a later date.
11.	**VMC's GST Number**	**24AAALV0111B1ZU**
12.	**VMC Contact email ID**	Email ID: it-projects@vmc.gov.in
13.	**RFP Document Available at**	https://vmc.gov.in & https://tender.nprocure.com
14.	**Publication of Pre-Bid Replies/Clarifications**	As decided by VMC

Vadodara Municipal Corporation (VMC) invites proposals for Supply, Installation, Testing, Commissioning, Operation & Management of Implementation of City Surveillance System in Vadodara City. The Authority will enter into agreement with the Successful Bidder selected in accordance with this RFP. The agreements will be in the format specified by the Authority.

Only the bidders matching the pre-qualification Criteria will be selected for further evaluation. The bidders are required to meet the minimum threshold limit for technical evaluation mentioned herein, post which their financial bid will be reviewed. The proposal with the lowest cost will be awarded the contract.

The Bidders are required to meet the minimum threshold technical and financial capability criteria, as stated in the RFP and Notice Inviting Tenders ("NIT") advertisement published in the local & National newspapers and as provided herein. Pursuant to that, the Bidders would be evaluated on the basis of detailed technical and financial proposals and qualify for undertaking the Project as set out in this RFP.

The RFP document contains information about the Project, bidding process, proposal submission, qualification and financial proposal requirement. Proposal in the form of BID is requested for the item(s) in complete accordance with the documents/attachments as per following guidelines.

Instruction to the bidders for bid submission:

- ✓ Tender documents are available only in electronic format which Bidders can download free of cost from the website www.Vadodaracity.gov.in and https://vmc.gov.com
- ✓ Bidder shall upload their financial bid on https://tender.nprocure.com.
- ✓ Bidder shall submit technical proposal (both at https://tender.nprocure.com and in hard copy) with eligibility documents, non-refundable bid processing fees bid fee and bid security EMD in separate sealed envelopes super scribed with title and packaged all together in big envelope with tender number to VMCOffice.
- ✓ The bid shall specify time schedule of various activities.
- ✓ Bids complete in all respects should be submitted on or before the BIDDUEDATE.
- ✓ Services offered should be strictly as per requirements mentioned in this Bid document.
- ✓ Please spell out any unavoidable deviations, Clause/ Article wise in your bid under the heading Deviations.
- ✓ Once quoted, the bidder shall not make any sub sequent price changes, whether resulting or arising out of any technical/commercial clarifications sought regarding the bid, even if any deviation or exclusion may be specifically stated in the bid. Such price changes shall render the bid liable for rejection. However, VMC reserve the right to revised financial offer.
- ✓ The duration of the Contract period for this activity will be of **Implementation Period + 5 years.**
- ✓ Bidders who wish to participate in this bid will have to register on https://tender.nprocure.com. Further bidders who wish to participate in online bids will have to procure Digital Certificate as per Information Technology Act 2000 using which they can sign their electronic bids. Bidders can procure the same from

  (n) code solutions – a division of GNFC Ltd., or any other bidder licensed by Controller of Certifying Authority, Govt. of India. Bidders who already have a valid Digital Certificate need not procure a new Digital Certificate.

  - In case of any clarifications required, please contact on email ID of VMC at it-projects@vmc.gov.in quoting RFP No. in the subject line in the following format only:

- ✓ Subject: Reg.: RFP Ref. No.: 262/26-27- <Topic of Concern> related.

Glossary & Definition

Terms	*Meaning*	*Terms*	*Meaning*
VMC	Vadodara Municipal Corporation	**RFP**	Request for Proposal
AMC	Annual Maintenance Contract	**RO**	Request Order
AP	Access Points	**SAN**	Storage Area Network
ATS	Annual Technical Support	**SI**	Bidder
BOM	Bill of Material	**SLA**	Service Level Agreement
CEO	Chief Executive Officer	**SOP**	Standard Operating Procedures
DC	Data Center	**SRS**	Software Requirement Specifications
DD	Demand Draft	**VMC**	Vadodara Municipal Corporation
DR	Disaster Recovery	**TQ**	Technical Qualification
EMD	Earnest Money Deposit	**UAT**	User Acceptance Testing
FRS	Functional Requirement Specifications	**GPRS**	Global Packet Radio Service
GIS	Geographical Information System	**GSM**	Global System for Mobile Communication
GPS	Global Position System	**BoQ**	Bill of Quantity
ICCC	Integrated Command Control Center	**COTS**	Commercial-off-the-shelf
ICT	Information and Communication Technology	**DC**	Data Centre
INR	Indian Rupee	**FPS**	Frames per second
LoI	Letter of Intent	**FR**	Face Recognition
MTTR	Mean Time to Repair	**FRS**	Functional Requirement Specification
OEM	Original Equipment Manufacture	**HLD**	High Level Design
O&M	Operations and Maintenance	**IA**	Implementation Agency (bidder/vendor/ implementation agency are the selected bidder)
PBG	Performance Bank Guarantee	**KPI**	Key Performance Indicators
POE	Power over Ethernet	**LLD**	Low Level Design
PQ	Pre-Qualification	**SI**	Bidder
CCC	Command and Control Centre	**O&M**	Operations and Maintenance
TRAI	Telecom Regulatory Authority of India	**RI**	Road Re-Instatement
UPS	Uninterruptible Power Supply	**RoW**	Right of Way
VMS	Video Management Software/System	**CCTV**	Closed Circuit Television
GUI	Graphical User Interface	**DIT**	Directorate of Information Technology
ICT	Information and Communication Technology	**EMS**	Enterprise Management System
IPS	Intrusion Prevention System	**GoG**	Government of Gujarat
LAN	Local Area Network	**IP**	Internet Protocol

SECTION: 1 PROJECT PROFILE

1.1 Objective of this Project

VMC intends to set up a system through which the VMC officers can be informed about the city surveillance alerts in real time. The key objective for the project is defined as below:

- Civic issues monitoring
- Alerts on traffic rule violation
- Crime prevention, Crime detection and investigation
- Public safety
- Traffic management
- Crowd control
- Protection of critical infrastructure
- Emergency response coordination, Evidence in legal proceedings
- Remote monitoring, Incident management

SECTION: 2 ELIGIBILITY CRITERIA

The bidder must possess the requisite experience, strength and capabilities in providing services necessary to meet the requirements as described in the RFP document. Keeping in view the complexity and volume of the work involved, following criteria are prescribed as the eligibility criteria for the bidder interested in undertaking the project.

The bidder must also possess technical know-how and financial ability that would be required to successfully do the implementation of City Surveillance System in Vadodara City as per VMC's requirement for the entire contract duration.

The bids must be complete in all respect and should cover entire scope of work as stipulated in the bid document. This invitation to bid is open to all bidders who qualify the eligibility Criteria as given below:

The Pre-Qualification Criteria for the selection of the Bidder are given below. For the purpose of evaluation, the total Project Value shall be considered as the sum of Capital Expenditure (CAPEX) Cost and Operation & Maintenance (O&M) Cost. OEM experience, project references, credentials, financial capability, or technical qualifications shall not be considered as the Bidder's experience for meeting the Pre-Qualification Criteria or Technical Evaluation requirements, unless the Bidder itself is the OEM of the offered products/solutions. Only the experience and credentials of the Bidder submitting the bid shall be considered for evaluation, subject to the provisions of this Tender Document.

2.1 Pre-Qualification (PQ) Criteria

Sr. No:	Eligibility Criteria	Supporting documents (Notary certified true copy / self-attested)
1	The Bidder shall be an Authorized System Integrator (SI) / Authorized Channel Partner of the respective OEM(s) for the proposed CCTV Cameras (Bullet, Dome & PTZ), Video Management System (VMS), Servers, Storage, Backup Software, SAN Switches, Core/Distribution/Access Switches (including PoE Switches), Active LED Video Wall, Video Wall Controller, EMS/NMS, UPS, and all other active IT and surveillance equipment proposed under this Tender. The Bidder shall submit valid Manufacturer's Authorization Form (MAF) , OEM Authorization Certificate with back-to-back OEM warranty and technical support for the entire five (5) year Contract Period after Go-Live for the above products along with the Technical Bid.	The Bidder shall submit a valid Manufacturer's Authorization Form (MAF) & back-to-back OEM warranty and support for the entire five (5) year Contract Period, as per the prescribed format, from the respective OEM(s) for the proposed IP CCTV Cameras (Bullet, Dome & PTZ), Video Management System (VMS), Servers, Storage, Backup Software, SAN Switches, Core/Distribution/Access Switches (including PoE Switches), Active LED Video Wall, Video Wall Controller, EMS/NMS, UPS, The MAF shall be enclosed with the Technical Bid.
2	Bidder should be an Indian firm ▪ Bidder should be registered under the Companies Act 1956 in India or a Limited Liability Partnership Firm under Limited Liability Partnership Firm Act 2008 at the time of the bidding ▪ Bidder should have a registered number of, GST, Income Tax / Pan number ▪ Bidder should be in operation in India for a period of at least 3 years as on publication of this tender.	▪ Copy of certification of incorporation issued by competent authority/ Registration Certificate/ Shop & Establishment certificate ▪ Copy of PAN card ▪ Copy of GST registration
3	▪ The Bidder should have average annual turnover of at least Rs. 17 Cr. last three audited financial years ending 31 March 2026, i.e. (FY 2023-24, FY 2024-25, FY 2025-26). ▪ The bidder must have positive net worth and should be profit making in each of the last three financial years as on 31st March 2026.	▪ Copy of audited Balance Sheet, audited Profit & Loss statements for each of the last 3 financial years as on 31st March 2025. ▪ Certificate from the statutory auditor / Charted Accountant (CA) clearly specifying the annual turnover for each of the last 3 financial years as on 31st March 2026. Original or Notarized Copy should be submitted for evaluation.
4	The proposed all CCTV Cameras OEM should possess valid STQC / BSI-Er complied certification as on the bid submission date. Also IP CCTV System OEMs should own the MAC Address of the proposed components &/or network devices.	Copy of the valid certification should be submitted.
5	Bidder should have successfully executed followings in Central / State Government / Municipal Corporations / Urban Local Bodies / PWD / Public Sector Undertaking/Private Organization companies within last 07 years ending on the tender floating date: ▪ One similar completed work costing not less than 80% of estimated cost i.e. Rs 40 Crore OR ▪ Two similar completed works each costing not less than the 50% of estimated cost i.e. Rs 25	For each project, bidder has to submit following: ▪ Copy of Work Order / Agreement ▪ Work completion certificate issued by client OR Work completion (Certified by CA / CS / independent auditor of the bidding entity) with Project Citation. Note:

	Crore. OR ▪ Three similar completed works each costing not less than the amount 40% of estimated cost i.e. Rs 20 Crore. The bidder should submit a certificate from client on completion status of on-going project and performance of bidder. Similar Work/Project Means "Supply, Installation, Testing, Commissioning and/or Maintenance of OFC enabled CCTV Surveillance Project involving DC Infrastructure, EMS/NMS etc.	▪ In cases where the proposed project includes civil and electrical works, the bidder shall consider only the cost of Cameras, along with Server with Storage, and associated IT/ITES components (excluding civil and basic electrical works) for the purpose of meeting the eligibility criteria. The bidder shall submit a Chartered Accountant (CA) certificate, duly bearing a valid UDIN, confirming that the project cost considered for qualification strictly comprises only the aforementioned equipment and components. ▪ In case of ongoing project, the proposed project must be commissioned as on the online bid submission date.
6	Bidder should have successfully executed followings in Central / State Government / Municipal Corporations / Urban Local Bodies / PWD / Public Sector Undertaking/Private Organization companies within last 07 years ending on the tender floating date: Bidder shall have demonstrable similar work experience of minimum Outdoor **650+ CCTV** cameras with Servers, Storage City wide UG fibre network of minimum **80 Km** in last seven (7) years in India on bid submission date. ▪ Similar Works Means " Supply, Installation, Testing, Commissioning and/or Maintenance of OFC enabled CCTV Surveillance Project involving DC Infrastructure, EMS/NMS etc. The bidder should submit a certificate from client on completion status of on-going project and performance of bidder.	For each project, bidder has to submit following: ▪ Copy of Work Order / Agreement ▪ Work completion certificate issued by client OR ▪ Work completion (Certified by CA / CS / independent auditor of the bidding entity) with Project Citation. Note: ▪ In cases where the proposed project includes civil and electrical works, the bidder shall consider only the cost of Cameras, along with Server with Storage, and associated IT/ITES components (excluding civil and basic electrical works) for the purpose of meeting the eligibility criteria. The bidder shall submit a Chartered Accountant (CA) certificate, duly bearing a valid UDIN, confirming that the project cost considered for qualification strictly comprises only the aforementioned equipment and components. ▪ In case of ongoing project, the proposed project must be commissioned as on the online bid submission date.
7	Proof of Office Location	Provide Professional Tax paid receipt, for local vendor and for non-local vendor any of below proof required Company Registration Proof or Profession Tax Proof or Shop & Establishment Proof or Property Tax or Electricity Bills or Telephone Bill or Valid Lease Agreement should be submitted with the bid for Proof at their office location.
8	The Bidder shall have a minimum of twenty (20) technically qualified employees on its payroll as on the date of bid submission, engaged in activities such as CCTV surveillance system implementation, networking, optical fibre (OFC), system integration, installation, testing, commissioning, operation & maintenance	Certificate from HR head confirming Compliance.

	(O&M), and technical support.	
9	The bidder should: ▪ Not have been blacklisted by Central Government / Any State Government / PSU in India as on the date of bid submission. ▪ Not be insolvent, in receivership, bankrupt or being wound up, not have its affairs administered by a court or a judicial officer, not have its business activities suspended and must not be the subject of legal proceedings for any of the foregoing reasons. ▪ The Bidder shall have an office in Gujarat. In case the Bidder does not have an office in Gujarat at the time of bid submission, an undertaking shall be submitted to establish an office in Gujarat within 45 days from the date of Work Order issued by VMC. ▪ Not have their directors and officers convicted of any criminal offence related to their professional conduct or the making of false statements or misrepresentations as to their qualifications to enter into a procurement contract within a period of three years preceding the commencement of the procurement process, or not have been otherwise disqualified. ▪ Not a subsidiary/affiliate/attached office of any border companies as may be banned by Government of India for doing business in India as per revision of GFR rules. ▪ Self-declaration of not supplying products from such a country or, if from such country, has been registered with the competent authority as per the procedure laid down in reference to government orders	The Bidder shall submit a self-declaration/undertaking, duly signed and stamped by the Authorized Signatory, in the format prescribed in this RFP.
10	The authorized signatories of the bidder should execute a Power of Attorney / Notarized board resolution copy to nominate one person as an authorized signatory regarding all communications and execution of this project. The authorized signatory should sign all communications including the bid.	Original Power of Attorney / Notarized board resolution copy should be submitted in order to support their authorization to sign the document. The power of attorney should be submitted on a stamping of INR 300/- (Rupees Three Hundred Only).
11	Bidder should have to submit affidavit physically on Stamp paper as per format mentioned in TQ_6.	ORIGINAL on Non-Judicial Stamping of Rs. 300/- duly attested by Magistrate / Notary.
12	The Bidder possess following Certifications which are valid at the time of bid submission: Mandatory: • CMMi Level 3 or above Any 2 of below Certifications • ISO/IEC 27001:2013 or above for Information Security Management System • ISO 20000-1:2011 or above for IT Service Management • ISO 14001:2015 or above for Environmental Management System • ISO 45001:2018 or above for Occupational Health and Safety (OH&S) Management Systems •ISO 9001: 2015 for Quality Management Systems	Copies of the valid certificate in the name of the Bidder

Note:

1) The Bidder must attach valid documents in support to their Pre-Qualification as mentioned above. Without proper supporting documents, the bid proposals are liable to be rejected. The technical proposal should be submitted in hard copy with soft version stored in pen drive.

2) For all cited projects under bidder's experience criteria; the bidders have to submit work order with full BoQ, contract agreement go-live/amount of work completion certificate and copy of invoice submitted to the client as a supporting document for each project. Sub-contracted projects by bidder will not be considered for evaluation.

3) The Successful Bidder shall use only the makes/brands specified in the Approved Makes List of this Tender. Before procurement and installation, the Bidder shall submit the final make, model, OEM datasheets, test certificates (where applicable), and other relevant technical documents for approval by the competent authority of VMC.

4) Wherever the make/brand is not specified for any item, equipment, material, or accessory, the Successful Bidder shall submit the details, technical specifications, OEM credentials, certifications, and product datasheets of the proposed make/brand for approval by VMC. The decision of VMC regarding acceptance or rejection of the proposed make/brand shall be final and binding. No additional cost or time extension shall be admissible on this account.

5) In addition to the makes specified in this Tender, VMC may permit the use of equivalent makes/brands approved by Vadodara Municipal Corporation, provided they fully comply with the technical specifications and quality requirements of this Tender.

6) VMC (or the nominated party) reserves the right to check/validate the authenticity of the information provided in the Pre-qualification and Technical evaluation criteria.

7) The bidder needs to provide contact detail (email & phone number) of senior official from client.

8) The bidder should submit authorization certificate of Original Equipment Manufacturer (OEM) (or multiple OEMs) specific to the bid. The bidder should have a support agreement/arrangement for services including supply of spare parts etc. which includes the post -sales support activities for the entire project period.

9) A Bidder shall submit only one bid against this Tender. Submission of multiple bids by the same Bidder, either directly or indirectly, shall lead to disqualification of all such bids. OEMs/Product Manufacturers may provide Manufacturer Authorization Forms (MAFs) to multiple bidders. However, the experience and credentials of the OEM shall not be considered as the Bidder's experience unless the Bidder itself is the OEM.

10) All above mentioned documents for pre-qualification criteria of bid must be notary certified true copy/ self-attested.

11) Work Order and Work Completion Certificate must be in English language only. In case, Work Order or Work Completion Certificate is in any other language, bidder has to submit notarized Work Order in English language only.

12) For Pre-qualification criteria of work experience, bidder's any work experience as lead member / consortium member / joint venture / sub-contractor will not be considered as supporting documents.

13) Above mentioned eligibility criteria should be necessarily met and adequate documentary proof be submitted for verification. If any / partial / all of the above mentioned eligibility criteria don't fulfill by bidder, in that case, such bid is disqualified for technical bid opening.

14) Physical bid must be submitted by Postal registered AD or Postal speed post only. Any other mode is not allowed.

15) In case bid / document size is large then bidder can divide it into parts. Each part of bid should have proper indexing and page number mentioned and all pages of RFP must be signed and sealed.

16) Tender will be awarded to bidder having lowest Price offer (Least Cost Based Selection (LCBS or L1). If more than one bidders quoted the lowest pricing (L1).

17) All CCTV cameras supplied under this tender Bullet Camera, Dome Camera, and PTZ Camera shall be of the same OEM (Original Equipment Manufacturer) and same product make/series across the entire project. Mixing of cameras from different manufacturers/brands for different camera types (e.g., Bullet from one OEM and Dome/PTZ from another) shall not be permitted under any circumstances.

This requirement is mandatory to ensure:

- Uniform build quality, firmware ecosystem, and cybersecurity posture across all deployed cameras;
- Single-point OEM warranty, support, and spares availability for the entire camera fleet, avoiding multi-vendor coordination issues during the O&M period.
- Seamless interoperability, firmware/security patch management, and long-term supportability through a single OEM relationship.
- Simplified troubleshooting, AMC/CMC accountability, and reduced integration risk with the central VMS.

18) The bidder shall clearly declare, in the technical bid, the single OEM/brand proposed for all three camera types (Bullet, Dome, and PTZ), along with the respective model numbers and datasheets. Bids proposing cameras of different makes/brands for different camera categories shall be treated as non-compliant and liable for rejection at the technical evaluation stage.

19) Proposed OEM/make shall hold valid STQC/BIS-ER certification for all three camera models (Bullet, Dome, and PTZ) offered under this tender.

SECTION: 3 EVALUATION OF BID

3.1 Evaluation Process

3.1.1 Pre-Qualification Evaluation

i. VMC shall validate the - "RFP Document fee & Bid Security/Earnest Money Deposit (EMD)".
ii. VMC shall open the bid and check the bidder's eligibility as per the "Pre-Qualification Criteria". Each of the Pre-Qualification condition mentioned in above Section is MANDATORY. In case, the Bidder does not meet any one of the conditions, the bidder shall be disqualified.
iii. Technical and Financial bids for those bidders who don't pre-qualify will not be opened.

3.1.2 On-field testing

i. VMC will review the technical bids of the short-listed bidders to determine whether the technical bids are substantially responsive. Bids that are not substantially responsive are liable to be disqualified at Authority's discretion.
ii. Bidders will be asked to give demonstration/Pilot of the proposed product & solution to VMC as per the use cases that shall be shared with the Bidders who qualify the Pre-Qualification Stage if require.
iii. Cost of on-field testing will be reimbursed to the unsuccessful bidders at the rate quoted by the L1 bidder discovered in this tender.
iv. If the quality is not up to the mark, VMC will disqualify the bidder.

3.1.3 Stage 3: Financial Evaluation

i. Financial bid will be evaluated only for the bidders who succeed in PQ
ii. All the technically qualified bidders will be notified to participate in Financial Bid opening process.
iii. The Financial bids for the technically qualified bidders shall be opened on the notified date and time. They will be reviewed to determine whether the financial bids are substantially responsive or not. Bids that are not substantially responsive are liable to be disqualified at Authority's discretion.
iv. The bidder quoting the lowest price to VMC shall be the Successful Bidder (L1) for the Project.
v. Bidders shall be ranked L1, L2, and L3 etc. in increasing order quoted by them in their financial bid.
vi. Any bidder who is quoting unreasonable price to become lowest bidder (abnormally low bids) shall be disqualified and EMD will be forfeited as per the procurement guideline of GoI, GoG and decision of commissioner, VMC.
vii. VMC has planned to install CCTV cameras at approx. 150 locations in Vadodara city.
The bidders may also sub-let the portion of installation & maintenance of field devices to another agency.
viii. The VMC would have the right to review the Proposals and seek clarifications where necessary. The response from the Bidder(s) shall only be in writing but no change in the substance of the Proposal would be permitted.
ix. Financial Proposals of Bidders who do not qualify the eligibility criteria will not be opened and will be returned unopened.
x. The Proposal should be unconditional and any conditionality attached with the proposal may result in the rejection of the Proposal.
xi. The Financial bids for the technically qualified bidders shall be opened on the notified date and time. They will be reviewed to determine whether the financial bids are substantially responsive or not. Bids that are not substantially responsive are liable to be disqualified at Authority's discretion.
xii. Financial Bids that are not as per the format provided in the RFP shall be liable for rejection.
xiii. If L1 bidder fails to submit the Performance Bank Guarantee (PBG) and align the agreement as per the timelines and requirement defined in this RFP, L2 bidder will be invited for negotiations. In this case, L2 bidder will be asked to match the prices of L1 bidder. Final decision of awarding the project to L2 bidder will depend upon the mutually agreed price between both the parties (bidder and Authority).

3.1.4 Rights to Accept/Reject any or all Proposals

VMC reserves the right to accept or reject any proposal, and to annul the bidding process and reject all Bids at any time prior to award of Contract, without thereby incurring any liability to the affected Bidder or Bidders or any obligation to inform the affected bidder or bidders of the grounds for VMC's action.

3.1.5 Consortium / Joint Venture Restriction

Consortium, Joint Venture (JV), association of firms, partnership bidding, or any other arrangement involving more than one entity for submission of the Bid shall not be permitted under this Tender.

The Bidder shall be a single legal entity and shall independently meet all eligibility, technical, financial, and contractual requirements specified in the Tender Document.

Any Bid submitted by a Consortium, Joint Venture, association of firms, or any similar arrangement shall be summarily rejected and shall not be considered for further evaluation.

3.1.6 Notifications of Award and Signing of Contract

i. Prior to the expiration of the period of proposal validity, the bidders will be notified in writing through FAX or email or letter that its proposal has been accepted.

ii. VMC shall facilitate signing of the contract within the period of 30 days of the notification of award. However, it is to be noted that the date of commencement of the project and all contractual obligations shall commence from the date of issuance of Purchase Order/Letter of Acceptance (LOA)/Letter of Intent (LOI), whichever is earlier. All reference timelines as regards the execution of the project and the payments to the Implementation Bidder shall be considered as beginning from the date of issuance of the Purchase Order/Letter of Acceptance, whichever is earlier.

iii. The notification of award Purchase Order(PO)/Work Order(WO) will constitute the formation of the Contract. Upon the Bidder's executing the contract with VMC, it will promptly notify each unsuccessful bidder and return their EMDs.

iv. At the time VMC notifies the successful Bidder that its bid has been accepted, VMC will send the Bidders the Pro forma for Contract, incorporating all clauses/agreements between the parties. Within 15 days of receipt of the Contract, the successful Bidder shall sign and date the Contract and return it to VMC.

Note

i. Any conditional bid would be rejected.

ii. Errors & Rectification: Arithmetical errors will be rectified on the following basis:
    a. "If there is a discrepancy between the unit price and the total price that is obtained by multiplying the unit price and quantity, the unit price shall prevail, and the total price shall be corrected.
    b. If there is a discrepancy between words and figures, the amount in words will prevail."

iii. If the Bidder does not accept the error correction, its Bid will be rejected and EMD may be forfeited.

iv. Bidder must attach valid documents in support to their Pre-Qualification and Financial, Technical capabilities/strength, as mentioned above. Without proper supporting documents, the Bid proposals are liable to be rejected.

v. The bidder should submit authorization certificate of Original Equipment Manufacturer (OEM) (or multiple OEMs) specific to the bid. The bidder should have a support agreement/arrangement for services including supply of spare parts etc. which includes the post -sales support activities for the entire project period.

vi. All the proposed equipment should not be declared **End-of-Support by the OEMs for next 6 years** from the date of bid submission.

vii. Kindly note that the indicative/estimated quantity provided in the RFP would be used for evaluation Purposes. However, the payment would be done on actual usage basis.

SECTION: 4
SCOPE OF WORK and
TECHNICAL SPECIFICATIONS

4.1 Detailed Scope of Work

1. The Bidder shall be responsible to carry out the detailed field survey for each location in order to finalize infrastructure requirement, network bandwidth requirement, operational & administrative challenges etc. and shall submit detailed Location Survey Report and Project Implementation Plan for each location to VMC or consolidate location.
2. Detailed feasibility study and actual site survey for Poles, CCTV Cameras (including Type and quantity), source of power supply, last mile cabling route, kind of coupling needed to mount cameras and other end devices etc.
3. The bidder is required to install 4 or 5 no. of cameras with required hardware and accessories at each location (i.e. Vadodara City Requier locations).
4. The scope will be considered completed only when the bidder supplies the structural safety Civil / Electrical certification of installation scope with as built drawing.
5. The Bidder shall carry out the detailed assessment of the functional requirements for VADODARA CITY.
6. SURVEILLANCE SYSTEM and prepare the System Requirements Specifications (SRS) for the systems, in consultation with VMC and Project management consultant.
7. Procurement, Supply, Installation and Commissioning of VADOARA CITY SURVEILLANCE SYSTEM components with perpetual license.
8. The VADOARA CITY SURVEILLANCE SYSTEM shall be capable of generation of on-demand pictures and videos from selected or/and all sites where the solution has been deployed.
9. The bidder should ensure storage/archiving of min 30 days of video feed recording at 25/30 FPS for Majority of the time. However certain feed may be required to be retained for court/investigation related matters beyond the standard period.
10. Videos and images marked by VMC, Traffic Police should be stored for entire project duration.
11. Procurement, Supply, Installation and Commissioning of components with perpetual license and for unlimited no. of users for viewing and reporting purpose.
12. All Structure wiring shall be done by the Bidder at its own cost.
13. Connectivity is the responsibility of the bidder - The sizing of the connectivity requirement should be done in a manner to ensure live, real time feed from cameras without blurring, bleeding or freezing of video feeds. The elements in video feed should be clearly visible especially the number plate of vehicles. Bidder needs to define the connectivity in a manner that the SLA commitments are met as defined in the RFP.
14. Obtaining the required permissions from the authorities/local bodies. However, VMC will provide adequate assistance/support as and when required.
15. Bidder will be responsible to obtain electric connection for the running of cameras.
16. Comprehensive Last mile connectivity (including supply, installation & commissioning of required material/hardware) between utilities and nearby Junction box (connectivity medium like CAT- 6/fiber as per the feasibility study report) including supply of active components and passive components within the junction box.
17. Bidder shall make all necessary arrangement during implementation phase for all required utilities (electricity, water waste disposal), ladder, man lifter etc. to complete the installation work and O&M.
18. Electricity and Utility Connection - The vendor shall be responsible for ensuring electricity connection and other utilities to make the device/hardware operational. Vendor responsibility involves making request for electricity/utility providing company, submission of requisite fee, test report, requisite certification, coordination, laying power cables and provisioning of requisite power points for all components in the scope of this project. Vendor shall be responsible for paying deposit and metering cost required for taking any electrical connections to concerned departments.

19. SI shall be responsible for laying last mile power cables and provisioning of requisite power points for all components in the scope of this project. SI shall be responsible for paying deposit and metering cost required for taking any electrical connections to concerned departments, this deposit shall be reimbursed by VMC. Since this component has dependency on approval from other agencies, SI should plan this requirement well in advance & submit the application to the electricity provider through VMC. Recurring charges (during the operational phase) for all the electricity consumption of field devices/components in the scope of this project will be paid by VMC to directly electricity provider.
20. VMC will appoint a nodal person to provide necessary guidance and support in this process. Authority will reimburse the cost on production of actual bill.
21. Since this component has dependency on approval from other agencies, vendor should plan this requirement well in advance & submits the application to MGVCL through VMC. Recurring charges (during operational phase) for all the electricity/utility consumption of components in the scope of this project will be borne by VMC but vendor will be responsible for coordination, informing authority to ensure timely payment of bills. Vendor will be responsible for maintenance of these electrical connections during the operational phase, and authority will reimburse material and Labour cost for breakdowns on submission of original bills.
22. Civil Work (Supply, Erection and commissioning) of Non-IT infrastructure like poles/frames/fixtures/housing etc. with proper electrical Earthing (as per IS-3043) wherever required based on the feasibility study report.
23. Bidder has to ensure that on completion of work, site has to be restored to its original condition as per rules / provisions of respective local bodies/authorities.
24. Integration of entire solutions with the existing Network Monitoring System (NMS), ICCC, VMS and various Corporation components/applications/solutions for achievement of overall objective of Vadodara Corporation project. An indicative Bill of Material (BoM) is defined in Financial Bid Format However, bidder has to provision all the required hardware/software/services which may have been inadvertently missed out to successfully run the entire system.
25. The Bidder will have to suitable size the PoE Switches based on the power rating of the camera (all Types) being proposed under the project.
26. OEMs connected with captioned Project found indulging in attempting to influence the processes by directly contacting the Authority officials, ostensibly in order to promote/lobby their own products for selection of vendors of any kind will be considered against the spirit of Ethical Practices and Good Industry Practices & will be considered for debarring your organizations from VMC as well as in other smart cities.
27. Integration with other monitoring systems - The Bidder shall provide an open API for integration of the proposed VADODAR CITY SURVEILLANCE SYSTEM with existing or any other system like GIS & Application should be integrated with GIS maps. Also Integration with ICCC and with the Police department to show feed of CCTV at Police commissioner office, connectivity and VMS integration statement shall be vendor responsibility.
28. Electricity Cost in O&M – Electricity cost during warranty phase will be reimbursed to bidders on actual basis. The bidder will have to make payment and submit the actual electricity bill for reimbursement.
29. Not Mandatory - VMC may provide rack space at its Control Centre for hosting infrastructure for backup of footages from VADODARA CITY SURVEILLANCE SYSTEM.
30. Capacity Building and Training - The Bidder shall conduct proper need-based training for all the concerned staff of VMC and other stakeholders and draw up a systematic training plan. Bidder shall provide hard copy & soft copy of the training materials to all the trainees. The training shall be held at various office/department locations as finalized by VMC.
31. The Successful bidder is also required to offer Enterprise and Network Management System for all the networking equipment.
32. Digging, back filling and ensuring to make the site to its original condition will be the responsibility

of the Successful bidder. The Successful bidder shall also abide by the rules / provisions of respective local authorities in this regard.

33. Successful bidder shall provide Technical Team for training, handholding support, operating the system and maintain the same for a period of five years at Vadodara
34. The Successful Bidder shall implement a structured cabling system for all Data Centre, CCC, Network Rooms, Rack Locations, Field Cabinets, Junction Boxes, and Surveillance Sites. All cables shall be properly routed, dressed, bundled, tagged, ferruled, and terminated in a neat and professional manner using industry best practices.
35. The Bidder shall ensure proper cable management through cable trays, raceways, ducts, cable managers, patch panels, and rack organizers. Loose, hanging, exposed, improperly terminated, or unlabeled cables shall not be acceptable. All power, network, fiber, and control cables shall be clearly identified at both ends with permanent labels indicating source and destination details.
36. All racks, cabinets, patch panels, switches, servers, and field equipment shall be properly organized and documented to facilitate maintenance, troubleshooting, and future expansion. The Bidder shall submit cable schedules, labelling details, rack layouts, and as-built documentation for approval by VMC. Any untidy, unsafe, or non-standard cabling installation observed during inspection shall be rectified by the Bidder at no additional cost to VMC.

4.2 Supply & Delivery

The bidder shall provide delivery Challan and all supporting document for delivery of required, hardware & software items. It is responsibility of bidder for supplying perpetual licenses for the proposed solution. All licenses supplied by the bidder for the purpose of this project shall be perpetual in nature and shall be in the name of VMC and will include maintenance and insurance for the entire duration of the contract. BIDDER will maintain an inventory of all software components procured, license renewals etc..

- The Successful Bidder shall use only the makes/brands specified in the Approved Makes List of this Tender. Before procurement and installation, the Bidder shall submit the final make, model, OEM datasheets, test certificates (where applicable), and other relevant technical documents for approval by the competent authority of VMC.
- Wherever the make/brand is not specified for any item, equipment, material, or accessory, the Successful Bidder shall submit the details, technical specifications, OEM credentials, certifications, and product datasheets of the proposed make/brand for approval by VMC. The decision of VMC regarding acceptance or rejection of the proposed make/brand shall be final and binding. No additional cost or time extension shall be admissible on this account.
- In addition to the makes specified in this Tender, VMC may permit the use of equivalent makes/brands approved by the Vendor Committee of Vadodara Municipal Corporation, provided they fully comply with the technical specifications and quality requirements of this Tender.
- The manufacturer for hardware shall have manufacturing and testing facility be accredited by national/state government recognized agency. The bidder will have to submit the accreditation, safety and quality certificate with delivery of hardware.
- The manufacturer calibration lab accreditation certificate should be submitted with technical proposal for review and approval prior to inspection along with QAP.
- The Purchaser (VMC) or its representative shall have the right to inspect and / or to test the Goods to confirm their conformity to the Contract. The special conditions of contract and / or the Technical Specification shall specify what inspections and tests the Purchaser requires and where they are to be conducted. The Purchaser shall notify the Supplier in writing of the identifying of any representatives retained for these purposes.
- The inspections and tests may be conducted on the premises of the Supplier or its sub IA(s), at point of delivery and / or at the Good's final destination. Should any inspected or tested Goods fail to conform to the Specifications, the Purchaser may reject them, and the Supplier shall either replace the rejected

Goods or make all alterations necessary to meet specification requirements free of cost to the Purchaser. The Purchaser's right to inspect, test and, where necessary, reject the Goods after the Good's arrival in India shall in no way be limited or waived by reason of the Goods having previously been inspected, tested and passed by the Purchaser or its representative prior to the Goods' shipment from the country of origin.

- Total delivered quantity from each line item of price bid shall be reviewed and inspected at warehouse of BIDDER at project location by Third Party Inspectors (TPI)/VMC team.
- Further, the Successful bidder must not bid/supply any equipment that is likely to be declared end of life/sale within seven years from the date of FAT. The Successful bidder would be required to replace all such equipment with latest and at least of equivalent configuration during project duration.
- For every unit supplied by the supplier, the conformance to the Specifications mentioned in the Tender shall be established by the supplier.
- All the expenses related to inspection at manufacturing and testing facility of supplier viz. lodging & boarding, transportation, all facilities including third party inspection (TPI) fees, testing charges and other related commercial expanses as and where required shall be borne by the IA.
- The supplier shall facilitate in-process and / or Pre-delivery inspection by the Representatives of the Purchaser, as and when, the same is required by the Purchaser.
- The Successful Bidder shall assess the existing server, storage, and IT infrastructure and carry out all necessary upgrades, configurations, and integration of existing applications (VMS/IVMS, Database, NMS, Antivirus Server, etc.) required to support the proposed CCTV cameras and licenses without any additional cost to VMC.
- Also all the instruments along with local panel indicators will be tested and verified against the approved datasheet, Catalogue, duly certified by manufactures for technical specification.
- Integration of instruments will be tested for data transmission and will be tested with data monitoring, trends, alarms and event generation.
- Materials shall be delivered after receipt of Release note from third party inspection authority and shall be delivered to site/ Store without any damage during transportation and handling.
- Material shall be dispatched for Vadodara in the store or at site within the Municipal limit as per instruction of Engineer In charge.
- Quantity mentioned in price bid is for the existing works, Bidders bound to supply additional quantity required by the VMC during the validity for 2 years of the contract at the same unit rate and terms & conditions.
- Work shall be carried out as per approved drawing/s and or instruction/s by engineer-in-charge.
- Bidder needs to provide installation plan in advance at the beginning of the week to Engineer-in-charge.
- Bidder needs to give weekly progress report about the work carried out of the by-gone week.
- Bidder shall be responsible in case of breaking wall, ground. Repairing of damage wall/ground will be in scope of bidder. No extra cost will be paid by VMC.
- Luminaries will be in conducting mode based on the astronomical guidelines & the individual poles housing LED luminaires will get electric supply as per the necessary and determined schedule.
- Controller repairing and replacement will be in the bidder's scope for the entire project duration including operation and maintenance.
- All the equipment deployed for the successful running of the project will be under warranty for the entire project duration including operation and maintenance.
- The Successful Bidder shall integrate the proposed solution with the existing systems deployed at VMC/ICCC, including but not limited to Videonetics VMS, Motadata, Trend Micro, existing Active Directory/LDAP (if available), network infrastructure, storage, servers, backup solution, EMS/NMS, and any other applications, software, or platforms identified by VMC during implementation.

- The Bidder shall ensure seamless interoperability, data exchange, alarm/event sharing, health monitoring, user authentication, and centralized management without affecting the existing operational environment. All hardware, software, APIs, connectors, licenses (if required), professional services, customization, configuration, testing, and commissioning required for successful integration shall be included in the Bid price. No additional payment shall be made by VMC for such integration activities.

4.3 Resource Deployment and Support Requirements

a) Bidder shall deploy adequate technical, operational, and management resources required for successful implementation, operation, maintenance, monitoring, and support of the CCTV Surveillance System throughout the Contract Period.

b) Bidder shall nominate a single Point of Contact who shall coordinate with VMC for all project-related activities during implementation and O&M phases.

c) Bidder shall provide 24x7x365 monitoring, operations, maintenance, incident management, and technical support services from the City Command and Control Centre and field locations as required to meet the prescribed SLA requirements.

d) Bidder shall ensure availability of qualified technical resources for fault diagnosis, preventive maintenance, corrective maintenance, system upgrades, reporting, and restoration of services within the timelines specified in the SLA.

e) Bidder shall ensure adequate deployment of resources, tools, equipment, vehicles, spares, and support infrastructure necessary for uninterrupted operation of the system. No additional payment shall be made by VMC for deployment of such resources.

f) Bidder shall ensure proper knowledge transfer and continuity of services in case of any change in deployed resources, without affecting project performance or SLA compliance.

g) VMC may review the adequacy of deployed resources at any time during the Contract Period and may direct the Bidder to augment resources if service levels are not being achieved.

h) Bidder shall remain solely responsible for achieving all performance parameters, uptime requirements, service levels, response times, restoration timelines, monitoring obligations, and other contractual requirements irrespective of the number of resources deployed.

i) Any failure to meet the prescribed SLA parameters due to inadequate resource deployment, lack of technical support, delayed response, or operational deficiencies shall attract penalties as specified in the SLA and Penalty Schedule of this Tender.

4.4 Installation, Integration, Testing & Commissioning

1) The Successful Bidder shall study the Scope of Work, site conditions, and technical requirements in detail and provide a complete end-to-end City Surveillance System as specified in this RFP.
2) The Bidder shall clearly indicate the proposed solution, architecture, OEM make and model, and compliance with the technical specifications in the Technical Bid.
3) The Successful Bidder shall be responsible for the supply, installation, configuration, integration, testing, commissioning, and successful Go-Live of all hardware, software, licenses, network equipment, OFC infrastructure, passive components, field devices, and accessories required under the scope of this Tender.
4) The Bidder shall supply all necessary software, perpetual/subscription licenses (where applicable), tools, accessories, mounting hardware, cables, connectors, consumables, and documentation required for successful implementation. No additional payment shall be made for any item required to complete the system.
5) The Bidder shall provide complete documentation including system architecture, IP addressing scheme, rack layout, cable schedules, OFC route diagrams, labeling details, asset inventory, configuration details, test reports, OEM warranty certificates, user manuals, operation manuals, and as-built drawings. All documents shall be submitted in both hard and soft copy and shall remain the property of VMC.

6) The Successful Bidder shall complete installation, integration, testing, commissioning, and Go-Live of the complete system within the timeline specified in the RFP. Any additional hardware, software, networking, or integration components required for successful implementation shall be deemed included in the Bid.
7) The Successful Bidder shall be responsible for end-to-end installation, configuration, testing, integration, and commissioning of all CCTV cameras, poles, OFC network, switches, servers, storage, video wall, workstations, UPS, VMS, EMS/NMS, backup solution, and all associated components.
8) The Successful Bidder shall obtain all necessary permissions, approvals, road-cut permissions, utility clearances, traffic permissions, and statutory approvals from the concerned authorities wherever required for execution of the work.
9) VMC may permit installation, testing, or commissioning of individual locations or groups of locations based on site readiness. However, the final Go-Live and acceptance shall be considered only after successful commissioning of the complete project.
10) If any delay is anticipated due to reasons beyond the Bidder's control, the Bidder shall notify VMC in writing with supporting documentary evidence within seven (7) days of the occurrence of such event. Delay attributable to the Bidder shall attract penalties as per the Contract.
11) The Successful Bidder shall provide comprehensive warranty, CAMC, preventive maintenance, corrective maintenance, software updates, security patches, technical support, and 24×7×365 operational support for the entire Contract Period.
12) The unit rates quoted by the Bidder for additional quantities, if required by VMC during the Contract Period, shall remain valid as specified in the RFP, subject to the terms and conditions of the Contract.
13) The performance of the system and services shall be monitored against the prescribed SLA. Failure to achieve the required performance levels shall attract penalties and other contractual actions, including termination, as specified in this Tender.
14) The Successful Bidder shall ensure that no damage is caused to roads, footpaths, underground utilities, pipelines, Government buildings, public or private properties, or any existing infrastructure during execution of the work. Any damage, including dents, scratches, or paint damage, shall be repaired, restored, and repainted by the Bidder at its own cost to the satisfaction of VMC.
15) The Successful Bidder shall provide and install standardized, weatherproof, UV-resistant, and vandal-resistant CCTV surveillance signage on all designated camera poles/locations as approved by VMC. The Bidder shall maintain, clean, repair, and replace the signage throughout the Contract Period without any additional cost to VMC..

4.5 Reporting, Database Administration and Trouble Shooting

i. The Successful Bidder shall submit Daily, Weekly, Monthly, Quarterly, and Incident Reports in the formats prescribed by VMC. The reports shall include system health, camera availability, network status, storage utilization, alarms, faults, preventive maintenance, corrective maintenance, SLA compliance, uptime, downtime, bandwidth utilization, and other MIS reports as required by VMC.
ii. The Successful Bidder shall maintain complete records of all hardware, software, firmware, licenses, warranties, AMC/CAMC details, OEM support agreements, asset inventory, configuration changes, upgrades, replacements, and maintenance history throughout the Contract Period.
iii. The Successful Bidder shall be responsible for end-to-end administration of all databases associated with the Video Management System (VMS), EMS/NMS, servers, storage, and other applications. This shall include database backup, restoration testing, optimization, performance tuning, health monitoring, user management, storage management, security hardening, and configuration management.
iv. The Successful Bidder shall perform regular backups of all system configurations, databases, application data, recordings, logs, and other critical information as per the approved backup

policy. Periodic restoration testing shall be carried out to ensure data integrity and recoverability.

v. The Successful Bidder shall provide comprehensive troubleshooting and resolution of hardware, software, network, storage, database, CCTV cameras, field devices, OFC network, VMS, EMS/NMS, and other system components to ensure uninterrupted operation and compliance with the prescribed SLA.
vi. The Successful Bidder shall provide VMC and/or its authorized representatives unrestricted access to system reports, logs, configurations, documentation, asset records, and other information required for monitoring, audit, review, inspection, or knowledge transfer during the Contract Period.
vii. The Successful Bidder shall promptly provide copies of all technical documentation, configuration files, database backups, asset registers, test reports, as-built documents, and any other project-related information requested by VMC during the Contract Period.
viii. The Successful Bidder shall ensure proper knowledge transfer, documentation updates, and technical assistance to VMC personnel throughout the Contract Period and during project handover, without any additional cost to VMC.

4.51 Testing, Acceptance & Go-Live of the City Surveillance System

VMC shall appoint its officials and/or a third-party agency to witness, verify, inspect, test, and certify the supplied equipment, software, network infrastructure, and overall City Surveillance System before Go-Live. The Successful Bidder shall provide all necessary assistance, test equipment, documentation, and technical support during the inspection, testing, and acceptance process. Any deficiencies observed during testing shall be rectified by the Successful Bidder at no additional cost to VMC.

The following tests shall be successfully completed before Go-Live:

a. Functional Testing: Verification of all CCTV cameras, PTZ operations, VMS, video recording, playback, analytics, alarms, event management, user management, EMS/NMS, video wall, workstations, servers, storage, backup system, and all integrated components.

b. Hardware & Infrastructure Testing: Verification of cameras, poles, OFC network, switches, UPS, racks, junction boxes, earthing, structured cabling, network connectivity, storage, servers, and all field devices.

c. Network & Integration Testing: Verification of end-to-end connectivity, bandwidth utilization, latency, redundancy, interoperability, ONVIF compliance, third-party integrations, and communication between field devices and ICCC.

d. Performance Testing: Verification of recording performance, live view, playback, failover, storage performance, network performance, camera availability, and overall system performance under peak load conditions.

e. Security Testing: Verification of cybersecurity controls, user authentication, role-based access control, password policies, firmware integrity, secure communication, audit logs, encryption, and compliance with the specified security requirements.

f. Vulnerability Assessment & Penetration Testing (VAPT): The Bidder shall carry out VAPT through a CERT-In empaneled agency (if required by VMC) and rectify all identified vulnerabilities before Go-Live.

g. OFC Testing: OTDR, insertion loss, end-to-end connectivity, splice loss, attenuation, and fiber performance testing for 100% of installed OFC links.

h. Field Acceptance Testing (FAT/SAT): Verification of camera positioning, field of view, image quality (day/night), PTZ operation, analytics accuracy, IR performance, recording quality, and incident identification capability at each surveillance location.

The Successful Bidder shall submit the following deliverables:

- Compliance Report along with rectification of all observations.
- Final Test and Acceptance Report.
- VAPT Report (where applicable).
- OFC Test Reports (OTDR & OLTS).
- Camera Acceptance Report.
- As-Built Drawings, Asset Register, Configuration Documents, and System Documentation.
- Go-Live Readiness Report.
- Final Acceptance Certificate issued by VMC.

Prior to Go-Live, the Successful Bidder shall:

Prepare and obtain approval of the Go-Live Plan from VMC.

Complete installation, integration, testing, commissioning, documentation, and user training.

Ensure that all hardware, software, licenses, databases, configurations, and network services are fully operational.

Complete migration of existing CCTV cameras and recordings, wherever applicable.

Rectify all observations raised during testing and obtain VMC approval.

The entire City Surveillance System shall operate successfully for a minimum stabilization period of four (4) weeks after Go-Live. During this period, the Successful Bidder shall provide continuous onsite and remote support, promptly rectify all defects, and ensure compliance with the prescribed SLA. Final Operational Acceptance shall be issued by VMC only after successful completion of the stabilization period and satisfactory system performance.

4.6 Operation & Maintenance

The selected vendor shall be required to provide operations and maintenance services for solution including, but not limited to, production monitoring, troubleshooting and addressing the functionality, availability and performance issues, etc.

The Bidder shall be responsible for Operations & Maintenance of CCTV Projects, IT Infrastructure for the period of 5 years' post Go-live and operational acceptance of the systems, which shall include but not limited to the following activities:

i. System Administration/Management support, offer technical support for VADODARA CITY SURVEILLANCE SYSTEM, Upkeep, configuration and operation of VADODARA CITY SURVEILLANCE SYSTEM applications, Install or upgrade software's/libraries, maintain security policies, troubleshoot, train/supervise staff etc.

ii. Generation of status and analytical reports based on incidents recorded in the system, if VMC purchased analytics

iii. Compressive Annual Maintenance Contract (CAMC) - All the VADODARA CITY SURVEILLANCE SYSTEM hardware & Software components supplied by the Bidder shall carry comprehensive on-site warranty.

iv. The Successful Bidder shall provide O&M support on a 24×7×365 basis, including during VMC working hours, non-working hours, weekends, and public holidays, to ensure uninterrupted operation of the entire City Surveillance System and compliance with the prescribed SLA.

v. The Bidder shall also be responsible for the following activities:
   - Adherence to latest Industry Standards, guidelines as specified by government authorities.
   - Standard Operating Procedures (Manuals, Forms, Process documents) to standardize the processes during O&M phase.
   - The Bidder shall ensure that the end of support for any of the component/equipment is not reached during the concurrency of the contract.
   - The Bidder shall ensure that all the peripherals, accessories, sub-components required for the

functionality and completeness of the solution, including but not limited to devices, equipment, accessories, cables, software, licenses, tools, etc. are provided according to the requirements of the solution.

- VMC shall not be responsible if the Bidder has not provisioned some components, sub-components, assemblies, sub-assemblies as part of Bill of Materials in the RFP. The Bidder shall have to provision these & other similar things to meet the solution requirements at no additional cost and time implications to purchaser.
- The Bidder shall ensure there is a comprehensive onsite support arrangement for duration of the contract.
- Considering the criticality of the infrastructure, Bidder is expected to design the solution considering the RFP requirement of no single point of failure with high level of redundancy and resilience to meet the network uptime requirements.
- Bidder shall be responsible for periodic updates & upgrades of all equipment, cabling and connectivity provided at all locations during the contract period.
- Bidder shall be responsible for setting up / building / renovating the necessary physical infrastructure including provisioning for network, power, rack, etc. at all the locations.
- Successful Bidder shall implement a structured cabling system for all Data Centre, CCC, Network Rooms, Rack Locations, Field Cabinets, Junction Boxes, and Surveillance Sites. All cables shall be properly routed, dressed, bundled, tagged, ferruled, and terminated in a neat and professional manner using industry best practices.
- Bidder shall ensure proper cable management through cable trays, raceways, ducts, cable managers, patch panels, and rack organizers. Loose, hanging, exposed, improperly terminated, or unlabeled cables shall not be acceptable. All power, network, fiber, and control cables shall be clearly identified at both ends with permanent labels indicating source and destination details.
- All racks, cabinets, patch panels, switches, servers, and field equipment shall be properly organized and documented to facilitate maintenance, troubleshooting, and future expansion. The Bidder shall submit cable schedules, labelling details, rack layouts, and as-built documentation for approval by VMC. Any untidy, unsafe, or non-standard cabling installation observed during inspection shall be rectified by the Bidder at no additional cost to VMC.

i. The system shall provide configurable MIS reports, dashboards, and logs on an hourly, daily, weekly, monthly, quarterly, and yearly basis, or as required by VMC. All software, licenses, and utilities required for report generation shall be included in the Bid.

ii. Any item, component, software, license, hardware, accessory, tool, service, or activity not specifically mentioned in the BOQ but required for successful completion of the Scope of Work or compliance with the Technical Specifications shall be deemed to be included in the Bid without any additional cost to VMC.

iii. The Successful Bidder shall execute the work in a professional, safe, and workmanlike manner using new, genuine, and OEM-approved materials, strictly in accordance with the specifications, approved drawings, applicable standards, and instructions issued by VMC.

iv. The Successful Bidder shall carry out the work strictly as per the approved drawings, layouts, designs, implementation plan, and instructions issued by the Engineer-in-Charge. No deviation shall be made without prior written approval from VMC.

v. The Successful Bidder shall prepare and implement a comprehensive preventive and corrective maintenance plan for all CCTV cameras, poles, OFC network, switches, servers, storage, VMS, EMS/NMS, video wall, UPS, workstations, and all associated infrastructure throughout the Contract Period.

vi. VMC shall not provide any manpower, tools, test instruments, vehicles, lifting equipment, consumables, spare parts, calibration equipment, cleaning materials, or accessories. The Successful Bidder shall provide all resources required for successful execution and maintenance of the project.

vii. The Successful Bidder shall maintain adequate stock of critical spare parts and consumables throughout

the Contract Period. Any failed component shall be repaired or replaced within the SLA timelines. In cases where procurement of OEM spare parts is required, the Bidder shall arrange the same on priority and restore the system within the applicable SLA.

viii. The Successful Bidder shall immediately report all critical faults, security incidents, equipment failures, fiber cuts, and service interruptions to VMC and maintain a detailed fault register, maintenance logbook, and incident history.

ix. Upon completion of the project, the Successful Bidder shall hand over all perpetual software licenses, OEM warranty certificates, configuration files, passwords, database backups, documentation, as-built drawings, asset registers, and other project deliverables to VMC.

x. The Trial Run/Stabilization Period shall form part of the Contract Period. Final Go-Live and Operational Acceptance shall be issued by VMC only after successful completion of the stabilization period and compliance with all acceptance criteria.

xi. The Successful Bidder shall provide all required vehicles, hydraulic platforms (Hydra), ladders, tools, testing instruments, safety equipment, fuel, and logistics for installation, preventive maintenance, breakdown maintenance, and emergency restoration throughout the Implementation and O&M Period.

xii. During the O&M Period, the Successful Bidder shall provide comprehensive support, including:

i. Preventive and corrective maintenance of all hardware, software, licenses, and network infrastructure.

ii. 24×7×365 monitoring and technical support from the CCC.

iii. Monitoring and maintenance of CCTV cameras, VMS, servers, storage, EMS/NMS, network equipment, and OFC infrastructure.

iv. Fault diagnosis, troubleshooting, repair, replacement, firmware updates, security patches, and system optimization.

v. Incident management, change management, configuration management, and SLA reporting.

vi. User support, technical assistance, and coordination with VMC for issue resolution.

vii. Submission of periodic MIS reports, health reports, uptime reports, asset reports, and SLA compliance reports.

xiii. The Project Manager and other designated representatives of the Successful Bidder shall remain available for review meetings at VMC premises or project sites whenever required and shall provide all necessary technical support and progress updates.

i. The successful bidder will ensure that in the event of change of project resources during the course of the project, prior intimation to the VMC and suitable knowledge transfer takes place. Also, the replacement of the resource should be of higher or similar skill-set, experience level and shall need to be approved by VMC.

ii. The Bidder shall deploy all required resources, technical personnel, tools, equipment, and support infrastructure during the implementation phase at no additional cost to VMC. No separate payment shall be made for resource deployment during the implementation period. Payments towards Operations & Maintenance (O&M) services shall commence only after successful Go-Live and acceptance of the complete system by VMC, and shall be released as per the approved O&M payment schedule and SLA compliance.

iii. The Successful Bidder shall issue photo identity cards, uniforms, and appropriate safety gear (PPE) to all personnel deployed for implementation, operation, and maintenance of the project. The identity cards shall be worn at all times while working at VMC premises or project sites.

iv. No person below the age of eighteen (18) years shall be deployed for any work under this Contract. All deployed personnel shall be physically fit, adequately trained, and possess the required qualifications and experience. Documentary proof shall be produced whenever requested by VMC.

v. The Successful Bidder shall deploy competent technical personnel as required under the Contract. VMC reserves the right to verify the qualifications, experience, and identity of any deployed personnel and may require replacement of any unsuitable resource.

vi. The Successful Bidder shall deploy separate and adequate teams for Project Management, Survey & Design, Installation & Commissioning, OFC & Civil Works, Network & System Integration, Field

Maintenance, FRT, and 24×7×365 CCC/O&M support to meet the project schedule and SLA requirements.

vii. The ownership of all hardware, software, licenses, configurations, customizations, documentation, drawings, databases, asset records, and other project deliverables procured or developed under this Contract shall vest with Vadodara Municipal Corporation (VMC). All software licenses shall be registered in the name of Vadodara Municipal Corporation (VMC). The Successful Bidder shall submit OEM licenses, warranty certificates, MAFs, ATS/support details, and related documents before claiming payment.

viii. VMC may issue written instructions for alterations, additions, deletions, or modifications to the work within the overall scope of the Contract. Such changes shall be executed by the Successful Bidder at the approved contract rates and in accordance with the terms and conditions of the Contract.

ix. The Successful Bidder shall maintain proper inventory and secure storage of all project materials, equipment, tools, spares, and consumables at the project site. The inventory shall be regularly updated and made available for inspection by VMC whenever required.

x. The Successful Bidder shall construct, maintain, and remove any temporary access, barricading, protection, traffic diversion, or temporary works required for execution of the project. Upon completion, the site shall be restored to its original condition or better at no additional cost to VMC.

xi. The Successful Bidder shall prepare, maintain, and hand over complete project documentation, including but not limited to:

- Project Execution Plan and Implementation Schedule.
- Approved Survey Reports and Site Layouts.
- Network Architecture and IP Addressing Scheme.
- As-Built Drawings (Civil, OFC, Electrical, and Network).
- Equipment Datasheets and OEM Manuals.
- Installation, Configuration, and Commissioning Reports.
- OTDR, OFC, Electrical, Earthing, and FAT/SAT Test Reports.
- Asset Register and Inventory Details.
- User Manuals, Administrator Manuals, and Maintenance Manuals.
- Standard Operating Procedures (SOPs).
- Preventive Maintenance Schedule.
- Warranty, License, and Support Documents.
- Backup.
- Final Project Completion Report.

xii. The Successful Bidder shall keep all documentation updated throughout the Contract Period and submit two (2) hard copies and one (1) soft copy of all documents to VMC. All documentation shall remain the exclusive property of Vadodara Municipal Corporation (VMC).

x. **Capacity Building**: The Successful Bidder shall provide comprehensive training and capacity building to VMC personnel and other stakeholders as directed by VMC. The training shall cover system operation, monitoring, incident handling, VMS, EMS/NMS, video retrieval, report generation, user administration, preventive maintenance, troubleshooting, backup & recovery, and cyber security awareness. The Bidder shall provide all necessary training materials, user manuals, audio/video training aids, and hands-on sessions. Soft copies of the training materials shall be handed over to VMC, and attendance records along with feedback forms shall be maintained and submitted.

xi. **Handover of the System**: Upon completion or termination of the Contract, the Successful Bidder shall hand over the complete City Surveillance System to VMC or its authorized agency without interruption of services. The handover shall include all hardware, software, perpetual licenses, source/configuration files (where applicable), passwords, databases, backups, documentation, SOPs, as-built drawings, asset inventory, warranty documents, OEM support details, training records, and all other project-related

information required for smooth transition and continued operation of the system.

xii. The Successful Bidder shall deploy and maintain adequate qualified technical, operational, field support, FRT, CCC monitoring, and supervisory resources, along with all required tools, vehicles, testing instruments, and equipment necessary for successful implementation, operation, maintenance, and support of the City Surveillance System throughout the Contract Period.

xiii. The Successful Bidder shall ensure adequate resources for preventive maintenance, corrective maintenance, emergency breakdown maintenance, OFC restoration, camera maintenance, network management, server and storage support, CCC operations, and emergency incident response to achieve the prescribed Service Level Agreements (SLAs).

xiv. The number, qualification, deployment pattern, and utilization of resources shall be the sole responsibility of the Successful Bidder. VMC shall monitor the services based on SLA compliance and system performance and shall not be responsible for monitoring individual resource attendance. The Successful Bidder shall remain fully responsible for meeting all uptime, response time, restoration time, image quality, recording availability, and other SLA requirements specified in the Contract.

xv. The Successful Bidder shall nominate a dedicated Project Manager/Single Point of Contact (SPOC) for coordination with VMC throughout the Contract Period. The SPOC shall be responsible for project coordination, escalation management, incident resolution, periodic reporting, and ensuring compliance with all contractual obligations. Failure to maintain the required service levels shall attract penalties as specified in the SLA and Penalty provisions of this Tender.

xvi. Additional Quantity

1. At the time of award of contract, the quantity of goods, works or services originally specified in the bidding documents may be increased or decreased. The successful bidder shall not object to the upward or downward variation in quantities of any item.
2. Payment for additional quantities within the variation limit shall be made at tender rates and the tender rates shall be valid for 2 years of the contract.
3. No claim shall be entertained or become payable for price variation of additional quantities.
4. Repeat orders for extra items or additional quantities may be placed within 2 years of the original request order. The Unit Rate mentioned in the Commercial bid formats shall be used for the purpose of "Repeat Orders" for respective items. However, based on the market trends, VMC retains the right to negotiate the Tender rate and/or request better specifications based on market and technological scenario. Delivery or completion period may be proportionally increased.

    a. VMC reserves its right to award additional quantity of work order of the original quantity (as per state and central govt. guideline) at the same price and terms and conditions quoted in original contract.
    b. VMC also reserves right to award change request work order of the original project cost for the work needed to execute the work defined in scope of work defined in original contract

- **IT Infrastructure Management**
    - The selected vendor would be responsible for managing all project IT Infrastructure and maintain required service level to keep IT Infrastructure up and running.
- **Problem Identification and Resolution**
    - Identification and resolution of application problem City Surveillance System (e.g. system malfunctions, performance problem City Surveillance System and data corruption etc.) shall be part of the selected vendor's responsibility.

4.7 Incident Management System

The Successful Bidder shall deploy and maintain a centralized Incident Management System (IMS)/Helpdesk for the entire Contract Period to record, monitor, track, escalate, resolve, and close all incidents related to the City Surveillance System. The system shall support the complete lifecycle of incidents arising during both the

Implementation (CAPEX) and O&M (CAMC) phases and shall comply with ITIL best practices and/or ISO/IEC 20000 Service Management principles.

The Incident Management System shall provide, at a minimum, the following functionalities:

Sr. No.	Process / Requirement
1	Every incident, complaint, alarm, service request, preventive maintenance activity, and breakdown shall be assigned a unique ticket number and recorded with complete details, including date & time, location, device ID, fault description, priority, assigned engineer, action taken, and closure time.
2	Every incident shall follow a standardized workflow from ticket creation, acknowledgement, assignment, diagnosis, escalation, resolution, verification, and closure, ensuring compliance with the prescribed SLA.
3	Incidents shall be categorized and prioritized based on severity, criticality, business impact, and urgency to ensure timely restoration of services.
4	The system shall support automatic functional and hierarchical escalation to the Project Manager, OEM, FRT Team, and VMC officials whenever incidents are not resolved within the prescribed SLA timelines.
5	The system shall provide real-time dashboards, ticket status, SLA monitoring, escalation matrix, preventive maintenance schedules, and periodic MIS reports.
6	The Bidder shall maintain a complete incident history, root cause analysis (RCA), corrective actions, preventive actions, and closure reports for all major incidents and shall make the same available to VMC whenever required.
7	The Incident Management System shall support web-based access, role-based user management, notifications through e-mail/SMS (where applicable), audit logs, and export of reports in standard formats.
8	Failure to resolve incidents within the prescribed SLA timelines shall attract penalties as specified in the SLA and Penalty provisions of this Tender.

4.8 Enterprise Management Systems (EMS)

The Successful Bidder shall provide and implement an integrated Enterprise Management System (EMS) for centralized monitoring, management, reporting, fault management, asset management, SLA monitoring, and operational support of the entire City Surveillance System. The EMS shall continuously monitor the health, availability, performance, and security of all IT and field infrastructure deployed under this project.

The EMS shall include, but not be limited to, the following modules:

- SLA & Contract Management System for monitoring SLA compliance, uptime, downtime, preventive maintenance, penalties, warranty, CAMC, and contractual obligations.
- Network Management System (NMS) for real-time monitoring of switches, routers, firewalls, OFC links, network bandwidth utilization, latency, packet loss, device availability, configuration changes, and network alarms.
- Server & Storage Monitoring System for monitoring the health, performance, utilization, availability, CPU, memory, disk, storage, backup status, virtualization (if applicable), and hardware components of all servers and storage systems.
- Helpdesk & Incident Management System for ticket logging, complaint registration, fault tracking, escalation, preventive maintenance scheduling, SLA monitoring, engineer assignment, incident closure, and MIS reporting.
- Asset Management System for maintaining the complete inventory of cameras, poles, OFC, switches, servers, storage, UPS, racks, licenses, software, accessories, warranties, AMC/CAMC details, and maintenance history.

- Dashboard & Reporting System for providing configurable real-time dashboards, alerts, notifications, system health status, uptime reports, SLA reports, asset reports, preventive maintenance reports, incident reports, and management information reports (MIS).

The proposed EMS shall provide a centralized, web-based, role-based dashboard with secure user authentication, audit logs, and configurable access privileges. The system shall support automated alerts through e-mail/SMS (where applicable), report scheduling, report export, and integration with the proposed VMS and other project components.

2. SLA & Contract management System

The SLA & Contract Management solution should enable the VMC to capture all the System based SLAs defined in this RFP and then calculate quarterly (or for any duration) penalty automatically. Measuring service performance requires incorporation of a wide variety of data sources of the Surveillance project. The SLA solution should support the collection data from various sources in order to calculate Uptime / Performance / Security SLAs. Various features required in this component to EMS are -

- It must be a centralized monitoring solution for all IT assets (including servers, network equipment etc.)
- The solution must have integrated dashboard providing view of non performing components / issues with related to service on any active components
- The solution must follow governance, compliance and content validations to improve standardization of service level contracts
- Application should be pre-configured so as to allow the users to generate timely reports on the SLAs on various parameters.
- The solution must support Service Level Agreements & Lifecycle Management including Version Control, Status Control, Effectively and audit Trail to ensure accountability for the project.
- The solution must have the ability to define and calculate key performance indicators from an End to End Business Service delivery perspective related to Surveillance Project under discussion.
- The solution should support requirements of the auditors requiring technical audit of the whole system
- The solution most have an integrated dashboard, view of Contract Parties & current SLA delivery levels and view of Services & current SLA performance
- The solution should support SLA Alerts escalation and approval process.
- Solution should support effective root cause analysis, support capabilities for investigating the root causes of failed service levels and must make it possible to find the underlying events that cause the service level contract to fail.
- Accept Data from a variety of formats; provide pre-configured connectors and adapters, Ability to define Adapters to data source in a visual manner without coding.
- Support for Defining and Calculating Service Credit and Penalty based on clauses in SLAs. Reporting
- Ability to generate reports on penalty and credit due, to check on non-compliance of SLAs for the surveillance project
- Monetary penalties to be levied for non-compliance of SLA, thus the system must provide Service Level Performance Report over time, contract, service and more.
- The solution should provide historical and concurrent service level reports for the surveillance project in order to ensure accountability of the service provider's performance
- Automatic Report creation, execution and Scheduling, must support variety of export formats including Microsoft Word, Adobe PDF etc.
- The solution must support Templates for report generation, Report Filtering and Consolidation and Context sensitive Drill-down on specific report data to drive standardization and governance of the surveillance

project

- The solution must support security for drill-down capabilities in dashboard reports ensuring visibility for only relevant personnel of the surveillance project
- Support real-time reports (like at-a-glance status) as well as historical analysis reports (like Trend, TopN, Capacity planning reports etc.)
  - Resource utilization exceeding or below customer-defined limits
  - Resource utilization exceeding or below predefined threshold limits

A List of SLAs that needs to be measured centrally by SLA contract management system are given in the RFP document. These SLAs must be represented using appropriate customizable reports to ensure overall service delivery.

3. Network Management System

Solution should provide fault & performance management of the server side infrastructure and should monitor IP\SNMP enabled devices like Routers, Switches, PA System, Emergency Call Boxes, Sensors, etc. Proposed Network Management shall also help monitor key KPI metrics like availability, in order to measure SLA's. Following are key functionalities that are required which will assist administrators to monitor network faults & performance degradations in order to reduce downtimes, increase availability and take proactive actions to remediate & restore network services.

- The proposed solution must automatically discover manageable elements connected to the infrastructure and map the connectivity between them. Solution should provide centralized monitoring console displaying network topology map.
- Proposed solution should provide customizable reporting interface to create custom reports for collected data.
- The system must use advanced root-cause analysis techniques and policy-based condition correlation technology for comprehensive analysis of infrastructure faults.
- The system should be able to clearly identify configuration changes and administrators should receive an alert in such cases.

4. Server Performance Monitoring System

- The proposed tool should integrate with network performance management system and support operating system monitoring for various platforms supplied as part of this Project.
- The proposed tool must provide information about availability and performance for target server nodes.
- The proposed tool should be able to monitor various operating system parameters such as processors, memory, files, processes, file systems, etc. where applicable.

Solution Architecture of OFC Network

The diagram below represents the high level network architecture diagram of proposed Fibre Optic Network in Vadodara City:

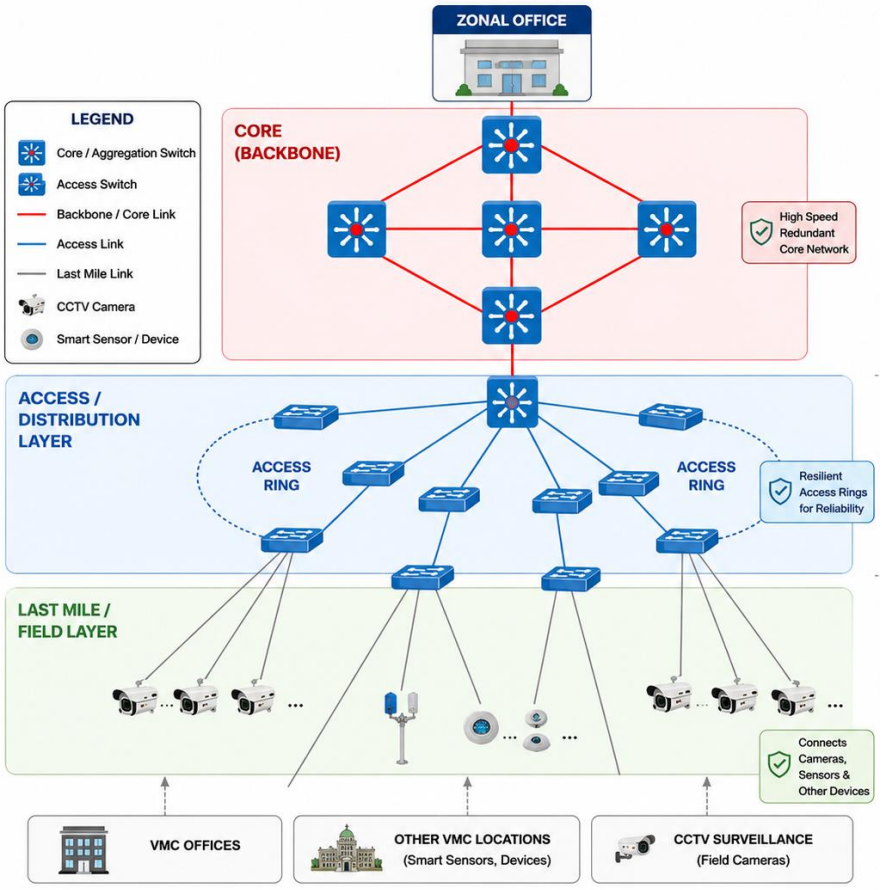


Figure 1 - Conceptual Architecture Diagram for FIBRE OPTIC NETWORK

The above diagram illustrates FIBRE OPTIC NETWORK hybrid topology comprising of a full mesh backbone design along with a ring design for the C0re and access layers. Below are the design considerations for Fibre Optic Network:

Core Layer

The Core backbone mesh topology shall meet the following minimum requirements:

- The Core backbone mesh topology shall be constructed using 48 Core Armoured Single Mode Optical Fibre Cables (OFC).
- The Core ring shall have 4 POP locations (at each of zonal offices of VMC) connecting to each other as backbone of the network for providing bandwidth to the entire network components. The main PoP shall be at CCC (DC Area) at Badamadi Baug.
- The Core network shall be highly scalable.

- The Core shall utilize a 10 Gigabit Ethernet/ Optical technology and shall support 40 Gigabit Ethernet/ Optical technologies.
- The Core ring shall have (1+1) configuration of 40mm High Density Polyethylene (HDPE) pipes. One duct shall be kept free, with pulling ropes/strings inserted and properly marked.
- The Core architecture shall be formed in full mesh topology to handle dual point of failures.
- 48 Core Fibre Optic Cable shall be used to build the core network. Out of 12 pairs, 1 pair shall be used to connect all 5 POP locations.
- 3 pairs from remaining pairs shall be used for providing redundancy in the network and for maintenance purposes.
- Remaining pairs shall be kept reserved for future expansion of the core network. It shall be used to add further POP locations in the network to increase the bandwidth capacity or to be utilised for other purposes, such as Wifi, Leasing etc.
- The maximum Fibre distance between two Core Switches in 2 cut redundant mesh, as measured by optical time domain meter (OTDR) shall not exceed 40 KMs.
- Core switch will be installed at all the Main POP location, the switching backplane needs to be as per overall current and future requirement, at every POP location.
- Distribution switches shall be installed at 4 Zonal POP locations
- The proposed ring based architecture shall support resiliency of sub 50ms ensuring high availability and faster convergence for service continuity.
- The proposed Core Ring shall consider 30% more Fibre length than surface length,
  a. 18% for looping in chambers,
  b. 7% for modulation,
  c. 1% for splice joint preparation &
  d. remaining 4% for wastage.
- The Core ring shall have chamber at approx. 200 meters and shall have OFC joint splice closure at approx. 1500 meters.
- Additional chambers to be constructed wherever there is a road crossings or at sharp turns

Access Layer

- The access layer comprising of ring topology design which is connected to each POP location. Each POP location shall have required number of which shall connecting to VMC offices, Junctions, Sensors/Display signs, VMC Hospitals etc. to provide connectivity.
- The Access ring shall be constructed using a 48 Core Armoured Single mode Optical Fibre Cables (OFC).
- It shall utilize a 1 Gigabit / 10 Gigabit Ethernet/Optical technology.
- It shall have (1 + 1) configuration of 40mm High Density Polyethylene (HDPE) pipes. Two ducts shall be kept free, with pulling ropes/strings inserted and properly marked.

- The access architecture shall be formed using ring topology.
- The access ring is designed so that it can support up to 1 edge switch at every 350 meters.
- The maximum distance of one access ring shall not exceed 80 kilometres.
- Not More than 16 Nos of switches should be considered within a single ring.
- The distance between two Hand Holes/RCC Chamber should not exceed 200 meters.
- A loop of 20 meters of OFC shall be left in every Hand Hole.
- 48 Core Optical Fibre Cable shall consist of 6 Fibre per tube and shall have total 8 numbers of tubes.
- 1 tube shall be utilized to provide connectivity to all smart elements/CCTV Cameras/Sensors/Displays to be connected as part of scope of work of this project.
- 2 tubes shall be kept reserved to provide redundancy in the network and maintenance purposes.
- Remaining tube shall be reserved to provide connectivity to future expansion/usages.
- The proposed Access ring length considered is 30% more than surface length i.e.
- 18% for looping in chambers,
- 7% for modulation,
- 1% for splice joint preparation &
- Remaining 4% for wastage.

Colour code and marking for 40mm HDPE

- The color code for 40mm High Density Polyethylene (HDPE) pipes shall be as follows:
- The core backbone mesh topology shall be identified by using Green, Navy and Blue colours. The Core backbone mesh Optical Fibre Cables (OFC) shall be placed in Green colour conduit.
- The Access ring shall be identified using Yellow and Pink colours. The Access ring Optical Fibre Cables (OFC) shall be place in a Yellow colour conduit
- All the pipes shall be clearly marked at intervals of 1 meters with the following data with 5 mm (or more) high letter : Client logo, Manufacture's name or trade mark
- i) Year of manufacturing ii) Type of HDPE pipe and size iii) Running length marking. The details of marking on pipe shall be approved by VMC before commencement of manufacturing.

Protection for OFC

- The Optical Fibre Network shall be built by considering following protection methods to protect the OFC from accidental damage.
- **GI Pipe**: GI pipe shall be used to provide additional protection for HDPE ducts at rail/road crossings, built-up area/city limits, on culverts, bridges wherever the depth is less than 0.3Mtr
- **DWC Pipe:** DWC pipe shall be used to provide additional protection for HDPE ducts at rail/road crossings, built-up area/city limits, on culverts, bridges wherever the depth is less than 0.6Mtr but more than 0.5Mtr

- **Concreting:** Concreting shall be used to provide additional protection on bridges, culverts and also on stretches wherever depth of excavation is less than 0.3Mtr.
- **RCC Half round:** Where-ever cables are laid in cities, towns and villages where human habitation exists or where construction activity can happen in near future,
- Full Round or Half Round, 100mm, NP2 Grade shall be put after 100 mm padding with excavated material over duct, wherever the depth is less than 0.5Mtr but more than 0.4Mtr
- **RCC Full round:** Where-ever cables are laid in cities, towns and villages where human habitation exists or where construction activity can happen in near future, Full Round or Half Round, 100mm, NP2 Grade shall be put after 100 mm padding with excavated material over duct, wherever the depth is less than 0.4Mtr but more than

Integration with existing OFC Network

The new OFC network being created under this project shall be integrated with existing OFC Network of Vadodara Eye Project and CCC project. During the integration time, there can be certain downtime for existing Vadodara Eye project and CCC project, which needs to be worked out jointly with existing SI of Vadodara Eye Project and CCC project. Re-configuration of existing switches/routers of Vadodara Eye and CCC Project shall be under scope of the successful bidder of this RFP.

Scope of Services for the City Wide OFC Project

The scope of work & services covers the design, procurement, supply, installation, construction, testing and commissioning of a city wide Optical Fibre NETWORK (Optical Fibre Cables (OFC) Network) in the *Vadodara city*, along with the associated active, passive components, civil, mechanical and power. The SI should ensure the successful implementation of the proposed System and provide capacity building support to city authorities as per the scope of services described below. Any functionality not expressly stated in this document but required to meet the needs of the VMC to ensure successful operations of the system shall essentially be under the scope of the SI and for that no extra charges shall be admissible.

SI Shall have to Conduct a detailed assessment, scoping study (Feasibility Study) and develop a comprehensive project plan, including detailed technical architecture and gap analysis.

Following services and applications can be incorporated under project SOW:

- CCTV Existing and new.
- All other smart elements/sensors/displays on field, as part of this project and additional similar installations
- Connectivity for VMC offices, hospitals, police station etc.

The scope of work for FIBRE OPTIC (OFC) NETWORK shall cover the following activities but not limited to:

1.1 **Inclusions**

The scope of project has been designed to include the implementation of following key components

1.2 **Laying of Underground Optical Fiber across the city**

This component of the project would involve laying underground optical Fiber across the City so that the identified project locations are connected with each though OFC. Optical Fiber shall be laid primarily along the road network of the city. This would involve:

- Verification and optimization of routes designed based on GIS data
- Laying of optical Fiber
- Relaying of road

Total length of underground Fiber to be laid across the City for creating the backbone and other rings is estimated to be around **100 km.** This would comprise Core Ring and multiple Access Rings. The same network may also be extended **for further 100 km** in future. The SI is required to follow the methodology for Optical Fiber Cable Execution as per RFP.

Optical Fiber Cable Execution Methodology

The SI shall adhere to the below guidelines while executing the work:

- The SI shall prepare the list of equipment, number of field employees deployed and the implementation schedule taking into consideration all the requirements of concerned Municipal and Government authorities.
- The SI shall submit the work plan and implementation schedule with list of equipment and personnel to be deployed on field for execution of works for approval of Client.
- The SI shall commence work post approval of work plan and schedule by the Client.
- If any requirement, direction, guideline, regulation, or instruction issued by the Central Government, State Government, statutory authority, regulatory body, utility agency, police authority, or any other competent authority conflicts with any instruction issued by VMC, the Bidder shall immediately notify VMC in writing, highlighting the nature of such conflict. VMC shall review the matter and issue appropriate directions. The Bidder shall comply with all applicable laws, rules, regulations, and statutory requirements at all times during the execution and maintenance of the project.
- The SI shall inform all concerned authorities and obtain NOC or permissions as required before starting the excavation and Hand Hole construction works.
- The SI shall register and get approval from concerned Government authorities to carry out the work as required.
- The SI shall adhere to guidelines issued by concerned Government authorities while executing the work.
- The SI shall lay the underground OFC by Open Cut/Open Trench or Horizontal Directional Drilling (HDD) method as permitted by the concerned Government Authority pertaining to specific road.
- The Optical Fiber Cable (OFC) shall be laid in HDPE/DWC pipe at a minimum depth of 1.20 meters from the finished ground level in unpaved areas and 1.50 meters below the finished road

level while crossing roads, National Highways, State Highways, major junctions, or as per the latest applicable standards, guidelines, and permissions issued by the concerned road-owning authority (MoRTH/NHAI/IRC/PWD/VMC or other statutory authorities). In rocky areas or where the specified depth cannot be achieved due to site constraints, the Bidder shall adopt an approved protection method (such as RCC encasement, GI pipe, HDPE casing, or any other approved protection) after obtaining prior written approval from VMC and the concerned authority.

- The SI shall lay OFC of 48core/96core through 40mm HDPE pipe.
- The SI shall construct Hand Holes at successive intervals of 200 Meters or less.
- The SI shall obtain all land permits, Permissions, NOC, any land lease rights or as such any licensing requirement that may be necessary to erect the POP's from all the concerned Government authorities.

Open Cut Method

Excavation

- The SI shall undertake the trenching & ducting activity for laying of OFC as per Survey drawings approved by the Client.
- The SI shall place route marker as per given alignment & maintaining offset distance from road centre as per norms set by concerned government authorities.
- The SI shall use barricading and signage board as per requirements of concerned Government authority pertaining to specific roads.
- The SI shall take trial pits before starting trenching / HDD, at every 100m distance to examine position of existing underground utilities. The SI shall align the trench as per observations from the trial pits.
- The SI shall take precaution during excavation to avoid any possible damage to other underground utilities and shall indemnify the Owner/Employer for damages if any. This includes but not limited to collecting as-build drawings of existing utilities, studying the survey drawings for location of utilities and visual inspection of site.
- The SI shall coordinate with the existing utility owners before starting the excavation work. If required, the SI shall ensure the presence of representative of existing utility owners.
- The SI shall achieve minimum depth of 700mm whenever the 40 mm HDPE pipes are to be laid under the road surface and minimum depth of 1000mm whenever 40 mm HDPE pipes are laid on exposed sand or soil surface.
- The SI shall construct the trench with width of not less than 300mm at the bottom of the trench.
- The SI shall be responsible for shoring and strutting the walls of the trench on either side of the trench as per the guidelines of concerned Government authorities.
- The SI shall follow below mentioned norms in case of any deviations in depth of trench.

Depth of Trench (Finished Ground Level)	Recommended Protection
1000 mm to 1200 mm	No additional protection required. OFC shall be laid in approved HDPE/DWC pipe with warning tape and route markers.
800 mm to less than 1000 mm	OFC shall be laid in 110 mm HDPE/DWC pipe with brick/RCC protection cover or as approved by VMC.
500 mm to less than 800 mm	OFC shall be laid inside 110 mm HDPE/DWC pipe with RCC protection slabs throughout the route.
200 mm to less than 500	OFC shall be laid inside 100 mm GI pipe with minimum 150

mm	mm PCC (Plain Cement Concrete) protection.
Less than 200 mm or where the required depth cannot be achieved	A 300 mm × 300 mm RCC box culvert shall be constructed, and the OFC shall be routed through a 100 mm GI pipe installed within the box culvert. The Bidder shall obtain prior approval from VMC before adopting this method.

Optical Fiber Cable (OFC) Testing Methodology

- All Optical Fiber Cable (OFC) testing shall be carried out in accordance with the latest applicable ISO/IEC 14763-3, ISO/IEC 11801, TIA-568, and relevant BIS/Telecommunication standards.
- The Bidder shall test 100% of all installed optical fiber links before commissioning and submit the test reports to VMC for approval.
- Testing shall include Optical Time Domain Reflectometer (OTDR) testing, Insertion Loss (Optical Loss Test Set - OLTS), continuity, polarity, and end-to-end connectivity verification.
- The attenuation of each optical fiber link shall be measured using the Insertion Loss Method with a calibrated Optical Light Source and Optical Power Meter.
- Single Mode Fiber (SMF) shall be tested at 1310 nm and 1550 nm wavelengths in both directions (Bidirectional Testing).
- OTDR testing shall be carried out at 1310 nm and 1550 nm to verify fiber length, attenuation, splice loss, connector loss, reflectance, and any events or faults along the fiber route.
- The measured attenuation, splice loss, connector loss, and OTDR trace results shall comply with the applicable industry standards and OEM recommendations. Any fiber link not meeting the specified acceptance criteria shall be rectified and retested by the Bidder at no additional cost to VMC.
- The Bidder shall submit detailed test reports indicating the fiber identification, route, cable length, attenuation values, splice loss, connector loss, OTDR traces, test date, test equipment details, and the name of the certified testing engineer.
- All testing shall be carried out using calibrated and certified test instruments. Calibration certificates of the test equipment shall be submitted to VMC upon request.
- The OFC testing shall be witnessed by the designated representative of VMC whenever required.
- The testing shall be performed by trained and certified engineers authorized by the OEM or the respective test equipment manufacturer.
- **The Bidder shall obtain and submit an OEM-backed warranty certificate for the Optical Fiber Cable system with a minimum warranty period of 25 years, covering the installed OFC and passive cabling infrastructure**.

4.9 Functional Requirement

Vadodara City Surveillance System
The Bidder shall carry out design, development, testing, supply, installation, integration, commissioning, operation and maintenance of CITY SURVEILLANCE SYSTEM.

The components of CITY SURVEILLANCE SYSTEM as envisaged in this project are:
- CCTV cameras
- Street pole, Junction Box
- Router, Switch, Rack, UPS
- Edge analytics-based processing
- Processing server, Storage, EMS
- Application for officers (SMS Alerts/Screen Pop Ups)
- Backend Application
- OFC

System Design Principles
The proposed CITY SURVEILLANCE SYSTEM planned to detection of Crime and civic issues, the CITY SURVEILLANCE SYSTEM shall be designed considering the following guiding principles:

Scalable: The system shall be scalable to future growth in number of locations. The proposed solutions shall support these scalability requirements without requiring any change. The additional hardware required to include locations beyond the mentioned locations under CITY SURVEILLANCE SYSTEM is to be mutually decided by VMC and the Bidder at an additional cost.

High Performance: The system design shall be in such a way as to be resilient to technological sabotage. To take care of remote failure, the systems shall be configured to mask and recover with minimum outage. The Bidder shall be required to make the provision for high availability for all the services of the systems.

Secure: The network shall have built-in security features as per industry practices in line with the requirement for CITY SURVEILLANCE SYSTEM. Access control shall be implemented at all levels. The Bidder shall make provisions for security of field equipment as well as protection of the software systems from hackers and other threats. The virus and worms' attacks shall be well defended with gateway level Antivirus system. Furthermore, all the system logs shall be properly stored & archived for future analysis and forensics whenever desired. The following guidelines shall be observed for security:

- Build a complete audit trail of all activities and operations using log reports, so that errors in system– intentional or otherwise – can be traced and corrected.
- Access controls shall be provided to ensure that the systems are not tampered or modified by the unauthorized persons or system operators.
- Implement data security to allow for changes in technology and business needs.
- The security of the field devices shall be ensured with system architecture designed in a way to secure the field devices in terms of physical damage & unauthorized access.

Interoperable: The systems shall have capability to take inputs from other third-party systems as per situational requirements. All products shall be open standards based and should be interoperable with different vendors' products following industry standards.

Fault Tolerance and Resilient: The systems shall have inbuilt redundancy features to provide high availability. Redundancy shall be proposed at various levels to ensure that single link failure does not affect the critical functionality

Information to be captured by Edge Devices

Cameras being the core of the entire Surveillance system, it is important that their selection is carefully done to ensure suitability & accuracy of the information captured on the field and is rugged, durable & compact. These cameras need to work on 24 X 7 basis and transmit quality video feeds to the centralized data centers and would capture the video feeds at 25/30 FPS. VMC shall take the regular review of the requirements for video resolution, FPS may change to suit certain specific requirements for example, and there could be a situation when certain cameras are required to be viewed at higher FPS for specific period. It is estimated that not more than 2% of the cameras would be required to be viewed at higher FPS at a given point of time. Cameras should maintain minimum bitrate of 4 Mbps for bullet/fixed camera per lens to transmit quality image/video feed (appropriately focused, clear, un-blurred, jitter free, properly lit etc.). Packet loss should be less than 0.5%.

Role Based Access to the Entire System

Various users should access the system using single sign on and should be role based. Different roles which could be defined (to be finalized at the stage of SRS) could be Administrator, Supervisor, Officer, Operator, etc. Apart from role-based access, the system should also be able to define access based on location. Other minimum features required in the Role Based Authentication Systems are as follows:

i. The Management Module should be able to capture basic details (including mobile number & email id) of the VMC Personnel & other personnel requiring Viewing / Administration rights to the system. There should be interface to change these details, after proper authentication.
ii. Rights to different modules / Sub-Modules / Functionalities should be role based and proper log report should be maintained by the system for such access.

Storage / Recording Requirements

It is proposed that the storage solution is modular enough to ensure compliance to the changes in storage / recording policy, to be evolved upon initial deployment of the system. Following storage requirements are proposed for the project:

a) The storage solution proposed is that the video feeds would be available for 30 days. After 30 days, the video feeds would be overwritten unless it is flagged or marked by the CSDL/Police / Vadodara City Police for investigation or any other purpose. The video feeds of all relevant cameras capturing the incident or flagged in question would be stored until the Police deem it good for deletion.

b) For incidents that are flagged by the Police or any court order, the video of the relevant portion from all relevant cameras should be stored/archived separately for investigation purposes and a committee at Police Department can decide when this video feed can be deleted.

c) Audit trail of the system to be maintained on permanent basis / as per the backup policy defined.

d) Retrieval time for any data stored should comply with the defined service level agreements.

e) The Recording Servers / System, once configured, shall run independently of the Video Management system and continue to operate if the Management system is off-line.

f) The system shall support the use of separate networks, VLANs or switches for connecting the cameras to the recording servers to provide physical network separation from the clients and facilitate the use of static IP addresses for the devices.

g) The system shall support H.265, H.264 with better compression, H.264 formats for all IP cameras connected to the system.

h) The system shall record the native frame rate and resolution supplied by the camera or as configured by the operator from the System Administration Server.

i) The system shall allow for the frame rate, bit rate and resolution of each camera to be configured independently for recording. The system shall allow the user to configure groups of cameras with the same

frame rate, bit rate and resolution for efficient set-up of multiple cameras simultaneously.

j) The system shall support Archiving or the automatic transfer of recordings from a camera's default database to another location on a time-programmable basis without the need for user action or initiation of the archiving process. Archiving shall allow the duration of the camera's recordings to exceed the camera's default database capacity. Archives shall be located on either the recording server or on a connected network drive. If the storage area on a network drive becomes unavailable for recording, the system should have the ability to trigger actions such as the automatic sending of email alerts to necessary personnel.

k) Bandwidth optimization

- The Video Management System (VMS)/Recording Server shall support multiple video compression standards, including H.265 (HEVC), H.264, and Smart Codec, with configurable resolution, frame rate, and bitrate settings to optimize bandwidth and storage utilization.
- The VMS shall support adaptive streaming, sub-streaming, motion-based streaming, and dynamic bandwidth optimization. Users shall have the option to continuously stream live video or receive video updates based on motion/events to minimize network bandwidth consumption.
- The Bidder shall calculate and submit the detailed bandwidth, storage, and network utilization for the proposed solution based on the specified camera parameters, recording policy, retention period, codec, frame rate, resolution, bitrate, and VMC's operational requirements. The calculations shall be supported by OEM sizing tools or detailed engineering calculations and shall form part of the technical bidThe Recording Server / System shall support IP cameras from various manufacturers.

l) The system shall support automatic failover for Recording Servers; this functionality shall be accomplished by Failover Server as a standby unit that shall take over if one of a group of designated Recording Servers fails. Recordings shall be synchronized back to the original Recording Server once it is back online.

m) The system shall support multiple Failover Servers for a group of Recording Servers.

n) The system shall support Simple Network Management Protocol (SNMP) for third-party software systems to monitor and configure the system.

o) The system shall act as an SNMP agent which can generate an SNMP trap because of rule activation in addition to other existing rule actions.

Alert Management Module

a) The alert management module shall allow for continuous monitoring of the operational status and event-triggered alarms from various system servers, cameras and other devices. The alarm management module shall provide a real-time overview of alarm status or technical prob City Surveillance System while allowing for immediate visual verification and troubleshooting.

b) The alarm management module shall provide interface and navigational tools through the client including;

   i. Graphical overview of the operational status and alarms from servers, network cameras and external devices including motion detectors and access control systems.

   ii. Intuitive navigation using a map-based, hierarchical structure with hyperlinks to other maps, servers and devices or through a tree-view format.

b) The module shall include flexible access rights and allow each user to be assigned several roles where each shall define access rights to cameras.

c) Basic VMS should be capable to accept third party generated events / triggers

Project execution requirements

Actual placement of pole & number of cameras at each location, type of cameras, fixation of height & angle for the cameras to ensure maximum coverage shall be done in consultation with VMC. A detailed survey shall be conducted, by the SI along with a team of VMC. SI shall prepare the detailed report for field level requirements e.g. Cameras (types & numbers), Camera Mounting requirements, Power Requirements, Connectivity Requirements etc. for perusal of Purchaser. Indicative list of the field level hardware to be provided by SI is as

follows:

- Cameras
- Network Switches
- Pole for cameras / Edge devices
- Junction box
- Networking and power cables and other related infrastructure.

Based on detailed field survey as mentioned above, SI shall be required to supply, install and commission the surveillance system at the identified locations and thereafter undertake necessary work towards its testing. SI shall use industry-leading practices during the implementation phase w.r.t positioning and mounting the cameras, poles and junction boxes. Some of the check-points that need to be adhered to by the SI while installing / Commissioning cameras are as follows:

1. Ensure surveillance objective is met while positioning the camera such that the required field of view is being captured as finalized in field survey
2. Ensure camera is protected from the on-field challenges of weather, physical damage and theft.
3. Make proper adjustments to have the best possible image / video captured.
4. Ensure that the pole is well placed for vibration resistance adhering to the road safety norms.
5. Collusion preventive barriers around the junction box & pole foundation in case it's installed in collision prone place.
6. Appropriate branding or color coding of poles and junction boxes, to warn mischief mongers against tampering with the equipment at the junction.
7. SI shall ensure that the poles erected to mount cameras are good, both qualitatively and aesthetically. Also, all the new poles should be futuristic to be able to house other ICT elements like Environment Sensors, Small Variable Message Display, Telecom Equipment, etc. so that their presence can be optimized by Police department and as well as monetized in near future if required.
8. SI shall use the industry leading practices while positioning and mounting the cameras and ensure that the pole / mast implementation is vibration resistant. Necessary arrangements for bird scare spikes on top of camera to be also thought of to prevent birds from sitting on top of camera box.
9. The poles shall be installed with base plate, pole door, pole distributor block and cover.
10. Base frames and screws shall be delivered along with poles and installed by the SI.
11. The poles should be installed along with cantilever with minimum length of 1.8 meter, so that there is clear line of sight.
12. SI shall be responsible to undertake required structural analysis regarding the regulated load conditions and considering the respective wind load while installing the poles / cantilevers for edge level components
13. SI shall provide structural calculations and drawings for the approval of Purchaser. The design shall match with common design standards as applicable under the jurisdiction of purchaser/authorized entity.
14. SI shall coordinate with concerned authorities / municipalities for installation.
15. Poles and cabinet shall be so designed that all elements of the field equipment could be easily installed and removed.
16. SI shall ensure that physical look of the installation area returns to neat & tidy conditions after installation of poles, cantilevers etc. The placement shall be designed keeping in mind the normal flow of vehicular traffic and pedestrian movement is not disturbed.

4.10 Technical Specification Require Equipment

The technical specifications of CITY SURVEILLANCE SYSTEM Devices shall be as set out in the table below:

Bullet 5MP Camera for Surveillance

Parameter	Technical Requirement / Specification	Compliance (YES/NO)
STQC / BIS-ER Status	Mandatory. Camera model must hold a valid MeitY IoTSCS certification as verified on the STQC portal, and/or BIS-ER (Enforcement of Registration) compliance. Test/registration certificate to be enclosed with the technical bid.	
Image Sensor	Minimum 1/2.8" Progressive Scan CMOS (Sony STARVIS or equivalent low-light sensor).	
Effective Resolution	Minimum 5 Megapixel (2592 x 1944) effective physical pixel array.	
Frame Rate	Main stream: full 5MP resolution at minimum 30 fps (PAL/50Hz compliant).	
Optical Lens	Motorized varifocal zoom lens, minimum range 5.0mm-50.0mm (+/-10% tolerance permitted); Auto-Iris/P-Iris; remote zoom and back focus.	
Video Compression	H.265+, H.265, H.264 High Profile, and MJPEG streaming profiles, with multi-stream (minimum triple stream) capability.	
Minimum Illumination	Color: 0.005 lux or better (F1.2, AGC ON); B/W: 0 lux with integrated Smart IR.	
Wide Dynamic Range	True 120dB WDR (minimum); Highlight Compensation (HLC) for headlight/glare reduction.	
Night Vision / IR Range	Minimum 50 meters, adaptive Smart IR array (auto-adjusted to avoid overexposure of near-field objects).	
Audio Interface	Minimum 1-channel audio input (Line-in/Mic-in) and 1-channel output; built-in G.711/AAC audio codec.	
Edge AI / Video Analytics	Built-in edge-based analytics covering, at minimum: Automatic Number Plate Recognition (ANPR) where deployed for traffic/junction use, Vehicle/Human Classification, Line Crossing Detection, Intrusion/Perimeter Detection, and Object Left/Removed Detection.	
Network Protocols	IPv4/IPv6, HTTP/HTTPS, TCP/IP, UDP, RTP/RTSP, DHCP, NTP, IGMP (multicast).	
Interoperability	ONVIF Profile S, G, T, and M compliant, for integration with multi-OEM Smart City VMS platforms.	
Cyber Security	HTTPS/TLS 1.3, IEEE 802.1X authentication, IP address filtering, digest authentication, complex password enforcement, and secure boot/firmware signing where available.	
Physical Housing	Rugged die-cast aluminum or equivalent metal enclosure; IP67 weatherproof rated; IK10 vandal-resistant rated.	
Operating Temperature	-30°C to +60°C operating window, with built-in internal thermal regulation, suitable for outdoor Indian climatic conditions.	
Power Input	PoE+ (IEEE 802.3at, Class 4) as primary; localized 12V DC/24V AC as fallback, with integrated surge protection.	
Local Storage	MicroSD / MicroSDHC / MicroSDXC card slot supporting up to 512GB,	

	supplied with minimum 512GB card for edge storage/fallback recording.	
Warranty	Minimum 5 years comprehensive OEM warranty from the date of GoLive.	

Dome Camera Indoor and out for Surveillance & Compliance Matrix – 5 MP IP Rugged Dome Camera

Parameter	Technical Requirement / Specification	Compliance (YES/NO)
STQC / BIS-ER Status	Mandatory. Camera model must hold a valid MeitY IoTSCS certification as verified on the STQC portal, and/or BIS-ER (Enforcement of Registration) compliance. Test/registration certificate to be enclosed with the technical bid.	
Image Sensor	Minimum 1/2.8" Progressive Scan CMOS (Starvis or equivalent low-light sensor).	
Effective Resolution	Minimum 5 Megapixel (2592 x 1944, or 2880 x 1620) effective physical pixel array.	
Frame Rate	Full 5MP resolution at minimum 30 fps.	
Lens	2.8 mm to 13.0 mm Motorized Varifocal Zoom; Auto-Iris / P-Iris; Remote Zoom/Focus. +-10% allowed	
Video Compression	H.265+, H.265, H.264 with multi-stream (minimum triple stream, or 4+ where supported) capability.	
Minimum Illumination	Colour: 0.01 lux or better (AGC ON); B/W: 0 lux with integrated Smart IR.	
Wide Dynamic Range	True 120dB WDR (minimum).	
Night Vision / IR Range	Minimum 30 metres Smart IR (auto-adjusted to avoid overexposure of near-field objects).	
Audio & Alarm I/O	Built-in microphone/audio input-output; minimum 1 alarm input and 1 alarm output port.	
Edge AI / Video Analytics	Built-in edge-based analytics covering, at minimum: Face Detection, Perimeter/Intrusion Detection, Line Crossing Detection, Loitering Detection, and Object Left/Removed Detection.	
Network Protocols	IPv4/IPv6, HTTP/HTTPS, TCP/IP, UDP, RTP/RTSP, DHCP, NTP, IGMP (multicast).	
Interoperability	ONVIF Profile S, G, T, and M compliant, for integration with multi-OEM Smart City VMS platforms.	
Cyber Security	HTTPS/TLS, IEEE 802.1X authentication, IP address filtering, digest authentication, complex password enforcement, and secure boot/firmware signing where available.	
Physical Housing	Metal/aluminum dome housing; IP66/IP67 weatherproof rated (as applicable to outdoor/indoor deployment); IK10 vandal-resistant rated.	
Operating Temperature	-30°C to +60°C operating window, suitable for outdoor Indian climatic conditions.	
Power Input	PoE (IEEE 802.3af/at) as primary; localised 12V DC as fallback, with integrated surge protection.	
Local Storage	MicroSD / MicroSDHC / MicroSDXC card slot supporting up to 512 GB, supplied with minimum 512GB card for edge storage/fall-back recording.	
Warranty	Minimum 5 years comprehensive OEM warranty from the date of GoLive.	

Tender Note:

The bidder shall submit valid documentary evidence, certificates, declarations, compliance reports, and OEM datasheets supporting all the specified technical features, protocols, certifications, and compliance requirements. Self-declarations without supporting documentary evidence shall not be considered for technical evaluation. Any certification under renewal shall be supported by valid proof issued by the respective certification authority. The offered camera shall support all specified protocols and standards required for seamless integration with ICCC infrastructure, Enterprise Video Management Systems (VMS), Network Management Systems (NMS), and third-party ONVIF-compliant platforms. The bidder shall clearly highlight compliance in the OEM datasheets. Any feature, protocol, analytics, or functionality requiring additional licenses shall be supplied and included in the quoted price without any additional cost to VMC.

Network Connectivity

Network connectivity is one of the most important components of the project and needs very careful attention in assessment, planning and implementation. It is important not only to ensure that the required connectivity is provisioned within the required timelines but also ensure that it is reliable, secure and supports the required SLA parameters of Latency, Jitter, Packet Loss and Performance. Throughout end to end (data center till edge level) network should support the software defined networking (SDN).

STQC/BIS-ER

STQC/BIS-ER Compliance for CCTV Cameras- All new CCTV cameras proposed under this contract, including replacement and additional cameras, shall comply with the latest applicable STQC or BIS Essential Requirements (BIS-ER) notified by the Government of India. The bidder shall ensure that all supplied CCTV cameras possess valid STQC certification and BIS registration, wherever applicable, at the time of supply and installation. Documentary evidence, including valid certificates and test reports, shall be submitted to VMC prior to deployment. Any camera found non-compliant with STQC/BIS-ER requirements shall be replaced by the bidder at no additional cost to VMC.

Single OEM - Make for All Camera Types

All CCTV cameras supplied under this tender Bullet Camera, Dome Camera, and PTZ Camera shall be of the same OEM (Original Equipment Manufacturer) and same product make/series across the entire project. Mixing of cameras from different manufacturers/brands for different camera types (e.g., Bullet from one OEM and Dome/PTZ from another) shall not be permitted under any circumstances.

This requirement is mandatory to ensure:

- Uniform build quality, firmware ecosystem, and cybersecurity posture across all deployed cameras;
- Single-point OEM warranty, support, and spares availability for the entire camera fleet, avoiding multi-vendor coordination issues during the O&M period.
- Seamless interoperability, firmware/security patch management, and long-term supportability through a single OEM relationship.
- Simplified troubleshooting, AMC/CMC accountability, and reduced integration risk with the central VMS.

The bidder shall clearly declare, in the technical bid, the single OEM/brand proposed for all three camera types (Bullet, Dome, and PTZ), along with the respective model numbers and datasheets. Bids proposing cameras of different makes/brands for different camera categories shall be treated as non-compliant and liable for rejection at the technical evaluation stage.

Proposed OEM/make shall hold valid STQC/BIS-ER certification for all three camera models (Bullet, Dome, and PTZ) offered under this tender.

Security Management

The City Surveillance System shall be designed, implemented, and maintained with comprehensive security controls to ensure confidentiality, integrity, availability, and protection of all system components, applications, databases, networks, servers, storage systems, CCTV cameras, edge devices, and communication infrastructure. The system shall be protected against both internal and external cyber threats, unauthorized access, malware, ransomware, denial-of-service attacks, data breaches, and other security vulnerabilities.

The Security Management framework shall include, but not be limited to, the following:

- The entire system shall be protected through enterprise-grade anti-virus, anti-malware, anti-ransomware, endpoint protection, and security monitoring solutions.
- Appropriate security policies, access controls, authentication mechanisms, audit trails, logging, monitoring, and incident response procedures shall be implemented and enforced throughout the contract period.
- All communication between system components shall be secured using industry-standard encryption protocols.
- Role-Based Access Control (RBAC) shall be implemented to ensure access is granted only to authorized users based on their roles and responsibilities.
- The system shall support secure password policies, account lockout mechanisms, audit logging, and periodic security reviews.
- All servers, storage systems, databases, operating systems, applications, and network devices shall be hardened as per industry best practices and OEM recommendations.
- The bidder shall ensure timely deployment of security patches, firmware updates, and vulnerability remediation throughout the contract period.
- Processing, recording, analytics, storage, and management of CCTV video feeds shall be hosted only within the infrastructure deployed under this project. No CCTV video stream, metadata, analytics data, or recording shall be processed or stored on any public cloud platform or on servers located outside Vadodara City without prior written approval of VMC.
- The bidder shall maintain detailed security logs and audit trails for all user activities, system events, and administrative actions.

PTZ Camera for Surveillance 5 MP

Parameter	Technical Requirement / Specification	Compliance (YES/NO)
STQC / BIS-ER Status	Mandatory. Camera model must hold a valid MeitY IoTSCS certification as verified on the STQC portal, and/or BIS-ER (Enforcement of Registration) compliance. Test/registration certificate to be enclosed with the technical bid.	
Image Sensor	Minimum 1/2.8" Progressive Scan CMOS (or better).	
Effective Resolution	Minimum 5 Megapixel (2592 x 1944) effective physical pixel array.	
Optical / Digital Zoom	Minimum 30x optical zoom with auto-focus; minimum 12x additional digital zoom.	
Focal Length	5.0 mm to 50.0 mm Motorized Varifocal Zoom; Auto-Iris / P-Iris; Remote Zoom/Focus. +- 10% allowed	
Video Compression	H.265+, H.265, H.264 with multi-stream (minimum triple stream, or 4+ where supported) capability.	
Minimum Illumination	Color: 0.01 lux or better; B/W: 0 lux with integrated Smart IR.	
Wide Dynamic Range	True 120dB WDR (minimum).	
Night Vision / IR Range	Minimum 150 meters Smart IR, auto-adjustable as per zoom level.	
Pan / Tilt Range	Pan: 0°-360° continuous (endless) rotation. Tilt: minimum -10° to +90°, with auto-flip.	
Preset Positions	Minimum 256 presets, with auto-patrol/tour and auto-scan/pattern-scan functions.	
Frame Rate	Full 5MP resolution at minimum 30 fps.	
Edge AI / Video Analytics	Built-in edge-based analytics covering, at minimum: AI Auto-Tracking, Face Detection, Line Crossing Detection, Intrusion/Perimeter Detection, and Object/Vehicle Classification.	

Audio / Alarm I/O	Audio input/output; minimum 2 alarm inputs and 1 alarm output port.	
Network Protocols	IPv4/IPv6, HTTP/HTTPS, TCP/IP, UDP, RTP/RTSP, DHCP, NTP, IGMP (multicast).	
Interoperability	ONVIF Profile S, G, T, and M compliant, for integration with multi-OEM Smart City VMS platforms.	
Cyber Security	HTTPS/TLS, IEEE 802.1X authentication, IP address filtering, digest authentication, complex password enforcement, and secure boot/firmware signing where available.	
Physical Housing	Metal/aluminium housing; IP66/IP67 weatherproof rated.	
Operating Temperature	-30°C to +60°C operating window (wider range acceptable), with built-in wiper/heater/fan as required for site condition.	
Power Input	PoE+ (IEEE 802.3at) as primary; 24V AC/12V DC as fallback, with integrated surge protection.	
Local Storage	MicroSD / MicroSDHC / MicroSDXC card slot supporting up to 512 GB, supplied with minimum 512GB card for edge storage/fall-back recording.	
Warranty	Minimum 5 years comprehensive OEM warranty from the date of GoLive.	

Blade Chassis Enclosure:

Features	Description	Compliance (YES/NO)
Form Factor	Chassis should be able to house at least 12 or more half height dual-socket blade servers in a single chassis enclosure. Chassis should support blades in the same enclosure, occupying a max of 10U rack height.	
Chassis Support	Chassis should house half height dual-socket blade servers	
Processor Support	Chassis should support latest generation Intel processor based blade servers	
Connectivity	Chassis should support housing of FCoE, Ethernet, FC and SAS interconnect fabrics offering redundancy as a feature. Also should support network switch/module with 50Gb downlinks and 100G uplink to DC switch.	
	Redundant Interconnect modules shall be integrated within the chassis such that uplinks from the chassis can be directly connected to core LAN switches	
	Interconnect should support 50Gbps downlinks to the Blades in redundancy supporting carving of each port into at least eight ports without use of any additional virtualization software.	
	Should support aggregation of multiple enclosures to consolidate data center network connections, reduce hardware and to scale network bandwidth across multiple enclosures. Atleast 36 servers should be supported per aggregation. Layer 2 network traffic should be switched within enclosure aggregation (without using top of the rack switch)	
	When multiple chasssis aggregated, switching latency between enclosures should not exceed 1.0 micro second for Ethernet	
	Should support Multi-module link aggregation (MLAG) for resiliency against interconnect failure	
	Chassis offered with redundant Switches/ Modules, Server to Server communication should be in 1:1 non-blocking across all aggregated enclosures.	
Ethernet Module	Chassis should support at least 6 x 100G external uplink ports uplink able to the Top-of-Rack Switches / Core Switches.	
I/O Path for all Fabrics	Chassis should support six interconnect bays to be configured in 3+3 redundancy	
Midplane and Backplane	Chassis direct- connect midplane should deliver at least 16 Tb/s of bandwidth	
Management Modules, Utilities and Software	Management/controlling softwares have to be from the Blade Chassis OEM itself. 3rd party not allowed.	
	Solution should configured with redundant physical management appliances each with 8 cores, 16Gb Memory and min. 400GB NVMe storage to provision required compute, storage and server side network, OS as single package within an enclosure or on multiple connected enclosures with failover and high-availability	
	Should support activity, Health and Power LEDs for immediate status	

	Should support software-defined intelligence and auto-discovery for configuring profiles to provision compute, storage, fabrics and images within an enclosure or on multiple connected enclosures.	
	Should support Firmware and OS Driver updates for the servers using profile templates to monitor, flag, and remediate	
	Should offer collaborative user interface which support logical resources to physical resources mapping, Smart Search, Activity Log, HTML5 mobile access, and Customizable Dashboard	
	Should provide a dedicated management network for multi-enclosure communications, separate from data plane. Management port must support both 1G and 10G.	
	Should support frictionless Firmware and OS Driver updates using profile templates to 'monitor, flag, and remediate	
	Should support reporting capabilities for 1) asset and inventory information for the devices in the enclosures 2) thermal and power information, including real-time actual power usage per server and per enclosure Reports should be exportable to csv or Microsoft Excel format	
Power Modules	The enclosure should be populated fully with power supplies. Power supplies should support N+N redundancy configuration, where N is greater than 1. Chassis should offer a single phase power subsystem enabled with technologies for lower power consumption and offering Platinum energy efficiency. OEM should provide documents certifying the claims. Power supplies supplied should be such that this can accommodate all blade servers with maximum configuration and also be able to accommodate future blade servers.	
Cooling	Each blade enclosure should have a cooling subsystem consisting of redundant hot pluggable fans or blowers enabled with technologies for improved power consumption and acoustics	
CD/Diskette/USB	Chassis should provide display port and USB port to connect Laptop/Monitor locally	

Blade Server and virtualization License:

Parameter	Description	Compliance (YES/NO)
Processor	Latest series/ generation of 64 bit x86 processor(s) with 8 or higher Cores Processor speed should be minimum 3.2 GHz. Minimum 2 processors per each physical server	
RAM	Minimum 32 GB ECC DDR5 Memory per upgradable up to 4 TB capable of operating at 5600 MT/s	
Internal Storage	2 x 480 GB NVMe SSD hot swap disks	
Network interface	2 X 20GbE LAN ports for providing Ethernet connectivity Optional: 1 X Dual-port 16Gbps FC HBA (or FCoE) for providing FC connectivity. If bidder is offering FCoE based solution, corresponding ports must be present in server as well as storage controllers.	
RAID support	As per requirement/solution	
Operating System	Licensed version of 64 bit latest version of Linux/Unix/Microsoft® Windows based Operating system)	
Form Factor	Blade	
Virtualization	Shall support Industry standard virtualization hypervisor like Hyper-V, VMWARE, Morpheus, Oracle VM etc. OEM of the blade chassis and servers offered should in "Validated Configuration" list and certified by OEM to run virtualization.	

Storage: Fit in enclosure

Features	Description	Compliance (YES/NO)
Solution/ Type	• Bidder shall provide Unified storage solution (or a Combination of NAS/Scale-out NAS/SAN) supporting all required protocols (IP Based/FC/iSCSI/FC/NFS/CIFS etc.) for the offered storage solution, meeting benchmark performance parameters specified in SLA • Solution proposed should yield low cost per TB, while meeting the	

	performance parameters • Licenses for the actual protocols used in the storage solution must be provided from day 1. • If bidder is offering FCoE based solution, corresponding ports must be present in server as well as storage controller.	
Primary and secondary Storage	• Storage Capacity should be minimum 900 TB (usable, after configuring in offered RAID configuration) • RAID solution offered must protect against double disc failure. • Disks should be preferably minimum of 1.2 TB capacity for SAS and 14 TB or higher for SATA (combination as per performance and SLA requirements of overall solution) • To store all types of data (Data, Voice, Images, Video, etc.) • Proposed Storage System should be scalable (vertically/horizontally)	
Hardware Platform	• Rack mounted form-factor • Modular design to support controllers and disk drives expansion	
Controllers	• At least 2 Controllers in active/active mode • The controllers / Storage nodes should be upgradable seamlessly, without any disruptions / downtime to production workflow for performance, capacity enhancement and software / firmware upgrades.	
RAID support	• Should support various RAID Levels	
Cache	• Minimum 64 GB of useable cache across all controllers. If cache is provided in any additional hardware for the storage solution, then cache must be over and above 64 GB.	
Redundancy and High Availability	• The Storage System should be able to protect the data against single point of failure with respect to hard disks, connectivity interfaces, fans and power supplies	
Management software	• All the necessary software (GUI Based) to configure and manage the storage space, RAID configuration, logical drives allocation, snapshots etc. are to be provided for the entire system proposed. • Licenses for the storage management software should include disc capacity/count of the complete solution and any additional disks to be plugged in in the future, up to max capacity of the existing controller/units. • A single command console for entire storage system. • Should also include storage performance monitoring and management software • Should provide the functionality of proactive monitoring of Disk drive and Storage system for all possible disk failures • Should be able to take "snapshots" of the stored data to another logical drive for backup purposes	
Data Protection	The storage array must have complete cache protection mechanism either by de-staging data to disk or providing complete cache data protection with battery backup for up to 4 hours	

48 (12 X 4) core Optical Fiber Cable (OFC): (FIBER LIFE 25 YEARS or Higher)

Parameters	Minimum Specifications	Compliance (YES/NO)
Maximum Cabled Attenuation db/Km Fibre	1310nm:0.38 and 1550nm:0.25	
Tensile Strength	2200N	
Fibre Polarization Dispersion (PMD) Mode	≤0.4 ps/√km	
Impact Strength	25Nm	
Operating Temperature	minc 20^0 C to 70^0 C	
Color Coding of Tubes and Fibres	EIA/TIA-598	
Inner Jacket thickness	≥1.8mm	

Outer Jacket Thickness	≥0.65mm	
Water Tightness	EIA/TIA-455-81B	
Minimum Continuous Length	2km±10%,	
Cable Design Life	More than 25 Years	
Others	The optical Fibre cable shall be made of Germanium doped silica glass or pure silica glass.	
	The mode field eccentricity shall be less than 1 µm	
	The cladding of the Optical Fibre shall be made of silica glass having lower refractive index. The outside diameter of the cladded Fibre shall be 125 µm with tolerance of ± 2.0 µm.	
	The non-circularity of cladding surface shall be 2%, maximum.	
	The nominal Fibre coating diameter shall be in the range of 245 to 400 microns.	
	Maximum continuous operating temperature without optical degradation shall be 65°C.	
	Maximum optical loss variation at temperature	
	range of -5°C to +70°C shall	
	be ±0.05dB/km.	
	Water swellable yarns shall be	
	added to prevent water	
	ingress in the core of cable	
	The loose tubes carrying the Fibre cores shall be made of thermoplastic or equivalent material which will kink during normal operation of the cable including laying or blowing of cable.	
	Armour shall be made of corrugated steel tape for 48Core OFC for stronger protection.	
	No steel armour to be used for the 48 core OFC for faster splice joint to reduce downtime.	
	Outer sheathing shall be made of UV proof black MDPE/HDPE. The sheath shall have smooth finish and shall be termite resistant.	
	All the OFC shall be clearly marked at intervals of 1 meter with the following data which is not less than 5 mm high. The details of marking on cable shall be approved by CLIENT before	
	commencement of manufacturing.	
	· Name of Client	
	· No of Fibres (48Core)	
	· Type of OFC	
	· Manufacture's name or trade mark	
	· Year of manufacturing	
	· Running length marking	
	All optical Fibre cable shall be supplied on strong wooden drums provided with lagging with adequate strength, constructed to protect the cabling against all damage and displacement during transit, storage and subsequent handling during installation	

The Bidder shall submit route maps, fiber core utilization details, OTDR test reports, and as-built drawings for verification and approval by VMC prior to billing.

High Density Polyethylene Pipe (HDPE) 40 mm:

Parameters	Minimum Specifications	Compliance (YES/NO)
HDPE pipe Material	High Density Polyethylene Pipe with inner layer of solid permanent lubricant	
Outside Diameter	40mm+0.4mm	
Wall Thickness	3.1mm ±0.4mm	
Standard Length	1000meters±50m	
Thickness of Permanent Lubricant	≥0.4mm	
Visual	Smooth inside and outside surface free from blisters, shrink hole scratches and roughness	
Ovality	Max. 1.4mm	
Operating	0°C to 70°C	

Temperature		
Underground Life Expectancy	≥25 Years	
	Suitable ultra violet stabilizers may be used for manufacture of the PLB HDPE pipe to protect against UV degradation when stored in open for a minimum period of 8 months.	
	In the inner layer of PLB HDPE	
	pipe, the friction reducing, polymeric material to be used	
	as the inner layer lubrication material shall be integral with HDPE layer. The lubricant materials shall have no toxic or skin hazards for safe handling	
	The finished pipe shall be of good workmanship such that the pipe is free from blisters, shrink holes, flaking, chips, scratches, roughness, break and other defects. The pipe shall be smooth, clean and in round shape, without eccentricity.	
	The ends shall be cleanly cut and shall be square with axis of the pipe.	
	HDPE pipe shall be supplied in circular coils of 1000m length with End caps fitted with both ends of pipe to prevent the entry of any unwanted elements such as dirt, water, moisture, insects/rodents during transportation and storage.	
	All the pipes shall be clearly marked at intervals of 1 meter with the following data which is not less than 5 mm high. The details of marking on pipe shall be approved by Client before commencement of manufacturing. · Name of Client with logo · Manufacture's name or trade mark · Year of manufacturing · Type of HDPE pipe and size · Running length marking	

40 mm HDPE Coupler:

Parameters	Minimum Specifications	Compliance (YES/NO)
Type of Coupler	Push-Fit type 40mm Dia. Coupler	
Pulling Force	≥330 kgf	
Construction Material	HDPE	
	The 40 mm HDPE Coupler shall be able to provide a durable airtight and watertight joint between two pipes without deteriorating the strength of the pipes.	

Edge Level Industrial Grade Switch (For Cameras & other edge device)

Parameter	Specifications	Compliance (YES/NO)
Total Ports	12 (8 PoE + 4 SFP)	
PoE Copper Ports	8x 10/100/1000 Mbps	
SFP Fiber Uplinks	4x Gigabit SFP with SFP modules	
PoE Standard	IEEE 802.3af / 802.3at	
PoE Power per Port	30W full-load	
Total PoE Budget	240W	
PoE Management	Centralized PoE mgmt, PD detection	
Ring Redundancy Protocol	Proprietary self-healing ring/Turbo Ring & Turbo Chain	
Node Self-healing Time	< 50ms	
Ring Self-healing Time	< 50ms	
Standard Protocols	STP / EAPS / ERPS or RSTP / STP / MSTP	
Management Interface	CLI / Web / SNMP	
VLAN Support	Port-based / Tag-based / QinQ / 802.1Q or Port-based / 802.1Q / GVRP	

ACL	Yes	
QoS	Yes	
IGMP Snooping	Yes	
Jumbo Frames	9K	
MAC Table Size	8K	
IPv6 Support	Yes	
Config Backup / Restore	Auto backup/Restore	
Secure Management	HTTPS / SSH / SNMPv3	
EtherNet/IP	Yes	
Modbus TCP	Yes	
IP Rating	IP30 or Higher	
Lightning Protection	Yes	
Operating Temp (Standard)	Minimum -10°C to 60°C	
EMC / Vibration / Shock	Yes (stated)	
Mounting	DIN-rail	
Dual Power Input	Yes	

Network Connectivity Scope of work

Network connectivity shall be one of the critical components of the City Surveillance System and shall be designed, supplied, implemented, tested, commissioned, operated, and maintained by the bidder. The bidder shall conduct detailed site surveys and network planning to ensure seamless end-to-end connectivity between field devices, edge devices, aggregation points, network infrastructure, and the Central Command & Control Center (CCC).

The network infrastructure shall be designed to provide reliable, secure, scalable, and high-availability communication for CCTV cameras, video analytics, edge devices, switches, servers, storage systems, video walls, public address systems, emergency call boxes, sensors, and all associated field equipment. The bidder shall ensure that the network meets the prescribed Service Level Agreements (SLAs) related to availability, latency, jitter, packet loss, throughput, and overall network performance throughout the contract period.

The complete network architecture from the Data Center/CCC up to the field level shall support Software Defined Networking (SDN) or equivalent technology to enable centralized management, traffic engineering, network segmentation, security policy enforcement, automated provisioning, and future scalability.

The bidder shall be responsible for provisioning all required network components, including fiber optic connectivity, switches, routers, media converters, network management software, accessories, and associated passive infrastructure required for successful implementation and operation of the project. The network design shall ensure redundancy at critical locations to minimize service disruption due to hardware, fiber, or communication failures.

Network Monitoring System

The bidder shall provide, install, configure, commission, and maintain an Enterprise-Class Network Monitoring System (NMS) for centralized monitoring and management of the complete surveillance network infrastructure. The proposed solution shall provide fault management, performance management, configuration monitoring, event correlation, reporting, alerting, and SLA monitoring capabilities.

The Network Monitoring System shall support monitoring and management of all IP/SNMP-enabled devices deployed under the project, including but not limited to CCTV Cameras, Edge Devices, Industrial Switches, Routers, Firewalls, Servers, Storage Systems, Video Management Systems (VMS), Video Walls, Public Address Systems, Emergency Call Boxes, Sensors, UPS Systems, and other network-connected equipment.

The Network Monitoring System shall provide the following minimum functionalities:

a. The system shall automatically discover all manageable network devices and infrastructure elements connected to the network and maintain an updated inventory database. The solution shall provide a centralized monitoring dashboard with graphical network topology maps displaying

connectivity, device status, and network health.

b. The solution shall support configurable dashboards and customizable reporting tools for generation of operational, performance, availability, utilization, fault, inventory, and SLA reports.
c. The system shall continuously monitor network bandwidth utilization, link health, packet loss, latency, jitter, uptime, and device availability and shall generate reports for user-defined periods.
d. The system shall provide intelligent fault management through event correlation, root cause analysis, dependency mapping, and policy-based fault detection mechanisms to identify infrastructure failures, communication issues, configuration changes, and potential cyber-security incidents.
e. The system shall generate real-time alerts and notifications through Dashboard, Email, SMS, Mobile Application, or other approved notification mechanisms whenever any critical event, fault, device failure, communication loss, security incident, threshold breach, or configuration change is detected.
f. The system shall maintain audit logs of all configuration changes, user activities, network events, and administrative actions for compliance and troubleshooting purposes.
g. The system shall support SLA monitoring and reporting for all critical infrastructure components and generate automated monthly availability and performance reports.
h. The solution shall support northbound integration with the proposed ICCC platform, Video Management System (VMS), Helpdesk/Ticketing System, and other third-party systems through standard APIs and protocols.
i. The Network Monitoring System shall support role-based access control (RBAC), secure authentication, encrypted communication, and audit logging in accordance with prevailing cyber security guidelines and industry best practices.
j. All required software licenses, dashboards, reports, integrations, database licenses, upgrades, patches, and support services for the Network Monitoring System shall be included by the bidder for the entire contract period.

Non-IT and Civil Infrastructure Specifications & Guidelines

1. Public Works Guidelines

The Bidder/System Integrator (SI) shall be responsible for obtaining all necessary permissions and carrying out all civil, electrical, and public infrastructure works required under the project. The SI shall comply with all applicable regulations, standards, and instructions issued by Vadodara Municipal Corporation (VMC), Vadodara City Police, Road & Building Department, Utility Agencies, and other statutory authorities.

The scope of public works shall include, but not be limited to, excavation, trenching, road cutting, pole foundation construction, pole erection, junction box installation, cable laying, OFC laying, restoration work, wiring, barricading, fencing, signage, labeling, safety markings, and all associated civil and electrical works. The SI shall restore all roads, footpaths, pavements, and public properties to their original condition after completion of work.

2. PVC Conduit and Cable Protection

All conduits, ducts, HDPE pipes, PVC pipes, cable protection systems, and associated accessories shall conform to relevant BIS/ISI standards and industry best practices.

PVC conduits used for power and communication cabling shall be heavy-duty, UV-resistant, fire-retardant, corrosion-resistant, and suitable for outdoor installations. All materials shall carry valid ISI/BIS certification wherever applicable.

The SI shall ensure adequate protection of all cables against mechanical damage, water ingress, rodents, environmental exposure, and unauthorized access.

3. Earthing and Lightning Protection

The SI shall provide complete earthling and bonding arrangements for all electrical and electronic equipment deployed under the project, including CCTV cameras, poles, junction boxes, switches, UPS systems, cabinets, servers, storage systems, and communication equipment.

All earthing works shall comply with the latest version of IS 3043 (Code of Practice for Earthing) and other applicable Indian Standards.

The SI shall ensure that all metallic structures and equipment likely to become energized are effectively earthed. Surge protection devices and lightning protection mechanisms shall be provided wherever required to safeguard field and central infrastructure.

No separate payment shall be made for earthing and related accessories, and the same shall be deemed included in the quoted price.

4. Passive Cabling and Structured Cabling

The SI shall provide complete passive cabling infrastructure required for successful operation of the project.

All structured cabling shall comply with applicable ISO/IEC, TIA/EIA, and BIS standards. The cabling solution shall support data, video, voice, network management, and future expansion requirements.

The SI shall use suitable categories of network cables, fiber optic cables, patch panels, connectors, modular sockets, termination hardware, cable trays, ducts, and accessories as per project requirements.

All cables shall be properly routed, labeled, terminated, documented, and tested. Cable layouts shall be neat, organized, and easily maintainable. Adequate spare capacity shall be provided for future scalability and expansion.

5. Electrical Wiring Guidelines

All electrical wiring shall be carried out using ISI-marked copper conductor cables conforming to relevant BIS and IEC standards.

The wiring system shall comply with the following minimum requirements:

- PVC insulated stranded copper conductor cables of appropriate rating shall be used for all electrical installations.
- Looping system of wiring shall be adopted. Direct twisting or unauthorized joints in conductors shall not be permitted.
- All cables, conductors, and circuits shall be properly identified using ferrules, labels, tags, and cable markers at both ends.
- Power cables, communication cables, and fiber optic cables shall be routed separately to avoid electromagnetic interference and facilitate maintenance.
- Circuits originating from different power sources, distribution boards, or phases shall not be routed through the same conduit unless specifically approved.
- All cable terminations and splicing shall be carried out using approved terminal blocks, connectors, glands, and accessories. Twisted wire joints shall not be permitted.
- All sockets, switches, distribution boards, junction boxes, and enclosures shall be industrial-grade and suitable for outdoor deployment wherever applicable.
- Proper phase balancing shall be ensured for three-phase electrical installations.
- All electrical installations shall comply with applicable Indian Electricity Rules, Central Electricity Authority (CEA) Regulations, National Electrical Code (NEC), BIS Standards, and IEC Standards.
- The SI shall be responsible for supply, installation, testing, commissioning, documentation, certification, and maintenance of all electrical infrastructure deployed under the project.

6. Quality, Safety and Compliance

- The SI shall follow all applicable safety regulations during execution of work. Necessary barricading, caution boards, warning signs, traffic diversion arrangements, personal protective equipment (PPE), and worksite safety measures shall be provided throughout the project duration.

All materials supplied under the project shall be new, genuine, and sourced from authorized manufacturers. The SI shall submit test certificates, compliance certificates, warranty certificates, and inspection reports whenever requested by VMC.

All civil, electrical, and non-IT infrastructure works shall be executed in a professional manner and shall comply with applicable BIS, IEC, IS, IEEE, TIA/EIA, ISO, and other relevant national and international standards.

- **Pole, Foundation & Structural Stability Certificate**

The Successful Bidder shall design, supply, install, and certify all CCTV poles, mounting structures, brackets, and foundations for structural adequacy and stability in accordance with the applicable Indian Standards and

good engineering practices. A Structural Stability Certificate, duly signed and stamped by a Chartered/Qualified Structural Engineer, shall be submitted to VMC after installation and before Go-Live.
The Structural Stability Certificate shall remain valid for the entire five (5) year Contract Period after Go-Live. In case the certificate issued by the Structural Engineer is valid for a shorter period (e.g., three (3) years), the Successful Bidder shall, at its own cost, obtain and submit a renewed/revalidated Structural Stability Certificate before expiry of the earlier certificate so that the structural stability certification remains valid continuously throughout the entire Contract Period. Any failure, tilting, settlement, corrosion, structural damage, or instability of the pole or foundation shall be rectified or replaced by the Successful Bidder at no additional cost to VMC.

- **Standardized Signage for CCTV Surveillance Locations**

In order to ensure transparency, public awareness, and compliance with applicable privacy and surveillance regulations, the Bidder/System Integrator (SI) shall provide, install, and maintain standardized CCTV surveillance signage at all camera locations covered under the project.
The purpose of the signage shall be to clearly inform citizens, visitors, and stakeholders that the area is under video surveillance and that activities within the coverage area may be monitored and recorded for public safety, security, traffic management, law enforcement, and incident investigation purposes.
The signage shall be prominently displayed at visible locations near CCTV installations, entry points, public access areas, intersections, public buildings, parks, gardens, swimming pools, town halls, crematoriums, municipal offices, sensitive locations, and other surveillance zones identified by VMC. The location, size, design, and placement of signage shall ensure maximum visibility to the public without obstructing pedestrian or vehicular movement.
The SI shall design, manufacture, supply, install, and maintain weatherproof, UV-resistant, vandal-resistant, and corrosion-resistant signage suitable for outdoor environmental conditions. The signage shall include appropriate pictorial symbols indicating CCTV surveillance along with messages in Gujarati, Hindi, and English languages as approved by VMC.
The signage shall contain, at a minimum, the following information:
"YOU ARE UNDER CCTV SURVEILLANCE"
"This area is monitored and recorded through CCTV Cameras for public safety, security, traffic management, and law enforcement purposes."
"Vadodara Municipal Corporation (VMC)"
The design, content, dimensions, material specifications, language, color scheme, reflectivity requirements, and installation methodology shall be submitted by the bidder for approval of VMC before implementation.
The bidder shall be responsible for replacement of damaged, faded, stolen, defaced, or missing signboards throughout the warranty and maintenance period at no additional cost to VMC.
All costs related to design, manufacturing, transportation, installation, maintenance, replacement, fixtures, clamps, mounting arrangements, permissions, and associated works shall be deemed included in the bidder's quoted price and no separate payment shall be made by VMC.

CCTV Signage

Item	Specifications	Compliance (YES/NO)
Size	Board Width = 8" / 12" (For type A and B) Board Width = 12" / 18" / 24" (For type C and D)	
Plate Material	Corrosion resistant Aluminum Alloy as per IRC 67:2001 (Code of Practice for Road signs)	
Plate Thickness	Minimum 1.5 mm	
Retro-Reflective sheeting for sign-plate	Weather-resistant, having color fastness	
Other Specifications	As per IRC 67:2001 (Code of Practice for Road signs)	
Mounting	Can be mounted on wall or pole (appropriate mounting brackets to be provided)	

Design	As per following signage diagrams	

UPS -1 KVA Online UPS System

Characteristic	Specification	Compliance (YES/NO)
Capacity	1KVA,800 Watt (1Øinput/ 1Ø output)	
Topology	True Online Double Conversion UPS	
Technology	IGBT Based	
INPUT		
Input Facility	1Ø, 2-wire & Ground, (Phase & Nuetral + Ground)	
Input Voltage Range	1Ph,L-N+PE (175 to 270 Vac on full load) (120 to 280 Vac on 50% load)	
Input Frequency Range	45 ~55Hz	
Input Power Factor	≥0.99	
Total Harmonic Distortion Current (THDi)	≤5 % on Full Linear Load	
OUTPUT		
Power Factor	0.8	
Nominal Output Voltage	220/230/240Vac shall be available with +/-1% regulation in battery mode	
Voltage regulation	± 1% static, ± 5% for Dynamic Load	
Nominal Frequency	50Hz±2% (Synrchonized to Mains)	
Waveform	Sine wave	
Crest factor	3:1	
Total harmonic distortion (THDv)	< 3% (linear load); <5% (nonlinear load)	
Output Short Circuit Protection	Electronic, trip	
Transfer time	Mains mode to battery mode: 0 ms; Inverter mode to bypass mode: < 4ms	
Maintenance Bypass	Automatic & Manual	
Output Connection	Terminal at bottom or Indian Socket	
DC Ripple factor	≤1%	
Efficiency	Overall (AC to AC) : ≥87%	
Overload Protection	>100% ~ 110%: for 5 mins;	
	>110% ~ 125%: for 1 min;	
SPD		
Technology	L-N: MOV Technology & N-PE: GDT (Gas Discharge Tube) Technology	
Plug ability	Pluggable per pole for Safe and Easy Replacement / Maintenance	
Nominal Voltage Un (Vn)	240/415 V AC (TN-S) / (TT)	
Nominal Discharge Current In (8 /20µs) (In)	20 KA	
Maximum Discharge Current Imax (8 / 20µs) (Imax)	40 KA	
Short Circuit Current rating ISCCR (Isccr)	25KArms	
Voltage Protection Level Up (Vp)	≤ 1.5 kV	
Temporary Over Voltage UT (L-N) (Vt)	415 V AC (5s / Withstand Mode) 440V (120min–safe failure mode)	
Temporary Over Voltage UT (N- PE) (Vt)	1200 V AC (200ms / Withstand Mode)	
Max. lead length in-between active conductors to SPD to Earth bus bar in a panel	≤0.5 meter	
BATTERY & BATTERY BANK CHARGER		
DC BUS voltage	36VDC	
Battery Configuration	Minimum 936 VAH for 30mins of Backup on system load	

Batteries Type	SMF BATTERY	
Battery re-charge time (After complete discharge) to 90% capacity	8-10 Hours , UPS Charger should be 10% of the Battery Ah Capacity	
Battery Housing	Should be compact and space saving MS Steel Open Racks complete with interconnectors	
Interfaces	Should support DOS/UNIX/WINDOWS OS, RS 232 Connector should be available SNMP should be provided for remote Monitoring of UPS Health	
Cold Start	should be provided	
Environmental		
Operating Temperature	0-50 deg C with full load Elevated Burn in Test Report from NABL test house to be submitted along with Bid	
Type of Cooling	Forced Air	
Noise Level	<55 dBA at 1 meter distance	
Auto Start, Cold Start & EPO	Shall be available	
Humidity	0%-95% non-condensing	
INGRESS PROTECTION	IP 20	
MEASUREMENTS (ON LCD)	Input & Output Voltage, Input & Output Frequency, Bypass: Voltage & Frequency, Battery Level Indicator, Load Level indicator, Fault codes, , Overload, Short circuit, Low Battery.	
CERTIFICATIONS		
BIS	BIS Certificate to be submitted for the quoted Model	
CERTIFICATIONS	ISO 9001:2015(Quality Management System), ISO14001:2015(Environmental Management System) , ISO45001:2018 (Occupational Health & Safety Management System) , ISO 20000-1:2018(Information Technology Service Management System), ISO 27001:2022(Information Security Management System), ISO50001:2018 (Energy Management System), IEC /EN 62040-1-1 (Safety) IEC/EN-50091-1-1 (General & Safety Requirements) , IEC/EN 60950 (Information Technology Equipment Safety) , IEC/EN-50091-2 (EMC Safety) IEC/EN 62040-2 (EMC Emission) IEC 62040-3 (Performance & Design) IEC 60146 (Design & Manufacture) IEC/EN 61000-4-1 level4 IEC/EN 61000-4-2 level4 IEC/EN 61000-4-3 level4 IEC/EN 61000-4-4 level4 IEC/EN 61000-4-5 level4 IEC/EN 61000-4-6 level4	
Test Certificates	Test Report of UPS should be enclosed conducted from reputed NABL accredited Laboratory for the quoted model complying all technical parameters conducted within last 5 years from the date of opening of Bid.	
SUPPORT		
Web/Window base Support	It is mandatory to provide online centralized call tracking system Like OVSD (Open view Service Desk) for monitoring of logged calls viewing through web / windows based access for at least 2 officials.	
UPS OEM Eligibility Criteria		
FACTORY License	UPS OEM should have their Factory License Certificate & the same has to be submitted during bid submission	
Previous Experience: UPS OEM should have successfully deployed in a single order, minimum 5000 nos. of online UPS System in any Govt. or PSU organization in PAN India	Purchase order copies for the above Purchase(s)/Work(s) shall be submitted by the bidder.	

during last 7 years.		
OEM should have Minimum Turnover of Rs.275 Crores in last three Financial Year. OEM should have earned Profit during the last 3 Financial Year & It should have Positive Net Worth during last 3 Financial Year	Audited Balance Sheet of Last 3 Years along with Auditors Report to be enclosed.	
UPS OEM should have registered office of in the state of Execution & Engineers Posted in the Pan India.	Documentary evidence of OEM's own registered office in the state of Execution should be submitted in form of GST Certificate. 'Documentary evidence of Engineers in Company's payroll to be submitted (ESI/PF List has to be submitted Along with P TAX Challan of current year.)	

Surge Protection Power Strips

Parameter	Minimum Specifications	Compliance (YES/NO)
Clamping Voltage	240 V	
Response time	<10 ns	
Energy Dissipation	Min 500 joules	
Max voltage Spike protection	Up to 400 Volts	
Max current Spike Protection	5000 Amps	
Cable	Heavy duty Cable of standard Length	
Power Outlet	Min. 4 Nos. 5 Amp Indian Standard Power Outlets	
Operating Temperature	0-55 ° C .The bidder has to take into consideration of the internal temperature adjustment of the junction box	

Earthing

Minimum Specifications
All electrical components are to be earthen by connecting two earth tapes from the frame of the component ring and will be connected via several earth electrodes. The cable arm will be earthen through the cable glands. Earthling shall be in conformity with provision of rules 32, 61, 62, 67 & 68 of Indian Electricity rules 1956 and as per IS-3043.
Earthing should be done for the entire power system and provisioning should be there to earth UPS systems, Power distribution units etc. so as to avoid a ground differential. Authority shall provide the necessary space required to prepare the earthing pits.
All metallic objects on the premises that are likely to be energized by electric currents should be effectively grounded.
The connection to the earth or the electrode system should have sufficient low resistance in the range of 0 to 25 ohm to ensure prompt operation of respective protective devices in event of a ground fault, to provide the required safety from an electric shock to personnel & protect the equipment from voltage gradients which are likely to damage the equipment.
Recommended levels for equipment grounding conductors should have very low impedance level less than 0.25 ohm.
The Earth resistance shall be automatically measured on an online basis at a pre-configured interval and corrective action should be initiated based on the observation. The automatic Earthing measurements should be available on the UPS panel itself in the UPS.
There should be enough space between data and power cabling and there should not be any cross wiring of the two, in order to avoid any interference, or corruption of data.
The earth connections shall be properly made .A small copper loop to bridge the top cover of the transformer and the tank shall be provided to avoid earth fault current passing through fastened bolts, when there is a lighting surge, high voltage surge or failure of bushings.

Poles for Mounting Cameras

Parameter	Minimum Technical Specification	Compliance (YES/NO)
Pole Type & Material	MS tubular/octagonal pole fabricated as per IS 2713:1980. Hot Dip Galvanized (HDG) after fabrication as per IS 4759:1996. Minimum zinc coating thickness: 86 microns (610 g/m²) as per IS 2629:1985. Metallic/silver finish of HDG is the final surface — no additional silver coating required.	

Pole Height	Height as per site and camera requirement: • Standard road/footpath: 5–6 m above FGL • Intersections/wide area: 7–9 m above FGL Exact height confirmed location-wise in BoQ. Bidder to provide height vs location schedule.	
Pole Diameter & Wall Thickness	Minimum OD: 100 mm for poles up to 6 m; 114 mm OD for poles above 6 m. Minimum wall thickness: 3.6 mm (100 mm OD); 4.0 mm (114 mm OD) as per IS 1161. Structural design calculation to be submitted by bidder confirming adequacy for IS 875 Part 3 wind zone.	
Cantilever Arm	Min 4 No (4 Fix and one PTZ camera)- cantilever arm, min 2.0 m projection. Material: MS hollow section min 60 mm OD × 3 mm WT, HDG after fabrication. 5°–15° downward adjustable tilt. Cable entry at base. Additional arms provided at multi-camera locations.	
Base Plate	Minimum 300 mm × 300 mm × 12 mm thick MS base plate. Full penetration weld to pole bottom as per IS 816. Minimum 4 holes for J-bolt fixing. HDG along with pole.	
Mounting Provisions	(a) Camera bracket at cantilever tip — compatible with approved camera model. (b) JB mounting bracket on pole at 1.5–1.8 m above FGL. (c) Earthing lug (M8 HDG) on pole body near base. (d) Cable entry gland plate at pole base for underground cable entry. All brackets: MS, HDG, with SS fasteners.	
Internal Cable Routing	All power and data cables routed internally through hollow pole — fully concealed, no external wiring visible. MS cable entry box (100×100×75 mm, HDG) with IP-rated PVC cable glands at pole base. Underground routing: HDPE/DWC conduit min 50 mm dia from foundation to pole base. All cable openings sealed with weather-proof silicone sealant after termination.	
Alternative: Anchor Fastener Base	Where RCC foundation not feasible (heritage zone, paved road, rocky ground, underground utilities): Surface-mount anchor fastener base plate system. Base plate: min 500×500×10 mm MS HDG. Fixed with min 4 Nos. M16 chemical/mechanical anchors (Fischer or equivalent). Feasibility report + structural adequacy certificate from licensed Structural Engineer required before execution. Only where approved by Engineer-in-Charge.	
Wind Load Design	Pole structural design as per IS 875 (Part 3):2015 for applicable wind zone. Bidder to submit structural design calculation signed by licensed Structural Engineer with bid or before fabrication approval. Pole to withstand design wind speed without permanent deformation.	
Standards	HDG Recommended Practice, Hot Dip Zinc Coatings on Structural Steel, Steel Tubular Poles, Steel Tubes for Structural Purposes	

CCTV Pole Foundation Specification

Parameter	Minimum Technical Specification	Compliance (Yes/No)
Type of Construction	Reinforced Cement Concrete (RCC) foundation with J-Bolt / Anchor Bolt arrangement. Minimum M20 grade concrete (as per IS 456). Concrete mix to be machine-mixed and vibrated for proper compaction. No manual mixing shall be permitted for RCC foundation works.	
Minimum Height Above Ground	Minimum 200 mm above finished ground level (FGL). Above-ground pedestal portion to be chamfered at top edges and provided with smooth finish. Pedestal to be provided with drip mould at top to prevent water ingress along pole base. In flood-prone areas, height to be increased to minimum 300 mm.	
Foundation Size (L × W × D)	Minimum Foundation Size: 500 mm × 500 mm × 1200 mm (L × W × Depth). Actual foundation size and depth to be determined based on: (a) Pole height and design wind load calculation (b) Soil bearing capacity from site soil investigation. (c) Weight of pole, camera, and associated equipment Larger/deeper foundations shall be provided for poles in high wind zones, loose soil, or waterlogged areas. Structural design calculations.	
J-Bolt / Anchor Bolt	Minimum 4 Nos. J-Bolts or Anchor Bolts, conforming to the pole manufacturer's base plate design. Bolt specification: M24 (minimum) × 600 mm length, IS 1367	

Arrangement	Grade 8.8, Hot Dip Galvanized. Double nuts and washers (HDG) to be provided for each anchor bolt. Bolt template / jig to be used during concreting to ensure correct bolt positioning and alignment.	
Mounting Arrangement	Base Plate Mounting. Pole manufacturer's approved base plate (minimum 12 mm thick MS plate, hot dip galvanized) to be used. Base plate to be set level using precision levelling instruments (spirit level / optical level). Grouting: Provide non-shrink cement grout (min 50 mm thick) between base plate and pedestal top surface after pole plumbing.	
Reinforcement	Minimum 12 mm dia TMT bars (Fe 415/Fe 500) in both directions at 150 mm c/c spacing for pedestal. Longitudinal bars: minimum 4 Nos. 12 mm dia, with links/ties of 8 mm dia at 150 mm c/c. Minimum clear cover: 50 mm (below ground), 40 mm (above ground). Reinforcement cage to be properly tied and positioned before concreting.	
Excavation & Backfilling	Excavation to be done to required foundation depth plus 150 mm extra for PCC bed. PCC bed: minimum 75 mm thick, M10 grade at bottom of excavation. Backfilling: After curing, backfill with good quality earth in layers of 150 mm.	
Cable Conduit & Earthing	Minimum 2 Nos. GI/PVC conduit (minimum 50 mm dia) to be embedded vertically through foundation for cable routing (power + data). Conduits to be capped/sealed during concreting and opened after curing. Provide minimum 1 No. earthing pit adjacent to pole foundation, connected to pole structure with 25 mm × 3 mm GI flat. Earthing pit to comply. Earth resistance to be measured and recorded — maximum 5 Ohms. Earthing conductor to be bonded to reinforcement cage.	
Stability Report	Structural stability report from a licensed Structural Engineer to be submitted after foundation construction and before pole erection. Report to confirm: (a) Foundation dimensions as per design, (b) Concrete grade achieved, (c) Anchor bolt positioning and projection, (d) Earth resistance measurement.	
Drawing Submission	Pole Foundation General Arrangement (GA) Drawing and Structural Design Calculation to be submitted to Engineer-in-Charge and approved BEFORE commencement of any civil/foundation work. Drawing to include: foundation plan, section, reinforcement details, bolt arrangement, conduit layout, earthing details. As-built drawings to be submitted within 15 days of completion of each foundation.	

Junction Box

Parameter	Minimum Technical Specification	Compliance (YES/NO)
Size / Dimensions	Equipment to be housed: UPS/battery, Edge PoE+ Switch, MCB/PDU, terminations) Minimum internal dimensions: • Ground-mounted JB: 800 mm (H) × 600 mm (W) × 400 mm (D) Actual size confirmed after equipment layout drawing approval	
Cabinet Material	Body: Cold Rolled Steel (CRS) minimum or Stainless Steel SS304. Surface finish: Hot Dip Galvanized or minimum 60 micron epoxy powder coat (exterior). Interior: white powder coat. GI sheet (pre-galvanized) with additional powder coat is acceptable for cost economy.	
Sheet Thickness	Body panels: minimum 1.5 mm CRS/SS. Door: minimum 2.0 mm. Base plate: minimum 2.0 mm. Ground-mounted (higher impact risk): body 2.0 mm, door 2.5 mm.	
Locking Arrangement	Ground-mounted JB: 2 locks — (1) 3-point locking handle (top/mid/bottom) + (2) external padlock lug with 8 mm shackle padlock. Pole-mounted JB: 1 lock — quarter-turn or 3-point locking handle with padlock provision. Lock material: SS or zinc alloy. Provide 3 sets of keys per JB.	
IP & IK Rating	IP Rating: Minimum IP55 (dust protected + water jets). IP65 preferred for ground-mounted JBs in high-rainfall / coastal areas. IK Rating: Minimum IK08 (5 Joules impact). Internal temperature to be maintained within 0°C to 50°C under worst-case 50°C ambient.	
Mounting	(a) Pole-mounted: Clamp/U-bolt bracket (HDG MS, matching pole OD). Max loaded JB weight: 25 kg. (b) Ground-mounted: MS HDG base frame with min 4 anchor bolt holes, fixed to	

	concrete plinth per JB Foundation Spec.	
Internal Layout (DIN Rail / Rack)	• Min 2 Nos. 35 mm standard DIN rails for switches, MCBs, terminal blocks. • 16" / 19" rack provision (optional) for rack-mount UPS or patch panel. • Horizontal + vertical cable management channels. • Copper earthing bus bar (25 mm × 3 mm) inside JB, pre-drilled.	
Rain Canopy	1 No. rain canopy over JB door for all outdoor installations. Material: 1.5 mm MS HDG or powder-coated. Projection: min 150 mm beyond door face on top, left, and right sides. Bolted to JB body (not welded) for serviceability.	
Ventilation & Cooling	Passive: Min 2 louvred vents with insect mesh (on non-rain-facing sides). Active (if passive insufficient): Thermostatically controlled fan (IP54, 24/48V DC). Fan filter to be cleaned/replaced as part of CAMC schedule.	
Cable Entry & Glands	Min 4 Nos. cable gland knockouts (M25 + M32) at JB bottom. Min 2 Nos. on side. Gland plates: 2.0 mm CRS, replaceable. Cable glands: IP68 rated, nylon or brass. Unused knockouts: IP-rated blanking plugs. All cable entries to maintain JB IP rating post-installation.	
Earthing	External: M8 SS earthing lug on JB exterior, connected to pole/structure earth. Internal: 25 mm × 3 mm Cu earth bus bar, bonded to external lug via 16 mm² G/Y cable. All equipment inside JB bonded to earth bus bar. Door bonded to body via braided earth strap.	
Internal Lighting	1 No. LED light (12V/24V DC, min 5W) with door-actuated switch inside JB. 1 No. 5A 230V AC socket All internal wiring color-coded (R=live, B=neutral, G/Y=earth) and labelled.	
Labelling	(a) External: JB ID / Camera ID number on door. (b) Internal: Equipment asset labels + single-line wiring diagram laminated inside door. (c) Warning label: 'DANGER — ELECTRICAL HAZARD' on exterior.	
Finish & Color	Exterior: RAL 7035 light grey (or as directed) — min 60 micron epoxy powder coat. Interior: White powder coat. All cut edges, welds, and drilled holes: deburred + zinc-rich primer before powder coat.	
Standards & Compliance	IP Rating, IK Impact Rating, Enclosure Protection Degrees, Earthing, 35 mm DIN Rail, LV Switchgear Assemblies, Local Electrical Inspector / Safety Standards	

Junction Box Foundation Specification

Parameter	Specification	Compliance (Yes / No)
Type of Construction	Brick masonry plinth with cement mortar (1:6 ratio). Base slab of Plain Cement Concrete (PCC) M10 grade (1:3:6 mix). Brick class designation 5 (locally available).	
PCC Base Slab	75 mm thick PCC (M10, mix 1:3:6) at bottom of excavation as levelling course. This is mandatory — do not skip even for small plinths.	
Minimum Height Above Ground	Minimum 600 mm above Finished Ground Level (FGL). In waterlogging / low-lying areas: increase to 750 mm as directed by Engineer-in-Charge.	
Plinth Size (L × W)	Minimum 650 mm × 650 mm plan size. Actual size to match junction box base plus 75 mm clearance on all four sides. No larger than necessary — size only as per approved JB dimensions.	
Foundation Depth (Below Ground)	Minimum 300 mm below Finished Ground Level. In normal soil: 300 mm is sufficient. No deep excavation required unless soft/expansive soil is encountered.	
Anchor / Fixing Arrangement	Minimum 4 Nos. anchor bolts, M10 × 100 mm, Mild Steel (MS) zinc-plated or galvanized. Embedded in plinth during construction. Bolt pattern to match JB/enclosure base frame as per supplier drawing.	
Cable Entry Conduits	Minimum 2 Nos. PVC rigid conduit (ISI marked, 25–32 mm dia) as cable entry sleeves. Embedded in plinth during brick laying — no core cutting after construction. Seal conduit ends with PVC caps during construction; open after cable laying.	
Earthing	1 No. GI earthing terminal (25 mm × 3 mm GI flat, 300 mm length) embedded in plinth. Bond to JB enclosure earth stud using GI wire/flat. Earthing to comply with IS 3043. No separate earthing pit required for JB plinth — use building/pole earth if available nearby.	

Drawing Submission	Contractor to submit simple sketch / GA drawing of plinth with dimensions before work start. Drawing need not be structurally designed — just dimensional GA for approval by Engineer-in-Charge. As-built record to be updated after completion.	
Inspection	Dimensional check by site supervisor before and after construction. Verify: height, plan size, anchor bolt position, conduit placement, plaster finish. No formal concrete cube test required for brick masonry plinth.	
Standards & Compliance	Practice for Brickwork. Earthing, Sand for Plaster and better	

Automate Tape Library

Parameter	Minimum Specifications	Compliance (YES/NO)
Category	SMB to Mid-size (Modular)	
Drive Technology Supported	LTO-9, LTO-8, LTO-7 (repurpose LTO-6)	
Max Number of Drives	Up to 48 HH	
Cartridge Slots (Base)	40 slots (base module)	
Max Cartridge Slots (Expanded)	Up to 640 slots (15 expansion modules)	
Max Native Capacity	~11.5 PB (LTO-9, 640 slots @ 18TB native)	
Max Compressed Capacity	28.8 PB (2.5:1, LTO-9, 640 slots)	
Max Native Data Transfer Rate	51.4 TB/hr (48x LTO-9 drives)	
Drive Interface	8 Gb FC / 12 Gb SAS (LTO-9) / 6 Gb SAS (LTO-8 & earlier)	
Form Factor	3U base, scalable up to 48U	
Dimensions (H x W x D)	3U base (approx. 13.4 x 48 x 90 cm)	
Weight (Base, approx.)	Varies by config	
Mail Slots	5 slots per module	
Logical Partitions	Up to 21	
Encryption	AES 256-bit + KMIP 1.2 + Secure Manager	
Security Features	MFA, MSL Vault Partition, Secure Manager, KMIP, Path Failover	
Management Software	HPE CVTL + Tape Assure Advanced + Data Verification	
Barcode Reader	Integrated, standard	
Investment Protection (Drive Reuse)	Reuses drives from 1/8, MSL2024, MSL6480	
Reliability (Robot Load/Unload Cycles)	2,000,000 cycles	
Warranty	1-year NBD onsite, 9x5 phone support	
Ideal Use Case	SMB to mid-sized, scalable growth	
Example / Base SKU	Q6Q62C (Scalable Base Module)	

Tape data cartridge

Technical specifications	Data Cartridge	Compliance (YES/NO)
Capacity	6.25 TB Compressed 2.5:1 Supported	
Media format	Re-writable	

Quantity per package	1	
Read speed, media	400 MB/sec	
Bit density	343 kbits/inch	
Archival life	30 years	
Product color	Purple	
Tape length	846 m	
Tape width	12.65 mm	
Tape thickness	6.4 um	
Base material	Metal Particle	
Data tracks	2176	
Magnetic coercively	2,800 - 2,900 Oe	
Product dimensions (metric)	11.3 x 2.79 x 11.1 cm	
Weight	0.27 kg	

Backup Software

Parameter	Minimum Specification Requirement	Compliance (YES/NO)
Purpose	The software shall be used for backup and restoration of CCTV video recordings, flagged video clips, incident-related evidences, databases, applications, operating systems, configurations, logs, and other critical data generated under the project.	
Backup Policy	Support policy-based, automated, and scheduled unattended backup operations.	
Supported Platforms	Support backup and recovery of Physical Servers, Virtual Machines, Operating Systems, Databases, Applications, and File Systems.	
Database Backup	Support online backup and restore of SQL, PostgreSQL, MySQL, Oracle, and other industry-standard databases.	
Backup Types	Support Full Backup, Incremental Backup, Differential Backup, Selective Backup, Point-in-Time Recovery, Progressive Incremental Backup, Synthetic Full Backup, and Snapshot-based Backup.	
Simultaneous Sessions	Support multiple concurrent backup and restore sessions without service interruption.	
Recovery Capability	Support granular file-level, application-level, database-level, VM-level, and system-level recovery.	
Video Backup	Support backup of user-selected CCTV recordings, event-based recordings, alarm-triggered recordings, and evidentiary video clips.	
Deduplication	Global source-side and target-side deduplication for storage optimization.	
Compression	Support backup data compression to optimize storage utilization.	
Encryption	AES-256 bit encryption for backup data at rest and TLS 1.2/1.3 encryption for data in transit.	
Backup Scheduling	Support hourly, daily, weekly, monthly, and custom backup schedules.	
Retention Policy	Configurable retention and archival policies based on user requirements.	
User Interface	Support GUI-based Console, Web-based Management Interface, and Centralized Dashboard.	
Monitoring & Reporting	Real-time backup monitoring, alerts, audit logs, dashboard reporting, and email notifications.	
Scalability	Solution shall support future expansion without replacement of existing licenses.	
High Availability	Support backup server redundancy, failover, and disaster recovery deployment.	
Virtualization Support	Support VMware, Hyper-V, KVM, and equivalent virtualization platforms.	
Cloud Readiness	Support integration with public/private cloud storage for archival and disaster recovery purposes.	
Audit & Compliance	Maintain audit trails, backup history, restore logs, and compliance reporting.	
Security	Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), secure authentication, and ransomware protection mechanisms.	
Compatibility	Compatible with the proposed CCTV VMS, Storage System, Servers, and ICCC infrastructure.	
Warranty & Support	Minimum 5 Years Comprehensive Support, Software Updates, Security Patches, and Version Upgrades.	

VIDEO MANAGEMENT SOFTWARE: VIDOENETICS

Sr.No	Minimum Specification Requirement	Compliance (YES/NO)
1	VMS shall be used for centralized management of all field camera devices, video servers and client users.	
2	VMS server shall be deployed in a clustered server environment or support inbuilt mechanism for high availability and failover.	
3	VMS shall support a flexible rule-based system driven by schedules and events.	
4	VMS shall be supported for fully distributed solution for monitoring and control function, designed for limitless multi-site and multiple server installations requiring 24/7 surveillance with support	
	for devices from different vendors.	
5	VMS shall support ONVIF compliant internet protocol (IP) cameras.	
6	The bidder shall clearly list in their proposal the make and models that can be integrated with the VMS, additionally all the offered VMS and cameras must have Open Network Video Interface Forum (ONVIF) compliance.	
	VMS shall be enabled for any standard storage technologies and video wall system integration.	
7	VMS shall be enabled for integration with any external Video Analytics	
	Systems both server & edge based.	
8	VMS shall be capable of being deployed in a virtualized server environment without loss of any functionality.	
9	All CCTV cameras locations shall be overlaid in graphical map in the VMS Graphical User Interface (GUI). The cameras selection for viewing shall be possible via clicking on the camera location on the graphical map. The graphical map shall be of high resolution enabling operator to zoom-in for specific location while selecting a camera for viewing.	
10	VMS shall have an administrator interface to set system parameters, manage codecs, manage	
	permissions and manage storage.	
11	VMS day to day control of cameras and monitoring on client workstations shall be controlled	
	through the administrator interface.	
12	Whilst live control and monitoring is the primary activity of the monitoring workstations, video replay shall also be accommodated on the GUI for general review and also for pre- and post-alarm recording display.	
13	The solution design for the VMS shall provide flexible video signal compression, display, storage and retrieval.	
14	All CCTV camera video signal inputs to the system shall be provided to various command control centre(s), viewing center etc., and the transmission medium used shall best suit the relative camera	
	deployments and access to the CCTV Network.	
15	VMS client shall have the capability to work with touch enabled multi- monitor workstations. It shall be capable of displaying videos in up to three (3) monitors simultaneously.	
a.	AVI files	
b.	Motion- Joint Photographic Experts Group (M-JPEG)	
c.	Moving Picture Expert Group-4 (MPEG-4)	
d.	MP4 Export or Latest	
16	All streams to the above locations shall be available in real-time and at full resolution. Resolution and other related parameters shall be configurable by the administrator in order to provide for network constraints.	
17	The VMS shall support field sensor settings. Each channel configured in the VMS shall have an	
	individual setup for the following settings, the specific settings shall be determined according to the encoding device.	
18	The VMS shall support the following operations:	
a.	Adding an IP device	
b.	Updating an IP device	
c.	Updating basic device parameters	
d.	Adding/removing channels	
e.	Adding/removing output signals	
f.	Updating an IP channel	
g.	Removing an IP device	
h.	Enabling/disabling an IP channel	
i.	Refreshing an IP device (in case of firmware upgrade)	
j.	Multicast at multiple aggregation points	
19	The VMS shall support retrieving data from edge storage. Thus when a lost or broken connection is restored, it shall be possible to retrieve the video from SD card and store it on central storage.	

	System should support to view the recordings available over cameras local storage device (such as	
	an SD card), and copy them to the server.	
20	The VMS shall support bookmarking the videos. Thus, allowing the users to mark incidents on live	
	and/or playback video streams.	
21	The VMS shall allow the administrator to distribute camera load across multiple recorders and be	
	able shift the cameras from one recorder to another by simple drag and drop facility.	
22	VMS shall support automatic failover for recording.	
23	VMS should also support dual recording or mirroring if required.	
24	VMS shall support manual failover for maintenance purpose.	
25	VMS shall support access and view of cameras and views on a smartphone or a tablet (a mobile	
	device).	
26	VMS shall support integration with the ANPR application.	
27	VMS shall support integration with other online and offline video analytic applications.	
28	VMS shall be able to accept alerts from video analytics built into the cameras, other third party	
	systems, sensors etc.	
29	The VMC Licenses should be valid for lifetime	
	Client System	
#	**Functionality**	
1	Viewing live video from cameras on the surveillance system.	
2	Browsing recordings from storage systems.	
3	Creating and switching between multiple of views.	
4	Viewing video from selected cameras in greater magnification and/or higher quality in a designated hotspot.	
5	Using digital zoom on live as well as recorded video.	
6	Using sound notifications for attracting attention to detected motion or events.	
7	Getting quick overview of sequences with detected motion.	
8	Getting quick overviews of detected alerts or events.	
9	Quickly searching selected areas of video recording for motion (also known as Smart Search).	
	Remote Web Client	
#	**Description**	
1	The web-based remote client shall offer live view of up to 9 / 16 (configurable) cameras, including PTZ control (if applicable) and event / output activation. The Playback function shall give the user concurrent playback of multiple recorded videos with date, alert sequence or time searching.	
2	User Authentication – The Remote Client shall support logon usingthe user name and password credentials	
	Mobile Client	
#	**Description**	
1	The bidder shall be required to provide a standardized MobileApplication to integrate smart phones and tablets for 2-way communication with the Video Management System in a secure manner. It will be responsibility of SI to configure such tablets / Smartphone with the Surveillance System and ensure that all the necessary access is given to these mobile users.	
2	Communication with mobile client and server shall be encrypted with Digital Certificate.	
	Matrix Monitor	
#	**Description**	
1	Matrix Monitor – The Matrix Monitor feature shall allow distributed viewing of multiple cameras on the system onany monitor.	
2	The Matrix Monitor feature shall access the H.264/MJPEG/MPEG4 stream from the connected	
	camera directly and not sourced through the recording server.	
	Alarm Management Module	
#	**Description**	
1	The alarm management module shall allow for continuous monitoring of the operational status and event-triggered alarms from various system servers, cameras and other devices. The alarm management module shall provide a real-time overview of alarm status or technical problems while allowing for immediate visual verification and troubleshooting.	
2	The alarm management module shall provide interface and navigational tools through the client including;	
3	Graphical overview of the operational status and alarms from servers, network cameras and	
	external devices including motion detectors and access control systems.	
4	Intuitive navigation using a map-based, hierarchical structure with hyperlinks to other maps,	

	servers and devices or through a tree-view format.	
5	The module shall include flexible access rights and allow each user to be assigned several roles where each shall define access rights to cameras.	
6	Basic VMS should be capable to accept third party generated events / triggers.	
	Management / Integration Functionality	
#	**Description**	
1	The Surveillance System shall offer centralized management of all devices, servers and users.	
2	The Surveillance System should not have any limit on the number of cameras to be connected for Surveillance, Monitoring and Recording. Any increase in the no. of cameras should be possible	
	by augmentation of Hardware components.	
3	The Surveillance System shall support distributed viewing of any camera in the system using Video walls or big screen displays.	
4	The Surveillance System shall support alarm management. The alarm management shall allow for the continuous monitoring of the operational status and event-triggered alarms from system servers, cameras and other external devices.	
5	It should be possible to integrate the Surveillance System with 3rd-party software, to enable the users to develop customized applications for enhancing the use of video surveillance solution.	
	For e.g., integrating alarm management to initiate SMS, E-Mail, VoIP call, etc.	
6	The Management system shall store the overall network elements configuration in central database, either on the management server computer or on a separate DB Server on the network.	
7	System should be able to be integrated with Event Management / Incident Management System.	
	System Administration Functionality	
#	**Description**	
1	The System Administration Server shall provide a feature- rich administration client for system configuration and day-to-day administration of the system.	
2	The System Administration Server shall support different logs related to the Management Server.	
	· The System Log	
	· The Audit Log	
	· The Alert Log	
	· The Event Log	
3	Rules: The system shall support the use of rules to determine when specific actions occur. Rules shall define what actions shall be carried out under specific conditions. The system shall support rule initiated actions such as:	
	· Start and stop recording	
	· Set non-default live frame rate	
	· Send notifications via email	
	· Pop-up video on designated Client Monitor recipients	
	Major Server components for VMS	
Video Management Server(s)	Video Management System Servers will maintain coherent operations between all servers and workstations. It will host Control Centre, where the system is administered, and System database. It will monitor one or more Recorder servers on separate dedicated computers, storage devices, IP- compatible devices, and one or more workstation. All network communication will also be is performed via the	
	Video Management servers.	
Video	The Video Recorder Server will be a dedicated server that will store and processes video with the help of Video Management System	
Recording Server(s)		
Web Server(s)	It will be used to launch the client application remotely through web browsers.	
	The VMC Licenses should be valid for lifetime	

48-PORTS 32GBPS FC SAN SWITCH

Parameter	Specifications	Compliance (YES/NO)
Technical Parameters	Up to 48 autosensing Fiber Channel ports capable of speeds of 4/8/16/32 Gbps	
	N-Port ID Virtualization (NPIV) technology to provide independent management for each virtual machine.	
	N-Port Virtualization (NPV) and fabric-port (F-port) channeling features to enable scaling of SANs without reaching Fibre Channel domain ID limits.	

	N-Port Virtualization (NPV) and fabric-port (F-port) channeling features to enable scaling of SANs without reaching Fiber Channel domain ID limits. VSANs for fault isolation and Port Channels for Inter-Switch Link (ISL) resiliency.	
	Up to 1536 Gbps of aggregate bandwidth in a 1 rack unit (RU)	
	Centralized management tool with task-based wizards that simplifies management of a standalone switch or multiple switches and fabrics. Reduced total cost of ownership.	
	The Quality of Service (QoS) feature allows traffic to be classified into four distinct levels for service differentiation. QoS can be applied to help ensure that Fibre Channel data traffic for latency-sensitive applications receives higher priority over throughput-intensive applications such as data warehousing.	

48-PORT SFP+ TOP-OF-RACK SWITCH

Parameter	Specifications	Compliance (YES/NO)
Interface Requirement	48 ports of 1GbE/10GbE (SFP/SFP+)	
	4 nos. of 40G/100G SFP28 uplink ports.	
	The switch Shall support 1G SX,LX, 10Gbase-SR, LR, QSFP SR4, LR4 and QSFP28-SR4, LR4 and required transceivers should be added in BOM for connectivity	
Architecture	Should have 16GB DRAM and 32GB Flash.	
	The switch will have at up to 1.7 Tbps switching capacity.	
	Forwarding rates: The switch should have 1300 Mpps forwarding rates.	
	IPv4 Routing entry support : 24K or more.	
	IPv6 Routing entry support : 12K or more.	
	IPv4 and IPv6 Multicast Routes : 4K or more.	
	MAC address table size support: 1,40,000 or more.	
	Packet buffer : 32 MB or more	
	The device should be IPv6 ready from day one.	
	Should support the ability to configure backup of the previous configuration automatically.	
Quality of Service (QoS)	The Switch should support Strict priority (SP) queuing and Deficit Weighted Round Robin (DWRR)	
	The Switch should support Data Center Bridging (DCB),Priority Flow Control (PFC),Enhanced Transmission Service (ETS) and DCB Exchange Protocol	
	The Switch should Prevent accumulation of excessive congestion with periodic flushing. Avoids packets buffering for an extended time period and support Flow-Control Guard	
	The Switch should able to Marks packets as ECN-CE (Congestion Experienced) to help TCP to reduce receive window size during congestion Advanced Lossless Pool Configuration	
Resiliency and High Availability	The proposed switch should support distributed and redundant architecture by deploying two switches with each switch maintaining independent control and synchronized during upgrades or failover and should support upgrades during live operation.	
	The switch should have modular operating system with micro-services or equivalent architecture providing superior fault tolerance and high availability	
	The Switch should have Hot swappable power supply and fan modules from day one	
	The Switch should support Virtual Router Redundancy Protocol (VRRP) and Bidirectional Forward Detection (BFD)	
	The Switch should support Unidirectional Link Detection (UDLD) and support IEEE 802.3ad LACP	
Management	The Switch should support REST API interface and support Industry-standard CLI with a hierarchical structure	
	The Switch should provide Restricts access to critical configuration commands; offers multiple privilege levels with password protection; ACLs provide SNMP access; local and remote Syslog capabilities allow logging of all access.	
	The Switch should support IP SLA,SNMP v2c/v3 and sFlow and support TFTP and SFTP	

	The Switch should support ping and traceroute for IPv4 and IPv6 and support Network Time Protocol (NTP)	
	The Switch should IEEE 802.1AB Link Layer Discovery Protocol (LLDP) and Dual flash images	
	The switch should be managed through the Cloud NMS from day-1, all other Data Center ,Campus Switches, Wireless Access Points and wireless controller are from same OEM and should be managed from the one single cloud console of NMS from same OEM	
	The switch should support Zero-Touch Provisioning (ZTP)	
Layer 2 Switching	The Switch should minimum 1000 port-based or IEEE 802.1Q-based VLANs, VLAN Translation, Bridge Protocol Data Unit (BPDU) tunneling and Port mirroring	
	The Switch should support IEEE 802.1D STP, IEEE 802.1w Rapid Spanning Tree Protocol (RSTP) for faster convergence, and IEEE 802.1s Multiple Spanning Tree Protocol (MSTP) and Rapid Per-VLAN spanning tree plus (RPVST+) or equivalent	
Layer 3 Services	The Switch should support Address Resolution Protocol (ARP)	
	The Switch should support directed broadcast on configured network subnets.	
	The Switch should support Dynamic Host Configuration Protocol (DHCP)	
	The Switch should support DHCP Relay enables DHCP operation across subnets.	
	The Switch should support DHCP Server	
	The Switch should support for DHCP smart relay.	
	Allows DHCP relay coexistence with DHCP server for both IPv4 and IPv6.	
	The Switch should support Domain Name System (DNS)	
Layer 3 Routing	The Switch should support Static IPv4 and IPv6 routing and VRF	
	The Switch should support Sub-Interface	
	The Switch should support Open shortest path first (OSPF) and OSPFv3	
	The Switch should support Border Gateway Protocol 4 (BGP-4), Multiprotocol BGP (MP-BGP)	
	The Switch should support Routing Information Protocol version 2 (RIPv2) and Routing Information Protocol Next Generation (RIPng)	
	The Switch should support Policy Based Routing (PBR) and support the tunneling of IPv6 traffic in an IPv4 network.	
	The Switch should support Equal-Cost Multipath (ECMP),Generic Routing Encapsulation (GRE) and Dual IP stack	
	The Switch should support Dynamic VXLAN with BGP-EVPN	
	The Switch should support PIM-SM/IGMP snooping in the VXLAN Overlay	
	The Switch should support IPv6 VXLAN/EVPN Overlay Support	
	The Switch should support IPv6 traffic over the VXLAN overlay	
	The Switch should support VXLAN distributed anycast gateway,VXLAN ARP/ND suppression and support VXLAN Group-Based Policy (GBP) and Role-based Policies	
Security	The Switch should support Access control list (ACL) Features	
	The Switch should support ACLs that can also protect control plane services such as SSH, SNMP, NTP or web servers.	
	The Switch should support 802.1x, Mac-auth, Port-Access Policy, Static Port Filtering.	
	The Switch should support Private VLAN	
	The Switch should support Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access-Control System (TACACS+)	
	The Switch should support RADIUS authentication and accounting data to be passed safely and reliably across insecure networks such as the internet.	
	The Switch should support Secure shell (SSHv2)	
Multicast	The Switch should support Internet Group Management Protocol (IGMP),IGMPv1, v2, and v3	
	The Switch should support PIM Multicast Boundary (v4)	
	The Switch should support Multicast Listener Discovery (MLD)	
	The Switch should support Protocol Independent Multicast (PIM), PIM-SSM,PIM-SM and PIM-Dense Mode,	
	The Switch should support Any cast Rendezvous Point (RP), Multicast Service Delivery Protocol (MSDP),MSDP Mesh Groups	
	The Switch should support IGMP/MLD Snooping, support Fast Leave (FL)	

	and Forced-FastLeave (FFL)	
Environmental Features	The Switch should support RoHS (EN 50581:2012) regulations.	
	The switch support Operating temperature of 0°C to 40°C	
Regulatory Compliance	Switch shall conform to UL 60950 or IEC 60950 or CSA 60950 or EN 60950 or equivalent Indian Standard like IS-13252:2010 or better for Safety requirements of Information Technology Equipment.	
	Switch shall conform to EN 55022/55032 Class A/B or CISPR22 Class A/B or CE Class A/B or FCC Class A/B or equivalent Indian Standard like IS 6873 (Part 7): 2012 or better for EMC (Electro Magnetic Compatibility) requirements.	
OEM Qualification, Warranty and Support	The switch shall be offered with minimum Three years hardware warranty with NBD Shipment and software updates/upgrades from OEM directly	
	Switch or Switch's Operating System on different hardware plateform should be tested for EAL 2/NDPP or above under Common Criteria Certification.	
	All switches, transceivers components shall be from the same OEM. The OEM shall be consistently present in Leaders quadrant in Gartner's Magic Quadrant for Wired and Wireless LAN Access Infrastructure for last 5 years	

INDOOR LED DISPLAY VIDEO WALL

Parameter	Detailed Specifications	Compliance (YES/NO)
Curved /banded Type	Indoor Fine Pitch LED Video Wall Up to 6 Degrees	
Display Technology	SMD 3-in-1 Black LED / Direct- View LED	
Pixel Pitch	P1.25 mm -2.25mm or Higher	
Screen Resolution	As per display size requirement & Modular tiling to any wall size	
Brightness	≥ 600 Nits or Higher	
Refresh Rate	≥ 3840 Hz or Higher	
Contrast Ratio	≥ 5000:1 or Higher	
Viewing Angle (H/V)	160° / 160° or Better	
Gray Scale	14 Bit or Higher	
Color Temperature	Adjustable from 3000K to 10000K	
Cabinet Material	Die-Cast Aluminium	
Cabinet Maintenance	Front Service Access	
Module Protection	Anti-static, Dust Resistant	
Power Supply	Dual (Power & Signal) AC 100~240V, 50/60Hz	
Maximum Power Consumption	≤ 600 W/m²	
Average Power Consumption	≤ 200 W/m²	
Operating Temperature	-10°C to +40°C	
Operating Humidity	10% to 90% RH Non-Condensing	
Life Span	≥ 100,000 Hours	
MTBF	≥ 10,000 Hours	
Input Interfaces	HDMI, DVI, VGA, DP, USB, Network	
Installation	Wall Mounted / Floor Standing Structure	
Certifications	CE, FCC, RoHS, BIS (if applicable)	
Warranty	5 Years or Higher	
Eye comfort / certification	Low-blue light coating (general LED)	
Control System	Compatible with Novastar-type / Equivalent	
Serviceability	Front access	

LED Display Video Wall Controller

Parameter	Minimum Technical Specification	Compliance (YES/NO)
Type	Video Wall Display Controller — PC-based or appliance-based, rack-mount, purpose-built for 24×7 ICCC / Command Centre operation. Controller, display panels, and software shall be from a single OEM to ensure compatibility, unified support, and warranty.	

Processor (CPU)	Minimum Intel® Xeon® Processor, 2.1 GHz base clock, minimum 12 cores (24 threads). Or equivalent server-grade processor with equivalent or better multi-threaded performance. Must support ECC memory.	
Memory (RAM)	Minimum 64 GB DDR4 ECC RAM. Expandable to minimum 256 GB without hardware changes (additional DIMM slots). ECC mandatory for 24×7 error-correction.	
Storage	Minimum 2× 480 GB SSD in RAID 1 (mirrored) configuration for OS and application. Additional storage as required for layout database and logs.	
Cooling	Minimum dual redundant cooling fans (N+1 fan configuration). Hot-swap fan replacement preferred for 24×7 operation.	
Network Interface	Minimum 2× 1 Gbps Ethernet LAN (RJ45) for management and source network. Additional network ports as required for video-over-IP sources.	
Power Supply	(1+1) Hot-swap redundant power supply units (PSUs). Auto-failover: if one PSU fails, second PSU takes full load with zero downtime. Input: 100–240V AC, 50/60 Hz universal input.	
Video Output Ports	Minimum 4 video output ports (DisplayPort 1.4 / HDMI 2.0 or better). Each output port shall support minimum 3840×2160 (4K) @ 60 Hz resolution. Number of outputs to match video wall configuration — bidder to confirm based on panel layout.	
Graphics Card	Minimum 1 No. professional-grade discrete graphics card (NVIDIA Quadro RTX or AMD Radeon Pro or equivalent). Minimum 16 GB dedicated GDDR6 GPU memory. Minimum 4 simultaneous 4K output channels. ECC GPU memory preferred.	
Input Sources	Universal input: minimum support for HDMI 2.0, DisplayPort 1.4, DVI-I, USB capture. IP/network sources: RTSP, ONVIF, NDI, RTP/UDP, HTTP streams. Web browser / URL sources. Remote desktop / screen capture (Windows workstations on LAN).	
Form Factor	19" Rack-mount unit. Maximum 4U height (preferred 2U). Rack ears and mounting hardware included.	
Noise Level	Maximum 49 dBA measured at 1 metre distance at 25°C ambient temperature. Tested under 100% load condition.	
Operating Conditions	Input: 100–240V AC, 50/60 Hz, universal. Operating temperature: 0°C to 35°C. Humidity: 5% to 80% non-condensing.	
Wireless Connectivity	Controller software UI to be accessible via WiFi from iPad (iOS) and Android tablet through standard browser or dedicated app. No proprietary dongle required — standard WiFi/LAN access.	
Keyboard & Mouse Extension	Keyboard and mouse (with wrist rest) to be provided. KVM extender or USB extender (minimum 10 m) to route keyboard, mouse, and USB from controller room to operator workstation desk.	
24×7 Operation	Controller shall be rated and designed for continuous 24×7×365 operation. MTBF: minimum 50,000 hours. No scheduled maintenance reboot required during operation.	
Regulatory Compliance	CE, CB, UL/cUL, FCC Class A, BIS (Bureau of Indian Standards) certification mandatory. Product to be listed on BIS portal at time of bid submission.	
Software Architecture	Software shall be from the same OEM as display panels and controller (single OEM). Software shall be browser-based (HTML5) — no Java plugin required. Concurrent multi-client access: minimum 5/10 simultaneous operator clients.	
Layout Management	Pre-configure unlimited display layouts. Launch any saved layout instantly via single mouse click. Change layouts in real-time during operation without black screen or interruption. Save and load layouts from local machine or remote machine over LAN/WAN.	
Multi-Source Display	Display multiple sources simultaneously anywhere on the video wall in any size and position. Source window: freely resizable, freely positionable (pixel-level). Overlap windows: configurable layer/Z-order. Minimum window layers: 2 (configurable).	
Supported Source Types	Analog video: RGB/component (via capture card). Digital: HDMI, DisplayPort, DVI-I inputs. Network/IP: RTSP, ONVIF, NDI, RTP/UDP, HTTP. Remote desktops: Windows PC screen capture over LAN (no encoder hardware required). Web sources: URL, live web pages, dashboards. Embedded and external audio formats. Application windows: individual app window capture from networked workstations.	
Scheduling / Auto-Launch	Schedule specific layout to launch automatically based on time range (start time / end time). Multiple schedules configurable. Timer-based auto-switching of layouts as per user-defined time events.	

Remote Control (LAN/WAN)	Display wall and all sources controllable from remote PC over LAN without KVM hardware. Layouts sharable and launch able on remote display walls (meeting rooms, remote ICCCs) over LAN/WAN. Bi-directional remote keyboard and mouse control between networked workstations.	
Tablet Control	Launch layouts, change layouts, and control wall in real-time using iPad or Android tablet. Supported via browser-based UI over standard WiFi/LAN — no app installation required (browser-only preferred).	
Alarm Display	Software shall support display of alarms on the video wall. Alarm source integration: trigger layout change, pop-up alarm window, or highlight source on wall upon alarm event. Alarm acknowledgement from video wall operator console.	
Authentication & Security	Minimum 2 layers of user authentication: Admin and Operator roles. Additional roles preferred: Viewer (read-only). Username + password login. Session timeout configurable.	
Ticker / Crawl Text	Minimum 2 independent ticker/crawl text bars. Each ticker: configurable position (top or bottom of wall). Configurable text content, font, speed, colour, background. Both tickers can run simultaneously and independently.	
Screen Scraping / Desktop Capture	Capture and display screen content from remote Windows workstations without dedicated hardware encoder. Single application window capture (not just full desktop). Pre-configure captured desktop as a named source usable in any layout.	
OSD Overlays / Source Labels	Software shall support display of: UMD (Under Monitor Display), IDC, Source name label, Time (time zone-aware), Date, Text overlay, Logo/branding, Message ticker, Source status indicator (online / offline / error).	
Central Config Database	Single central configuration database for all sources, displays, layouts, and system variables. Online (live) reconfiguration without system restart. Auto-detection of new network sources.	
Audio Support	Integrated embedded and external audio format support. Audio decoding of video streams (HDMI embedded audio, IP stream audio). Audio routing configurable per source/layout.	
OEM & Warranty	Display panels, controller hardware, and software shall be supplied by a single OEM. OEM Authorizations Letter (MAF) from display controller and software manufacturer to be submitted with bid. Minimum 3 years comprehensive on-site OEM warranty. OEM shall have registered office and technical support center in India.	

EMS/NMS: MOTADATA

#	Enterprise Management System - Technical Specifications	Compliance (YES/NO)
1	For effective operations and management of IT Operations, there is a need for an industry-standard EMS (Enterprise Management System). Given the expanse and scope of the project, EMS becomes very critical for IT Operations and SLA Measurement. Some of the critical aspects that need to be considered for operations of IT setup of are: Network Monitoring, Network Configuration Management, ServiceDesk along with HelpDesk & IT Asset Management.	
2	The OEM of the proposed solution should be "Make In India" company with 15 years of presense in India and the solution should be deployed in min. 5 projects in Government/PSU segment in India. The documentary proof should be submitted at time of the bid submission.	
3	The solution should be capable of running in Linux platform and should be 64-bit application to fully utilize the server resources on which it is installed.	
4	The proposed EMS solution must comply with the following standards: CIS certification, ISO 27034-1 for application security, ISO 27001:2022 for information security management, GDPR and POPIA for data privacy, and SOC 2 Type 2 for service organization controls. This ensures robust security, regulatory compliance, and comprehensive data protection across all managed network operations. Valid supporting documents must be submitted with the bid.	
5	The system must not use any open-source third-party relational or time-series databases for reporting purposes. It should use either a native proprietary reporting database or an enterprise-licensed database designed for high-speed, high-volume observability workloads and aligned with required security constraints. The OEM must provide an undertaking, on their official letterhead, confirming the use of either an enterprise-supported or proprietary database as part of the bid submission	
6	OEM Should provide support for all upgrades and updates during the contract Period. The OEM should have a support center with Dedicated Support mail id and toll free number in INDIA. OEM Should provide escalation matrix along with bid submission.	
7	OEM should provide configuration and training for minimum 3 days along with training material.	
8	The OEM must have average annual turnover of atleast INR 30 Cr. or above in last 3 financial	

	years excluding the current financial year with positive net worth and solution should be "Make In India" company with 100% local content.	
	Network/Infra Monitoring	
9	The Monitoring Solution should provide Unified Architectural design offering seamless common functions including but not limited to: Event and Alarm management, Auto-discovery of the Network environment, Correlation, and root cause analysis, Reporting and analytics	
10	The solution must provide discovery & inventory of heterogeneous physical network devices like Layer-2 & Layer-3 switches, Routers and other IP devices and do mapping of LAN connectivity with granular visibility up to individual ports level.	
11	The operator should be able to build correlation rules in a simple GUI based environment where the Operator should be able to correlate cross domain events	
12	The solution shall provide future scalability of the whole system without major architectural changes.	
13	The system must not use any open-source third-party relational or time-series databases for reporting data. It should use a native, proprietary-built reporting database for high-speed, high-volume observability workloads and security constrains.	
14	All the required modules should be from same OEM and should be tightly integrated for single pane of glass view of enterprise monitoring	
15	The OEM of the proposed monitoring solution shall be present in latest Gartner's Market Guide Report of year 2025. The documentary proof must be enclosed with the bid.	
16	The solution must offer interactive end-to-end network path mapping with hop-by-hop performance analysis, threshold-based alerting, and historical trend insights. It should correlate node-level metrics with service impact, enrich paths with metadata (e.g., ISP info, geo/IP), and support policy enforcement with contextual visualizations.	
17	The solution must support custom dashboards for different role users such as Management, admin, and report users	
18	The proposed solution should allow the application of advanced statistical transformations and functions over selected KPIs, including but not limited to: Anomaly Detection Forecasting Moving Average Median Function Logarithmic Scale Transformation Delta/Derivative Functionality	
19	The solution should provide superior view of infrastructure health across system, networks, application and other IT Infrastructure components into a consolidated, central console	
20	The proposed solution must provide comprehensive and integrated management of IT infrastructure components to maximize the availability of IT services and SLA performance.	
21	The design functionality shall facilitate creation of templates used for monitoring key network resources, devices, and attributes. Default templates and best practice designs are provided for quick out-of-the- box implementation automating the work required to use OEM validated designs and best practices.	
22	The proposed solution must provide Health Monitoring reports of the network with settable periodicity -@24 Hrs, 1 week, 1 month.	
23	The proposed solution must provide the graphical layout of the network element with modules drawn using different colors to indicate their status	
24	The proposed solution should have multiple alerting feature to get the notification via email, sms and third-party systems	
25	The proposed solution should provide alert console with alert summary such as no. of correlated alert, network alert, server alert, virtualization alert, cloud alert, application alert etc.	
26	The proposed solution should be ITIL v4 certified with 12 practices. Documentary proof should be provided at the time of bidding.	
27	Widgets should allow advanced visualization customizations, such as: Toggling legends, adding rotation and adding threshold markers Conditional formatting (icon/color based on thresholds) Column visibility toggles and value formatting Metric/unit conversion and display customization Text Widget to bifurcate sections	
28	The proposed solution should offer a wide variety of intuitive widget visualizations to represent telemetry data effectively, including: Line Chart, Area Chart, Stacked Bar, Horizontal Bar, Vertical Bar, Pie, Donut, Top-N, Grid/Table, Sparkline, Treemap, Sankey, Bubble, Heatmap, Geo-Map, and Gauge	
29	The proposed NMS solution must provide agentless as well as agent-based monitoring for server infrastructure. The agents should be able to set polling interval as low as 1 second with low	

	overhead on target server infrastructure.	
30	The platform should include a dynamic runbook automation engine, allowing users to create, modify, and execute automation runbooks from the UI across a variety of incident response and administrative tasks.	
31	The solution should support custom plugin development using Go and Python, allowing tailored data collection logic and protocol support extensions.	
	Network Configuration Management	
32	The proposed solution should be able to take back up of running and startup configuration of network devices. It should also provide versioning for backup to track changes.	
33	The proposed system should be able to administer configuration changes to network elements by providing toolkits to automate the following administrative tasks of effecting configuration changes to network elements: a) Capture running configuration. b) Capture startup configuration c) Upload configuration; d) Write start-up configuration; e) Upload firmware	
34	The solution must support diverse compliance evaluation methodologies including standard checks, advanced custom configuration block evaluations, and CLI output-based validations.	
35	The proposed solution must support baseline configuration management, proactively detecting and alerting on deviations from established baselines for configuration compliance.	
36	The solution must enable dynamic template assignment for key operational activities including backup, restoration, synchronization, firmware upgrades, and hardware details retrieval.	
37	The solution should generate out-of-the-box reports catering to diverse operational, compliance, and audit requirements.	
38	The solution must support the creation of customized compliance benchmarks based on combinations of out-of-the-box and custom-defined rules.	
	Helpdesk/Service Desk Management	
39	The proposed solution should be aligned with ITIL framework principles and certified with ITIL4 with minimum 12 practices like Monitoring & Event Management, Incident Management, Service Request Management, Problem Management, Service Level Management, Capacity & Performance Management, Availability Management, Change Enablement, Release Management, Knowledge Management, Measurement & Reporting Management, IT Asset Management. (a documentary proof of the same should be provided at the time of bidding).	
40	The proposed helpdesk system shall provide flexibility of logging, viewing, updating and closing incident manually via web interface	
41	Each incident shall be able to associate multiple activity logs entries via manual update or automatic update from other enterprise management tools.	
42	The proposed helpdesk system shall be able to provide flexibility of incident assignment based on the workload, category, location etc.	
43	The proposed solution should automatically provide suggested knowledge base articles based on Incident properties with no programming	
44	The proposed solution should automatically suggest available technicians based on workload, average ticket closure time assigning tickets with no programming	
45	The proposed solution should tightly integrate with monitoring system to provide two way integration - E.g. when system down alarm created, it should automatically create ticket and assign it to technician, in case system comes up before ticket is resolved by technician, it should automatically close the ticket to minimize human efforts	
46	The proposed system must not create more than one ticket for same recurring alarm to avoid ticket flooding from Monitoring system	
47	The proposed helpdesk system shall support tracking of SLA (service level agreements) for call requests within the help desk through service types.	
48	The proposed solution should support defining periodic workflows using cron expressions, enabling precise scheduling of automated tasks with flexible recurrence patterns tailored to diverse operational needs	
49	The platform should provide a robust plugin or extension framework to facilitate rapid integration with existing enterprise systems such as ERP, Active Directory, and network monitoring solutions	
50	The Orchestrator should provide a visual workflow designer, enabling users to create, modify, and manage automated workflows without requiring deep coding knowledge.	
51	The proposed solution should support the use of dynamic expressions within workflow flow logic and actions, enabling runtime evaluations, conditional branching, and on-the-fly data transformations based on contextual parameters	
52	The proposed solution should allow referencing of field values from preceding steps or related records within workflows, facilitating seamless data propagation and manipulation across multi-	

	stage processes	
53	The solution should provide workflow administration settings, allowing administrators to configure orchestration policies and operational parameters	
54	The proposed solution should support webhook-triggered workflows with the ability to parse and utilize incoming payload data as variables, allowing for context-aware actions and intelligent branching within the orchestration logic	
55	The proposed solution should provide Active Directory–integrated self-service, enabling end users to reset passwords and unlock their AD accounts via the portal by answering configurable security questions—without technician intervention	
56	Users shall be able to submit and track requests via a self-service portal and mobile app.	
	Asset Management	
57	A configuration management database shall be established which stores unique information about each type Configuration Item CI or group of CI.	
58	The proposed solution allow scheduling periodic report to check current software and hardware inventory	
59	The proposed solution must allow attaching CI record to generated service tickets	
60	The proposed solution should support customizable discovery patterns and multi-method scans (agent, agentless, credential-less) across protocols like WMI, SSH, SNMP, and VMware, to automatically detect and onboard IT services and assets into the CMDB with precise control over discovery schedules and parameters	
61	The Proposed solution should provide end to end Asset Life Cycle Management: Makes it easier to handle the complete life cycle of an asset, that is, all stages/modules from procurement to disposal	
62	The Proposed solution should support maintaining AMC/Warranty Information with Alerting when about to expire also provide Asset Deletion capabilities enabled with workflow engine	
63	The Proposed solution should support Software License Metering: Helps to understand the software license compliance and the use of unauthorized software in the organization and helps to act proactively to curb illegal usage and problems associated with it.	
64	The proposed solution should provide Asset Dashboards/Reporting: Graphical representation all the assets based on Category, location, aging of the asset, customer, which can be further level down to the incident record ID	
65	The proposed solution should provide out of the box purchase and contract management modules to support end to end asset life cycle	
66	The proposed solution must provide asset baselining to manage and track asset effectively.	
67	The Proposed Solution should also provide Consumables Asset Tracking as well	
68	There should be facility to generate QR and Barcode for each asset and using Mobile Application (iOS and Android) it should provide details of that assets.	
69	The proposed Asset Management tool should have option to provide workflow for asset movement operation and tracking of asset Movement throughout the asset lifecycle.	
70	The should have facility to capture financial details along with depreciation cost for each assets.	
71	The proposed Asset Management tool must include Remote desktop feature and does not require any separate license for Remote Desktop Functionalities	
72	The proposed solution should have in built remote desktop capability for technicians to take remote control of for troubleshooting. Remote desktop should also support audio & video calling and file sharing.	

ANTIVIRUS FOR SEVERS: Trend Micro

Parameter	Detailed Specifications	Compliance (YES/NO)
Product	Server level antivirus	
Purpose	Hybrid cloud security for physical, virtual, cloud and container environments	
Security Modules	Anti-malware, Machine Learning, Web Reputation, Firewall, Intrusion Prevention, Integrity Monitoring, Application Control, Log Inspection	
Deployment	Software, AWS Marketplace, Azure Marketplace, As-a-Service	
Management	Single centralized dashboard / Deep Security Manager	
Cloud Platforms	AWS, Microsoft Azure, VMware Cloud	
Virtualization Support	VMware vSphere, NSX, VDI environments	
Multi-Tenancy	Supported	
Intrusion Prevention	Virtual patching, protection against SQL Injection, XSS and exploits	
Firewall	Bidirectional host-based firewall	
Integrity Monitoring	Real-time monitoring of critical files and registry changes	
Application Control	Whitelist-based application control for Windows and Linux	
Log Inspection	Analyzes 100+ log formats and SIEM integration	

Threat Intelligence	Trend Micro Smart Protection Network	
Container Security	Docker host and container protection	
Compliance	PCI DSS, HIPAA, SSAE 16, NIST support	
Directory Integration	Microsoft Active Directory	
SIEM Integration	Splunk, IBM Radar, HP Arc Sight and others	
Deployment Tools	Chef, Puppet, AWS OpsWorks, SCCM	
Certifications	Common Criteria EAL 2+, SAP Certified, VMware validated	

Tender Technical

- The bidder shall be responsible for the design, supply, installation, integration, testing, commissioning, operation, maintenance, warranty support, and successful functioning of all hardware, software, licenses, accessories, passive components, electrical infrastructure, networking components, and associated services required for implementation of the City Surveillance System. The technical specifications mentioned in this RFP for UPS Systems, Surge Protection Devices, Earthing Systems, Camera Mounting Poles, Junction Boxes, Tape Data Cartridges, Video Management Software (VMS), SAN Switches, Top-of-Rack Switches, Indoor Active LED Display Video Walls, Video Wall Controllers, EMS/NMS, and all other related equipment shall be considered as the minimum acceptable specifications.
- The bidder shall ensure that all proposed components are enterprise-grade, commercially available, currently supported by the OEM, and suitable for continuous 24x7x365 mission-critical surveillance operations. All equipment supplied under the project shall be new, unused, and sourced through authorized channels. The bidder shall ensure complete interoperability and seamless integration among all proposed components, irrespective of the OEMs involved.
- Any component, accessory, software license, mounting arrangement, cable, connector, patch cord, rack accessory, power distribution unit, surge protection device, earthing material, management software, monitoring tool, or any other item required for successful implementation and operation of the project, whether specifically mentioned in the Bill of Quantities (BoQ) or not, shall be deemed to be included within the bidder's scope without any additional cost to VMC.
- The bidder may offer specifications higher than those prescribed in this RFP; however, under no circumstances shall specifications lower than those specified be accepted. All proposed products shall comply with applicable BIS, IEC, ISO, IEEE, ONVIF, UL, CE, FCC, RoHS, NDAA, and other relevant national and international standards wherever applicable. The bidder shall submit OEM datasheets, technical literature, compliance statements, certifications, and test reports supporting compliance with the specified requirements.
- The bidder shall provide all necessary software licenses, firmware, subscriptions, updates, patches, security upgrades, management tools, and integration services required for successful operation of the system throughout the contract period. The complete solution shall be scalable and capable of supporting future expansion without requiring replacement of existing infrastructure.
- All equipment supplied under the project shall be covered under comprehensive OEM warranty and support for the period specified in the RFP. During the warranty and maintenance period, the bidder shall be responsible for replacement, repair, software updates, security patches, configuration support, and restoration of services to ensure compliance with the prescribed Service Level Agreements (SLAs).
- The successful bidder shall be solely responsible for the performance, reliability, availability, security, maintainability, and operational readiness of the complete City Surveillance System and all associated infrastructure deployed under the project.
- The proposed solution shall be based on open architecture and open industry standards to ensure interoperability, scalability, and future expansion.
- All hardware, software, licenses, accessories, tools, consumables, and services required for successful implementation of the project shall be deemed to be included in the Bid.
- The proposed solution shall be compatible with the existing ICT infrastructure, surveillance systems, network, servers, storage, security solutions, and third-party applications available at VMC.
- The solution shall support integration through standard APIs, ONVIF, SNMP, REST APIs, Syslog, LDAP/Active Directory, TCP/IP, HTTPS, and other industry-standard protocols wherever applicable.
- All integration, customization, configuration, migration, testing, commissioning, and interoperability activities required for successful implementation shall be carried out by the Successful Bidder without any additional cost to VMC.
- The proposed system shall be modular, scalable, and capable of supporting future expansion without replacement of the core infrastructure.

- All supplied equipment shall be new, unused, genuine, and sourced directly from the respective OEM or its authorized channel.
- The Successful Bidder shall ensure compatibility among all supplied hardware, software, networking equipment, and existing infrastructure.
- All software updates, firmware upgrades, security patches, bug fixes, and OEM-recommended updates shall be provided during the Warranty and CAMC period without additional cost.
- The Successful Bidder shall conduct Factory Acceptance Test (FAT), Site Acceptance Test (SAT), Integration Testing, Performance Testing, and User Acceptance Testing (UAT) before Go-Live.
- The complete system shall comply with the applicable BIS, TEC, IEEE, ISO, IEC, ONVIF, TCP/IP, and other relevant Indian and international standards.
- The Successful Bidder shall provide complete system documentation, as-built drawings, network diagrams, IP addressing scheme, asset register, SOPs, user manuals, administrator manuals, and maintenance manuals.
- The Successful Bidder shall maintain complete asset inventory and configuration records throughout the Contract Period.
- The solution shall support centralized monitoring, health monitoring, alarm management, event logging, audit trails, reporting, and SLA monitoring.
- The proposed solution shall provide role-based access control, password policy, audit logging, and cyber security features as per industry best practices.
- All required licenses shall be genuine, legally procured, and supplied in the name of Vadodara Municipal Corporation (VMC), wherever applicable.
- The Successful Bidder shall ensure that the supplied solution does not affect the performance or operation of any existing VMC system during implementation and integration.
- The Successful Bidder shall rectify any defects, failures, configuration issues, or interoperability issues identified during the Contract Period without additional cost to VMC.
- The Successful Bidder shall be responsible for obtaining all approvals, clearances, permissions, and coordination required for execution of the work wherever applicable.
- Any item, component, accessory, software, cable, connector, mounting hardware, tool, or service not specifically mentioned in the BOQ but required for successful installation, commissioning, integration, testing, operation, and maintenance of the complete system shall be deemed to be included in the Bid and shall be provided without any additional cost to VMC.
- The Successful Bidder shall ensure compliance with all applicable Government of India, Government of Gujarat, VMC, and statutory regulations throughout the Contract Period.
- The entire solution shall be designed to achieve the SLA, uptime, performance, availability, recording quality, cybersecurity, and operational requirements specified.

SECTION: 5
DELIVERABLE, TIMELINE & PAYMENT PROCESS

5.1 Project Deliverables, Time line and Payment Process

Deliverables and Timelines

Sr. No.	Deliverable	Timeline
1	Project Kick-off Meeting and Submission of Detailed Project Plan	Within 15 Days from Work Order
2	The Successful Bidder shall carry out a detailed site survey of all locations provided by VMC within fifteen (15) days from the date of handover of the locations by VMC and Detailed Site Survey and Submission of Site Survey Report	Within 30 Days from Work Order
3	Submission of Location-wise BOQ, Drawings, Network Architecture and Implementation Plan	Within 45 Days from Work Order
4	Procurement and Delivery of Equipment at Site	As per Approved Project Schedule
5	Installation of Poles, Junction Boxes, Earthing, Electrical Infrastructure and OFC Network	As per Approved Project Schedule
6	Installation of CCTV Cameras, PTZ Cameras, Edge Devices, Industrial Switches, UPS and Associated Field Equipment	As per Approved Project Schedule
7	Installation and Configuration of Servers, Storage Systems, VMS, EMS/NMS, Video Wall and Central Infrastructure	As per Approved Project Schedule
8	Integration with Existing ICCC Infrastructure and Other Applications	As per Approved Project Schedule
9	Testing, Commissioning and System Validation	As per Approved Project Schedule
10	Submission of As-Built Drawings, Asset Register, Test Reports and Documentation	As per Approved Project Schedule
11	User Acceptance Testing (UAT) and Final Acceptance Testing (FAT)	As per Approved Project Schedule
12	Go-Live of Complete City Surveillance System	As per Approved Project Schedule
13	Submission of Final Project Completion Report and Handover Documentation	Within 30 Days from Go-Live

The bidder shall prepare and submit a detailed Project Implementation Schedule within 15 days of issuance of the Work Order. The schedule shall clearly define milestones, dependencies, deliverables, resource deployment plan, and completion timelines for approval by VMC. All activities under the project shall be executed strictly as per the Approved Project Schedule. Any delay in achieving the approved milestones shall attract applicable penalties and liquidated damages as specified in the tender.

Project Implementation Timeline

The Successful Bidder shall complete the supply, installation, integration, testing, commissioning, and Go-Live of the complete City Surveillance System, including all field infrastructure, network connectivity, central infrastructure, software applications, licenses, integrations, training, documentation, and acceptance testing,

The project shall be considered successfully completed only after successful Go-Live of the entire system and issuance of Final Acceptance Certificate by VMC.

The bidder shall submit a detailed Project Implementation Schedule within fifteen (15) days from the date of Work Order for approval by VMC. However, the overall project completion and Go-Live timeline shall remain ____ Months from the date of Work Order, irrespective of any internal activity schedules submitted by the bidder.

Any delay beyond the stipulated Go-Live date shall attract penalties and liquidated damages as specified in this tender.

Go-Live Date: Within 90 Days (3 Months) from the date of issuance of Work Order.

Note: **one single Go-Live date** and do not want phase-wise.

a) No advance payment will be made.

b) All payments shall be made in Indian rupees.

c) Mode of Billing and Payments

i. The Bidder shall submit invoices in duplicate on the firm's printed bill forms along with all supporting documents, inspection reports, installation certificates, commissioning reports, and acceptance certificates as required by VMC.

ii. Payment shall be released only after successful supply, installation, integration, testing, commissioning, and acceptance of the respective items/services by VMC and submission of all required documents.

iii. The successful bidder shall submit a Performance Bank Guarantee (PBG) as specified in the tender conditions, which shall remain valid for the prescribed contract period.

iv. Operation & Maintenance (O&M)/CAMC payments shall be released on a **quarterly basis in arrears** after satisfactory performance of services during the respective quarter and after adjustment of applicable SLA penalties, if any.

v. VMC reserves the right to withhold part or full payment in case the work progress, quality of work, deliverables, or service levels are not found satisfactory as per the contract conditions.
vi. The bidder shall promptly rectify all defects, deficiencies, observations, and non-compliances pointed out by VMC. Payments withheld on account of such deficiencies shall be released after satisfactory rectification and verification by VMC.
vii. All payments shall be subject to deductions of applicable taxes, statutory recoveries, penalties, liquidated damages, and other recoveries as per the provisions of the contract.
viii. No payment shall be admissible for incomplete, unapproved, non-functional, or non-compliant work.

5.2 Milestone-Based Delay Penalty for CAPEX Implementation & payment schedule is

The Successful Bidder shall complete each milestone within the timelines specified in the Approved Project Schedule. Delay in achieving any milestone attributable to the Bidder shall attract penalties as defined below.

Sr. No.	Milestone	Payment (%)	Delay Penalty
1	Supply of Equipment at Site, Verification and Submission of Documents	60% of CAPEX Value	0.5% of Milestone Value per week of delay
2	Successful Installation, Integration, Testing, Commissioning and Acceptance by VMC	30% of CAPEX Value	1.0% of Milestone Value per week of delay
3	Successful Go-Live of Complete System and Operational Acceptance	10% of CAPEX Value	1.5% of Milestone Value per week of delay
4	O&M/CAMC Services	Quarterly Payments	SLA-based Penalties as defined in the Contract

5.3 CAPEX Implementation Delay Penalty

The Successful Bidder shall complete the entire CAPEX scope, including supply, installation, integration, testing, commissioning, documentation, training, and Go-Live of the complete City Surveillance System within the stipulated timeline specified in the Approved Project Schedule and in any case not later than the final Go-Live date specified in the Work Order.

In the event of delay attributable to the Bidder, the following penalties shall be applicable:

Delay Period	Penalty
Up to 15 Days	0.5% of total CAPEX Contract Value per week or part thereof
16 to 30 Days	1.0% of total CAPEX Contract Value per week or part thereof
31 to 60 Days	1.5% of total CAPEX Contract Value per week or part thereof
Beyond 60 Days	2.0% of total CAPEX Contract Value per week or part thereof

The maximum cumulative penalty for delay in completion of CAPEX implementation shall be limited to **10% of the Total CAPEX Contract Value**.

For milestone-based monitoring, VMC may also impose penalties for non-achievement of approved milestones. The penalty shall be calculated as follows:

Penalty = (Value of Delayed Milestone × Applicable Penalty Percentage × Number of Weeks Delayed)

If the delay in completion exceeds **90 days** from the approved Go-Live date, VMC reserves the right to:

- Invoke Performance Bank Guarantee (PBG).
- Procure remaining work through an alternative agency at the risk and cost of the Bidder.
- Recover additional costs incurred by VMC from pending payments and/or security deposits.
- Terminate the contract after giving due notice.

No penalty shall be applicable for delays arising due to force majeure events, written instructions from VMC, statutory approval delays not attributable to the Bidder, or any other reasons expressly approved by VMC in writing.

Illustration:

- Total CAPEX Contract Value = INR 5 Crore
- Delay = 4 Weeks

- Applicable Penalty = 1% per week
- Penalty = INR5 Crore × 1% × 4 = INR 20 Lakh

However, the total penalty shall not exceed 10% of the CAPEX Contract Value, i.e., INR 5 Crore.

Note: Penalty for implementation delay shall be independent of SLA penalties applicable during the OPEX period. Both penalties may be levied separately wherever applicable.

A. All payments under this contract shall be made to the accounts of the Bidders with the Bidders Banker.

B. The prices mentioned in the Price Bid should include all applicable taxes. Any deviations due to change in the rate of taxes and duties or any introduction of new taxes and duties would be bear by the Bidder and VMC shall not pay any additional amount for such tax rate deviation. The VMC shall be entitled to deduct tax at source or any other taxes/cess as may be applicable.

C. Suspension of payments: VMC may at any time during the term of the Contract, issue a written notice of suspension and suspend all payments to the qualified Bidder under the Contract, if the qualified Bidder fails to perform any of his obligations under this Contract. The VMC shall, prior to the suspension of the payments, request the qualified Bidder to remedy such failure within a period of 15 days from the date of' issue of such notice of suspension. The notice of suspension shall specify the nature of the failure. In the event the Bidder does not rectify the failure within the specified time prescribed, the VMC shall have the right to terminate the Contract.

- Total Project Duration – Implementation Period +5 Years of Compressive O&M
- VMC may issue commissioning and G0-Live approval in parts for allotted work.
- Penalties shall be levied only for the reasons attributable to the implementing Bidder. Any risks/ issues foreseen by the implementing Bidder shall be brought to the notice of the VMC immediately. If no such issues/risks are highlighted by the implementing IA, then it is expected that no delays will be there in the implementation schedule
- Part commissioning and Go-Live will be based on recommendations of VMC for purpose of better use of deployed Total CCTV Project solutions.
- The Successful Bidder shall complete the entire project, including supply, installation, integration, testing, commissioning, and Go-Live, within twelve (12) weeks from the date of the Work Order. In case of delay attributable to the Bidder, a penalty of 1% of the value of the delayed work per week or part thereof, subject to a maximum of 10% of the value of the delayed work, shall be levied.
- If items not installed or remain pending of hardware supply, then per week Rs. 10,000 penalties.
- In case of delay in project implementation phase on vendor side, VMC at its discretion may or may not increase the total duration of the project.
- The Bidder shall submit invoices along with all supporting documents, delivery challans, inspection reports, installation reports, commissioning certificates, and acceptance certificates as required by VMC.
- (60%) of the CAPEX value shall be released after successful supply and delivery of equipment at designated locations, verification by VMC, and submission of all required documents.
- (30%) of the CAPEX value shall be released after successful installation, integration, testing, commissioning, and acceptance of the supplied equipment by VMC.
- remaining Ten Percent (10%) of the CAPEX value shall be released after successful Go-Live of the complete City Surveillance System and issuance of Final Acceptance Certificate by VMC.
- O&M/CAMC charges shall be paid quarterly in arrears after satisfactory completion of services during the respective quarter and after adjustment of SLA penalties, if any.
- The Successful Bidder shall furnish a Performance Bank Guarantee (PBG) equivalent to 10% of the Contract Value valid throughout the implementation period and O&M/CAMC period as specified in the contract.
- VMC shall have the right to withhold payment wholly or partly in case of non-performance, non-compliance with specifications, delays, defects, or SLA violations until such deficiencies are rectified.
- All payments shall be subject to deductions towards applicable taxes, statutory recoveries, penalties,

liquidated damages, and other recoveries as per contract provisions.

Non-submission of declaration from all respective OEMs will allow VMC to cancel issued WORK ORDER, forfeiting of security deposit and proceed ahead for assigning the contract with next the lowest bidder (L2 Bidder).

NOTE: In case, end of service and support declared by OEM for any product at this juncture, successful bidder has to submit declaration letter of third party service provider confirming to provide back to back onsite service and support with labor and availability of spare parts to VMC throughout the project's contract period. However, successful bidder has to submit documentary proof issued by OEM confirming end of service and support of that particular product.

- Successful Bidder will be responsible for mentioned indicative list of activities applicable for successful Comprehensive Annual Maintenance Contract (CAMC) of all the components/sub-components of the project.
- Preventive maintenance of CCTV surveillance system (like cleaning of camera lenses etc.) shall be responsibility of CCTV operator on quarterly basis. It is the responsibility of CCTV operator to make sure that the CCTV room shall be properly aligned and organized in a decent manner at any point of time.
- Shifting of Camera pole / tower if any during contract period required / intimated by VMC will be carried out by successful bidder without any additional cost implication to VMC.
- STQC/BIS-ER Compliance for CCTV Cameras- All new CCTV cameras proposed under this contract, including replacement and additional cameras, shall comply with the latest applicable STQC or BIS Essential Requirements (BIS-ER) notified by the Government of India. The bidder shall ensure that all supplied CCTV cameras possess valid STQC certification and BIS registration, wherever applicable, at the time of supply and installation. Documentary evidence, including valid certificates and test reports, shall be submitted to VMC prior to deployment. Any camera found non-compliant with STQC/BIS-ER requirements shall be replaced by the bidder at no additional cost to VMC.
- All supplied hardware, software, licenses, and services shall be supported under comprehensive warranty and maintenance for a period of five (5) years from the date of Go-Live and final acceptance by VMC.
- The quoted prices shall be inclusive of supply, transportation, insurance, installation, testing, commissioning, integration, configuration, training, documentation, warranty, maintenance, and all other activities required for successful implementation of the project.
- The Bidder shall clearly specify the OEM, Make, Model Number, and Part Number of all proposed equipment, software, and licenses.
- Any hardware, software, accessory, component, fitting, mounting material, cable, connector, license, service, or item required for successful completion and operation of the project, whether explicitly mentioned in the BOQ or not, shall be deemed included in the Bidder's scope and quoted price without any additional cost to VMC.
- The Financial Bid shall clearly provide separate breakup of CAPEX and OPEX/CAMC components.
- Comprehensive Warranty and CAMC support obligations shall be clearly defined by the Bidder. CAMC rates shall be quoted separately for each year of the O&M period.
- The Bid shall remain valid for a minimum period of 180 days from the last date of bid submission.
- Software licenses, Video Management Software (VMS) licenses, analytics licenses, operating system licenses, database licenses, and other application licenses shall be perpetual unless specifically stated otherwise in the tender.
- The proposed solution shall be scalable and capable of supporting a minimum of 900 CCTV cameras, including future expansion without requiring replacement of the core infrastructure.
- The quantities indicated in the BOQ are tentative and may increase or decrease based on site conditions, survey findings, operational requirements, and VMC directives. Payment shall be made strictly on the basis of actual quantities supplied, installed, tested, commissioned, and accepted by VMC at approved unit rates.
- The proposed Storage System, Chassis, Servers, Video Management System, and associated central infrastructure shall support integration of approximately 250 existing CCTV cameras of the legacy project and shall be sized with a minimum 20% spare capacity for future expansion without requiring major hardware replacement.

5.4 Site Survey and BOQ Verification

The Successful Bidder shall carry out a detailed site survey of all locations provided by VMC within fifteen (15) days from the date of handover of the locations by VMC and submit the Site Survey Report for VMC's review and approval.

Detailed Site Survey and Submission of Site Survey Report shall be completed from the date the locations are provided by VMC.

The quantities provided in the tender BOQ are indicative and intended for evaluation purposes. The bidder shall verify the actual site requirements, installation conditions, power availability, network connectivity, mounting arrangements, civil requirements, and other infrastructure requirements during the site survey.

Based on the approved site survey, the bidder shall submit the location-wise detailed Bill of Quantities (BOQ) and implementation plan. Billing shall be made based on the actual quantities supplied, installed, tested, commissioned, and accepted by VMC at the approved unit rates quoted in the Financial Bid.

In case any variation in quantity, addition, deletion, or modification of items is identified during the site survey, the same shall be brought to the notice of VMC immediately for approval before execution. No additional work or quantity variation shall be executed without prior written approval from VMC.

The bidder shall be responsible for completing the project objectives and ensuring full functionality of the City Surveillance System. Any minor items, accessories, fittings, mounting hardware, cabling, connectors, consumables, software components, or other materials required for successful implementation of the project, but not specifically mentioned in the BOQ, shall be deemed included in the bidder's scope and quoted rates, and no separate payment shall be made for such items.

Payment shall be released based on the actual executed and accepted quantities as certified by VMC. However, the bidder shall not be entitled to claim any compensation, damages, idle charges, or loss of profit on account of variation in quantities from the indicative BOQ provided in the tender.

VMC reserves the right to increase, decrease, add, delete, or modify the quantities of any item during the contract period based on actual site requirements, budget availability, and project needs, and payment shall be made at the approved contract unit rates.

5.5 Existing Hardware Assessment and Reuse

The Successful Bidder shall conduct a detailed site survey and technical assessment of the existing hardware, infrastructure, and associated equipment available at VMC before execution of the work. Wherever technically feasible and approved by VMC, the existing hardware shall be reused and integrated into the proposed solution.

The BOQ provided in this Tender is indicative. The final quantity of new equipment shall be finalized based on the joint site survey, assessment of reusable existing hardware, and approval of VMC. Payment shall be made strictly on the actual quantities supplied, installed, commissioned, and accepted, after considering the reuse of existing hardware. The Bidder shall have no claim for variation in quantities arising out of such assessment.

The Successful Bidder shall submit a detailed Existing Hardware Assessment Report, indicating reusable, repairable, and non-reusable equipment, along with recommendations and supporting photographs, for approval by VMC before procurement and execution of the work.

5.6 Refurbished/Used Equipment

The Successful Bidder shall supply only new, unused, genuine, and currently manufactured hardware, software, and accessories. Refurbished, repaired, reconditioned, remanufactured, second-hand, or End-of-Life (EOL)/End-of-Support (EOS) products shall not be permitted under this Contract. The Bidder shall submit OEM authorization and warranty/support documents, if required by VMC, as proof of compliance.

5.7 Structured Cabling and Cable Management Requirements

The Successful Bidder shall implement a structured cabling system for all Data Centre, CCC, Network Rooms, Rack Locations, Field Cabinets, Junction Boxes, and Surveillance Sites. All cables shall be properly routed,

dressed, bundled, tagged, ferruled, and terminated in a neat and professional manner using industry best practices.

The Bidder shall ensure proper cable management through cable trays, raceways, ducts, cable managers, patch panels, and rack organizers. Loose, hanging, exposed, improperly terminated, or unlabelled cables shall not be acceptable. All power, network, fiber, and control cables shall be clearly identified at both ends with permanent labels indicating source and destination details.

All racks, cabinets, patch panels, switches, servers, and field equipment shall be properly organized and documented to facilitate maintenance, troubleshooting, and future expansion. The Bidder shall submit cable schedules, labelling details, rack layouts, and as-built documentation for approval by VMC. Any untidy, unsafe, or non-standard cabling installation observed during inspection shall be rectified by the Bidder at no additional cost to VMC.

5.8 Service Level Agreement & Penalty for O&M Phase

The minimum scope of work (but not limited) is mentioned herewith as below:

The purpose of this Service Level Agreement (hereinafter called SLA) is to clearly define the levels of service which shall be provided by the successful Bidder to End Customer and VMC for the duration of the contract. The SLA is intended to establish a clear set of measurable parameters against which the performance of the Implementation Bidder can be measured.

- The bidder needs to maintain a reliable database to monitor the project progress and their performance against the SLAs.
- Five Years' Onsite Comprehensive Maintenance Contract of IT Infrastructure & CCTV based system installed at identified establishment and prominent locations and CCC of VMC.
- Selection of an Agency for Comprehensive Operation, Maintenance, Facility Management and Upgradation of City Surveillance, IT Infrastructure and Command & Control Centre (CCC) in Vadodara for a Period of Five (5) Years
- Bidders need to report the compliance/non-compliance to SLAs on weekly and monthly basis.
- Maximum penalty is restricted to 10% of the project cost. Non-compliance to SLA beyond the penalty limit may lead to termination of contract on discretion of VMC.
- Penalties shall be levied only for the reasons attributable to the implementing Bidder. Any risks/ issues foreseen by the implementing Bidder shall be brought to the notice of the VMC immediately. If no such issues/risks are highlighted by the implementing Agency, then it is expected that no delays will be there in the implementation schedule
- Storage/archiving of video and picture stream for 30 days recording at 25/30 FPS for Majority of the time and 15 FPS on the lean period onH.264 compressions, 1080p resolution. However certain feed may be required to be retained for court/investigation related matters beyond the standard period.
- Resolve of problem/issues, trouble shooting and correction, service packs, device driver & Firmware Upgrades, Systems support like OS reinstallation, registration of problems / issues to Original Equipment Manufacturer (OEM) by raising the tickets and follow up till the problem resolution.
- Monitoring of day to day operation and provide technical support for resolving network related issues pertaining to antivirus, firewall, CCTV surveillance systems, router, switches, software, OFC link availability etc.
- Monitoring of Wired Network and provide technical support for resolving issues pertaining to them for effective use of network services.
- Hardening of "Operating System" (OS), blocking of all unnecessary opened ports, access and authorization control.
- Assist in IT Asset verification
- The comprehensive maintenance of hardware includes repair / replacement of all faulty systems / parts including fusser assemblies and internal plastic parts of the devices.

- Patch Management: Evaluation of suitability / requirement of Microsoft Windows Servers patches and application of the same on all servers if required.
- The successful bidder will be responsible for providing virus free computing environment at all locations covered in the project.
- Poles require regular preventive maintenance for the purpose of early detection of deterioration and thereby to prevent breakdowns and subsequent consequences. Therefore, the successful bidder's engineer should visit and conduct health checkup of all poles in every six months or as and when required. If anything is found to be faulty should be replaced / repaired without any additional cost to VMC.
- The parts / components / sub-assemblies used for repair / replacement by the successful bidder will be of the same make and model. Successful bidder has to give declaration that the parts changed are new one.
- In case of model discontinued declared by OEM then in such case, the successful bidder has to supply parts / components of same make with same / higher technical specifications of same OEM post approval of VMC. In case of systems that are not serviceable by the successful bidder due to obsolescence of technology or non-availability of parts / components / assemblies, the successful bidder has to provide the system of same / alternate make with same / higher technical configuration on concurrence of VMC decision in this regard shall be final and bind to the successful bidder.
- In case the system cannot be repaired within the office premises and is required to be taken to the successful bidder's workshop, then all cost of material's bi-directional transportation will be borne by the successful bidder. Stand-by arrangement will be provided by the successful bidder without any additional cost to VMC. Non supply of Stand-by arrangement will be treated as downtime and penalty shall be applicable as mentioned in the SLA (Service Level Agreement).
- Successful Takeover of all project components including IT & Non-IT hardware and software from existing agency of VMC.
- The bidder is responsible for the specified line items and Bill of Quantities (BOQ), which are associated with a Comprehensive Annual Maintenance Contract (CAMC) extending over a period of 05 (Five) years.
- The parts / components / sub-assemblies used for repair / replacement by the successful bidder will be of the same make and model. Successful bidder has to give declaration that the parts changed are new one.
- In case of model discontinued declared by OEM then in such case, the successful bidder.
- has to supply parts / components of same make with same / higher technical specifications of same OEM post approval of VMC Information Technology department of corporation office.
- The bidder is responsible integration cost is included as part of the Comprehensive Annual Maintenance Contract (CAMC) cost for the Command and Control Center (CCC). There are no additional charges for integrations, as outlined in the Statement of Work (SOW).
- The bidder is responsible for ensuring that all equipment within the Command and Control Center (CCC) is included in the monitoring application. Furthermore, the vendor is accountable for all aspects of the Comprehensive Annual Maintenance Contract (CAMC) for the CCC over a period of 5 years.
- Operation and Maintenance of all the hardware (IT & Non-IT) and software inclusive of cables and all other applicable accessories required for successful Comprehensive Annual Maintenance Contract (CAMC) and maintain SLA.
- Rectification / replacement (as per requirement) of faulty cables (CAT, power cables etc.) and all types of patch cords in field and data center along with all ancillary accessories. For cables installed in field, selected bidder will be responsible to rectify / replace the cables with the same methodology through which it was installed earlier. For ex.: underground (UG) laid cables will be maintained with UG method only. VMC may allow temporary change of installation method (for ex. Aerial) considering the field restrictions / challenges in terms of permission availability, technical feasibility etc. at its own discretion for maximum time of 10 days. SI will have to update temporary work with permanent method within 10 days failing which penalty of INR 5,000/- per day will be levied to SI.

- Replacement of faulty IT & Non-IT equipment along with all ancillary accessories with the existing / new make (brand) with matching / higher specifications (that must be approved by VMC before installation) which must be compatible with other network hardware & software components in all technical & functional aspects of integration to ensure successful working of the entire network / system with at least earlier level of accuracy. All cameras provided as a replacement of faulty camera should be ONVIF supported.
- All new CCTV cameras proposed under this contract, including replacement and additional cameras, shall comply with the latest applicable STQC and BIS Essential Requirements (BIS-ER) notified by the Government of India. The bidder shall ensure that all supplied CCTV cameras possess valid STQC certification and BIS registration, wherever applicable, at the time of supply and installation. Documentary evidence, including valid certificates and test reports, shall be submitted to VMC prior to deployment. Any camera found non-compliant with STQC/BIS-ER requirements shall be replaced by the bidder at no additional cost to VMC.
- The bidder shall ensure that all newly supplied CCTV cameras are ONVIF-compliant, cybersecurity-compliant, and compatible with the existing Video Management System (VMS) and Command & Control Centre (CCC) infrastructure.
- All Electrical Power involves the MGVCL connections and billings, which are paid by the vendor and then claimed by VMC, all Vadodara City Surveillance project (CAMC) MGVCL connections by payment to the vendor.
- The bidder bears the responsibility of guaranteeing that in the event a device or Original Equipment Manufacturer (OEM) is determined to be non-repairable, swift action is taken to replace it within a specified timeframe of 72 hours. This entails prompt identification of the non-repairable item, coordinating procurement or sourcing of a replacement, and ensuring its installation or integration into the system within the stipulated 72-hour window. This proactive approach aims to minimize downtime, maintain operational efficiency, and fulfill contractual obligations to provide uninterrupted service to the client.
- During the contract period, any civil work required to execute in the data center by the successful bidder for CAMC purpose, it has to bring into the notice of DIRECTOR (IT) – VMC and post approval, the successful bidder can perform the task. However, it is the bidder's responsibility to restore such area in the original position as per direction and satisfaction of DIRECTOR (IT) – VMC. Non-compliance of the same will allow VMC to complete the work from the third party and the cost for the same shall be recovered from henceforth contract payment.
- Preventive maintenance of CCTV surveillance system (like cleaning of camera lenses etc.) shall be responsibility of Support team member on regular basis. It is the responsibility of Support team member to make sure that the CCTV room shall be properly aligned and organized in a decent manner at any point of time.
- During the contract period, any civil work required to execute in CCC/ Field side by the successful bidder for maintenance purpose, it has to bring into the notice of DIRECTOR (IT) – VMC and post approval, the successful bidder can perform the task. However, it is successful bidder's responsibility to restore such area in the original position as per direction and satisfaction of DIRECTOR (IT) – VMC. Non-compliance of the same will allow VMC to complete the work from the third party and the cost for the same shall be recovered from henceforth quarterly payment.
- Successful bidder shall manage NMS which include a dashboard showing live status of all CCTV Camera with CCC equipment under the contract including daily issues, spares, breakdown status etc.
- Daily, weekly, monthly, quarterly and yearly uptime reports of all CCTV Camera and major CCC equipment shall be generated through this NMS. Generated reports in any format (excel, pdf etc.) shall be compulsory matched with system data. This system generated quarterly uptime reports shall be a mandatory document required while processing quarterly payment. Successful bidder is bound to provide uptime reports in any format whenever required by VMC.
- This software shall be configured to provide user based access (like location wise viewing / full control etc.) as per requirement defined by VMC.
- In the event that VMC procures and commissions new equipment during the contract period, the O&M/CAMC charges for such newly procured equipment shall become applicable from the date of commissioning. If the new equipment is procured against an existing asset covered under the

O&M/CAMC scope, the O&M/CAMC charges applicable to the existing asset shall cease from the date of commissioning of the new equipment. The bidder shall accordingly revise and adjust the O&M/CAMC billing.

- Exclusions: Consumables items like batteries and physical damaged to equipment (to be proved by successful bidder with proof) and burnt parts are excluded from this contract.
- Based on past years' experience, average supply of new field equipment and OFC with supporting accessories pertaining to project is as stated below:
- The successful bidder shall restore up to 15 fiber cuts per year, aggregating to a maximum of 75 fiber cuts during the five (5) year contract period. Accordingly, the successful bidder shall carry out the following indicative works/services without any additional cost implication to VMC during the contract period.
- The minimum trench depth for laying Optical Fiber Cable (OFC) shall be 1.65 meter from the finished ground level and near ground level. Depending on site conditions, road crossings, utility crossings, terrain, statutory requirements, or directions of VMC/concerned authorities, the trench depth shall be increased wherever required. Under no circumstances shall the trench depth be less than 1.6 meters, unless otherwise specifically approved in writing by VMC.
- All LT power/electrical cables from the CCTV pole to the Junction Box (JB) shall be laid at a minimum trench depth of 0.8 meters from the finished ground level. Wherever required due to site conditions, road crossings, utility crossings, or statutory requirements, the trench depth shall be increased up to 1.0 meter or more. Suitable protection such as DWC/HDPE pipe, GI pipe, or RCC/concrete encasement shall be provided wherever the minimum depth cannot be achieved or as directed by VMC.
- Where the required trench depth cannot be achieved due to site constraints, the cables shall be protected using HDPE/DWC pipes, GI pipes, RCC/concrete encasement, or any other approved protection method, as directed by VMC.
- After completion of cable laying, the trench shall be backfilled, compacted, and the road/footpath/pavement restored to its original condition or better, to the satisfaction of VMC and the concerned authority.
- Adherence to required SLA / uptime for all hardware & software components as described in this RFP. Selected SI will have to submit reports from NMS application in pdf and excel / csv formats.
- Obtaining the required RoW/Permissions from the authorities/local bodies including preparation of all required documentation for the same and submission to the office of the authorities. However, VMC will provide adequate assistance/support as and when required.
- Coordination with other vendors and internal / external stakeholders of the project for successful Comprehensive Annual Maintenance Contract (CAMC) O&M of the infrastructure.
- Bidders are required to collaborate and support in terms of sharing of data, infrastructure, technology and other details with other agencies appointed by VMC.
- Update in existing dashboards as per the requirement of the project and guideline provided by VMC. Selected bidder will have to deploy concern team as mentioned in requirement.
- Integration of any new APIs to integrate the existing systems with any new proposed system in future (Timeline for the development & testing will be mutually decided by VMC, selected bidder & new bidder).
- Preparation of daily, weekly, monthly project reports as per the requirement of the project and guideline provided by VMC.
- Maintaining / updating the existing project documents (network design documents / drawings, as-build drawings, reports, databases etc.) as per any changes happens in the network infrastructure.
- Preparation of new project integration documents (network design documents / drawings, as-build drawings, reports, databases etc.) as per project requirement.
- Attending all the meetings by competent project team members of the selected bidder at the prescribed office location as conveyed by VMC.

- In pursuit of best technological result, bidders are encouraged to provide same / new make of the existing products by ensuring that the functional / technical specifications are either same or better. Further, all the changed make shall be approved by VMC before they are deployed. The bidders have to ensure that the quoted make is compliant to the existing technology and affiliate infrastructure and capable of delivering the mentioned SLAs.
- Bidder will have to provide security check certificate for new equipment installed to confirm that the equipment is hacked proof.
- The refurbished / repaired items shall not be supplied & installed either on temporary or permanent basis. Repairs are only permitted for the currently installed items.
- The entire financial liability in respect of Technical Team deployed for this contract shall be of the successful Bidder and the VMC will in no way be liable.
- Successful Bidder shall be solely responsible for the Redressal of grievances / resolution of disputes relating to persons deployed. VMC shall, in no way, be responsible for settlement of such issues whatsoever.
- VMC shall not be responsible for any financial loss or any injury to any person deployed by the bidder in the course of their performing the functions/duties, or for payment towards any compensation.
- The persons deployed by successful bidder shall not claim or be entitled to pay, perks and other facilities admissible to regular / confirmed employees of VMC during or after expiry of the Contract.
- Bidders are requested to consider quantity of respective equipment and Fiber with supporting accessories to be supplied throughout the contract period and accordingly, bidders have to quote Part of Price Bid.

In case, if more number of above mentioned quantity of respective equipment OR Fiber with supporting accessories required during the contract period, in such case, successful bidder has to submit a quotation (comprised of supply, installation, commissioning and maintenance of equipment) to DIRECTOR (IT) – VMC. The rates of equipment OR services are already derived from Price Bid quoted by the successful bidder. DIRECTOR (IT) – VMC will issue an e-mail confirmation to carry out the work post verification of quoted price.

Successful bidder has to complete work within 48 working hours counted from an e-mail confirmation receipt by DIRECTOR (IT) – VMC. This timeline subject to VMC concerned department's approval for carrying out work in case of public premises. Time period consumed in process of concerned department's approval shall be excluded while counting 48 working hours' timeline of work completion. In case, successful bidder failed to complete work within 48 working hours' timeline then penalty shall be applicable as defined in SLA for nos. of CCTV Camera whose feed not available at CCC.

However, VMC will issue work order in this regard post approval of internal management. Successful bidder will receive 100% payment post equipment verification, commissioning and acceptance by VMC.

5.9 Resource Deployment Requirement

- Bidder shall deploy adequate qualified technical, operational, field support, and supervisory resources required for successful implementation, operation, monitoring, maintenance, and support of the CCTV Surveillance System throughout the Contract Period.
- Bidder shall ensure availability of sufficient resources for preventive maintenance, corrective maintenance, fault rectification, network restoration, fiber maintenance, camera maintenance, CCC operations, and emergency response to achieve the prescribed SLA requirements.
- Exact number, qualification, deployment location, and utilization of resources shall be the sole responsibility of the Bidder. VMC shall not prescribe or verify attendance of individual

resources. The Bidder shall remain fully responsible for meeting all uptime, performance, response time, restoration time, and other SLA obligations defined in the Contract.

- Bidder shall nominate a Single Point of Contact (SPOC) for coordination with VMC throughout the Contract Period. Failure to maintain adequate resources resulting in SLA violations shall attract penalties as per the SLA and Penalty Schedule of this Tender.

5.10 Back-to-Back OEM Support

The Successful Bidder shall obtain and maintain **back-to-back support** from the respective OEM(s) for all supplied hardware, software, licenses, and solutions throughout the Warranty and CAMC/Contract Period. The Bidder shall submit valid OEM support/authorization certificates, service entitlement documents, warranty certificates, or OEM confirmation letters as documentary proof to VMC before Go-Live and whenever renewed during the Contract Period. Failure to maintain valid OEM back-to-back support shall be treated as a material breach of the Contract and appropriate action, including SLA penalties or other contractual remedies, may be initiated by VMC.

5.11 Asset Management Services:

The Successful Bidder shall establish, maintain, and update a comprehensive Asset Management System for all hardware, software, licenses, networking components, field infrastructure, and associated assets deployed under this project throughout the contract period.

The scope of Asset Management Services shall include, but not be limited to, the following:

- Maintaining a centralized inventory database of all project assets including CCTV Cameras, PTZ Cameras, Edge Devices, Poles, Junction Boxes, Industrial Switches, UPS Systems, Sensors, Servers, Storage Systems, Video Wall Components, Networking Equipment, Fiber Infrastructure, Software Licenses, and other associated equipment.
- Recording detailed information of each asset including OEM Name, Make, Model, Serial Number, Asset ID, Configuration Details, Firmware Version, IP Address, Physical Location, Installation Date, Warranty Details, License Details, and Operational Status.
- Maintaining a complete software and license inventory including software versions, license validity, license keys, subscription details, support status, security patches, upgrades, and renewal history.
- Maintaining records of installation, commissioning, relocation, repair, replacement, preventive maintenance, breakdown maintenance, decommissioning, and disposal of all project assets.
- Recording all entry, exit, transfer, temporary replacement, and movement of assets during the contract period and submitting monthly asset inventory and movement reports to VMC.
- Assigning a unique Asset Identification Number/Tag to each asset supplied and installed under the project.
- Maintaining location-wise, asset-wise, OEM-wise, warranty-wise, and license-wise inventory records throughout the contract period.
- Ensuring 24x7x365 monitoring and availability of all CCTV cameras, poles, edge devices, networking equipment, servers, storage systems, video walls, sensors, software applications, and associated infrastructure as per the prescribed SLA requirements.
- Maintaining complete fault history, maintenance history, replacement history, and health status records for each asset.
- Providing periodic reports including Asset Inventory Reports, Warranty Reports, License Compliance Reports, Preventive Maintenance Reports, Breakdown Reports, and Asset Health Reports as and when required by VMC.
- Immediately notifying the designated VMC authority regarding any theft, vandalism, accidental damage, fire incident, natural calamity, or any other event affecting project assets

and generating an incident report for the same.

- Coordinating with VMC for lodging complaints/FIRs and completing necessary documentation in case of theft, vandalism, or loss of equipment.
- Providing standby equipment and restoring affected services within the prescribed SLA timelines in case of equipment failure, damage, or replacement requirements.
- Ensuring that all asset records are updated within 24 hours of any installation, replacement, relocation, repair, or configuration change.
- The complete Asset Management Database, inventory records, configuration details, maintenance records, and related documentation shall remain the property of VMC and shall be handed over in editable digital format at the end of the contract period.

5.12 Preventive Maintenance Services:

Successful bidder has to provide preventive maintenance services for all equipment's (IT as well as non-IT) covered in project at least once in a quarter or as per requirement. The preventive maintenance shall include (but not limited) as stated below:

- Cleaning and removal of dust and dirt from interior and exterior of equipment (like camera lenses etc.)
- It is the responsibility to make sure that CCC shall be properly aligned and organized in a decent manner at any point of time.
- Conduct inspection (check for loose contacts in cable and connections etc.), testing, satisfactory execution of diagnostics and necessary troubleshooting of equipment.
- Successful bidder has to intimate and take approval of DIRECTOR (IT) – VMC before carrying out preventive maintenance activity.
- Successful bidder has to maintain item-wise register of completed preventive maintenance activity. As well as corrective maintenance services
- Successful bidder has to maintain proper file / register of all incidents and problems occurred in ICCC and at field locations and subsequent corrective steps taken. Documentation of problems, isolation, and cause and rectification procedures for building knowledge base for known problems shall be maintained by successful bidder.
- Electrical Power involves the MGVCL connection and billing, which are paid by the vendor and then claimed by VMC, MGVCL connections by payment to the vendor.
- Prompt attendance and compliance or any complain of any CCTV Camera/ Sensors/Junction Box etc given by VMC Staff or from Traffic Police Staff.
- Site visit by technician of all the junction/field locations on specified scheduled intervals and he should get the signature of VMC staff in his diary. The dairy shall be submitted to the Engineer-in-charge of the VMC as and when asked.
- In case of emergency, the SI should attend the site immediately.
- The SI has to carry out Replacement/Maintenance of the field devices within specified SLA duration, from the receipt of the complaints.
- Periodical checking and cleaning of all field equipment such as Camera, systems, Sensors, etc its accessories specifically the cleaning of reflector and polycarbonate glass should be done fortnightly.
- Checking & tightening of cable terminals, replacement of wiring of field poles and to attend cable faults.
- Painting of all poles, signal heads and junction boxes etc. once a year with approved/specified colors as directly by Engineer-in-charge without any chargers.
- Muffing of poles should be done as and when suggested by Engineer-in-charge.

- Date wise fault attend, material replaced and replacement of accessories statement shall be prepared fort nightly and submit monthly in the Performa prepared by the department.
- Non availability of electric power at signals should be immediately informed to Engineer-in-charge and concerned sub division of electric utility company.
- The Project Manager or any authorized Senior-Level Officer of the SI/Bidder shall conduct a joint inspection of CCTV cameras, network infrastructure, poles, junction boxes, and other field equipment along with the designated VMC officials at least once every two (2) months or as and when directed by VMC. The SI/Bidder shall prepare and submit the inspection report, observations, identified issues, and corrective action plan within seven (7) days of such inspection. No Extra charge will be paid to SI on account of any additional material and labor.
- The successful bidder must take all precautionary measures to safe guard life of residents/citizens/pedestrian.
- Poster, Banner or any advertisement will not be allowed on any field device or poles (unless, the contractor shall take enough care in this regard.
- The scope of work described herein is indicative and not exhaustive. The System Integrator / Bidder shall be required to carry out all such activities, tasks, software changes and services etc as may be necessary for the successful implementation, operation, and maintenance of the project, even if not explicitly mentioned in this document. Any additional requirements, modifications, or enhancements as directed by VMC from time to time shall be deemed to be within the scope of work, and the System Integrator shall comply with the same without any additional cost implications, unless otherwise specified.
- If the Contractor fails to complete the work within the stipulated time as mentioned in the location-wise Work Order, a penalty shall be levied at the rate of 0.75% (zero point Seven five percent) of the value of the respective location-wise Work Order per week, or part thereof, subject to a maximum limit as prescribed in the tender conditions, until the work is completed to the satisfaction of VMC.
- Any damage caused to existing assets, structures, utilities, roads, pavements, ducts, poles, cabinets, cables, IoT devices, fiber network, or any other municipal or third-party property during the execution of work shall be entirely at the risk and cost of the Contractor. The Contractor shall be liable to repair, restore, or replace the damaged asset to its original condition or better, at no additional cost to VMC. (Subject to On Ground Survey).
- If the Contractor fails to carry out repair or replacement within the time specified by VMC, the Corporation shall have the right to carry out the repair/replacement through a third party, and the entire cost incurred along with applicable administrative charges shall be recovered from the Contractor's pending bills, Performance Security, or any other dues payable.
- Payments for services during the warranty period shall be released on a quarterly basis, subject to satisfactory performance of the System Integrator/Vendor and compliance with the applicable Service Level Agreements (SLAs). The payment shall be made after verification of services rendered during the respective quarter and submission of relevant reports, invoices, and supporting documents. Any applicable penalties or deductions as per the contract shall be adjusted prior to release of payment.
- The Bidder shall ensure that all CCTV cameras provide clear and usable video footage during both day and night conditions, including effective IR/Night Vision performance as per the specified camera coverage area. If any incident footage is found unclear, blurred, distorted, overexposed, underexposed, or not suitable for identification of persons, vehicles, registration numbers, or incident activities due to camera malfunction, IR/Night Vision failure, poor image quality, recording failure, storage failure, or network failure attributable to the Bidder, the same shall be treated as a Critical SLA Violation. VMC may impose a penalty of □5,000 per camera per incident and direct immediate rectification at the Bidder's cost.
- The System Integrator/Vendor shall maintain a complete and updated asset register for all supplied equipment, including cameras, batteries, routers, all components, and other field devices. Each asset shall be uniquely identified and recorded with make, model, serial number, location, and date of installation. In the event of any repair, replacement, or new supply, the System Integrator/Vendor shall ensure proper documentation and traceability. The details of the old (removed) item, repaired item, and new/replacement item shall be recorded with respective serial numbers, date of replacement/repair, reason for replacement, and current status. All replaced or faulty equipment shall

be tagged and returned/handed over to VMC or stored as per VMC directions. The System Integrator/Vendor shall ensure that any replacement equipment provided is new or OEM-certified, meeting the same or higher specifications. The complete record of such activities shall be maintained and submitted to VMC on a monthly/quarterly basis or as required, ensuring full transparency, auditability, and lifecycle tracking of all assets under the project.

5.13 SLA for Uptime - Availability of CITY SURVEILLANCE SYSTEM Pole, camera with Edge Device

During the Operation & Management period the implementing Agency is expected to keep the Technology and Performance levels above the specified level. It will be the responsibility of the implementing IA to design and code the software to ensure application is measurable in terms of uptime and performance

Bidder shall deploy and maintain a Ticket-Based Incident Management System for logging, tracking, escalation, resolution, and closure of all incidents related to the City Surveillance System.

The measurement of service availability and SLA compliance shall be based on:

- Reports generated from the Central Management Software, Video Management System (VMS), Network Management System (NMS), Ticketing System software deployed under the project.
- Inspection, verification, and audit conducted by VMC or its authorized representatives.
- Reports generated through licensed monitoring tools deployed by the Bidder.
- Reports generated through monitoring tools, applications, or platforms deployed by the Authority (VMC), Independent Agency, or Client during the contract period.

The records and reports generated from the above systems shall form the basis for calculation of uptime, downtime, incident response time, incident resolution time, and SLA compliance. In case of any dispute, the records maintained by VMC or its authorized monitoring platform shall prevail and be considered final for SLA assessment and penalty calculation. Bidder shall ensure continuous availability, operation, maintenance, and support of all components supplied, installed, integrated, and commissioned under the Project. The SLA framework is intended to ensure timely fault rectification, system availability, and uninterrupted surveillance services throughout the Contract Period.

The SLA shall apply to all field and central infrastructure including but not limited to:

- Fixed CCTV Cameras
- PTZ Cameras
- Edge Devices
- Junction Boxes
- Poles and Mounting Structures
- UPS Systems
- Network Switches
- OFC Connectivity
- Servers
- Storage Systems
- Video Management System (VMS)
- Active Video wall
- Recording Services
- Complete Surveillance Nodes

5.14 SLA Monitoring Mechanism

The Bidder shall deploy and maintain a Ticket-Based Incident Management System for registration, tracking, escalation, resolution, and closure of incidents.

The service availability shall be measured based on:

- VMS/NMS/EMS generated reports.
- Ticket-Based Incident Management System reports.
- Device health monitoring reports.
- Server and storage monitoring reports.
- Network monitoring reports.
- Active video wall
- Field verification conducted by VMC.

Availability Calculation

Availability shall be calculated on a **24x7x365 basis**.

$$\text{Availability (\%)} = \frac{\text{Total Time - Downtime}}{\text{Total Time}} \times 100$$

- Total Time = Total hours during the month.
- Downtime = Total period during which the component/service remained unavailable.

Sr. No.	Service Level Parameter	SLA Requirement	Measurement Basis
1	Fixed CCTV Camera Availability	Minimum 98.5%	Per Camera
2	PTZ Camera Availability		Per Camera
3	Edge Device Availability		Per Device
		Minimum 99.5%	
4	Network Switch Availability		Per Switch
5	Server Availability		Per Server
6	Storage Availability		Per Storage System
7	VMS Availability	Minimum 99.9%	Centralized System
8	Active Video Wall Availability (Including Controller, Display Units, Video Wall Management Software and Associated Hardware)		As per Quarterly SLA Penalty Matrix (Maximum 10% of Quarterly Contract Value)

***Total project Value/Cost**

The reports generated from the above systems shall be considered final for SLA measurement and penalty computation and Monitoring Software Logs, VMS Reports, Ticketing System and Verification by VMC.

5.15 Quarterly SLA Penalty Matrix

SLA performance shall be monitored monthly; however, penalties shall be accumulated and recovered from the Quarterly Invoice.

SLA Achievement During Quarter	Penalty Applicable
≥ SLA Requirement	Nil
SLA shortfall up to 2%	2% of Quarterly O&M/CAMC Charges
SLA shortfall > 4%	4% of Quarterly O&M/CAMC Charges
SLA shortfall > 6%	6% of Quarterly O&M/CAMC Charges
SLA shortfall > 8%	8% of Quarterly O&M/CAMC Charges
SLA shortfall >10%	10% of Quarterly O&M/CAMC Charges

Component-wise Penalty Calculation Methodology

For distributed components such as Cameras, Poles, Edge Devices, UPS, Switches, Junction Boxes, and OFC Links, penalty shall be calculated proportionately based on the affected quantity.

$$Penalty = QuarterlyO\&MCharges \times ApplicablePenalty\% \times \frac{AffectedQuantity}{TotalInstalledQuantity}$$

Example.:

Particular	Value
Quarterly O&M Charges	INR 30,00,000
Total Cameras Installed	900
Cameras Not Meeting SLA	18
Applicable Penalty	2%

Central Infrastructure Penalty

The Bidder shall ensure continuous availability and smooth operation of all centralized infrastructure components, including but not limited to the Video Management System (VMS), Recording Servers, Application Servers, Database Servers, Storage Systems, Central Monitoring Software, Active Video Wall, Video Wall Controllers, and associated software and hardware components. Since these systems are critical and support the overall surveillance ecosystem, any failure or unavailability of such centralized infrastructure shall be considered a major service disruption impacting the entire project.

The availability of centralized infrastructure shall be measured on a 24x7x365 basis through system-generated logs, monitoring tools, health reports, ticketing systems, and verification by VMC. In the event that any centralized infrastructure component fails to meet the prescribed SLA requirements, the applicable penalty shall be levied on the respective Quarterly O&M/CAMC charges. The Bidder shall restore the affected service within the stipulated resolution timeline and ensure that surveillance, recording, storage, monitoring, playback, and video wall display functionalities remain operational.

Any penalty applicable for centralized infrastructure failures shall form part of the overall SLA penalty calculation. However, the total cumulative SLA penalty recoverable during any quarter, including penalties for centralized infrastructure and field infrastructure, shall not exceed 10% (Ten Percent) of the Quarterly O&M/CAMC Invoice Value. Repeated failures of centralized infrastructure affecting surveillance operations may result in additional contractual actions, including replacement of faulty equipment, invocation of Performance Security, or termination of the Contract as per the tender conditions. Penalty shall be calculated directly on the Quarterly O&M Charges as these systems impact the entire surveillance network.

example:

$$Penalty = QuarterlyO\&MCharges \times ApplicablePenalty\%$$

Theft, Vandalism and Damage

- In case of theft, vandalism, accidental damage, fire, pole damage, equipment loss, or any other physical damage:
- The Bidder shall immediately lodge complaints/FIRs with the appropriate authorities.
- The Bidder shall coordinate insurance claims.
- The Bidder shall deploy temporary or standby equipment and restore services within SLA timelines.
- Delay in restoration beyond stipulated timelines shall attract SLA penalties.

Fault Rectification Time

Severity	Description	Resolution Time
Critical	Pole/Camera/Edge Device completely down	Within 12 Hours
Major	Video unavailable, recording failure, Edge Device malfunction	Within 24 Hours
Minor	Camera angle, focus, cleaning, configuration issue	Within 24 Hours

Active Video Wall Resolution Time

Severity	Incident Type	Resolution Time
Critical	Complete Video Wall Failure / Controller Failure	Within 4 Hours
Major	More than 20% Display Panels Not Functional	Within 8 Hours
Minor	Single Panel Failure / Display Layout Issue	Within 24 Hours

In the event of failure of the Active Video Wall System, including Video Wall Controller, Video Wall Management Software, Display Panels, Switching Components, or associated infrastructure, the Bidder shall restore the services within the stipulated resolution timelines. Failure to comply shall be considered for SLA calculation and penalty deduction. The penalty shall be calculated and accumulated on a quarterly basis and deducted from the respective Quarterly O&M/CAMC invoice. The cumulative penalty, including all SLA parameters under the Project, shall not exceed 10% of the Quarterly O&M/CAMC Invoice Value.

If the Active Video Wall remains unavailable for more than 24 continuous hours or if the same incident occurs more than three (3) times in a quarter, VMC may direct the Bidder to replace the faulty hardware & software at no additional cost and may invoke contractual remedies as deemed appropriate.

SLA for Operations & Maintenance (O&M) Services

Service Level Requirement:
The Successful Bidder shall ensure uninterrupted 24x7x365 operation, monitoring, maintenance, fault rectification, and support of the entire CCTV Surveillance System, including cameras, poles, network, fiber, switches, storage, servers, VMS, video wall, and associated infrastructure. The Bidder shall deploy adequate resources required to meet the prescribed SLA requirements throughout the Contract Period.

Measurement of Service Level Parameters:
The following shall be considered SLA non-compliances:

- Failure to meet system uptime and availability requirements.
- Failure to meet response and restoration timelines.
- Non-availability of CCTV cameras, network, storage, servers, video wall, VMS, or any associated infrastructure beyond permissible limits.
- Failure to provide 24x7 monitoring and technical support services.
- Repeated incidents indicating inadequate resource deployment affecting service levels.
- Failure to submit periodic reports, preventive maintenance records, or incident reports as required by VMC.

Penalty for Non-Achievement of SLA Requirements:

Service Parameter	Penalty
Failure to meet SLA response/restoration timelines	As per SLA Penalty Matrix
CCTV Camera/Node uptime below prescribed SLA	As per uptime penalty schedule
Central Infrastructure (VMS, Storage, Servers, Video Wall, NMS, Database, etc.) downtime	As per critical infrastructure penalty schedule
Failure to provide 24x7 monitoring and support services	INR 5,000 per occurrence
Repeated SLA violations (3 or more instances in a quarter)	Additional 10% deduction from quarterly O&M payment

Note:

- All works under execution shall be open for inspection and review by VMC or its authorized representatives.
- Any damage caused to VMC assets during execution or maintenance shall be rectified by the Bidder at its own cost.
- Penalties shall be recovered from running bills, quarterly O&M payments, Performance Security Deposit, or any amount payable to the Bidder.
- If uptime of any surveillance node (Pole, Camera, Edge Device, Switch, Network Link, etc.) remains

below 90% for three consecutive months, VMC may initiate corrective action, invoke Performance Security provisions, or terminate the contract as per tender conditions.

- Deployment and management of adequate manpower, vehicles, tools, spares, and support infrastructure shall be the sole responsibility of the Bidder and no separate payment shall be admissible for the same.
- Penalties shall be levied only for the reasons attributable to the implementing IA. Any risks/ issues foreseen by the implementing BIDDER shall be brought to the notice of the VMC immediately. If no such issues/risks are highlighted by the implementing IA, then it is expected that no delays will be there in the implementation schedule
- All works under or in course of execution or executed in pursuance of the contract shall at all times, be open to the inspection and supervision of VMC or appointed representatives.
- If the BIDDER or his workmen, or servants shall break, deface injure or destroy any part of the building or work in question or work in question in/on which they may be working or any building, road, fence, enclosure or grass-land or cultivated ground contiguous to the premises on which the work or any part-thereof is being executed or if any damage shall be done to the work from any cause whatever before completion of the work. The BIDDER shall make good the same at own expenses or in default, the VMC may cause the same to be made good by other BIDDER and deduct the expenses (of which the certificate of the Engineer-in-charge shall be final) from any sums that may thereafter become due to the BIDDER or from his security deposit or the proceeds of sale thereof or a sufficient portion thereof.
- The penalties as above will be recovered from any payment due to the Successful IA.
- If uptime of any surveillance node (Pole, Camera, Edge Device) remains below 90% for three consecutive months, the Purchaser reserves the right to issue a notice for corrective action, invoke Performance Security provisions, or terminate the contract as per tender conditions.

5.16 Spare Inventory

The Successful Bidder shall maintain an adequate inventory of critical spare parts throughout the Contract Period to ensure compliance with the prescribed SLA. As a minimum, the Bidder shall maintain 5% to 10% spare quantity (or at least one unit, whichever is higher) for all critical components supplied under this Contract, including but not limited to CCTV Cameras, PTZ Cameras, Switches, SFP Modules, Power Supplies, UPS Batteries, Hard Disks, Media Converters, Video Wall LED Display Modules/Panels, Video Wall Controller Modules, Receiving Cards, Sending Cards, Power Supply Units, and other critical active components. The spare inventory shall be readily available at the project site, CCC, or another location approved by VMC for immediate replacement in the event of failure. The cost of maintaining the required spare inventory shall be deemed included in the Bid price, and no separate payment shall be made by VMC.

5.17 Partial / Full Damage, Theft, and Loss of Hardware

- Hardware installed through this Project would become an important public asset. During the contract period of the Project the BIDDER shall be required to repair / replace any equipment if stolen / damaged/faulty. Appropriate insurance cover must be provided to all the equipment supplied under this project.
- In case of Partial/ Full damage or loss of the equipment due to reasons beyond the control like Theft, intentional damage by mishandling, Fire etc., the BIDDER should be in a position to supply working standby equipment with same configuration or higher with all services restored, as if it is a normal breakdown.
- Service Engineers/ Representatives of BIDDER shall invariably carry their identity cards with them, without which they will not be allowed to access department Systems. Service Engineers of the BIDDER shall have access to the Computer Systems/ Backup Solution only after obtaining clearance from department authorized officials. No component of the System/ data/ log information will be taken out of department premises without clearance from authorized Officials of the department.

- Hardware, poles, CCTV cameras, edge devices, networking equipment, OFC, switches, junction boxes, power systems, and all other assets installed under this Project shall be treated as critical public assets. During the entire contract period, the BIDDER shall be responsible for repair, replacement, restoration, and recommissioning of any equipment that becomes damaged, faulty, vandalized, lost, or stolen. The BIDDER shall obtain and maintain comprehensive insurance coverage for all equipment supplied and installed under the Project.
- In case of partial/full damage, theft, loss, vandalism, fire, accidental damage, riots, civil disturbances, natural calamities, mishandling, or any other unforeseen event, the BIDDER shall deploy and commission equivalent or higher-configuration standby equipment and restore all services within the stipulated SLA timelines. Such incidents shall not relieve the BIDDER of its obligation to maintain the prescribed service availability and uptime requirements.
- The BIDDER shall coordinate with VMC and concerned authorities for lodging FIRs, insurance claims, and related documentation in the event of theft, vandalism, or loss of assets. However, restoration of services shall not be delayed pending settlement of insurance claims.
- Any replacement equipment supplied under this clause shall be of equivalent or higher specifications and shall be fully compatible with the existing surveillance ecosystem, software, and management platforms.
- Service Engineers/Representatives of the BIDDER shall invariably carry valid identity cards and authorization letters while attending service calls. Without proper identification, they shall not be permitted access to any project infrastructure or facilities.
- Service Engineers of the BIDDER shall access the systems only after obtaining approval from authorized officials of VMC. No equipment, component, storage media, configuration data, logs, recordings, or project-related information shall be removed from the premises without prior written approval from VMC.
- Failure to restore services within the stipulated SLA timelines due to theft, damage, vandalism, or loss of equipment shall attract applicable SLA penalties as defined in the Contract.

Insurance

1. The successful Bidder/Contractor shall, at their own cost, obtain and maintain a valid Contractor's All Risk (CAR) Insurance Policy in the joint name of VMC and the Contractor for the entire duration of the contract, including any extensions. The policy shall cover all risks of loss or damage to works, materials, equipment, machinery, tools, IoT devices, OFC, HDPE pipes, poles, foundations, and associated installations from commencement of work until final acceptance by VMC, and shall also include third-party liability for bodily injury, death, or property damage arising during execution of the work. The sum insured shall not be less than the total contract value and shall cover risks such as fire, theft, flood, earthquake, storm, riot, strike, malicious damage, handling and erection risks, and other insurable risks relevant to the scope of work. A copy of the valid CAR policy along with proof of premium payment shall be submitted to VMC prior to commencement of work, failing which work shall not be permitted. In the event of any loss or damage, the Contractor shall immediately notify VMC and pursue the insurance claim, and any loss not covered or rejected by the insurer shall be borne entirely by the Contractor. Failure to maintain a valid CAR policy throughout the contract period shall be treated as a material breach of contract, and VMC shall have the right to suspend or terminate the contract without any liability.
2. The existing goods maintained and new goods supplied & installed under this Contract shall be fully insured by the selected bidder, against any loss or damage at the Tenderer's designated location for entire contract period. The insurance taken should cover risks such as flood, theft, fire, earthquake and cyclone (for wind speeds up to 80 kmph). The bidder shall submit to the VMC, certificate of insurance issued by the insurance company, indicating that such insurances have been taken against entire solution & certificate has to be submitted by bidder to VMC. Bidder shall be designated as the 'loss payee' in

such insurance policies; bidder shall be liable to pay premium for the insurance policy & shall ensure that each & every policy shall keep updated from time to time till end of contract period. In case of delayed procurement of insurance from selected bidder, all responsibilities of all supplied and/or installed equipment will be on bidder before taking the insurance.

3. Defect Liability Period (DLP) for the work executed under this contract shall be (60) months from the date of final work order by VMC, unless otherwise specified in the tender. During the DLP, the selected Bidder/Contractor shall, at its own cost and risk, rectify, repair, or replace any defects, deficiencies, damages, or failures in workmanship, materials, equipment, IoT devices, OFC, HDPE pipes, poles, foundations, or any part thereof that arise or are noticed during this period, within the time frame stipulated by VMC. In case the Contractor fails to attend or rectify such defects within the specified time, VMC shall have the right to get the same rectified through a third party at the risk and cost of the Contractor, and recover the expenses from the Security Deposit or any other dues payable. The DLP obligations shall be in addition to and not in substitution of the warranty obligations under the contract.
4. The bidder during the term of this contract undertakes to ensure that it has taken or shall take up all appropriate insurances for the delivery of goods that it is required to undertake under law as well as to adequately cover its obligations under this Contract: shall take out and maintain, at his own cost insurance with IRDA approved insurers against the risks, and for the coverage, as specified below: shall pay all premium in relation thereto and shall ensure that nothing is done to make such insurance policies void or voidable at the Tenderer's request, shall provide certificate of insurance to the Tenderer showing that such insurance has been taken out and maintained. Insurance will include employer's liability and workers' compensation insurance in respect of the Personnel of the bidder / bidder's Team, in accordance with the relevant provisions of the Applicable Law, as well as, with respect to such Personnel, any such life, health, accident, travel or other insurance as may be appropriate; and Insurance against loss of or damage to (i) equipment or assets procured in full or in part for fulfillment of obligations under this Contract (ii) the bidder's assets and property used in the performance of the Services (iii) third party insurance for life, health, accident or other insurance as may be appropriate as per the applicable law.
5. In case of theft of any asset, successful bidder(s) will have to carry out all required process to lodge police complain / FIR. VMC will assist bidder(s) wherever required.
6. Identifying any material breach of contract by Bidder, shall give Bidder a cure period of 90 days to correct the breach. If Bidder fails to cure the breach in the said time duration and accept its inability to correct, purchaser may terminate the part of the contract that is breached and employ a third party to do the work on behalf of purchaser.
7. Bidder shall not be liable for any compensation for the work executed this way. This procurement from a third party will be done at the selected bidder's (who has failed to perform its obligations & thus defaulted) risk, cost and responsibility. Any incremental cost borne by the Tenderer in procuring such Hardware / Services / Deliverables shall be borne by the selected bidder (who has failed to perform its obligations & thus defaulted). Any such incremental cost incurred in the procurement of such Hardware
8. / Services / Deliverables from other source will be recovered from the pending due and payable Payments / Security Deposit / Bank Guarantee provided by the selected bidder (defaulted bidder) under this scope of this RFP and if the value of the Hardware
9. / Services / Deliverables under risk purchase exceeds the amount of pending payable payments / Security Deposit and / or Bank Guarantee, the same may be recovered, if necessary, by due legal process.
10. Bidder shall execute the balance part of work as agreed under the contract.

5.18 Successful Bidder's Responsibility for Service Levels and Contract

- The Successful Bidder shall be solely responsible for providing, operating, monitoring, managing,

maintaining, and supporting the entire CCTV Surveillance System and associated infrastructure throughout the Contract Period.

- The Successful Bidder shall ensure deployment of adequate technical resources and shall provide 24x7x365 monitoring and support services from the City Command and Control Centre (CCC), including round-the-clock manpower for system monitoring, incident management, alarm/event monitoring, fault reporting, and technical issue resolution.
- The Successful Bidder shall ensure compliance with all Service Level Agreements (SLAs), uptime requirements, performance parameters, preventive maintenance schedules, fault rectification timelines, and operational obligations defined in the Contract.
- The Bidder shall deploy adequate technical resources, tools, spares, logistics, support arrangements, and management mechanisms necessary to achieve the prescribed SLAs. No additional payment shall be made by VMC for deployment of resources required to meet contractual obligations.
- Any failure to meet the prescribed SLA parameters, system availability requirements, response times, restoration timelines, image quality standards, recording requirements, monitoring obligations, or other contractual performance requirements shall attract penalties as specified in the SLA and Penalty Schedule of this Tender.
- The Successful Bidder shall be responsible for compliance with all applicable laws, statutory requirements, licenses, permits, taxes, duties, labour regulations, and other legal obligations associated with execution of the Contract.
- VMC shall not be responsible for any operational, financial, statutory, contractual, or legal liabilities arising out of the Bidder's performance of the Contract.
- The Successful Bidder shall remain fully responsible for the performance of the entire system and services irrespective of any subcontractors, suppliers, OEMs, service partners, or third parties engaged by the Bidder.
- The Bidder shall ensure uninterrupted operation, monitoring, maintenance, and support of the deployed system throughout the Contract Period and shall take all necessary corrective and preventive actions to maintain the required SLA levels.

Exclusion from SLA:

11. Down time due to failure / malfunctioning of any equipment / services provided by any third party agency appointed by VMC. However, it is responsibility of successful bidder has to submit supporting documents along with quarterly invoice proving that such outage is solely attributable to failure / malfunctioning of any equipment / services provided third party agency appointed by VMC.
12. Successful bidder will require to plan for 'scheduled outages / planned maintenance' in advance with prior approval of VMC. This will be planned during non - working hours. In exceptional circumstances, VMC may allow successful bidder to plan scheduled downtime in working hours.

Review of SLA Parameters

The SLA design is based on the scope of services and operational aspects of Implementation IA. Due to evolving nature of the proposed project, a review of SLAs will be conducted at the end of three months from the date of go-live of the Implementation IA. The purpose of this review is to reassess the SLAs based on the first three months' call volumes. VMC, after thorough analysis of the monthly statistics monitored as per the above mentioned SLA parameters, may consider revision of the SLA parameters and update this agreement. And if the revision occurs, then the revised SLAs would be final and binding for the rest of the term of the contract.

SECTION: 6

GENERAL TERMS & INSTRUCTIONS TO THE BIDDERS

6.1. Instructions to the bidder

- Bidders are advised to study all instructions, forms, terms, requirements and other information in the Bid Documents carefully.
- Every page of the proposal should necessarily be numbered and signed with seal.
- The bidder should not tamper the templates, and file types. If any additional information should be provided in a separate sheet where required.
- Submission of bid shall be deemed to have been done after careful study and examination of the Bid Document with full understanding of its implications.
- The response to this Bid Document should be full and complete in all respects. Failure to furnish all information required by the Bid Documents or submission of a proposal not substantially responsive to the Bid Documents in every respect will be at the bidder's risk and may result in rejection of its Proposal.
- Additionally, proposals of only those Bidders who satisfy the Conditions of Eligibility, stated herein, will be considered for evaluation by VMC.
- Any past projects where the bidder executed other works in addition to what is asked in Pre - Qualification Criteria in a single order, the bidder should consider only the value of asked project part and highlight only those components.

6.1.1 Purpose of Bid Document

The purpose of this Tender is to select a Bidder for the Implementation of the City Surveillance System in Vadodara City. This Tender Document provides the information necessary for prospective bidders to understand the scope, requirements, terms and conditions of the project and to prepare and submit their bids.

Only individual firms/entities meeting the eligibility criteria specified in this Tender Document are permitted to participate in the bidding process. **Consortium, Joint Venture (JV), association of firms, or any similar arrangement shall not be permitted.**

The term "Bidder" shall mean the firm/entity submitting the bid in response to this Tender.

The manner in which the Bid is required to be prepared, submitted, evaluated and accepted is explained in this Tender Document.

6.1.2 Proposal Preparation Cost

The bidder is responsible for all costs incurred in connection with participation in this process, including, but not limited to, costs incurred in conduct of informative and other diligence activities, participation in meetings/discussions/presentations, preparation of proposal, in providing any additional information required by VMC to facilitate the evaluation process, and in negotiating a definitive Contract or all such activities related to the bid process. The department will in no case be responsible or liable for those costs, regardless of the conduct or outcome of the bidding process.

This Bid Document does not commit the VMC to award a contract or to engage in negotiations. Further, no reimbursable cost may be incurred in anticipation of award. All materials submitted by the Bidder shall become the property of VMC and may be returned at its sole discretion.

6.1.3 Late Bids

The Bids received after the due date and the specified time (including the extended period if any) shall not be entertained. The Bids submitted by telex/telegram/fax/e-mail etc. shall not be considered. No correspondence will be entertained on this matter. PURCHASER shall not be responsible for any delay or non-receipt/ non-delivery of the documents. No further correspondence on the subject will be entertained. PURCHASER reserves the right to modify and amend any of the above-stipulated condition/Criteria depending upon project priorities vis-à-vis urgent commitments.

For the online bid submission, it is advised that the Bidder takes all necessary precaution for the same, including submitting the Bid well in advance to avoid any last-minute hassles, ensuring that the names/ formats

of the files to be uploaded are as per the e-tendering portal requirements, using the prescribed browser for upload etc. PURCHASER shall not entertain any bids which could not be uploaded or uploaded properly in the portal for whatsoever reasons.

6.1.4 Pre-bid Queries (Online)

A prospective Bidder requiring any clarification on the RFP Document may submit his queries, Bidder email. Email Id for submission of queries: it-projects@vmc.gov.in

The queries should necessarily be submitted in the following format:

Bidders Request for Clarification			
Name and Address of the Organization submitting request		Name and Position of Person submitting request	Contact Details of the Organization / Authorized Representative
			Tel: Mobile: Fax: Email:
S #	RFP Document Reference (Section No., Page No.)	Content of the RFP requiring clarification	Clarification Sought
1			

Queries submitted post the above mentioned deadline or which do not adhere to the above mentioned format may not be considered.

6.1.5 Amendment of RFP Document

At any time before the deadline for submission of bids, the VMC, may, for any reason, whether at its own initiative or in response to a clarification requested by a prospective Bidder, modify the RFP Document by an amendment.

The bidders are advised to visit the, https://tender.nprocure.com, www.vmc.gov.in without any notice being published in the newspaper. Thus, Bidder must check above websites periodically for necessary updates during tender process VMC also reserves the rights to amend the dates mentioned in this RFP for bid process.

In order to afford prospective Bidders reasonable time in which to take the amendment into account in preparing their bids, the VMC may, at its discretion, extend the last date for the receipt of Bids.

6.1.6 VMC's rights to terminate the selection process

VMC may terminate the RFP process at any time and without assigning any reason. VMC makes no commitments, express or implied, that this process will result in a business transaction with anyone.

This RFP does not constitute an offer by VMC.

The bidder's participation in this process may result in VMC selecting the bidder to engage in further discussions and negotiations toward execution of a contract. The commencement of such negotiations does not, however, signify a commitment by the VMC to execute a contract or to continue negotiations. VMC may terminate negotiations at any time without assigning any reason.

6.1.7 Right to reject any proposal

Notwithstanding anything contained in this RFP, VMC reserves the right to accept or reject any Proposal and to annul the Selection Process and reject all Proposals, at any time without any liability or any obligation for such acceptance, rejection or annulment, and without assigning any reasons therefore.

Besides other conditions and terms highlighted in the Tender Document, bids may be rejected under following circumstances:

General Rejection criteria:

1. Conditional Bids - If the information provided by the Bidder is found to be incorrect / misleading / fraudulent at any stage / time during the Tendering Process;
2. Any effort on the part of a Bidder to influence the bid evaluation, bid comparison or contract award decisions;
3. Bids received after the prescribed time & date for receipt of bids;
4. Bids without signature of person (s) duly authorized on required pages of the bid;
5. Bids without power of attorney/ board resolution or its certified true copy.
6. Technical Rejection criteria
7. Bid not sealed and packaged in envelope as asked in RFP;
8. Bidders not complying with the Eligibility Criteria given in this Tender
9. Technical Bid containing commercial details;
10. Revelation of Prices in any form or by any reason before opening the Commercial Bid;
11. Failure to furnish all information required by the Tender Document or submission of a Bid not substantially responsive to the Tender Document in every respect;
12. Bidders not quoting for the complete scope of work as indicated in the Tender Documents, addendum /corrigendum (if any) and any subsequent information given to the Bidder;
13. Bidders not complying with the Technical and General Terms and conditions as stated in the Tender Documents;
14. The Bidder not confirming unconditional acceptance of full responsibility of providing services in accordance with the scope of work and Service Level Agreements of this Tender;
15. Attempt by bidders to influence the decision of bid evaluation committee in unlawful manner or proved cause to obstruct the process of fair and transparent bid evaluation by authority.
16. During the bid process, if a Bidder indulges in any such deliberate act as would jeopardize or unnecessarily delay the process of bid evaluation and finalization.
17. During the bid process, if any information found wrong / manipulated / hidden in the bid.

6.1.8 Discount

The Bidders are advised not to indicate any separate discount in the Financial Bid. Discount, if any, should be merged with the quoted prices. Discount of any type, indicated separately, shall not be considered for evaluation purpose. However, in the event of such an offer is found to be the lowest without considering the discount, the Purchaser shall avail such discount at the time of award of contract.

6.1.9 Commercial Rejection Criteria

1. Incomplete price Bid;
2. Price Bids that do not conform to the Tender's price bid format;
3. Not submitting commercial bid online at https://tender.nprocure.com
4. Total price quoted by the Bidder does not include all statutory taxes and levies applicable;
5. If there is an arithmetic discrepancy in the commercial Bid calculations the Technical Committee shall rectify the same. If the Bidder does not accept the correction of the errors, its Bid may be rejected.
6. Misrepresentation/ improper response by the Bidder may lead to the disqualification. If such
7. disqualification / rejection occurs after the Proposals have been opened and the highest ranking Bidder gets disqualified / rejected, then VMC reserves the right to consider the next best Bidder, or take any other measure as may be deemed fit in the sole discretion of VMC, including annulment of the Selection Process.

6.1.10 Bid Fee and Earnest Money Deposit (EMD) and amount

- The bidder should pay non-refundable Bid Fee of Rs. 20,000 /- (Rupees Twenty Thousands only) in favor of "Municipal Commissioner, Vadodara" payable at Vadodara, from Nationalized or Scheduled Banks except Co-operative Banks (as approved by Gujarat State Govt. and VMC), payable at

Vadodara. The Bid fees shall be in the form of a Demand Draft.

The bidder should also pay bid security Earnest Money Deposit (EMD) of Rs. 1,00,00,000 /- (Rupees One Crore) shall be either in form of Demand Draft in favour of "Municipal Commissioner, Vadodara" from any nationalized/scheduled banks, payable at Vadodara for the period of 6 years.

OR

Bank Guarantee issued by Nationalized Bank including IDBI Bank/Private Banks – Axis Bank, HDFC Bank and ICICI Bank along with banks mentioned in the GR of Finance Department of Government of Gujarat (FD/MSM/e-file/4/2024/2859/DMO dated 01/05/2025) only in favor of "Commissioner, Vadodara Municipal Corporation" for the period of 6 years.

- No interest will be payable by the VMC on the Earnest Money Deposit (EMD).
- In case bid is submitted without EMD or Bid fees as mentioned above then VMC reserves the right to reject the bid without providing opportunity for any further correspondence to the bidder concerned.
- The EMD of unsuccessful Bidders will be returned by the Authority, without any Interest, as promptly as possible on acceptance of the Proposal of the Selected Bidder or when the Authority cancels the Bidding Process.
- The Selected Bidder's EMD will be returned, without any interest, upon the Selected Bidder signing the Agreement and furnishing the Security Deposit / Performance Guarantee in accordance with the provision thereof
- The decision of VMC regarding forfeiture of the EMD and rejection of bid shall be final & shall not be called upon question under any circumstances.
- The EMD may be forfeited:
    - If a Bidder withdraws their bid or increases their quoted prices during the period of bid validity or its extended period, if any; or
    - In the case of a successful bidder, if the Bidder fails to sign the Contract or to furnish Performance Bank Guarantee within specified time
    - During the bid process, if a Bidder indulges in any such deliberate act as would jeopardize or unnecessarily delay the process of bid evaluation and finalization.
    - During the bid process, if any information found wrong / manipulated / hidden in the bid.

6.1.11 Sealing, Marking and Submission of Technical Bid

The price bid must be submitted online on https://tender.nprocure.com. It should not be sent physically, if submitted physically the bid shall be rejected.

No Bidder shall submit more than one Bid for the Project. A Bidder bidding individually or as a member of a Consortium shall not be entitled to submit another bid either individually, as the case may be.

Bidders are required to submit their technical in separate sealed envelopes as per instructions given below:

- The Technical Bid must be sent to VMC, **strictly by Postal Speed Post or Registered Post only.**

  Documents received in any other manner or mode (like email) will not be considered.

    - Envelope 1: Tender fee (original DD) & Bid Security/Earnest Money Deposit (EMD) (originalDD/BG)
    - Envelope 2: Eligibility documents for Pre-Qualification criterion with all annexures described in RFP
    - Envelope 3: Support documents for Technical Qualification evaluation with all annexures described in RFP
    - Envelope 4: Soft copy of Technical Bid in a Pen Drive
- All four envelopes should be packaged all together in big envelope with tender number.
- Each page of the technical bid above should bear the initials of the applicant along with the seal of the Applicant in token of confirmation of having understood the contents. Also, each page of technical

proposal should be numbered with a proper index.

Each Bidder shall submit only one proposal containing documents as below.

a. Original DD/BG of the Bid fee & EMD

b. Pre-qualification Criteria related documents

c. Each page of the above should bear the initials of the Applicant along with the seal of the Applicant in token of confirmation of having understood the contents.

d. Details of EMD & Tender fee shall be submitted in electronic format in a pen drive (by scanning) while uploading the bid. This submission shall mean that EMD & tender fees are received for purpose of opening the bid. Accordingly, offer/ tenders of those bidders whose EMD & tender fee is received electronically, shall be opened. However, for the purpose of realization of EMD and Tender fee, bidder shall send the EMD as well as Tender fee in required format in original through Postal RPAD/ Postal Speed Post so as to reach to registered office of VMC. Punitive action shall be initiated for non - submission of EMD & Tender fees in original to registered office by bidder including abeyance of registration and cancellation of E-tendering code for one year.

e. In case of a trust, the identification of beneficial owner(s) shall include identification of the author of the trust, the trustee, the beneficiaries with fifteen percent or more interest in the trust and any other natural person exercising ultimate effective control over the trust through a chain of control or ownership.

f. An Agent is a person employed to do any act for another, or to represent another in dealings with third person.

g. The successful bidder shall not be allowed to sub-contract works to any selected vendor from a country which shares a land border with India unless such selected vendor is registered with the Competent Authority (Registration committee constituted by the Department for Promotion of Industry and Internal Trade (DPIIT)).

u) Resolution of Dispute

The VMC and the Bidder shall make every effort to resolve amicably, by direct informal negotiation, any disagreement or dispute arising between them under or in connection with the contract. If after thirty days from the commencement of such informal negotiations, the VMC and the Bidder have been unable to resolve amicably a contract dispute; either party may require that the dispute be referred for resolution by formal arbitration.

(i) In the event of any dispute arising with regards to the terms & conditions or interpretation of any clause of the agreement signed with the selected Bidder, the "Municipal Commissioner, VMC" shall be the sole arbitrator under the Indian Arbitration Act 1940 and whatever decision given by the "Municipal Commissioner, VMC" shall be final and binding on both the parties.

(ii) Any dispute arises with regards to any terms and conditions of the agreement signed with the selected Bidder; it shall be solely subject to the jurisdiction of the Vadodara courts.

v) Dismantling and Reinstallation

In case of dismantling and re-installation of hardware at project site due to reasons not attributed to vendor's performance, authority will reimburse the amount for dismantling and reinstallation work on submission of original bills and as per the rate discovered in commercial bid. Vendor will have to take prior approval from authority before executing such tasks. For the items, which is not discovered in price bid and is essential for project requirement, authority may take reference from other on-going projects at VMC and market rate to finalize the rate for such items. However, it is the responsibility of vendor to execute the task of Dismantling and Reinstallation on request of authority.

w) Electricity and Utility Connection

The vendor shall be responsible for ensuring electricity connection and other utilities to make the device/hardware operational. Vendor responsibility involves making request for electricity/utility providing company, submission of requisite fee, test report, requisite certification, coordination, laying

power cables and provisioning of requisite power points for all components in the scope of this project. Vendor shall be responsible for paying deposit and metering cost required for taking any electrical connections to concerned departments.

SI shall be responsible for laying last mile power cables and provisioning of requisite power points for all components in the scope of this project. SI shall be responsible for paying deposit and metering cost required for taking any electrical connections to concerned departments, this deposit shall be reimbursed by VMC. Since this component has dependency on approval from other agencies, SI should plan this requirement well in advance & submit the application to the electricity provider through VMC.

VMC will appoint a nodal person to provide necessary guidance and support in this process.

Authority will reimburse the cost on production of actual bill.

Since this component has dependency on approval from other agencies, vendor should plan this requirement well in advance & submits the application to MGVCL through VMC. Recurring charges (during operational phase) for all the electricity/utility consumption of components in the scope of this project will be borne by VMC but vendor will be responsible for coordination, informing authority to ensure timely payment of bills. Vendor will be responsible for maintenance of these electrical connections during the operational phase, and authority will reimburse material and Labour cost for breakdowns on submission of original bills.

x) Documentation of Project Outcomes

The vendor shall be responsible for project documentation including case study, impact assessment report, success report, explanatory video and stakeholder benefit video, training material with project brief ppts (detail of requirement is mentioned in scope of Work section). This will help in educating project stakeholders about functions and benefit of the project. Also such material will be helpful in replicating the projects at other locations and submitting the documentation at national and international forum.

6.2 Back-to-Back OEM Support

The Successful Bidder shall obtain and maintain back-to-back support from the respective OEM(s) for all supplied hardware, software, licenses, and solutions throughout the Warranty and CAMC/Contract Period. The Bidder shall submit valid OEM support/authorization certificates, service entitlement documents, warranty certificates, or OEM confirmation letters as documentary proof to VMC before Go-Live and whenever renewed during the Contract Period. Failure to maintain valid OEM back-to-back support shall be treated as a material breach of the Contract and appropriate action, including SLA penalties or other contractual remedies, may be initiated by VMC.

6.3 Performance Bank Guarantee

1. The successful bidder shall at his own expense, deposit with department, within 30 days of the notification of award (done through issuance of the Purchase Order/Letter of Acceptance/Letter of Intent), an unconditional and irrevocable Performance Bank Guarantee (PBG) from Nationalized or Scheduled Banks (approved by Gujarat State Govt. and VMC) except Co-operative Banks in favor of "Municipal Commissioner, Vadodara" for the due performance and fulfillment of the contract by the bidder.
    - If the Bidder, fails to furnish the Performance Guarantee within stipulated time period, it shall be lawful for the Authority to forfeit the EMD and cancel the contract or any part thereof
    - Selected Bidder has to submit 5 PBGs of 5% each of total Value of WO

6.4 Additional Quantity/Change Request

a. VMC reserves its right to award additional quantity of work order at the same or lower price and terms and conditions quoted in original contract.
b. VMC also reserves right to award change request work order for the work needed to execute the work defined in scope of work defined in original contract

c. Payment for additional items
   - Based on the final project plan or during the project duration if there is need for additional items and Based on the final project plan or during the project duration if there is need for additional items and there is variation in the quantities as defined in the financial bid, implementation bidder will be required to arrange additional items. Payment for such additional items will be done separately by VMC.

d. In such case of any additional requirement, payment will be done based on the rate provided in the financial bid of the Implementation Bidder. Payment mechanism to be followed for additional items is given below:
e. CAPEX COMPONENT - 30% CAPEX payment on the successful delivery of additional item, post approval from VMC and 70% CAPEX payment on the successful installation of the additional item, post approval on the installation report.
f. OPEX COMPONENT - OPEX part will be equally divided in remaining part of the payment during O&M. OPEX payment will be prorated based on the duration left in O&M. That is, if additional item is procured in 4th Quarter then OPEX will be paid for 17 quarters only (starting from 4th Quarter till 20th Quarter).

6.5 Termination of Contract

Termination of Contract VMC may, without prejudice to any other remedy under this Contract and applicable law, reserves the right to terminate for breach of contract by providing a written notice of 30 days stating the reason for default to the SI and as it deems fit, terminate the contract either in whole or in part in the following ways.

- Termination by Default: for failing to perform obligations under the Contract of if the quality is not upto the specification or in the event of non-adherence to time schedule.
- Termination for Convenience: VMC by written notice sent to the SI, may terminate the Contract, in whole or in part, at any time for its convenience. The notice of termination shall specify that the termination is for VMC's convenience, the extent to which performance of the SI under the Contract is terminated, and the date upon which such termination becomes effective.
- Termination for Non-Performance: If the SI fails to deliver any or all of the project requirements / operationalization / go-live / performance parameters (PERFORMANCE STANDARDS) of the project within the time frame specified in the contract; or
  If the SI fails to perform any other obligation(s) under the contract.
- Termination for Insolvency/NCLT proceedings: The Department may at any time terminate the contract by giving written notice to the bidder(s), if the bidder(s) becomes bankrupt or otherwise insolvent. In this event, termination will be without compensation to the bidder(s), provided that such termination will not prejudice or affect any right of action or remedy which has accrued or will accrue thereafter to the Department. Bidder agrees to handover all requisite data in terms of code, documents, etc. along with requisite to the office of the VMC identified team in this case.

Prior to providing a notice of termination to the SI, VMC shall provide the SI with a written notice of 30 days instructing the SI to cure any breach/ default of the Contract, if VMC is of the view that the breach may be rectified.
On failure of the SI to rectify such breach within 30 days, VMC may terminate the contract by providing a written notice of 30 days to the SI, provided that such termination will not prejudice or affect any right of action or remedy which has accrued or will accrue thereafter to VMC. In such event the SI shall be liable for penalty/liquidated damages imposed by the VMC. The performance Guarantee shall be forfeited by the VMC.

The Tenderer may, terminate this Contract in full or in part by giving the SELECTED BIDDER a prior and written notice indicating its intention to terminate the Contract under the following circumstances:

- Where the Tenderer is of the opinion that there has been such Event of Default on the part of the SELECTED BIDDER which would make it proper and necessary to terminate this Contract and may include failure on the part of the SELECTED BIDDER to respect any of its commitments with regard to any part of its obligations under its Bid, the RFP or under this Contract including the following:
- failure to perform the obligations under the Contract;
- the SELECTED BIDDER and its team have failed to conform with any of the service specifications as

set out in the RFP and the Contract;

- the SELECTED BIDDER has failed to demonstrate or sustain any representation or warranty made by it in this Contract, with respect to any of the terms of its Proposal, the RFP and this Contract;
- The SELECTED BIDDER has failed to comply with any terms and conditions of this RFP & the Contract;
- There is an undue delay in achieving the agreed timelines for delivering the services under this Contract due to reasons solely attributable to the SELECTED BIDDER;
- If it comes to knowledge of the Tenderer that the SELECTED BIDDER or any of their personnel have been involved in any fraudulent or corrupt practices or any other practice of similar nature;
- Where it comes to the Tenderer's attention that the SELECTED BIDDER (or the SELECTED BIDDERs' Team) is in a position of actual conflict of interest with the interests of the Tenderer, in relation to any of the terms of the SELECTED BIDDER's Bid, the RFP or this Contract.
- Where the SELECTED BIDDER's ability to survive as an independent corporate entity is threatened or is lost owing to any reason whatsoever, including inter-alia the filing of any bankruptcy proceedings against the SELECTED BIDDER, any failure by the SELECTED BIDDER to pay any of its dues to its creditors, the institution of any winding up proceedings against the SELECTED BIDDER or the happening of any such events that are adverse to the commercial viability of the SELECTED BIDDER. In the event of the happening of any events of the above nature, the Tenderer shall reserve the right to take any steps as are necessary, to ensure the effective transition of the project to a successor SELECTED BIDDER and to ensure business continuity.
- Where there has been an occurrence of such Event of Defaults, inter alia, as stated above, the VMC shall issue a notice of default to the SELECTED BIDDER, setting out specific defaults / deviances / omissions and providing a period of up to thirty
- (30) days to enable the SELECTED BIDDER to remedy the default/ deviances / omissions committed. It shall be the responsibility of the SELECTED BIDDER to maintain the agreed Quality of Service, even during the period when the notice for termination of agreement is pending and if the Quality of Performance of Solution is not maintained, during the said notice period, it shall be treated as material breach liable for termination at risk and consequent of SELECTED BIDDER and Performance Bank Guarantee shall be forfeited, without any further notice.
- Termination for Insolvency: The Tenderer may at any time terminate the Contract by giving written notice to the SELECTED BIDDER, without compensation to the SELECTED BIDDER, if the SELECTED BIDDER becomes bankrupt or otherwise insolvent, provided that such termination shall not prejudice or affect any right of action or remedy which has accrued or shall accrue thereafter to the Tenderer.
- Termination for Convenience: The Tenderer, may, by prior written notice sent to the SELECTED BIDDER at least 3 months in advance, terminate the Contract, in whole or in part at any time for its convenience. The notice of termination shall specify that termination is for the Tenderer's convenience, the extent to which performance of work under the Contract is terminated, and the date upon which such termination becomes effective.
- The SELECTED BIDDER may, subject to approval by the Tenderer, terminate this Contract before the expiry of the term by giving the Tenderer a prior and written notice at least 6 months in advance indicating its intention to terminate the Contract.

Consequences of Termination

- In the event of termination of this contract, VMC is entitled to impose any such obligations and conditions and issue any clarifications as may be necessary to ensure an efficient transition and effective continuity of the services which the SI shall be obliged to comply with and take all available steps to minimize the loss resulting from that termination/ breach, and further allow and provide all such assistance to VMC and/ or succeeding vendor, as may be required, to take over the obligations of the SI in relation to the execution / continued execution of the requirements of this contract.
- In the event of termination of this contract, VMC shall have ownership over entire volume of delivered & installed software and hardware irrespective payment made to successful bidder.
- In the event of the SI being unable to service the contract for whatever reason, VMC would evoke the PBG. Notwithstanding and without prejudice to any rights whatsoever of department under the Contract in the matter, the proceeds of the PBG shall be payable to department as compensation for any loss resulting from the Bidder's failure to complete its obligations under the Contract. Department shall notify the Bidder in writing of the exercise of its right to receive such compensation within 14 days,

indicating the contractual obligation(s) for which the Bidder is in default.

- VMC shall also be entitled to make recoveries from the Bidder's bills, performance bank guarantee, or from any other amount due to him, the equivalent value of any payment made to him due to inadvertence, error, collusion, misconstruction or misstatement

- All plans, drawings, specifications, designs, reports, other documents prepared by the Vendor and materials for which part delivery payment is made in the execution of the contract shall become and remain the property of VMC and before termination or expiration of this contract the SI shall deliver all such documents, prepared under this contract along with a detailed inventory thereof, to VMC.

6.6 Confidentiality:

The qualified Bidder, their personnel and their contractors hereby agree and undertake that during the term of the Contract, and thereafter, the Bidder agrees as follows:

- To maintain as confidential all Information (hereinafter defined), using such high degree of care as is appropriate to prevent any unauthorized use or disclosure, in any manner;
- Not to disclose, directly or indirectly, in any manner whatsoever, any Information to anyone;
- Not to make use of any Information which includes the results of any activities performed by the Bidder or any other data garnered by the Bidder, in any manner, directly or indirectly, for his own purposes or the benefit of anyone or any other entity other than the VMC;
- That the Information is the sole property of the VMC and that all rights, title and interest to the Information are held and owned by the VMC (and/or its affiliates) and disclosure or obtaining or procurement of the Information by him does not give him any rights or license as to the Information nor does it create any interest in his favour in such Information.

  Any proprietary or confidential Information, including but not limited to what is stated above, relating to the tender document and/or Contract and/or VMC's business or operations shall be disclosed only with the prior written consent of VMC. For the purposes of this tender document and the Contract, the term "Information" shall be interpreted and construed as including any trade secret, technical information, know-how or confidential, or proprietary information in the possession or control of the VMC, whether such information pertains to the VMC or any third party and whether such information are protected under any laws or not and shall include the following:

  - any information relating to the VMC's technology, processes, use of materials, data, archives, resources of any nature whatsoever and in any form whatsoever including inventions, discoveries, concepts, know-how, trademarks, copyrights, techniques, designs, schematics, specifications, drawings, diagrams, layouts, methods of processing and operational procedures;
  - all information and other data that may be developed or designed by the VMC and/or any of its affiliates;
  - information that describes the VMC's services, products and strategies, including but not limited to, processes, research and experimental work product administration and management;
  - information relating to the VMC's (and/or any of its affiliates) existing or planned businesses or business initiatives, organization, marketing activities and materials, reports, studies, organizational restructuring plans, and financial information;
  - confidential information and material of any third parties with which or whom the VMC conducts business or in respect of which the VMC has confidentiality and non-disclosure obligations;
  - any information, as described above, procured, obtained or contained in any proposals, plans, information memoranda or correspondences maintained by the VMC or provided to any personnel including the Bidder;
  - any document, agreement, correspondences, letters, records, working papers, specifications, lists of customers, lists of suppliers, drawings, notes, studies, manuals, schedules, test results or other documents, computer print-outs, computer readable information or emails relating to the business of the VMC, whether internal or with any third party, which are not available for public view;
  - any other information pertaining to or related to the VMC or its affiliates, associates or group companies, customers and suppliers, including, but not limited to, information related to overseas operations, including transaction processes, operations information.

- Rights in the Confidential Information
- All ownership and Intellectual Property Rights in the confidential information shall remain vested in the VMC.
- All passwords should be handed over to the Director (IT) or an officer who will be responsible as and when he asks.
  The ownership of all passwords, hardware base locking system (if any developed by the Bidder), keys will be solemnly of the Vadodara Municipal Corporation. There will not be any right of the Bidder on it in any form.

6.7 Insurance

1) The successful Bidder/Contractor shall, at their own cost, obtain and maintain a valid Contractor's All Risk (CAR) Insurance Policy in the joint name of VMC and the Contractor for the entire duration of the contract, including any extensions. The policy shall cover all risks of loss or damage to works, materials, equipment, machinery, tools, IoT devices, OFC, HDPE pipes, poles, foundations, and associated installations from commencement of work until final acceptance by VMC, and shall also include third-party liability for bodily injury, death, or property damage arising during execution of the work. The sum insured shall not be less than the total contract value and shall cover risks such as fire, theft, flood, earthquake, storm, riot, strike, malicious damage, handling and erection risks, and other insurable risks relevant to the scope of work. A copy of the valid CAR policy along with proof of premium payment shall be submitted to VMC prior to commencement of work, failing which work shall not be permitted. In the event of any loss or damage, the Contractor shall immediately notify VMC and pursue the insurance claim, and any loss not covered or rejected by the insurer shall be borne entirely by the Contractor. Failure to maintain a valid CAR policy throughout the contract period shall be treated as a material breach of contract, and VMC shall have the right to suspend or terminate the contract without any liability.
2) The existing goods maintained and new goods supplied & installed under this Contract shall be fully insured by the selected bidder, against any loss or damage at the Tenderer's designated location for entire contract period. The insurance taken should cover risks such as flood, theft, fire, earthquake and cyclone (for wind speeds up to 80 kmph). The bidder shall submit to the VMC, certificate of insurance issued by the insurance company, indicating that such insurances have been taken against entire solution & certificate has to be submitted by bidder to VMC. Bidder shall be designated as the 'loss payee' in such insurance policies; bidder shall be liable to pay premium for the insurance policy & shall ensure that each & every policy shall keep updated from time to time till end of contract period. In case of delayed procurement of insurance from selected bidder, all responsibilities of all supplied and/or installed equipment will be on bidder before taking the insurance.
3) Defect Liability Period (DLP) for the work executed under this contract shall be (60) months from the date of final work order by VMC, unless otherwise specified in the tender. During the DLP, the selected Bidder/Contractor shall, at its own cost and risk, rectify, repair, or replace any defects, deficiencies, damages, or failures in workmanship, materials, equipment, IoT devices, OFC, HDPE pipes, poles, foundations, or any part thereof that arise or are noticed during this period, within the time frame stipulated by VMC. In case the Contractor fails to attend or rectify such defects within the specified time, VMC shall have the right to get the same rectified through a third party at the risk and cost of the Contractor, and recover the expenses from the Security Deposit or any other dues payable. The DLP obligations shall be in addition to and not in substitution of the warranty obligations under the contract.
4) The bidder during the term of this contract undertakes to ensure that it has taken or shall take up all appropriate insurances for the delivery of goods that it is required to undertake under law as well as to adequately cover its obligations under this Contract: shall take out and maintain, at his own cost insurance with IRDA approved insurers against the risks, and for the coverage, as specified below: shall pay all premium in relation thereto and shall ensure that nothing is done to make such insurance policies void or voidable at the Tenderer's request, shall provide certificate of insurance to the Tenderer showing that such insurance has been taken out and maintained. Insurance will include employer's liability and workers' compensation insurance in respect of the Personnel of the bidder / bidder's Team, in accordance with the relevant provisions of the Applicable Law, as well as, with respect to such Personnel, any such life, health, accident, travel or other insurance as may be appropriate; and Insurance against loss of or damage to (i) equipment or assets procured in full or in part for fulfillment of obligations under this Contract (ii) the bidder's assets and property used in the performance of the Services (iii) third party insurance for life, health, accident or other insurance as may be appropriate as per the applicable law.

5) In case of theft of any asset, successful bidder(s) will have to carry out all required process to lodge police complain / FIR. VMC will assist bidder(s) wherever required.
6) Identifying any material breach of contract by Bidder, shall give Bidder a cure period of 90 days to correct the breach. If Bidder fails to cure the breach in the said time duration and accept its inability to correct, purchaser may terminate the part of the contract that is breached and employ a third party to do the work on behalf of purchaser.
7) Bidder shall not be liable for any compensation for the work executed this way. This procurement from a third party will be done at the selected bidder's (who has failed to perform its obligations & thus defaulted) risk, cost and responsibility. Any incremental cost borne by the Tenderer in procuring such Hardware / Services / Deliverables shall be borne by the selected bidder (who has failed to perform its obligations & thus defaulted). Any such incremental cost incurred in the procurement of such Hardware
8) / Services / Deliverables from other source will be recovered from the pending due and payable Payments / Security Deposit / Bank Guarantee provided by the selected bidder (defaulted bidder) under this scope of this RFP and if the value of the Hardware
9) / Services / Deliverables under risk purchase exceeds the amount of pending payable payments / Security Deposit and / or Bank Guarantee, the same may be recovered, if necessary, by due legal process.
10) Bidder shall execute the balance part of work as agreed under the contract.

6.8 Damages for Breach of Agreement:

1) In the event of Bidder being in material default and such default is cured before Termination, the Bidder shall pay to VMC as Damages, an amount of INR 2,00,000/- (Two Lacs only) per week or for part thereof, till such default is cured, in one instance within 30 (thirty) days of receiving the demand.
2) In the event of the Bidder being in default in the due and faithful performance of its obligations and failing to remedy such default within the specified period, VMC shall be entitled to forfeit the performance security and appropriate the same as damages for such default. Upon such encashment and appropriation of the performance security, VMC shall grant a period of 15 (fifteen) days to the Bidder to provide fresh Performance Security and the Bidder shall within the time so granted furnish to VMC such Performance Security failing which VMC shall be entitled to Terminate this Agreement.

6.9 Defect Liability Period (DLP)

Defect Liability Period (DLP) for the work executed under this contract shall be (60) months from the date of final work order by VMC, unless otherwise specified in the tender. During the DLP, the selected Bidder/Contractor shall, at its own cost and risk, rectify, repair, or replace any defects, deficiencies, damages, or failures in workmanship, materials, equipment, IoT devices, OFC, HDPE pipes, poles, foundations, or any part thereof that arise or are noticed during this period, within the time frame stipulated by VMC. In case the Contractor fails to attend or rectify such defects within the specified time, VMC shall have the right to get the same rectified through a third party at the risk and cost of the Contractor, and recover the expenses from the Security Deposit or any other dues payable. The DLP obligations shall be in addition to and not in substitution of the warranty obligations under the contract.

6.10 Confidential Policy

All information and other materials disclosed, furnished, communicated or supplied by the VMC to Successful bidder, including Successful bidder's directors, officers, employees, affiliates, or its expressly authorized representatives or agents are strictly confidential and shall not be divulged by receiving party to any third party during the term of contract or thereafter without VMC prior written consent.

For avoidance of doubt, "Confidential Information" shall be deemed to include all information including but not limited to any technical, price and financial information, improvement, inventions, know how, innovations, technology, trade secrets, professional secrets, copyrights and any other intellectual property, discoveries, ideas, concepts, papers, software in various stages of development, techniques, models, data, source code, object code, documentation, manuals, flow charts, research, process, procedures, functions, VMC customer names and other information related to VMC customers, price lists and pricing policies. However, the both Parties hereto acknowledge that Confidential Information shall not include any information that: -

- is now or subsequently becomes publicly known or available without breach of contract;
- was previously in the possession of successful bidder without any obligation of confidentiality and which was not acquired from, provided, given, sold or otherwise disclosed (directly or indirectly) by VMC not through contract.

Successful bidder further agrees to require its employees and its sub-contractors and their employees to execute an appropriate non-disclosure letter / agreement prior to performing any work under contract and ensure that only such authorized representatives who are expressly authorized and whose duties require them to possess the Confidential Information shall have access to the Confidential Information on a need-to-know basis.

Save as expressly provided hereunder, nothing contained herein shall be construed, implicitly or otherwise, as being the granting of a license to use the Confidential Information disclosed by VMC. Successful bidder shall ensure that the Confidential Information will not be copied or reproduced or transmitted by any means and in any form whatsoever (including in an externally accessible computer or electronic information retrieval system) by successful bidder or its representatives without the prior written permission of VMC.

Successful bidder and its personnel shall not, either during or after expiration of contract, disclose any proprietary or confidential information relating to services, contract or VMC's business or operations without prior written consent of VMC.

Successful bidder has to follow all rules and regulations of guidelines and has to follow strictly to all rules and privacy. Some of guidelines are as follows, however it is not limiting to the same. It is responsibility of successful bidder to contact VMC authority and get all guidelines to carry out work in VMC premises.

- All paper works prepared to project are very confidential property of VMC authority. If required, successful bidder has to submit privacy agreement with VMC as per mutually agreed format. All responsibility lies with successful bidder of privacy of documents handed over to him for execution.
- The staff of successful bidder should not involve with any activity related to VMC.
- No activity is allowed in VMC premises by any staff of successful bidder that affects to VMC security and peace.

6.11 Limitation of Liability

Final payment by VMC in pursuance of tender terms shall not mean release of successful bidder from all his liabilities under the contract. Successful bidder shall be liable and committed under the contract to fulfil all his liabilities and responsibilities, till such time Security Deposit Guarantee is released by VMC.

Provided that successful bidder shall continue to remain responsible for all liabilities which arise on account of Gross Negligence, Willful Misconduct, breach or fraud, legal noncompliance even beyond release of Security Deposit.

Successful bidder shall continue to remain responsible for all liabilities which arise on account off,

- Any obligation of successful bidder for performance / warranty / defect liability, or
- Any obligation of successful bidder to indemnify VMC with respect to infringement of intellectual property rights, or
- Failure of successful bidder to pay taxes, duties as required under this tender / Contract.
- Breach of confidentiality requirements set forth in this bid or Applicable Law, or
- In case of negligence, breach, fraud, Gross Negligence or Willful Misconduct of successful bidder.

6.12 Data Security

Successful bidder and its team have to maintain the confidentiality and security of data in true spirit. In no case, breach in security of data will be acceptable. Violation of this clause may lead to severe penalties including to the termination of contract as decided by VMC. In this regard, successful bidder has to sign

Non-Disclosure Agreement (NDA) with VMC. It will be successful bidder's responsibility to get this agreement signed along with contract agreement.

6.13 Pole, Foundation & Structural Stability Certificate

The Successful Bidder shall design, supply, install, and certify all CCTV poles, mounting structures, brackets, and foundations for structural adequacy and stability in accordance with the applicable Indian Standards and good engineering practices. A Structural Stability Certificate, duly signed and stamped by a Chartered/Qualified Structural Engineer, shall be submitted to VMC after installation and before Go-Live. The Structural Stability Certificate shall remain valid for the entire five (5) year Contract Period after Go-Live. In case the certificate issued by the Structural Engineer is valid for a shorter period (e.g., three (3) years), the Successful Bidder shall, at its own cost, obtain and submit a renewed/revalidated Structural Stability Certificate before expiry of the earlier certificate so that the structural stability certification remains valid continuously throughout the entire Contract Period. Any failure, tilting, settlement, corrosion, structural damage, or instability of the pole or foundation shall be rectified or replaced by the Successful Bidder at no additional cost to VMC.

6.14 Integration with Existing Systems

The Successful Bidder shall integrate the proposed solution with the existing systems deployed at VMC/ICCC, including but not limited to Videonetics VMS, Motadata, Trend Micro, existing Active Directory/LDAP (if available), network infrastructure, storage, servers, backup solution, EMS/NMS, and any other applications, software, or platforms identified by VMC during implementation.

The Bidder shall ensure seamless interoperability, data exchange, alarm/event sharing, health monitoring, user authentication, and centralized management without affecting the existing operational environment. All hardware, software, APIs, connectors, licenses (if required), professional services, customization, configuration, testing, and commissioning required for successful integration shall be included in the Bid price. No additional payment shall be made by VMC for such integration activities.

6.15 Critical Spare Inventory

The Successful Bidder shall maintain an adequate inventory of critical spare parts throughout the Contract Period to ensure compliance with the prescribed SLA. As a minimum, the Bidder shall maintain **5**% to 10% spare quantity (or at least one unit, whichever is higher) for all critical components supplied under this Contract, including but not limited to CCTV Cameras, PTZ Cameras, Switches, SFP Modules, Power Supplies, UPS Batteries, Hard Disks, Media Converters, Video Wall LED Display Modules/Panels, Video Wall Controller Modules, Receiving Cards, Sending Cards, Power Supply Units, and other critical active components. The spare inventory shall be readily available at the project site, CCC, or another location approved by VMC for immediate replacement in the event of failure. The cost of maintaining the required spare inventory shall be deemed included in the Bid price, and no separate payment shall be made by VMC.

6.16 Other Terms and Conditions

1. The Successful Bidder shall repaint all installed CCTV poles, mounting structures, and fabricated structures (where applicable) at least once after 2 years from the Go-Live date, so that repainting is carried out two (3) times during the entire Contract Period, including the initial Color painting at the time of installation. The repainting shall include surface preparation, rust removal (where required), primer, and weather-resistant paint/color, ensuring that the structures remain free from corrosion and in good condition throughout the Contract Period.
2. Period of Validity of Bids:

   Bids shall be valid for 120 days after the date of financial bid opening prescribed by VMC. Any Bid valid for a shorter validity period shall be rejected by VMC as non-responsive bid.

   In exceptional circumstances, VMC may solicit the all Bidder's consent to an extension of the period of validity. The request and the responses thereto shall be made in writing. A Bidder may refuse the request without forfeiting its EMD. A Bidder granting the request will not be permitted to modify its bid/rates.

3. The selected Bidder for award of the assignment, (a) will sign an agreement on the Govt. stamp paper to be purchased by the bidder as per the prevailing rules and regulations of VMC and (b) will have to pay the security deposit, @ 3 % of the order value by D.D.
4. The Bidder shall bear all the costs associated with the preparation and submission of its bid, and VMC in no case will be responsible or liable for these costs, regardless of conduct or outcome of bidding process.
5. VMC is not responsible for delay, loss or non-receipt of tender documents sent by the post.
6. The Bidder is expected to examine all instructions, forms, terms and specifications in the bidding documents. Failure to furnish all information required by the bidding documents or submission of a bid not substantially responsive to the biding documents in every respect will be at the Bidder's risk and may result in rejection of its bid.
7. The Security Deposit will be retuned only after successful supply of the items and the E.M.D. will be returned only one year after completion of the warranty period.
   Security Deposit:

   The successful bidder will be required to place Security Deposit @ 3% of the consideration of the Contract by Demand Draft or Banker's Cheque Payable at Vadodara in favour of the "Municipal Commissioner, Vadodara" of any scheduled/nationalized bank within 15 days from the date of notice of award of contact, failing which a penalty @ 0.065% of the amount of security deposit will be imposed for delay of each day. The EMD placed may be considered for conversion towards the security deposit and amount falling short of the required amount shall be payable.
8. Quantity Variation:
   1. At the time of award of contract, the quantity of goods, works or services originally specified in the bidding documents may be increased or decreased. The successful bidder shall not object to the upward or downward variation in quantities of any item within the variation limit of ± 50%.
   2. Payment for additional quantities within the variation limit shall be made at tender rates.
   3. No claim shall be entertained or become payable for price variation of additional quantities.
   4. Repeat orders for extra items or additional quantities may be placed within 3 years of the original request order. The Unit Rate mentioned in the Commercial bid formats shall be used for the purpose of "Repeat Orders" for respective items. However, based on the market trends, VMC retains the right to negotiate the Tender rate and/or request better specifications based on market and technological scenario. Delivery or completion period may be proportionally increased.
9. Fraud and Corruption
   In case of Fraud and Corruption by bidder and/or manpower deputed for this service, VMC shall retain all rights to lodge Police Complaint/ FIR against bidder under various provisions of Indian Penal Code and any other penal law inforce including Information Technology act and Prevention of Corruption act.
10. The Successful Bidder shall inspect all installed CCTV poles and fabricated structures at least once every year throughout the five (5) year Contract Period after Go-Live. For MS painted poles, repainting shall be carried out whenever required to maintain protection against corrosion and proper appearance. For Hot-Dip Galvanized (GI) poles, repainting shall not be mandatory, however, any damaged, scratched, corroded, or deteriorated galvanized surface shall be restored using an approved cold galvanizing compound or equivalent protective coating, or as per the OEM/manufacturer's recommendation, to ensure structural integrity and corrosion protection throughout the five (5) year Contract Period. The selected bidder should provide tape data cartridges for backup. The backup should be taken automatically.
11. The approx. quantity will be based on budget but it may be increase or decrease based on requirement.
12. The scope of warranty by the bidder also includes reinstallation and reconfiguration of the item in case of breakdown.
13. VMC will purchase CCTV Cameras' from the date of contract, with the same rate as purchase order within 2 years.
14. All the installation should be finished within 4 months from the work order.
15. A vendor is unconditionally bound to VMC's All Terns & General Conditions
16. Support (telephonic, email, onsite) to be provided for resolution of any issues. The response time to be 4 hours with the resolution time of 2 working days.

17. The selected bidder shall depute representatives (a) to supervise the unloading and transportation, so as to prevent any damages due to mishandling in these operations and (b) for installation of equipment's at the specified locations. The supervisory representatives shall be fully responsible for the respective operations and will give test before handling over the equipment's. The bidder free of all charges shall provide this service of the supervisory representatives.
18. Hand-over of the system at the end of contractual period along with all documentation required to operate and maintain the system in healthy condition.
19. No supplied hardware put on roadside to cause an accident and Damage of supplied parts. It is completely under vendor scope to prevent before final installation.
20. Bidder to ensure compliance to the entire cyber security standard required safeguarding the edge devices; connectivity & desktop to ensure cyber attach resilient operations
21. If required, the maintenance work & technical support will have to be carried out beyond VMC's & Bidder's normal working hours and on holidays too, without any extra charges. The maintenance engineers should be available & approachable round-the-clock (24X7) through telephone.
22. For Selected bidder supplied items are not fulfill the specification of RFP/Solution then bidder needs to supply higher specification items without any kind of additional cost.
23. The Successful Bidder shall provide comprehensive Operations & Maintenance (O&M) support for a period of five (5) years from the date of Go-Live. The Bidder shall ensure 24x7x365 monitoring, operation, maintenance, fault rectification, and support of the entire CCTV Surveillance System, including cameras, poles, junction boxes, OFC network, switches, servers, storage, video wall, VMS, and associated infrastructure.
24. The Bidder shall carry out Quarterly Preventive Maintenance (PM) of all field and central infrastructure and submit PM reports to VMC. Daily fault rectification, camera cleaning, focus adjustment, angle correction, OFC restoration, network troubleshooting, and equipment replacement shall be included within the scope of work.
25. The Bidder shall arrange Hydra/Crane services, vehicles, OFC splicing resources, and Field Response Teams (FRT) comprising technicians, splicers, supervisors, drivers, and necessary equipment for timely restoration of services. No separate payment shall be made for manpower, vehicles, tools, Hydra services, preventive maintenance activities, or fault rectification services, and the cost thereof shall be deemed included in the quoted O&M/CAMC charges.
26. The quoted prices shall be inclusive of supply, transportation, insurance, installation, testing, commissioning, integration, configuration, training, documentation, warranty, maintenance, and all other activities required for successful implementation of the project.
27. The Bidder shall clearly specify the OEM, Make, Model Number, and Part Number of all proposed equipment, software, and licenses.
28. Any hardware, software, accessory, component, fitting, mounting material, cable, connector, license, service, or item required for successful completion and operation of the project, whether explicitly mentioned in the BOQ or not, shall be deemed included in the Bidder's scope and quoted price without any additional cost to VMC.
29. Comprehensive Warranty and CAMC support obligations shall be clearly defined by the Bidder. CAMC rates shall be quoted separately for each year of the O&M period.
30. All software licenses, Video Management Software (VMS) licenses, analytics licenses, operating system licenses, database licenses, and other application licenses shall be perpetual unless specifically stated otherwise in the tender.
31. The proposed solution shall be scalable and capable of supporting a minimum of 900 CCTV cameras, including future expansion without requiring replacement of the core infrastructure.
32. The quantities indicated in the BOQ are tentative and may increase or decrease based on site conditions, survey findings, operational requirements, and VMC directives. Payment shall be made strictly on the basis of actual quantities supplied, installed, tested, commissioned, and accepted by VMC at approved unit rates.
33. The proposed Storage System, Chassis, Servers, Video Management System, and associated central infrastructure shall support integration of approximately 250 existing CCTV cameras of the legacy project and shall be sized with a minimum 20% spare capacity for future expansion without requiring major hardware replacement.
34. The Bidder shall arrange and manage Hydra/Crane vehicles, tools, testing equipment, safety equipment, consumables, and all resources required for maintenance and restoration activities at no

additional cost to VMC.

35. The Bidder shall deploy and maintain a dedicated Field Response Team (FRT) consisting of adequate skilled manpower, including technicians, OFC splicers, supervisors, drivers, vehicles, and required tools, for timely fault rectification and maintenance activities throughout the Contract Period.
36. The Bidder shall maintain sufficient inventory of spare equipment and consumables to ensure compliance with SLA requirements and minimize downtime of surveillance infrastructure.
37. Patent Rights

    In the event of any claim asserted by a third party of infringement of copyright, patent, trademark or industrial design rights arising from the use of the Goods or any part thereof in VMC's country, the Bidder shall act expeditiously to extinguish such claim. If the Bidder fails to comply and VMC is required to pay compensation to a third party resulting from such infringement, the Bidder shall be responsible for the compensation including all expenses, court costs and lawyer fees. VMC will give notice to the Bidder of such claim, if it is made, without delay.

Annexure 1: Technical Bid Format

Checklist for Technical Qualification Document

<<To be printed on bidder company's letterhead and signed by Authorized signatory>>

Sr. No.	Documents to be submitted	Submitted (Y/ N)	Documentary Proof (Page No.) of Proposal
1	Bid Processing Fee (DD) as per RFP		
2	Bid Security EMD (DD/BG) as per RFP		
3	Technical Proposal Packaged in Envelope with Pen Drive as per RFP		
4	Power of attorney / board resolution to the authorized Signatory of the RFP		
5	Copy of Certificate of Incorporation/Registration certificate/ Shop& Establishment Certificate – Bidder		
6	Copy of Audited Balance Sheet and Profit and loss statement for last three financial years Bidder		
7	Supporting Documents like Rent Agreement/ Electricity Bill / Self-Declaration on Company's Letter head to be submitted for Office Location. OR Undertaking from authorized signatory to open the local office within 60 days from issuance of LOI to be submitted – Bidder		
8	Copy of GST registration – Bidder		
9	Copy of PAN registration – Bidder		
10	Bid Covering Letter		
11	Particulars of the Bidders		
12	Details of Annual Turnover for last three financial years		
13	Certificate from the Statutory auditor / CA clearly specifying the annual turnover for the specified years		
14	Declaration letter that the firm is not blacklisted by Central Government or any State Government organization / PSU in Indiaat the time of submission of the Bid, in the format given in the RFP		
14	Affidavit on Non-judicial Rs 300 stamp paper		
15	Details of the similar projects executed		
16	Solution Design		
18	Format for Authorization Letters from OEMs		
19	Format for Bid EMD		
20	Make & Model of all Hardware & Software components		
21	Compliance to Technical Specifications as mentioned in the RFP on OEM's Letterhead.		
22	Bill of Material without prices		
23	Datasheets highlighting the Functional and Technical Specification parameters in each datasheet for compliances		
24	Self-declaration by the Bidder duly signed and stamped by the authorized signatory - Not be insolvent, in receivership, bankrupt or being wound up, not have its affairs administered by a court or a judicial officer, not have its business activities suspended and must not be the subject of legal proceedings for any of the foregoingreasons		
25	Self-declaration by the Bidder duly signed and stamped by the authorized signatory - Not have their directors and officers convicted of any criminal offence related to their professional conduct or the making of false statements or misrepresentations as to their qualifications to enter into a procurement contract within a period of three years preceding the commencement of the procurement process, or not have been otherwise disqualified.		

Note:

All technical bid document(s)/ details should be duly sealed & signed as required.

In case of the deviation in the authorization letter by the manufacturer & forwarding letter; the price bid of such bidder will not be opened.

Any conditional mention regarding any technical details or prices in any document(s)/ forwarding letter; price bid of such bidder will not be opened.

TQ_1: Technical Bid Cover Letter

<<To be printed on bidder company's letterhead and signed by Authorized signatory>>

Date:

dd/mm/yyyy

To
Director IT
Information Technology Department
Vadodara Municipal Corporation
Khanderao Market, Vadodara – 390001

Subject: RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city.

Reference: Tender No :<No> Dated<DD/MM/YYYY>

Dear Sir/ Madam,
Having examined the Bid Document (and the clarification / corrigendum issued thereafter, if any), the receipt of which is hereby duly acknowledged, we, the undersigned, offer to provide the professional services as required and outlined in the Bid Document for the "RFP Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city.
We attach hereto our responses to Technical-Qualification & Commercial proposals as required by the Bid Document. We confirm that the information contained in these responses or any part thereof, including the exhibits, and other documents and instruments delivered or to be delivered to VMC is true, accurate, verifiable and complete. This response includes all information necessary to ensure that the statements therein do not in whole or in part mislead VMC in its shortlisting process.

We fully understand and agree to comply that on verification, if any of the information provided here is found to be misleading the selection process, we are liable to be dismissed from the selection process or termination of the contract during the project, if selected to do so.

We agree for unconditional acceptance of all the terms and conditions set out in the Bid Document (& subsequent clarification / corrigendum, if any) document and also agree to abide by this tender response for a period of 180 days from the Bid Opening date. We hereby declare that in case the contract is awarded to us, we shall submit the contract performance guarantee bond in the form prescribed the Bid Document.

We agree that you are not bound to accept any tender response you may receive. We also agree that youreserve the right in absolute sense to reject all or any of the products/ services specified in the tender response.

It is hereby confirmed that I/We are entitled to act on behalf of our company/ corporation/ firm/ organization and empowered to sign this document as well as such other documents, which may be required in this connection.

Signature of Authorized
Signatory (with official seal)
Name :
Designation :
Address :
Telephone& Fax :
E-mail address
:

TQ_2: Bidder Information Format

<<To be printed on bidder company's letterhead and signed by Authorized signatory>>

To whomsoever it may concern,

Please find below the details of bidder for participation in Selection of Bidder for RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city.

#	Particulars	Bidder
1	Name of the organization	
2	Type of Organization (Pvt. Ltd/ Public Limited)	
3	Address of Registered office	
4	Company Registration Details	
5	Date of Registration	
6	PAN	
7	GST	
8	Number of years of operations in India	
9	Authorized Signatory Name Authorized	
10	Signatory Designation Authorized	
11	Signatory Contact Details	
12	Authorized Signatory email ID	

Yours Sincerely,

Signature of Authorized Signatory (with official seal)Name:
Designation :
Address :
Telephone& Fax :
E-mail address :

Note: To be submitted with any other supporting details specified as Document Proof in Section 3

TQ_3: Bidders Annual turnover over in last 3 financial years

<<To be printed on bidder company's letterhead and signed by Authorized signatory>>

Date:

dd/mm/yyyy

To
Director IT
Information Technology Department
Vadodara Municipal Corporation
Khanderao Market, Vadodara – 390001

Subject: RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city

Sir/ Madam,
I have carefully gone through the Terms & Conditions contained in the RFP Document Selection of Bidder for RFP for Selection of Bidder for Implementation of City Surveillance System in Vadodara City.

I hereby declare that below are the details regarding Overall turnover over last 3 financial years for our organization

#	Details	FY 2022-23 (i)	FY 2023-24 (ii)	FY 2024-25 (iii)	Average Turnover [(i)+(ii)+(iii)/3]
1	Overall Annual Turnover- Bidder				

Contact Details of officials for future correspondence regarding the bid process:

Details	Authorized Signatory	Secondary Contact
Name		
Title		
Company Address		
Mobile		
Fax		
Email Id		

I further certify that I am competent officer in my company to make this declaration.

Yours Sincerely,

Signature of Authorized Signatory (with official seal)
Name :
Designation :
Address :
Telephone& Fax :
E-mail address :

Note: To be submitted with any other supporting details specified as Document Proof in Section 3 of RFP.

TQ_4: Auditor's/CA Certificate for turnover for bidder

Date:
dd/mm/yyyy
To
Director IT
Information Technology Department
Vadodara Municipal Corporation
Khanderao Market, Vadodara – 390001

This is to certify that the Annual Turnover as per books and records of ___for thefollowing financial years are as under.

#	Financial Year Ending	Annual Turnover (INR)
1.	31st March, 2023	
2.	31st March, 2024	
3.	31st March, 2025	
	Average Turnover	

I further certify that I am competent officer in my company to make this declaration.

Yours Sincerely,

Signature of Auditor (with official seal)Name :
Designation :
Address :
Telephone& Fax :
E-mail address :

TQ_5: Self-Declaration – No Blacklisting

<<To be printed on company's letterhead and signed by Authorized signatory>>

Date: dd/mm/yyyy

To
Director IT
Information Technology Department
Vadodara Municipal Corporation
Khanderao Market, Vadodara – 390001

Sir/Madam,
In response to the Tender Ref. No. _____ dated _____ for RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city, as an owner/ Director of , I/ We hereby declare that presently our Company/ firm _____ is having unblemished record and is not declared ineligible for corrupt and fraudulent practices either indefinitely or for a particular period of time by any State/ Central Government/ PSU.

We further declare that presently our Company/ firm _____ is not blacklisted and not declared ineligible for reasons other than corrupt and fraudulent practices by any State/ Central Government/ PSU during last five years, from date of this bid submission.

If this declaration is found to be incorrect then without prejudice to any other action that may be taken, my/ our security may be forfeited in full and the tender if any to the extent accepted may be cancelled.

Name of the Bidder :
Authorized Signatory :
Seal of the Organization: Business
Address :
Date :
Place :

TQ_6: Affidavit

(The affidavit format as indicated below to be furnished on non-judicial stamp paper of Rs: 300 and duly notarized)

Name of work: RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city.

1. I, the undersigned, do hereby certify that all the statements made in the required attachments are true and correct.
2. The undersigned hereby certifies that neither our Company/Firm, M/s _____, nor any of its Directors has abandoned any work or had any contract for similar works rescinded or terminated due to default during the last five (5) years as on the date of submission of this Bid.
3. The undersigned hereby authorize(s) and request(s) any bank, person, authorities, government or public limited institutions, firm or corporation to furnish pertinent information deemed necessary and requested by the VMC to verify our statements or our competence and general reputation.
4. The undersigned understands and agreed that further qualifying information may be requested, and agrees to furnish any such information at the request of the VMC.
5. The VMC and its authorized representative are hereby authorized to conduct any inquiries or investigations to verify the statements, documents, and information submitted in connection with this application and to seek clarification from our bankers and clients regarding any financial and technical aspects. This Affidavit will also serve as authorization to any individual or authorized representative of any institution referred to in the supporting information, to provide such information deemed necessary and requested by you to verify statements and information provided in the tender or with regard to the resources, experience and competence of the Applicant.
6. My/ our offer shall not be considered in case of fake/ forged document(s) found during verification at any stage or at any stage of contract. I/ We are agreed to whatever action (s) taken by competent authority of corporation in the aforesaid circumstances such as forfeiture of security deposit and debarring from participation in future tenders for the period/ years as deemed fit by the corporation and informing the same to all other state/ central level Government/ semi government organizations.

_____
Signed by the Authorized Signatory of the firm

Title of the office: _____ N

Date: _____

TQ_7: Details of experience

Date:

dd/mm/yyyy

To
Director IT
Information Technology Department
Vadodara Municipal Corporation
Khanderao Market, Vadodara – 390001

Sir/Madam,

I have carefully gone through the Terms & Conditions contained in the RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city".
I hereby declare that below are the details regarding relevant work that has been taken up by our company.

Assignment Name:	
Location:	Approx. Value of Services:
Name of Client:	Duration of assignment (months):
Client Contact Person, Title/Designation, Tel. No./Address:	Total No. of Beneficiary from thesystem: Total No of client end users:
Start date (month/year):	No. of professional staff-months provided by your firm/organization for the proposed Solution:
Completion date (month/year):	
Description of Project:	
Description of Actual Services provided:	
Mandatory Supporting Documents:	
Work order / Contract for the project/ Purchase Order	
Client Certificate giving present status of the project and view of the quality of services by the Bidder	

I further certify that I am competent officer in my company to make this declaration.
Yours Sincerely,
Signature of Authorized Signatory (with official seal)Name :
Designation :
Address :
Telephone& Fax :
E-mail address :

TQ_8: Solution Design

S.No.	Chapters of Technical Proposal	Description
Chapter 1: About Bidder		
1.1		
1.2		
1.3		
1.4		
1.5		
1.6		
Chapter 2: Understanding of Scope of Work and Approach of Bidder for Implementation		
2.1		
2.2		
2.3		
2.4		
2.5		
2.6		
2.7		
2.8		
2.9		
2.10		
2.11		
2.12		
Chapter 3: Approach & Methodology for Deliverables		
3.1		
3.2		
3.3		
3.4		
3.5		
3.6		
3.7		
3.8		
3.9		
3.10		
3.11		
3.12		
	Chapter 4: Expectation of Bidder from Authority	
	Chapter 5: Annexures	

TQ_9: Curriculum Vitae (CV) of Project Team

Sr. No.	Details	Response
1	Proposed Position & Skill Set	
2	Name of Firm	
3	Name of Staff [Insert full name]	
4	Date of Birth	
5	Education [Indicate college/university and other specialized education of staff member, giving names of institutions, degrees obtained, and dates of obtainment]	
6	Membership of Professional Associations / Societies	
7	Summary of key Training and Certifications	
8	Countries of Work Experience: [List countries where staff has worked in the last ten years]	
9	Language Proficiency	(Read/Write/Speak) - (Excellent/Good/Fair)
10	Number of years of experience	
11	Employment Record [Starting with present position, list in reverse order every employment held by staff member since graduation, giving for each employment as per format provided]	From [Year]:_____ To [Year]: _____ Employer: Positions held:
12	Detailed Tasks Assigned [List all tasks to be performed under this assignment]	
13	Highlights of assignments handled and significant accomplishments. [Among the assignments in which the staff has been involved, indicate the following information for those assignments that best illustrate staff capability to handle the tasks listed under point 12.]	Name of assignment or project: Year: Location: Client: Main project features: Positions held: Activities performed:

SIGNATURE: ……..……... DATE OF SGNING: Day Month Year

Format for Un-priced Bill of Material

The Bidder should provide the proposed Bill of Material (BoM) here. Bidder should refer to the Indicative BoM provided in the Volume II of this tender and should reproduce the same here. Kindly note that any additional items required should be clearly mentioned under 'additional line items' category towards the end of this table. Also note that details of the make/brand and model against each line item, wherever applicable, should be mentioned. The Bidder may add any additional line item (with adequate details) in the proposed BoM table below (towards the end of the table), that may be required to fulfill the tender and project requirements in totality. Kindly note that the indicative/estimated quantity provided in the RFP would be used for evaluation purposes; however, the payment would be done on actual usage basis.

Un-priced Bill of Material for all components

Un-priced Bill of Material						
#	BoM Line Item	Unit of measurement	Quantity Proposed (repeat the indicative quantity as provided in RFP)	Make / Brand	Model Details	Full Compliance with RFP Requirements (Yes / No)
1						
n						

TQ 10: Format for Authorization Letters from OEMs

<<To be printed on letter head of OEM and signed by Authorized signatory of OEM>>
Date: dd/mm/yyyy
To
Director IT
Information Technology Department
Vadodara Municipal Corporation
Khanderao Market, Vadodara – 390001

Ref.: Tender No: <No> Dated <DD/MM/YYYY>

Subject: Authorization of <company name of Bidder> to Supply and Provide Services of our Products/Equipment/Solution for RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city

Sir,

1. This is to certify that I/We am/are the Original Equipment Manufacturer in respect of the Products/ Equipment/ Solution listed below.
2. I/We confirm that <name of Bidder> have due authorization from us to supply our Products/ Equipment/ Solution and provide services to PURCHASER, for our Products/ Equipment/ Solutions listed below as per Request for Proposal (tender) document relating to Selection of Master Bidder (MSI) for Implementation and Maintenance of ICT Initiatives for Vadodara City We further endorse the warranty and contracting terms provided by bidder to PURCHASER.
3. I/We also undertake that we will provide support to PURCHASER in quality of deliverables and in ensuring that the solution is implemented in the best of ways by exploiting all the capabilities offered by the solution, to meet the requirements of PURCHASER.
4. We herewith certify that the Products/ Equipment/ Solutions quoted by us are not end of the life and we hereby undertake to support this Products/ Equipment/ Solutions for the contract duration of min 5 years 6 months from the date of Submission of the Bid).

Sr. No.	Name of Products/ Equipment/ Solutionas per RFP	Proposed Products/ Equipment/ Solutionwith brand, model & part number	OEM warranty	Remarks

CC: Bidder's corporate name

Yours faithfully,

(Signature of the Authorized Signatory from OEM)
Name
Designation
Seal.
Date: Place:
Business Address:

(Signature of the Authorized SignatoryBidder)
Name
Designation
Seal.
Date: Place:
Business Address:

TQ 11: Bank Guarantee format for EMD

FORMAT OF THE UNCONDITIONALAND IRREVOCABLE BANK GUARANTEE FOR EARNEST MONEY DEPOSIT

(**To be on non-judicial stamp paper of appropriate value as per Stamp Act relevant to place ofexecution.**)

In consideration of the .....(*Insert name of the Bidder*) submitting the Bid *inter alia*
For RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city, for meeting the terms and conditions in response to the RFP DOCUMENT---------- ---- dated ----------issued by **Vadodara Municipal Corporation** ("VMC"), and VMC agreeing to consider the Bid of *[Insert the name of the Bidder]* in accordance with the terms of the E-BID DOCUMENT, the …… (Insert name and address of the bank issuing the Bid Bond, and address of the head office) (Here in after referred to as "Guarantor Bank") hereby agrees unequivocally, irrevocably and unconditionally to pay to VMC or its authorized representative at..... [*Insert Name of the Place from the address of VMC*] forthwith on demand in writing from VMC or any representative authorized by it in this behalf an amount not exceeding Rupees..... on behalf of M/s .....[Insert name of the Bidder].

This guarantee shall be valid and binding on the Guarantor Bank up to and including..... (*Insert date of validity of Earnest Money Deposit in accordance with the terms of reference of the E-BID DOCUMENT*) and shall not be terminable by notice or any change in the constitution of the Guarantor Bank or by any other reasons whatsoever and our liability hereunder shall not be impaired or discharged by any extension of time orvariations or alternations made, given, or agreed with or without our knowledge or consent, by or between concerned parties.

Our liability under this Guarantee is restricted to Rupees …….. (Rs .....).
VMC or its authorized representative shall be entitled to invoke this Guarantee until [*Insert Date, which is six months after the date in the preceding sentence*]. The Guarantor Bank hereby expressly agrees that it shall not require any proof in addition to the written demand from VMC or its authorized representative, made in any format, raised at the above mentioned address of the Guarantor Bank, in order to make the said payment to VMC or its authorized representative.

The Guarantor Bank shall make payment hereunder on first demand without restriction or conditions and notwithstanding any objection, disputes, or disparities raised by the Bidder or any other person. The Guarantor Bank shall not require VMC or its authorized representative to justify the invocation of this BANK GUARANTEE, nor shall the Guarantor Bank have any recourse against VMC or its authorized representative in respect of any payment made hereunder.

This BANK GUARANTEE shall be payable at Vadodara.

This BANK GUARANTEE shall be interpreted in accordance with the laws of India and the courts at……. shall have exclusive jurisdiction.

The Guarantor Bank represents that this BANK GUARANTEE has been established in such form and with such content that it is fully enforceable in accordance with its terms as against the Guarantor Bank in the manner provided herein.
This BANK GUARANTEE shall not be affected in any manner by reason of merger, amalgamation, restructuring, liquidation, winding up, dissolution or any other change in the constitution of the Guarantor Bank.

This BANK GUARANTEE shall be a primary obligation of the Guarantor Bank and accordingly, VMC or its authorized representative shall not be obliged before enforcing this BANK GUARANTEE to take any action in any court or arbitral proceedings against the Bidder, to make any claim against or any demand on the Bidder or to give any notice to the Bidder to enforce any security held by VMC or its authorized representative or to exercise, levy or enforce any distress, diligence or other process against the Bidder.

The Guarantor Bank hereby agrees and acknowledges that VMC shall have a right to invoke this Bank Guarantee either in part or in full, as it may deem fit.

Notwithstanding anything contained hereinabove, our liability under this Guarantee is restricted to Rupees --- ----------and it shall remain in force until .....[*Date to be inserted on the basis of Terms of Reference of the E-BID DOCUMENT*], with an additional claim period of 6 (six) months thereafter.We are liable to pay the guaranteed amount or any part thereof under this BANK GUARANTEE only if ……VMC or its authorized representative serves upon us a written claim or demand.

In witness whereof the Bank, through its authorized officer, has set its hand and stamp on this ……..day of …… at …….

Witness:

Signature	Signature
Name	Name
Address	Address
Designation with Bank Stamp Signature	Designation with Bank Stamp Signature

Name and address

Attorney as per power of attorney No. ……

For..... [Insert Name of the Bank]

Banker's Stamp and Full Address:

Dated this …… day of ..... 2026.

Note: The Stamp Paper should be in the name of the Executing Bank

Annexure 2: Commercial Bid Format & Instructions

2.1 Commercial Bid Cover Letter

<<To be printed on letter head of Bidder and signed by Authorized signatory>>

Date: dd/mm/yyyy

To
Director IT
Information Technology Department
Vadodara Municipal Corporation
Khanderao Market, Vadodara – 390001

Subject: RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city

Reference: Tender No :< No> Dated<DD/MM/YYYY>

Dear Sir/ Madam,
We, the undersigned Bidders, having read and examined in detail all the bidding documents in respect of "Selection of Bidder for RFP Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city" do hereby propose to provide services as specified in the Bid Document referred above.

PRICE AND VALIDITY
All the prices mentioned in our Tender are in accordance with the terms as specified in the Tender documents. All the prices and other terms and conditions of this Tender are valid for entire contract duration. We hereby confirm that our Tender prices include all taxes. Taxes are quoted separately under relevant sections, as specified in the Bid Document formats.
We have studied the clause relating to Indian Income Tax and hereby declare that if any income tax, surcharge on Income Tax, Professional and any other corporate Tax in altercated under the law, we shall pay the same.

DEVIATIONS
We declare that all the services shall be performed strictly in accordance with the Bid Documents and there are no deviations
Further we agree that additional conditions, if any, found in our bid documents, shall not be given effect to.

QUALIFYING DATA
We confirm having submitted the information as required by you in your Instruction to Bidders. In case you require any other further information/documentary proof in this regard before evaluation of our Tender, we agree to furnish the same in time to your satisfaction.

BID PRICE
We declare that our Bid Price is for the entire scope of the work as specified in the Bid Document. The bidprice at which the contract is awarded shall hold good for entire tenure of the contract. These prices are indicated in the subsequent sub-sections of this Section.

CONTRACT PERFORMANCE GUARANTEE BOND
We hereby declare that in case the contract is awarded to us, we shall submit the contract Performance Bank Guarantee in the form prescribed in the Bid Document.

We hereby declare that our Tender is made in good faith, without collusion or fraud and the information contained in the Tender is true and correct to the best of our knowledge and belief.
We understand that our Tender is binding on us and that you are not bound to accept a Tender you receive. We confirm that no Technical deviations are attached here with this commercial offer.

Thanking you,

Yours faithfully,

(Signature of the Authorized Signatory)
Name
DesignationSeal.
Date:
Place:
Business Address:

2.2 General instructions for Commercial Bid

1) Bidder should provide all prices as per the prescribed format under this Annexure.
2) All the prices are to be entered in Indian Rupees (INR) only
3) Prices indicated in the schedules shall be inclusive of all taxes, Levies, duties including GST.
4) All prices should be inclusive of all required accessories/parts, installation charges and 5 years onsite repair warranty and comprehensive annual maintenance.
5) It is mandatory to provide breakup of all Taxes, Duties and Levies wherever asked for.
6) The bidder will have to arrange for storage of the goods at his own cost for storage of any items supplied as part of this scope of work.
7) VMC reserves the right to ask the SI to submit proof of payment against any of the taxes, duties, levies indicated.
8) The Unit Rate as mentioned in the following formats may be used for the purpose of 'Change Order' (limited to thirty percent of quantity described in commercial bid) for respective items including license fee, if any. The unit-rates discovered shall be valid for duration for 5 years 6 months from the date of opening of the financial bid.
9) VMC also intends to utilize various rates obtained through this tender for requirements across various departments. Bidders are requested to factor this potential demand and give the best possible rate to VMC.
10) No escalations of prices will be considered under any circumstances.
11) Warranty Period is defined as 5 years from the date of Go-Live of the complete system in working condition to the end user. The bidder shall be responsible to maintain the system up to 5 years 6 months which includes 5 year of onsite repair warranty and comprehensive Annual maintenance.
12) The bidders may visit the site and obtain additional information at their own cost and responsibility.
13) VMC to scale up and scale down resources based on requirements. Any line items may be removed
14) from the scope based on VMC's discretion.
15) During the payment stage, VMC reserves the right to ask the Bidder to submit proof of payment against any of the taxes, duties, levies indicated.
16) The Bidder needs to account for all Out of Pocket expenses due to Boarding, Lodging and other related items. No additional/separate payment shall be made regarding the same.
17) For the purpose of evaluation of Commercial Bids, VMC shall make appropriate assumptions to arrive at a common bid price for all the Bidders. This however shall have no co-relation with the Contract value or actual payment to be made to the Bidder.
18) In case of repeat of same item in commercial bid, the least price quoted by bidder shall be considered while placing additional order.
19) The Contract Price shall be firm and not subject to any alteration.
20) The Implementation agency should be deemed to have satisfied itself as to the correctness and sufficiency of the contract price, which shall, except as otherwise provided for in the contract, cover all its obligations under the contract.
21) Please note invitation of price discovery rate for future requirements does not imply guarantee of any additional work or any increase in scope. The price discovery rates are being invited to meet any exigency requirements if a need emerges during the period of contract with respect to deployment of Technical Team resources.
22) The rate/cost quoted for VMC and price discovery elements would be considered for additional procurement/payment in future.
23) Wherever present, the items mentioned as Lump Sum in above table will have quantity as 1 in the BoQ, the bidder should consider the same as Lump Sum and submit the commercials.

2.3 Commercial Bid Format

Quantities shown in the tender are approximate and no claim shall be entertained for quantities of work executed being either more or less than those entered in the tender of estimate.

This price bid has been prepared for price identification approx. 150 locations. Based on mutual agreement and discovered price more locations can be allotted to the bidders.

Electricity Cost in O&M – Electricity cost during O&M phase will be reimbursed to bidders on actual basis. The bidder will have to make payment and submit the actual electricity bill for reimbursement.

Assumptions -:

- Dismantling & Reinstallation cost has been kept for cost discovery purpose for future requirement (bidders are advised to put reasonable and justifiable pricing)

Note: (Rate should be inclusive of all the taxes, VMC may ask for bifurcation)

Field Infrastructure & CCTV Components (considering approx. 600 new CCTVs with 150 poles and other devices) with additional server and storage space for existing 250 CCTVs i.e. Total 900 CCTVs

Part A : Edge Devices				
Sr. No.	**Item**	**Qty**	**Unit**	**Base Rate/Unit With Tax**
1	Pole with Foundation, Electrification, Earthing and Civil Work	150	LOT	
2	PTZ CCTV Camera with Mounting Accessories	150		
3	Fixed CCTV Camera with Mounting Accessories	450		
4	Video Management Software (VMS) Licenses – Videonetics	600	Licenses	
5	Field Junction Box (Outdoor)	150	Nos.	
6	Industrial Managed PoE Switch with SFP Ports	150	Nos.	
7	UPS with Battery Backup	150	Nos.	
8	CCTV Signage Board	150	Nos.	
9	Wiring, Cabling, Conduit, Earthing and Accessories	150	Nos.	
B. OFC Network Infrastructure				
10	Optical Fiber Cable- 48 core (OFC) including Two (2) HDPE Duct + OFC Laying, Blowing, Splicing, Termination and Testing + LIU, Patch Panels, Patch Cords, SFPs and OFC Accessories	75	KM	
C. Data Centre (DC) Infrastructure				
11	Dome CCTV with mounting accessories	40	Nos.	
12	Server Chassis	1	Nos.	
13	Blade / Rack Servers (900 CCTVs & 20% open space after installation of all applications & Operating System) **	12	Nos.	
14	Storage System with Capacity to Cater 900 CCTV Cameras with minimum 20% empty space after installation	Lu ms um	TB	
15	Virtualization Software Licenses	1	Lot	
16	SAN Switch	2	Nos.	
17	ToR Switch	2	Nos.	
18	42U Network Rack with Accessories	1	Lot.	
19	Backup Software Included Automate Tap Library	1	Lot	
20	Tape Drive / Backup Media	1	Lot	
21	EMS / NMS Licenses	1	Licenses	
22	Antivirus / Endpoint Security Licenses for Server	1	Licenses	
23	Integration with Existing Applications / IoT Platform / CCC Applications	1	Lot	
D. Command & Control Centre Infrastructure				
24	Active LED Video Wall + Video Wall Controller + Civil, Electrical,	1	Set + Lot	

	Furniture and Installation Work for Video Wall with Existing Video Wall Buy back			
E. OPEX / CAMC (Post Go-Live)				
25	CAMC for all the above items for a total period of 5 years after Go-Live (including 1 year comprehensive warranty/support after Go-Live), along with all required manpower, 24x7 support services, and resources necessary.	5	Year	
F. Miscellaneous				
26	Any Other Item Required for Complete Functionality of Solution	1	Lot	

Buyback

Sr. No.	Description	Qty.	Unit	Buyback Value INR
1	Existing Video Wall System (Panels x 28, Controller, Mounting Structure, Cabling & Accessories)	1 Lot	Lot	
			Total Buyback Value	**INR**

- All A, B, C, D, E, F items/products/licenses with 5 years' comprehensive support
- In some products makes and models are fixed due to existing items/products/licenses with same make & model is available
- HP server/storage should be fixed in existing chassis /rack
- In future, in case of new requirement then will purchase based on price given of single item/products
- Final outcome by L1 vendor is to incorporate/integrate and smooth functionality of all new products/licenses
- Actual quantity may be differ based on actual requirement, budget and integration with existing items/products/licenses.
- CAMC for all the above items shall be provided for a period of five (5) years from the date of Go-Live. The CAMC scope shall include all required manpower, 24x7 support, preventive and corrective maintenance, spares, tools, vehicles, and resources necessary to meet the SLA requirements.
- The existing Video Wall system shall remain the property of VMC until successful commissioning of the new Video Wall system.
- **The Successful Bidder shall assess the existing server, storage, and IT infrastructure and carry out all necessary upgrades, configurations, and integration of existing applications (VMS/IVMS, Database, NMS, Antivirus Server, etc.) required to support the proposed CCTV cameras and licenses without any additional cost to VMC.
- The Successful Bidder shall dismantle and remove the existing Video Wall only after obtaining written approval from VMC.
- Any damage caused to the existing infrastructure during dismantling or removal shall be repaired by the Successful Bidder at its own cost.
- The quoted Buyback Value shall remain valid throughout the Contract Period and shall be adjusted in the final CAPEX payment.
- **L1 (Lowest Evaluated Bid) = Total of (Part A + Part B + Part C + Part D + Part E + Part F) – Buyback Value.**
- The **Lowest Evaluated Bid (L1)** shall be determined based on the above calculation. The accepted **Buyback Value** quoted by the Bidder shall be deducted from the total quoted bid value for the purpose of financial evaluation.
- The Buyback Value quoted by the Bidder shall be considered final and shall be deducted from the total accepted CAPEX value payable by VMC. No additional payment shall be made towards dismantling, handling, transportation, or disposal of the existing equipment.

Annexure 3: Agreement Format

ANNEXURE- I: Undertaking

(The Undertaking as per following format on non-judicial stamp paper of Rs. 300/- is to be furnished by selected vendor / successful bidder when asked by VMC after opening of tenders/Price bid.)

Name of Work: Selection of Bidder for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city.

Ref: Tender Notice No. _____

Respected Sir

We had participated for the above-mentioned work and we were qualified for the Criteria mentioned in the subjected works tender.

Upon asked by VMC, we hereby give our consent with this undertaking to carry out and execute the works of Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city , if the same is awarded to above/below the estimated rates.

We hereby agree and abide ourselves and assure VMC that we will not deny to carry out and execute the work if the same is awarded to us by the competent authority of VMC at above mentioned rates and terms mentioned in the subjected work's tender, failing which, competent authority of VMC may take penalty actions like disqualifying or debarring us for future works of VMC for the period as decided by VMC upon their discretion, or any other action as decided by competent authority of VMC.

This undertaking will remain in force up to …….

Sign and Stamp of Selected vendor / Successful Bidder.

ANNEXURE- II: Master Service Agreement

(To be printed on Rs. 300/- Stamp Paper)

This **AGREEMENT** is made at_____, Vadodara, Gujarat on this_____day of_____,_____,BETWEEN ----------, hereinafter referred to as **"Client**", or ***"VMC" (which expression unless repugnant to the context therein shall include its administrator and permitted assignees)*** of the **FIRST PART**;
AND
----------, a company registered under the Companies Act, 1956 or partnership firm registered under Indian Partnership Act 1932 or Proprietary Firm having its registered office at ----------
----------, hereinafter referred to as "**Service Provider**" or "**SP**" or "**Vendor**" or "**Bidder**", (which expression unless repugnant to the context therein, shall include its successors, administrators, executors and permitted assignees), of the **SECOND PART**.

Whereas VMC has envisaged the Selection of Bidder for (Hereinafter referred to as the "**Project**"); and whereas VMC published the RFP for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city;

And whereas M/s. ---------- has submitted its proposal for "Selection of Bidder for Selection of Bidder for Supply, Installation, Testing, Commissioning (SITC) and Operation & maintenance of Surveillance Cameras in Vadodara city.";

AND whereas VMC has selected M/s.....as successful bidder and issued Letter of Intent/Work Order dated ……..to the successful bidder who in turn signed and returned the same as a token of acceptance of Letter of Intent.

And whereas VMC and M/s---------- have decided to enter into this Agreement on the terms and conditions stipulated hereinafter.
NOW, THEREFORE, in consideration of the premises covenants and promises contained herein and other good and valuable considerations, the receipt and adequacy of which is hereby acknowledged, the parties intending to be bound legally, IT IS HEREBY AGREED between the Parties as follows:

1. Definitions

In this Agreement, the following terms shall be interpreted as indicated, -

(a) "VMC" means Vadodara Municipal Corporation;
(b) "Contract" means this Agreement entered into between VMC and the Bidder including all attachments and annexure thereto and all documents incorporated by reference therein;
(c) "Bidder" means M/s ---------- interchangeably referred to as "Bidder" in the contract; and
(d) "RFP" means the Tender Published by VMC (Ref. No----------) and the subsequent Corrigenda / Clarifications issued.
(e) "Go Live or successful completion of implementation of the project" date means the day after the date on which the proposed project stream becomes operational after successful conclusion of all acceptance tests to the satisfaction of VMC.
(f) "Deliverable" means any action / output generated by the Bidder while discharging their contractual obligations. This would include information and all the other services rendered as per the scope of work.

2. Interpretation

The documents forming this Agreement shall be deemed to be mutually explanatory of one another. In the event of any conflict, inconsistency, or discrepancy between the documents constituting this Agreement, the following order of precedence shall apply:

This Agreement;
Scope of Services as specified in the RFP.
Accepted Commercial Proposal of the Successful Bidder.
Clarifications, Addenda, and Corrigenda issued by VMC subsequent to the RFP.
The RFP Document issued by VMC.
Letter of Intent (LoI)/Letter of Acceptance (LoA)/Work Order issued by VMC to the Successful Bidder.
Technical Proposal and Commercial Proposal submitted by the Successful Bidder in response to the RFP and Payment Terms as specified in the RFP.

In case of any ambiguity or inconsistency, the document having the higher order of precedence shall prevail.

3. Term of the Agreement

The term of this agreement shall be a period of 5 years (inclusive 5 years of warranty) post Go-Live of the project.

In the event of implementation period getting extended beyond implementation timelines, for reasons not attributable to the Service Provider, VMC reserves the right to extend the term of the Agreement by corresponding period to allow validity of contract for 5 years post 3 months of successful running of complete network. (Note: Delay caused due to any reason not in control of the *Bidder c*ould not be attributed to the project period.)

VMC also reserves the right to extend the contract at its sole discretion for additional duration, beyond the 5 years of post-implementation period. Terms and conditions of such an extension shall be prepared by VMC and finalized in mutual discussion with the *Bidder*.

4. Payment Schedule & Milestone

The Payment Schedule & Milestone is as per the RFP and subsequent Addendum & Corrigendum released, if any. The same will form as Annexure of the Contract

5. Scope Extension

VMC reserves right to extend the scope of services for the price & timelines as given in Annexure I & Annexure to this Agreement.

6. Use & Acquisition of Assets during the term

Service Provider shall

- take all reasonable & proper care of the entire hardware & software, network or any other information technology infrastructure components used for the project & other facilities leased/owned by the bidder exclusively in terms of the delivery of the services as per this Agreement (hereinafter the "Assets" which include all the hardware / Software / furniture / data / documentations / manuals / catalogs / brochures / or any other material procured, created or utilized by the *Bidder* or VMC) in proportion to their use & control of such Assets which will include all upgrades/enhancements & improvements to meet the needs of the project arising from time to time; Note: Hardware upgrades outside the RFP scope would not be part of the original contract and would be catered through change request. Assets would be owned by the VMC however, the Service Provider would be custodian of the same during the entire contract period and would take care of all damage, insurance, theft etc.
- Maintain sufficient spare inventory at all times, for all items of importance;
- keep all the tangible Assets in good & serviceable condition (reasonable wear & tear excepted) &/or the intangible Assets suitably upgraded subject to the relevant standards as stated in of the RFP
- ensure that any instructions or manuals supplied by the manufacturer of the Assets for use of Assets & which are provided to the bidder will be followed by the Bidder & any person who will be responsible for the use of the Asset;
- take such steps as may be recommended by the manufacturer of the Assets & notified to the bidder or as may be necessary to use the Assets in a safe manner;
- To the extent that the Assets are under the control of the bidder, keep the Assets suitably housed & in conformity with any statutory requirements from time to time applicable to them,
- Provide and facilitate access to VMC or its nominated agencies & any persons duly authorized by him/her to enter any land or premises on which the Assets are for the time being sited so as to inspect the same, subject to any reasonable requirements;
- Not, knowingly or negligently use or permit any of the Assets to be used in contravention of any statutory provisions or regulation or in any way contrary to law;
- Use the Assets exclusively for the purpose of providing the Services as defined in the contract;
- Obtain a sign off from VMC or its nominated agencies at each stage as is essential to close each of the above considerations.

Ownership of the Assets shall vest with VMC on Go Live of the project. Ownership of any asset, created during the contractual period after go Live, shall also vest with VMC upon creation of such asset. Bidder shall not use VMC data to provide services for the benefit of any third party, as a service bureau or in any other manner. If damage to the assets is found unacceptable to the VMC, then corresponding penalty/liquidated damages shall be recovered from Bidder from the fees payable.

7. Concessions permissible under statutes

Bidder, while quoting against this tender, must take cognizance of all concessions permissible, if any, under the statutes and ensure the same is passed on to VMC, failing which it will have to bear extra cost. In case Bidder does not avail concessional rates of levies like customs duty, exciseduty, sales tax, etc. VMC will not take responsibility towards this. However, VMC may provide necessary assistance, wherever possible, in this regard.

8. Site visit and verification of information

Bidders are encouraged to submit their respective Bids after doing a thorough survey of project site and ascertaining for themselves the site conditions, traffic, location, surroundings, climate, availability of power, water and other utilities for construction, access to site, handling and storage of materials,

weather data, applicable laws and regulations, and any other matter considered relevant by them.
It shall be deemed that by submitting a Bid, the Bidder has made a complete and careful examination of the Bidding Documents;

- received all relevant information requested from the Authority
- accepted the risk of inadequacy, error or mistake in the information provided in the Bidding Documents or furnished by or on behalf of the Authority relating to any of the matters;
- satisfied itself about all matters, things and information including matters herein above necessary and required for submitting an informed Bid, execution of the Project in accordance with the Bidding Documents and performance of all of its obligations thereunder;
- acknowledged and agreed that inadequacy, lack of completeness or incorrectness of information provided in the Bidding Documents or ignorance of any shall not be a basis for any claim for compensation, damages, extension of time for performance of its obligations, loss of profits etc. from the Authority, or a ground for termination of the Concession Agreement by the Concessionaire
- acknowledged that it does not have a Conflict of Interest; and agreed to be bound by the undertakings provided by it under and in terms hereof.

9. Taxes

The Prices mentioned in the Price Bid should include all applicable taxes & duties as applicable.

GST
GST (Goods & Service Tax) has come in existence from 1st July, 2017. Selected vendor/Successful Bidder is bound to pay any amount GST prescribed by the Govt. of India per the terms of Contract agreed upon during the course of execution of this Contract.

During the course of execution of Contract, if there is any change in Rate of GST (Goods & Service Tax) by the Government, the same shall be reimbursed/recovered separately by VMC, subject to the submission of Original Receipt/Proof for the amounts actually remitted by the Successful Tendered/Selected vendor to the Competent Authority along with a Certificate from Chartered Accountant of Selected vendor/Successful bidder certifying that the amount of GST paid to the Government and the same shall be intimated/submitted/claimed within 30 (Thirty) Days from the date of payment. Remittance of GST within stipulated period shall be the sole responsibility of the Successful bidder/selected vendor, failing which, VMC may recover the amount due, from any other payable dues with VMC and decision of VMC shall be final and binding on the Selected vendor/Successful Bidder in this regard. Further the non- payment of GST to the Government may lead to the termination of contract and forfeiture of Security Deposit/Performance Guarantee Amount.

If imposition of any other new Taxes/Duties/Levies/Cess or any other incidentals etc. or any increase in the existing Taxes/Duties/Levies/Cess or any other incidentals etc. (excluding GST) are imposed during the course of the contract, the same shall be borne by the Selected vendor/Successful Bidder Only, in no case VMC shall be liable for the same.

y) Firm Prices and Bid Currency

Prices quoted must be firm and final and shall not be subject to any upward modifications, on any account whatsoever. Prices shall be expressed in Indian Rupees (INR) only.

z) Right to vary the scope of the work at the time of award

VMC reserves its right to make changes to the scope of the work at the time of execution of the resultant Agreement. If any such change causes an increase or decrease in the cost of, or the time required for the SI's performance of any part of the work under the Agreement, whether changed or not changed by the order, an equitable adjustment (if required) shall be made in the Contract Value or time schedule, or both, and the Agreement shall accordingly be amended. Any claims by the SI for adjustment under this Clause must be asserted within thirty (30) days from the date of the SI's receipt of the VMC changed order.

aa) **Modification or Withdrawal of Bids**

A Bidder wishing to withdraw its bid shall notify VMC by e-mail prior to the deadline prescribed for bid submission. A withdrawal notice may also be sent by electronic means such as e-mail, but it must be followed by a signed confirmation copy, postmarked at least one-day prior the deadline for submission of bids.

The notice of withdrawal shall:

Be addressed to VMC at the address named in the bid Data Sheet, Bear the Contract name, the <Title> and < bid No.>, and the words "bid Withdrawal Notice." Bid withdrawal notices received after the bid submission deadline shall be ignored, and the submitted bid shall be deemed to be a validly submitted bid.

No bid may be withdrawn in the interval between the bid submission deadline and the expiration of the specified bid validity period. Withdrawal of a bid during this interval may result in the forfeiture of the Bidder's EMD.

10. Work Order/ Purchase Order

aa) For all installations to be carried out in the VMC under this contract, the contract will be signed with VMC and the work order will be issued by the VMC.

bb) Validity Period of the Contract

Upon selection of the bidder and the contract is made, the validity of the Bidder contract period would be valid till 5 Years after go-live, unless revoked for whatever reasons. If at any stage during the tenure of the period, it comes to the notice of VMC, directly or through some other complaint, that the Bidder had misinterpreted the facts or submitted any false information or hidden any information, which could have affected the signing of this agreement with the Bidder, this agreement shall stand terminated immediately under intimation to the Bidder.

The contract would be subjected to review at the end of its validity period for renewal. If any need, necessities for such review during the validity period would be considered by VMC on its merit.

cc) Quantity variation

The quantity mentioned in the Commercial Bid format (Section 9) is only for indicative in nature. The actual quantity depends on the actual site survey conducted by the bidder and after same is approved by VMC for installations at various locations.

dd) Price Variation

During the validity of the contract including the extended period if any, if the Bidder supplied any item included in BoM of this RFP to any other department / organization / individual at a price lower than the price fixed in the contract, the bidder must voluntarily pass on the price difference with immediate effect.

ee) Governing Law

The Bidding Process shall be governed by, and construed in accordance with, the laws of India and the Courts at Vadodara shall have exclusive jurisdiction over all disputes arising under, pursuant to and/or in connection with the Bidding Process.

11. Restriction on Transfer of Agreement

The Bidder shall not assign or transfer its right in any manner whatsoever under this agreement to a third party or enter into any agreement for sub-contracting and/or partnership relating to any subject matter to the agreement to any third party or any sister-concerned firm within a group either in whole or in any part i.e., partnership/third party interest shall be created.

Failure to agree with the Terms & Conditions of the Bid Document/ Contract
Failure of the bidder to agree with the Terms & Conditions of the Bid Document/Contract shall constitute sufficient grounds for the annulment of the award of contract, in which event the contract may be awarded to the next most responsive bidder.

12. Terms and Conditions of the Tender

Bidder is required to refer to the draft Contract Agreement, attached as Annexure in this Bid Document, for all the terms and conditions to be adhered by the successful bidder during Project Implementation and Post implementation period.
Please note that one needs to read the Contract Agreement as a whole document; and the Annexure mentioned there-in may not correspond to the Bid Document Annexure. Please refer to the Interpretation Section of the Draft/Master Service Agreement.

13. Liability

Except as provided in this Agreement, hereinabove, neither party shall be liable to other party or any other party by virtue of termination of this Agreement for any reason whatsoever for any claim for loss or profit or on account for any expenditure, investment, leases, capital improvements or any other commitments made by the other party in connection with their business made in reliance upon or by virtue of this Agreement.

14. Force Majeure

In the event that any Damages to items due to Vandalism (physical Majeure attack by public, tampering of equipment by VMC staff and damage due to accidents) or due to Force Majeure events (such as earthquake, fire, natural calamities, war, act of God) of any kind during Warranty Period and Maintenance Period shall be the liability of VMC. In such case, VMC shall request the successful Bidder to repair/replace the damaged unit and reinstall the same. All costs towards the same shall be reimbursed by VMC to the successful Bidder less of insurance proceeds if need of replacement so arise then replacement shall be on tender rates only.

The Service Provider shall not be liable for forfeiture of its Performance Guarantee, imposition of liquidated damages or termination for default, if and to the extent that it's delay in performance or other failure to perform its obligations under the contract is the result of an event of Force Majeure. For purposes of this Clause, "Force Majeure" means an event beyond the "reasonable" control of the Service Provider , not involving the Service Provider 's fault or negligence and not foreseeable. Such events may include Acts of God & acts of Government of India in their sovereign capacity.

For the Bidder to take benefit of this clause it is a condition precedent that the Bidder must promptly notify the VMC, in writing of such conditions and the cause thereof within 2 calendar days of the Force Majeure event arising. VMC, or the consultant / committee appointed by the VMC shall study the submission of the Bidder and inform whether the situation can be qualified one of Force Majeure. Unless otherwise directed by the VMC in writing, the Bidder shall continue to perform its obligations under the resultant Agreement as far as it is reasonably practical, and shall seek all reasonable alternative means for performance of services not prevented by the existence of a Force Majeure event. In the event of delay in performance attributable to the presence of a force majeure event, the time for performance shall be extended by a period(s) equivalent to the duration of such delay. If the duration of delay continues beyond a period of 30 days, VMC and the Bidder shall hold consultations with each other in an endeavor to find a solution to the problem.

Notwithstanding anything to the contrary mentioned above, the decision of the VMC shall be final and binding on the Bidder

The Bidder shall not be liable for forfeiture of its Performance Guarantee, imposition of liquidated damages or termination for default, if and to the extent that it's delay in performance or other failure to perform its obligations under the contract is the result of an event of Force Majeure. For purposes of this Clause, "Force Majeure" means an event beyond the "reasonable" control of the Bidder, not involving the Bidder's fault or negligence and not foreseeable. Such events may include Acts of God & acts of Government of India in their sovereign capacity.

In the event of delay in performance attributable to the presence of a force majeure event, the time for performance shall be extended by a period(s) equivalent to the duration of such delay. If the duration of delay continues beyond a period of 30 days, VMC and the SI shall hold consultations with each other in an endeavor to find a solution to the problem.
Notwithstanding anything to the contrary mentioned above, the decision of the VMC shall be final and binding on the SI.

15. Safety Regulation, Accident and Damage

The Bidder shall be responsible at his own cost in and relative to performance of the work and bidder to observe and to ensure observance by his Sub-selected vendors, agents and servants of the provisions of Safety Code as hereinafter appearing and all fire, Safety and security regulations as may be prescribed by the Owner from time to time and such other Precautions, measures as shall be necessary and shall employ / deploy all equipment necessary to protect all works, materials, properties, structures, equipment, installations, communications and facilities whatsoever from damage, loss or other hazard whatsoever (including but not limited to fire and explosion) and shall during construction and other operations minimize the disturbance and inconvenience to the Owner, other bidders, the public and adjoining land and property owners and occupiers, and crops, trees and vegetation and shall indemnify and keep indemnified the One from and against all losses and damages and costs, charges and expenses and penalties, actions, claims, demands and proceedings whatsoever suffered or incurred by or against the Owner, as thecase may be, virtue of any loss, alteration, displacement, disturbance or destruction or accident to any works materials, properties, structures, equipment, installations communications and facilities and land and property owners and occupiers and crops, trees and vegetation as aforesaid, with the intent that the Bidder shall be exclusively responsible for any accident, loss, damage, alteration, displacement, disturbance or destruction as aforesaid resultant directly or indirectly from any breach by the Bidder of his obligation aforesaid or upon any operation, act or omission of the bidder his Sub-selected vendor(s) or agent(s) or servant(s).

The Bidder's liabilities under Clause (a) and otherwise under the Contract shall remain unimpaired notwithstanding the existence of any storage cum erection or other insurance covering any risk, damage, loss or liability for which the Bidder is liable to the Owner in terms of the foregoing Sub-Clause or otherwise and / or in respect of which the Bidder has indemnified the Owner with the intent that notwithstanding the existence of such insurance, the Bidder shall be and remain fully liable for all liabilities and obligations under the contract and indemnified to the Owner, and the Owner shall not be obliged to seek recourse under such policy(ies) in preference to recourse against the Bidder or otherwise to exhaust any other remedy in preference to the remedies available to in under the Contract prior written approval of VMC. However, even if the work is sub-contracted / outsourced, the sole responsibility of the work shall lie with the SI. The SI shall be held responsible for any delay/error/non-compliance etc. of its sub-contracted vendor. The details of the sub-contracting agreements (if any) between both the parties would be required to be submitted to VMC.

16. Ownership and Licenses

The ownership of all hardware/software developed/customized/ configured/ procured as part of the project and related documentation for the project would always lie with the VMC. All licenses for software procured related to project have to be in the name of Vadodara Municipal

Corporation. The bidder will be required to produce the Licenses/ATS/Warranty and other documents from the respective OEMs clearly mentioning the product name, quantity, duration, type of support, etc. The payment for the respective item will be subject to submission of the aforesaid documents to VMC.

17. Conflict of Interest

A "Conflict of Interest" is any situation that might cause an impartial observer to reasonably question whether Bidder actions are influenced by considerations of your firm's interest at the cost of Government. The Bidder agrees that it shall hold the VMC's interest paramount, without any consideration for future work, and strictly avoid any Conflict of Interest with other assignments of a similar nature. In the event the Bidder foresees a Conflict of Interest, the Bidder shall notify VMC forthwith and seek its approval prior to entering into any arrangement with a third party which is likely to create a Conflict of Interest.

Bidders shall not have a conflict of interest that may affect the Selection Process or the scope (the "Conflict of Interest"). Any Bidder found to have a Conflict of Interest shall be disqualified. VMC requires that the Bidder provides professional, objective, and impartial advice and at all times hold the VMC's interests paramount, avoid conflicts without her assignments or its own interests, and act without any consideration for future work. The Bidder shall disclose to VMC in writing, all actual and potential Conflicts of Interest that exist, arise or may arise (either for the Bidder or its team) during the term of the Agreement as soon as it becomes aware of such a conflict.

18. General Clause related to Any Bidders/Sub-Selected vendor from a Country which shares a Land Border with India

I. Any bidder from a country which shares a land border with India will be eligible to bid in this tender only if the bidder is registered with the Competent Authority.

II. "Bidder" (including the term 'tenderer', 'consultant' or 'service provider' in certain contexts) means any person or firm or company, including any member every artificial juridical person not falling in any of the descriptions of bidders stated hereinbefore, including any agency branch or office controlled by such person, participating in a procurement process.

III. "Bidder from a country which shares a land border with India" means:

h. An entity incorporated, established or registered in such a country; or
i. A subsidiary of an entity incorporated, established or registered in such a country; or
j. An entity substantially controlled through entities incorporated, established or registered in such a country; or
k. An entity whose beneficial owner is situated in such a country; or
l. An Indian (or other) agent of such an entity; or

IV. The beneficial owner for the purpose of (iii) above will be as under:

1. In case of a company or Limited Liability Partnership, the beneficial owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person, has a controlling ownership interest or who exercises control through other means.
   a. Explanation—
   b. "Controlling ownership interest" means ownership of or entitlement to more than twenty- five per cent. of shares or capital or profits of the company;
   c. "Control" shall include the right to appoint majority of the directors or to control the management or policy decisions including by virtue of their shareholding or management rights or shareholders agreements or voting agreements;
2. In case of a partnership firm, the beneficial owner is the natural person(s) who, whether acting alone or together, or through one or more juridical person, has ownership of entitlement to more than fifteen percent of capital or profits of the partnership;
3. In case of an unincorporated association or body of individuals, the beneficial owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person, has ownership of or entitlement to more than fifteen percent of the property or capital or profits of such association or

body of individuals;

4. Where no natural person is identified under i or ii or iii above, the beneficial owner is the relevant natural person who holds the position of senior managing official;
5. In case of a trust, the identification of beneficial owner(s) shall include identification of the author of the trust, the trustee, the beneficiaries with fifteen percent or more interest in the trust and any other natural person exercising ultimate effective control over the trust through a chain of control or ownership.

V. An Agent is a person employed to do any act for another, or to represent another in dealings with third person.

VI. The successful bidder shall not be allowed to sub-contract works to any selected vendor from a country which shares a land border with India unless such selected vendor is registered with the Competent Authority (Registration committee constituted by the Department for Promotion of Industry and Internal Trade (DPIIT)).

19. Resolution of Dispute

The VMC and the Bidder shall make every effort to resolve amicably, by direct informal negotiation, any disagreement or dispute arising between them under or in connection with the contract. If after thirty days from the commencement of such informal negotiations, the VMC and the Bidder have been unable to resolve amicably a contract dispute; either party may require that the dispute be referred for resolution by formal arbitration.

All questions, disputes or differences arising under and out of, or in connection with the contract, shall be referred to two Arbitrators: one Arbitrator to be nominated by the VMC and the other to be nominated by the Bidder. In the case of the said Arbitrators not agreeing, then the matter will be referred to an umpire to be appointed by the Arbitrators in writing before proceeding with the reference. The award of the Arbitrators, and in the event of their not agreeing, the award of the Umpire appointed by them shall be final and binding on the parties. The Arbitration and Reconciliation Act 1996 shall apply to the arbitration proceedings and the venue of the arbitration shall be Vadodara. Cost of arbitration shall be borne by each party proportionately. However, expenses incurred by each party in connection with the preparation, presentation shall be borne by the party itself. The provisions of this clause shall survive termination of this Agreement.

20. Dismantling and Reinstallation

In case of dismantling and re-installation of hardware at project site due to reasons not attributed to vendor's performance, authority will reimburse the amount for dismantling and reinstallation work on submission of original bills and as per the rate discovered in commercial bid. Vendor will have to take prior approval from authority before executing such tasks. For the items, which is not discovered in price bid and is essential for project requirement, authority may take reference from other on-going projects at VMC and market rate to finalize the rate for such items.

However, it is the responsibility of vendor to execute the task of Dismantling and Reinstallation on request of authority.

21. Electricity and Utility Connection

The vendor shall be responsible for ensuring electricity connection and other utilities to make the device/hardware operational. Vendor responsibility involves making request for electricity/utility providing company, submission of requisite fee, test report, requisite certification, coordination, laying power cables and provisioning of requisite power points for all components in the scope of this project. Vendor shall be responsible for paying deposit and metering cost required for taking any electrical connections to concerned departments.

SI shall be responsible for laying last mile power cables and provisioning of requisite power points for all components in the scope of this project. SI shall be responsible for paying deposit and metering cost required for taking any electrical connections to concerned departments, this deposit shall be reimbursed by VMC. Since this component has dependency on approval from other agencies, SI should

plan this requirement well in advance & submit the application to the electricity provider through VMC. Recurring charges (during the operational phase) for all the electricity consumption of field devices/components in the scope of this project will be paid by VMC to directly electricity provider.
VMC will appoint a nodal person to provide necessary guidance and support in this process. Authority will reimburse the cost on production of actual bill.
Since this component has dependency on approval from other agencies, vendor should plan this requirement well in advance & submits the application to MGVCL through VMC. Recurring charges (during operational phase) for all the electricity/utility consumption of components in the scope of this project will be borne by VMC but vendor will be responsible for coordination, informing authority to ensure timely payment of bills. Vendor will be responsible for maintenance of these electrical connections during the operational phase, and authority will reimburse material and Labour cost for breakdowns on submission of original bills.

22. Documentation of Project Outcomes

The vendor shall be responsible for project documentation including case study, impact assessment report, success report, explanatory video and stakeholder benefit video, training material with project brief ppts (detail of requirement is mentioned in scope of Work section). This will help in educating project stakeholders about functions and benefit of the project. Also such material will be helpful in replicating the projects at other locations and submitting the documentation at national and international forum.

23. Bid Participation

Only individual firms/entities shall be eligible to participate in this Tender. Consortium, Joint Venture (JV), partnership between multiple firms for the purpose of this bid, or any similar arrangement shall not be permitted.

A Bidder shall submit only one bid in response to this Tender. Submission of multiple bids by the same Bidder, either directly or indirectly, shall result in rejection of all such bids.

The Bidder shall be solely responsible for execution, implementation, operation, maintenance, performance obligations, and compliance with all terms and conditions of the Contract. No change in ownership, management control, or legal status of the Bidder that materially affects contract performance shall be permitted without prior written approval of VMC.

24. Additional Quantity/Change Request

a. VMC reserves its right to award additional quantity of work order of the original quantity of total project cost (as per state and central govt. guideline) at the same price and terms and conditions quoted in original contract.
b. VMC also reserves right to award change request work order of the original project cost for the work needed to execute the work defined in scope of work defined in original contract
c. Payment for additional items
   – Based on the final project plan or during the project duration if there is need for additional items and Based on the final project plan or during the project duration if there is need for additional items and there is variation in the quantities as defined in the financial bid, implementation bidder will be required to arrange additional items. Payment for such additional items will be done separately by VMC.
d. In such case of any additional requirement, payment will be done based on the rate provided in the financial bid of the Implementation Bidder. Payment mechanism to be followed for additional items is given below:
e. CAPEX COMPONENT - 30% CAPEX payment on the successful delivery of additional item, post approval from VMC and 70% CAPEX payment on the successful installation of the additional item, post approval on the installation report.
f. OPEX COMPONENT - OPEX part will be equally divided in remaining part of the quarterly payment during O&M. OPEX payment will be prorated based on the duration left in O&M. That is, if additional item is procured in 4th Quarter then OPEX will be paid for 17 quarters only (starting from 4th Quarter till 20th Quarter).

25. Termination of Contract

Termination of Contract VMC may, without prejudice to any other remedy under this Contract and applicable law, reserves the right to terminate for breach of contract by providing a written notice of 30 days stating the reason for default to the SI and as it deems fit, terminate the contract either in whole or in part in the following ways.

- Termination by Default: for failing to perform obligations under the Contract of if the quality is not up to the specification or in the event of non-adherence to time schedule.
- Termination for Convenience: VMC by written notice sent to the SI, may terminate the Contract, in whole or in part, at any time for its convenience. The notice of termination shall specify that the termination is for VMC's convenience, the extent to which performance of the SI under the Contract is terminated, and the date upon which such termination becomes effective.
- Termination for Non-Performance: If the SI fails to deliver any or all of the project requirements / operationalization / go-live / performance parameters (PERFORMANCE STANDARDS) of the project within the time frame specified in the contract; or
  If the SI fails to perform any other obligation(s) under the contract.
- Termination for Insolvency/NCLT proceedings: The Department may at any time terminate the contract by giving written notice to the bidder(s), if the bidder(s) becomes bankrupt or otherwise insolvent. In this event, termination will be without compensation to the bidder(s), provided that such termination will not prejudice or affect any right of action or remedy which has accrued or will accrue thereafter to the Department. Bidder agrees to handover all requisite data in terms of code, documents, etc. along with requisite to the office of the VMC identified team in this case.

Prior to providing a notice of termination to the SI, VMC shall provide the SI with a written notice of 30 days instructing the SI to cure any breach/ default of the Contract, if VMC is of the view that the breach may be rectified.

On failure of the SI to rectify such breach within 30 days, VMC may terminate the contract by providing a written notice of 30 days to the SI, provided that such termination will not prejudice or affect any right of action or remedy which has accrued or will accrue thereafter to VMC. In such event the SI shall be liable for penalty/liquidated damages imposed by the VMC. The performance Guarantee shall be forfeited by the VMC.

Consequences of Termination

- In the event of termination of this contract, VMC is entitled to impose any such obligations and conditions and issue any clarifications as may be necessary to ensure an efficient transition and effective continuity of the services which the SI shall be obliged to comply with and take all available steps to minimize the loss resulting from that termination/ breach, and further allow and provide all such assistance to VMC and/ or succeeding vendor, as may be required, to take over the obligations of the SI in relation to the execution / continued execution of the requirements of this contract.
- In the event of termination of this contract, VMC shall have ownership over entire volume of delivered & installed software and hardware irrespective payment made to successful bidder.
- In the event of the SI being unable to service the contract for whatever reason, VMC would evoke the PBG. Notwithstanding and without prejudice to any rights whatsoever of department under the Contract in the matter, the proceeds of the PBG shall be payable to department as compensation for any loss resulting from the Bidder's failure to complete its obligations under the Contract. Department shall notify the Bidder in writing of the exercise of its right to receive such compensation within 14 days, indicating the contractual obligation(s) for which the Bidder is in default.
- VMC shall also be entitled to make recoveries from the Bidder's bills, performance bank guarantee, or from any other amount due to him, the equivalent value of any payment made to him due to inadvertence, error, collusion, misconstruction or misstatement

26. Indemnity

The Service Provider agrees to indemnify and hold harmless VMC its officers, employees and agents(each

a "Indemnified Party") promptly upon demand at any time and from time to time, from and against any and all losses , claims, damages, liabilities, costs (including reasonable attorney's fees and disbursements) and expenses (collectively, "Losses") to which the Indemnified Party may become subject, in so far as such losses directly arise out of, in any way relate to, or result from

i. any mis-statement or any breach of any representation or warranty made by the Bidder or

ii. The failure by the Service Provider to fulfill any covenant or condition contained in this Agreement, including without limitation the breach of any terms and conditions of this Agreement by any employee or agent of the Service Provider. Against all losses or damages arising from claims by third Parties that any Deliverable (or the access, use or other rights thereto), created Service Provider pursuant to this Agreement, or any equipment, software, information, methods of operation or other intellectual property created by Service Provider or sub-selected vendors pursuant to this Agreement, or the PERFORMANCE STANDARDSs (I) infringes a copyright, trade mark, trade design enforceable in India, (II) infringes a patent issued in India, or

(III) constitutes misappropriation or unlawful disclosure or use of another Party's trade secretes under the laws of India(collectively, "Infringement Claims"); provided, however, that this will not apply to any Deliverable (or the access, use or other rights thereto) created by (A) "Implementation of Project by itself or through other persons other than Service Provider or its sub-selected vendors; (B) Third Parties (i.e., other than Service Provider or sub-selected vendors) at the direction of VMC, or

iii. any compensation / claim or proceeding by any third party against VMC arising out of any act, deed or omission by the Service Provider or Claim filed by a workman or employee engaged by the Service Provider for carrying out work related to this Agreement. For the avoidance of doubt, indemnification of Losses pursuant to this section shall be made in an amount or amounts sufficient to restore each of the Indemnified Party to the financial position it would have been in had the losses not occurred.

iv. Any payment made under this Agreement to an indemnity or claim for breach of any provision of this Agreement shall include applicable taxes.

27. Third Party Claims

a. Subject to Sub-clause (b) below, the Service Provider (the "Indemnified Party") from and against all losses, claims litigation and damages on account of bodily injury, death or damage to tangible personal property arising in favor or any person, corporation or other entity (including the Indemnified Party) attributable to the Indemnifying Party's performance or non-performance under this Agreement or the PERFORMANCE STANDARDSs.

b. The indemnities set out in Sub-clause (a) above shall be subject to the following conditions:

(i) The Indemnified Party, as promptly as practicable, informs the Indemnifying Party in writing of the claim or proceedings and provides all relevant evidence, documentary or otherwise;

(ii) The Indemnified Party shall, at the cost and expenses of the Indemnifying Party, give the Indemnifying Party all reasonable assistance in the defense of such claim including reasonable access to all relevant information, documentation and personnel. The indemnifying party shall bear cost and expenses and fees of the Attorney on behalf of the Indemnified Party in the litigation, claim.

(iii) if the Indemnifying Party does not assume full control over the defense of a claim as provided in this Article, the Indemnifying Party may participate in such defense at its sole cost and expense, and the Indemnified Party will have the right to defend the claim in such manner as it may deem appropriate, and the cost and expense of the Indemnified Party will be borne and paid by the Indemnifying Party.

(iv) The Indemnified Party shall not prejudice, pay or accept any proceedings or claim, or

compromise any proceedings or claim, without the written consent of the Indemnifying Party;

(v) bidder hereby indemnify & hold indemnified the VMC harmless from & against any & all damages, losses, liabilities, expenses including legal fees & cost of litigation in connection with any action, claim, suit, proceedings as if result of claim made by the third party directly or indirectly arising out of or in connection with this agreement.

(vi) all settlements of claims subject to indemnification under this Article will: (a) be entered into only with the consent of the Indemnified Party, which consent will not be unreasonably withheld & include an unconditional release to the Indemnified Party from the claimant for all liability in respect of such claim; & (b) include any appropriate confidentiality agreement prohibiting disclosure of the terms of such settlement;

(vii) the Indemnified Party shall take steps that the Indemnifying Party may reasonably require to mitigate or reduce its loss as a result of such a claim or proceedings; &

(viii) In the event that the Indemnifying Party is obligated to indemnify an Indemnified Party pursuant to this Article, the Indemnifying Party will, upon payment of such indemnity in full, be subrogated to all rights & defenses of the Indemnified Party with respect to the claims to which such indemnification relates;

(ix) in the event that the Indemnifying Party is obligated to indemnify the Indemnified Party pursuant to this Article, the Indemnified Party will be entitled to invoke the Performance Bank Guarantee, if such indemnity is not paid, either in full or in part, & on the invocation of the Performance Bank Guarantee, the Indemnifying Party shall be subrogated to all rights & defenses of the Indemnified Party with respect to the claims to which such indemnification relates.

28. Publicity

Any publicity by the bidder in which the name of VMC is to be used should be done only with the explicit written permission from VMC.

29. Warranties

a. The Service Provider warrants and represents that:
   i. It has full capacity and authority and all necessary approvals to enter into and to perform its obligations under this Agreement;
   ii. This Agreement is executed by a duly authorized representative of the Service Provider;
   iii. It shall discharge its obligations under this Agreement with due skill, care and diligence so as to comply with the service level agreement.

b. In the case of the PERFORMANCE STANDARDSs, the Service Provider warrants and represents, that:
   - the Service Provider has full capacity and authority and all necessary approvals to enter into and perform its obligations under the PERFORMANCE STANDARDSs and to provide the Services;
   - The PERFORMANCE STANDARDSs have been executed by a duly authorized representative of the Service Provider;
   - The Service Provider is experienced in managing and providing works similar to the Services and that it will perform the Services with all due skill, care and diligence so as to comply with service level agreement;
   - The Services will be provided and rendered by appropriately qualified, trained and experienced personnel as mentioned in the RFP;
   - Service Provider has and will have all necessary licenses, approvals, consents of third Parties free

from any encumbrances and all necessary technology, hardware and software to enable it to provide the Services;

- The Services will be supplied in conformance with all laws, enactments, orders and regulations applicable from time to time;
- Service Provider will warrant that the goods supplied under the contract are new, unused, of the most recent higher version /models and incorporate all recent improvements in design and materials unless provided otherwise in the contract. The Service Provider further warrants that the goods supplied under this contract shall have no defects arising from design, materials or workmanship.
- The overall system design shall be such that there is no choking point / bottleneck anywhere in the system (end-to-end) which can affect the performance / PERFORMANCE STANDARDSs.

Subject to the fulfillment of the obligations of the Service Provider as provided for in sub clause (viii) above, in the event that such warranties cannot be enforced by VMC, the Service Provider will enforce such warranties on behalf of VMC and pass on to VMC, the benefit of any other remedy received in relation to such warranties.

c. Notwithstanding what has been stated elsewhere in this Agreement and the Schedules attached herein, in the event the Service Provider is unable to meet the obligations pursuant to the implementation of the Project, Operations and Maintenance Services and any related scope of work as stated in this Agreement and the Schedules attached herein, VMC will have the option to invoke the Performance Guarantee after serving a written notice of thirty (30) days on the bidder.

30. Limitation of Liability towards VMC

The Bidder's liability under the resultant Agreement shall be determined as per the Law in force for the time being. The Bidder shall be liable to the VMC for loss or damage occurred or caused or likely to occur on account of any act of omission on the part of the Bidder and its employees, including loss caused to VMC on account of defect in goods or deficiency in services on the part of Bidder or his agents or any person / persons claiming through or under said Bidder. However, such liability of Bidder shall not exceed the contract value. This limitation of liability shall not limit the Bidder's liability, if any, for damage to Third Parties caused by the Bidder or any person or firm acting on behalf of the Bidder in carrying out the scope of work envisaged herein.

31. Data Ownership

All the data created as the part of the project shall be owned by VMC The Bidder shall take utmost care in maintaining security, confidentiality and backup of this data. Access to the data / systems shall be given by the Bidder only as per the IT Security Policy, approved by VMC. VMC its authorized representative(s) shall conduct periodic / surprise security reviews and audits, to ensure the compliance by the Bidder Vendor to data / system security.

32. Intellectual Property Rights

Deliverables provided to VMC by Service Provider during the course of its performance under this Agreement, all rights, title and interest in and to such Deliverables, shall, as between Service Provider

and VMC , immediately upon creation, vest in VMC . To the extent that the Service Provider Proprietary Information is incorporated within the Deliverables, Service Provider and its employees engaged hereby grant to VMC a worldwide, perpetual, irrevocable, non-exclusive, transferable, paid-up right and license to use, copy, modify (or have modified), use and copy derivative works for the benefit of and internal use of VMC.

33. Fraud and Corruption

VMC requires that Bidder must observe the highest standards of ethics during the execution of the contract. In pursuance of this policy, VMC Libraries defines, for the purpose of this provision, the terms set forth as follows:

a. "Corrupt practice" means the offering, giving, receiving or soliciting of anything of value to influence the action of VMC libraries in contract executions.
b. "Fraudulent practice" means a mis-presentation of facts, in order to influence a procurement process or the execution of a contract, to VMC libraries, and includes collusive practice among bidders (prior to or after Proposal submission) designed to establish Proposal prices at artificially high or non-competitive levels and to deprive VMC of the benefits of free and open competition.
c. "Unfair trade practices" means supply of services different from what is ordered on, or change in the Scope of Work which is given in the contract.
d. "Coercive Practices" means harming or threatening to harm, directly or indirectly, persons or their property to influence their participation in the execution of contract.

If it is noticed that the Bidder has indulged into the Corrupt / Fraudulent / Unfair / Coercive practices, it will be a sufficient ground for VMC for termination of the contract and initiate black-listing of the vendor.

34. Exit Management

(i) Exit Management Purpose

This clause sets out the provisions, which will apply during Exit Management period. The Parties shall ensure that their respective associated entities carry out their respective obligations set out in this Exit Management Clause.

The exit management period starts, in case of expiry of contract, at least 6 months prior to the date when the contract comes to an end or in case of termination of contract, on the date when the notice of termination is sent to the Bidder. The exit management period ends on the date agreed upon by the VMC or Six months after the beginning of the exit management period, whichever is earlier.

(ii) Confidential Information, Security and Data

Service Provider will promptly on the commencement of the exit management period, supply to the VMC or its nominated agencies the following:

a) Information relating to the current services rendered and performance data relating to the performance of the services; Documentation relating to Surveillance Project, Project's Intellectual Property Rights; any other data and confidential information related to the Project;
b) Project data as is reasonably required for purposes of the Project or for transitioning of the services to its Replacing Successful Bidder in a readily available format.
c) All other information (including but not limited to documents, records and agreements) relating to the services reasonably necessary to enable the VMC and its nominated agencies, or its Replacing Vendor to carry out due diligence in order to transition the

provision of the Services to VMC, or its nominated agencies, or its Replacing Vendor (as the case may be).

d) Information relating to the current services rendered and customer and performance data relating to the performance of sub-contractors in relation to the services;
e) Documentation relating to Intellectual Property Rights;
f) Documentation relating to sub-contractors;
g) All current and updated data as is reasonably required for purposes of purchaser or its nominated agencies transitioning the services to its Replacement SI in a readily available format nominated by the purchaser, its nominated agency;
h) All other information (including but not limited to documents, records and agreements) relating to the services reasonably necessary to enable purchaser or its nominated agencies, or its Replacement SI to carry out due diligence in order to transition the provision of the Services to purchaser or its nominated agencies, or its Replacement System integrator (as the case may be).
i) Before the expiry of the exit management period, the SI shall deliver to the purchaser or its nominated agency all new or up-dated materials from the categories set out in schedule above and shall not retain any copies thereof, except that the SI shall be permitted to retain one copy of such materials for archival purposes only.

(iii) Employees

Promptly on reasonable request at any time during the exit management period, the Successful Bidder shall, subject to applicable laws, restraints and regulations (including in particular those relating to privacy) provide to VMC, a list of all employees (with job titles and communication address) of the Successful Bidder, dedicated to providing the services at the commencement of the exit management period; To the extent that any Transfer Regulation does not apply to any employee of the Successful Bidder, VMC, or Replacing Vendor may make an offer of contract for services to such employee of the Successful Bidder and the Successful Bidder shall not enforce or impose any contractual provision that would prevent any such employee from being hired by the VMC, or any Replacing Vendor.

(iv) Rights of Access to Information

At any time during the exit management period, the Successful Bidder will be obliged to provide an access of information to VMC and / or any Replacing Vendor in order to make an inventory of the Assets (including hardware / Software / Active / passive), documentations, manuals, catalogs, archive data, Live data, policy documents or any other material related to the Surveillance Project.

(v) Exit Management Plan

Successful Bidder shall provide VMC with a recommended exit management plan ("Exit Management Plan") within 90 days of signing of the contract, which shall deal with at least the following aspects of exit management in relation to the PERFORMANCE STANDARDS as a whole and in relation to the Project Implementation, the Operation and Management PERFORMANCE STANDARDS and Scope of work definition.

a) A detailed program of the transfer process that could be used in conjunction with a Replacement Vendor including details of the means to be used to ensure continuing provision of the services throughout the transfer process or until the cessation of the services and of the management structure to be used during the transfer;

b) Plans for the communication with such of the Successful Bidder, staff, suppliers, customers and

any related third party as are necessary to avoid any material detrimental impact on Project's operations as a result of undertaking the transfer;

c) Plans for provision of contingent support to the Surveillance Project and Replacement Vendor for a reasonable period (minimum one month) after transfer.
d) Successful Bidder shall re-draft the Exit Management Plan annually to ensure that it is kept relevant and up to date.
e) Each Exit Management Plan shall be presented by the Successful Bidder to and approved by VMC or its nominated agencies.
f) The terms of payment as stated in the Terms of Payment Schedule include the costs of the Successful Bidder complying with its obligations under this Schedule.
g) During the exit management period, the Successful Bidder shall use its best efforts to deliver the services.
h) Payments during the Exit Management period shall be made in accordance with the Terms of Payment Schedule.

35. Miscellaneous

a) Confidentiality

"Confidential Information" means all information including Project Data (whether in written, oral, electronic or other format) which relates to the technical, financial and operational affairs, business rules, citizen information, video footages, alert information, any police department data, products, processes, data, crime / criminal secrets, design rights, know-how and personnel of each Party and its affiliates which is disclosed to or otherwise learned by the other Party or sub selected vendors (whether a Party to the contract or to the PERFORMANCE STANDARDS) in the course of or in connection with the contract (including without limitation such information received during negotiations, location visits and meetings in connection with the contract or to the PERFORMANCE STANDARDS) or pursuant to the contract to be signed subsequently.

Except with the prior written permission of VMC, the Service Provider and its Personnel shall not disclose such confidential information to any person or entity not expected to know such information by default of being associated with the project, nor shall the Service Provider and it's Personnel make public the recommendations formulated in the course of, or as a result of the Project.

a. The Service Provider recognizes that during the term of this Agreement, sensitive data will be procured & made available to it, its Sub selected vendors & agents & others working for or under the Service Provider. Disclosure or usage of the data by any such recipient may constitute a breach of law applicable causing harm not only to VMC whose data is used but also to its stakeholders. Service Provider, its Sub selected vendors & agents are required to demonstrate utmost care, sensitivity & strict confidentiality. Any breach of this Article will result in VMC & its nominees receiving a right to seek injunctive relief & damages from the Service Provider.
b. Each Party agrees as to any Confidential Information disclosed by a Party to this Agreement (the "Discloser") to the other Party to this Agreement (the "Recipient") &
    i. to take such steps necessary to protect the Discloser's Confidential information from unauthorized use, reproduction & disclosure, as the Recipient takes in relation to its own Confidential Information of the same type, but in no event less than reasonable care;
    ii. to use such Confidential Information only for the purposes of this Agreement or as otherwise expressly permitted or expressly required by this Agreement or as

otherwise permitted by the Discloser in writing; &

iii. not, without the Discloser's prior written consent, to copy the Confidential Information cause or allow it to be copied, directly or indirectly, in whole or in part, except as otherwise expressly provided in this Agreement, or as required in connection with Recipient's use as permitted under this Article, or as needed for the purposes of this Agreement, or as needed for the purposes of this Agreement, provided that any proprietary legends & notices (whether of the Discloser or of a Third Party) are not removed or obscured; &

iv. Not, to disclose, transfer, publish or communicate the Confidential Information in any manner, without the Discloser's prior written consent, to any person except as permitted under this Agreement.

c. The restrictions of this Article shall not apply to confidential Information that:

i. is or becomes generally available to the public through no breach of this Article by the Recipient; &

ii. Was in the recipient's possession free of any obligation of confidence prior to the time of receipt of it by the Recipient hereunder; &

iii. Is developed by the Recipient independently of any of discloser's Confidential Information;

iv. Is rightfully obtained by the Recipient from third Parties authorized at that time to make such disclosure without restriction; &

v. is identified in writing by the Discloser as no longer proprietary or confidential; or vi. Is required to be disclosed by law, regulation or Court Order, provided that the recipient gives prompt written notice to the Discloser of such legal & regulatory requirement to disclose so as to allow the Discloser reasonable opportunity to contest such disclosure.

d. to the extent that such disclosure is required for the purposes of this Agreement, either Party may disclose Confidential Information to:

i. its employees, agents & independent selected vendors & to any of its affiliates & their respective independent selected vendors or employees; &

ii. its professional advisors & auditors, who require access for the purposes of this Agreement, whom the relevant Party has informed of its obligations under this Article & in respect of whom the relevant Party has informed of its obligations under this Article has used commercially reasonable efforts to ensure that they are contractually obliged to keep such Confidential Information confidential on terms substantially the same as set forth in this Article. Either Party may also disclose confidential Information or any entity with the other Party's prior written consent.

e. The provisions of this Article shall survive three years post expiration or any earlier termination of this Agreement.

f. confidential Information shall be & remain the property of the Discloser & nothing in this Article shall be construed to grant either Party any right or license with respect to the other Party's confidential Information otherwise than as is expressly set out in this Agreement.

g. Subject as otherwise expressly provide in this Agreement all Confidential information in tangible or electronic form under the control of the Recipient shall either be destroyed, erased or returned to the Discloser promptly upon the earlier of: (i) the written request of the Disclose, or, (ii) termination or expiry of this Agreement or, in respect of the PERFORMANCE STANDARDSs, the termination or expiry of the PERFORMANCE STANDARDSs. Notwithstanding the forgoing, both Parties may retain, subject to the terms of

this Article, reasonable number of copies of the other Party's Confidential Information solely for confirmation of compliance with the confidentiality obligation s of this Agreement.

h. Neither Party is restricted by the provisions of this clause from using (including using to provide products or perform services on behalf of third Parties) any ideas, concepts, know- how & techniques that are related to the Recipient's employees or agents (and not intentionally memorized for the purpose of later recording or use) (collectively, the "residuals"). This Article shall not permit the disclosure or use by either Party or any financial (including business plans), statistical, product, personnel or customer data or the other Party. Each party agrees not to disclose the source of the Residuals.

i. Both Parties agree that monetary damages would not be a sufficient remedy for any breach of this clause by the other Party & that VMC & bidder, as appropriate, shall be entitled to equitable relief, including injunction & specific performance as a remedy for any such breach. Such remedies shall not be deemed to be the exclusive remedies for a breach by a Party of this clause, but shall be in addition to all other remedies available at law or equity to the damaged Party.

j. In connection with the Services, Service Provider may from time to time undertake one or more quality assessment reviews for the purpose of improving the Project. In order for such reviews to be frank & candid, for the greatest benefit to both VMC & Service Provider, they shall be kept confidential to the greatest extent possible. The Parties agree that any documentation created in connection with such quality assessment reviews shall be confidential Information of Service Provider which is licensed to VMC for any internal use except that in no event shall such documentation or the results of such reviews be discoverable or admissible (or used for any purpose) in any arbitration or legal proceedings against Bidder related to this Agreement or the Services.

b) Standards of Performance

The Bidder shall provide the services and carry out their obligations under the Contract with due diligence, efficiency and professionalism/ethics in accordance with generally accepted professional standards and practices. The Bidder shall always act in respect of any matter relating to this contract. The Bidder shall abide by all the provisions/Acts/Rules/Regulations, Standing orders, etc. of Information Technology as prevalent in the country.
The Bidder shall also conform to the standards laid down by VMC or Government of Gujarat or Government of India from time to time.

c) Sub Contracts

Sub-contracting / out sourcing would be allowed only in O&M phase for work like

- Provisioning of technical man-power for operations and maintenance of application, hardware and cloud.

The bidder is expected to provide details of the sub-selected vendors for the work which is allowed as mentioned in the clause. Use of personnel not on payroll of the Bidder shall be considered as sub-contracting. The Bidder shall solely responsible for the work carried out by subcontracting under the contract. Bidder shall be the sole point of contact for the entire project throughout the project period.

d) Care to be taken while working at Public Place

Bidder should follow instructions issued by *concerned Competent Authority and* VMC from time to time for carrying out work at public places. Bidder should ensure that there is no damage caused to any private or public property. In case such damage is caused, Bidder shall immediately bring it to the notice of concerned organization and VMC in writing and pay necessary charges towards fixing of the damage. Bidder should also ensure that no traffic *congestion*/public inconvenience is caused while carrying out work at public places.

Bidder shall ensure that its employees/representatives don't breach privacy of any citizen or establishment during the course of execution or maintenance of the project.

e) Compliance with Labor regulations

The Bidder shall pay fair and reasonable wages to the workmen employed by him, for the contract undertaken by him and comply with the provisions set *forth* under the Minimum wages Act and the Contract Labor Act 1970.

f) Independent Selected vendor

The Successful Bidder shall act solely as an independent contractor under this Agreement. Nothing contained herein shall be construed as creating any employer–employee, agency, representative, or fiduciary relationship between VMC and the Successful Bidder. The Successful Bidder shall have no authority to act on behalf of, bind, or incur any obligation or liability in the name of VMC unless specifically authorized in writing by VMC. Similarly, VMC shall not be responsible for any acts, omissions, liabilities, or obligations of the Successful Bidder except as expressly provided under this Agreement.

g) Waiver

A waiver of any provision or breach of this Agreement must be in writing and signed by an authorized official of the Party executing the same. No such waiver shall be construed to affect or imply a subsequent waiver of the same provision or subsequent breach of this Agreement.

h) Performance Guarantee

The Bidder shall submit performance guarantee which is unconditional & irrevocable bank guarantee equal to an amount equivalent of 10% of the Project Value (CAPEX). However, for Operation & Maintenance phase, the bidder will submit the yearly bank guarantee of amount equivalent of 10% of the Project Value (OPEX). in the format prescribed in RFP issued by any of the Nationalized Banks Only. The performance bank guarantee shall be from Nationalized or Scheduled Banks except Co-operative Banks in favor of "CEO, Vadodara Municipal Corporation".
The performance guarantee shall be valid for the term agreement & shall be renewed & maintained by the Bidder for the term of the agreement & extension, if any. The performance guarantee shall be forfeited / liquidated by the VMC as a penalty in the event of failure to complete obligations or breach of any of the conditions by the Bidder.

i) Personnel/Employees

i. Personnel/employees assigned by Service Provider to perform the services shall be employees of Service Provider or its sub-selected vendors, & under no circumstances will such personnel be considered as employees of VMC. Service Provider shall have the sole responsibility for supervision & control of its personnel & for payment of such personnel's employee's entire compensation, including salary, legal deductions withholding of income taxes & social security taxes, worker's compensation, employee & disability benefits & the like & shall be responsible for all employer obligations under all laws as applicable from time to time. The VMC shall not be responsible for the above issues concerning to personnel of Service Provider.

ii. Service Provider shall use its best efforts to ensure that sufficient Service Provider personnel are employed to perform the Services, & that, such personnel have appropriate qualifications to perform the Services.

iii. Each Party shall be responsible for the performance of all its obligations under this Agreement & shall be liable for the acts & omissions of its employees & agents in connection therewith.

j) Variations & Further Assurance

a. No amendment, variation or other change to this Agreement or the PERFORMANCE STANDARDSs shall be valid unless made in writing & signed by the duly authorized representatives of the Parties to this Agreement.

b. Each Party to this Agreement or the PERFORMANCE STANDARDSs agree to enter into or execute, without limitation, whatever other agreement, document, consent & waiver & to do all other things which shall or may be reasonably required to complete & de liver the obligations set out in the Agreement or the PERFORMANCE STANDARDS.

k) Severability & Waiver

a) if any provision of this Agreement or the PERFORMANCE STANDARDSs, or any part thereof, shall be found by any court or administrative body of competent jurisdiction to be illegal, invalid or unenforceable the illegality, invalidity or unenforceability of such provision or part provision shall not affect the other provisions of this Agreement or the PERFORMANCE STANDARDSs or the remainder of the provisions in question which shall remain in full force & effect. The relevant Parties shall negotiate in good faith in order to agree to substitute any illegal, invalid or unenforceable provision with a valid & enforceable provision which achieves to the greatest extent possible the economic, legal & commercial objectives of the illegal, invalid or unenforceable provision or part provision within 7 working days.

b) No failure to exercise or enforce & no delay in exercising or enforcing on the part of either Party to this Agreement or the PERFORMANCE STANDARDSs of any right, remedy or provision of this Agreement or the PERFORMANCE STANDARDSs shall operate as a waiver of such right, remedy or provision in any future application nor shall any single or partial exercise or enforcement of any right, remedy or provision preclude any other or further exercise or enforcement of any other right, remedy or provision.

l) Entire Agreement

This MSA, the PERFORMANCE STANDARDSs & all schedules appended thereto & the contents of the RFP subsequent corrigenda issued thereon & clarification (undertakings) accepted by the VMC constitute the entire agreement between the Parties with respect to their subject matter.

m) Survivability

The termination or expiry of this Agreement or the PERFORMANCE STANDARDSs for any reason shall not affect or prejudice any terms of this Agreement, or the rights of the Parties under them which are either expressly of by implication intended to come into effect or continue in effect after such expiry or termination.

n) The stamp duty payable for the contract shall be borne by the Service Provider**.**

36. Applicable Law

The contract shall be governed by the laws and procedures prescribed by the Laws prevailing and in force in India, within the framework and enactment made from time to time concerning such commercial dealings/processing. All legal disputes are subject to the jurisdiction of Vadodara courts only.

IN WITNESS whereof the parties hereto have signed this on the day, month and year first herein above written.

Signed, sealed and delivered
By ----------
---------- --,--
For and on behalf of the ***VMC. Vadodara***

Signed, sealed and delivered
By ----------
For and on behalf of the "Service Provider",

Witnesses:
(1)
(2)

Attachments to the Agreement:

1) Scope of Services for the Service Provider
2) Detail Commercial proposal of the Service Provider accepted by VMC
3) Corrigendum Document published by VMC subsequent to the RFP for this work
4) RFP Document of VMC for this work
5) LoI issued by the VMC to the successful bidder
6) The successful bidder's "Technical Proposal" and "Commercial Proposal" submitted in response to the RFP
7) Payment Schedule and Milestones

ANNEXURE- III: Format for Performance Bank Guarantee

<< To be printed on Rs. 300/- Stamp Paper >>

IN CONSIDERATION OF …….. Through ……. VMC, Vadodara (Vadodara Municipal Corporation) for Selection of Bidder for Implementation of City Surveillance System in Vadodara City. (hereinafter referred to as the "said work") on the terms and conditions of the AGREEMENT dated the ……..day of..... 2026 executed between VMC on the one part and the Company (Name of the Company) on the other part (hereinafter referred to as "the said AGREEMENT) and on the terms and conditions specified in the Contract, Form of Offer and Form of acceptance of Offer, true and complete copies of the offer submitted by the Company, the said Acceptance of Offer and the said AGREEMENT are annexed hereto.

The Company has agreed to furnish VMC in Guarantee of the Nationalized Bank for the sum of Rs …….. (Agreement in Words and Figures) only which shall be the Security Deposit for the due performance of the terms covenants and conditions of the said AGREEMENT. We ..... Bank Registered in India under Act and having one of our Local Head Office at…….. do hereby guarantee to VMC in …….. Department.

i. Due performance and observances by the Company of the terms covenants and conditions on the part of the Company contained in the said AGREEMENT, AND
ii. Due and punctual payment by the Company to VMC of all sum of money, losses, damages, costs, charges, penalties and expenses that may become due or payable to VMC by or from the Company by reason of or in consequence of any breach, non-performance or default on the part of the Company of the terms covenants and conditions under or in respect of the said AGREEMENT.

AND FOR THE consideration aforesaid, we do hereby undertake to pay to VMC on demand without delay demur the said sum of Rs ..... (Rupees..... only) together with interest thereon at the rate prescribed under .....from the date of demand till payment or such lesser sum, as may be demanded by VMC from us as and by way of indemnity on account of any loss or damage caused to or suffered by VMC by reason of any breach, non-performance or default by the Company of the terms, covenants and conditions contained in the said AGREEMENT or in the due and punctual payment of the moneys payable by the Company to VMC there under and notwithstanding any dispute or disputes raised by the Company in any suit or proceeding filed before the Court relating thereto our liability hereunder being absolute and unequivocal and irrevocable AND WE do hereby agree that –

a) The guarantee herein contained shall remain in full force and effect during the subsistence of the said AGREEMENT and that the same will continue to be enforceable till all the claims of VMC are fully paid under or by virtue of the said AGREEMENT and its claims satisfied or discharged and till VMC certifies that the terms and conditions of the said AGREEMENT have fully and properly carried out by the Company.
b) We shall not be discharged or released from liability under this Guarantee by reason of
    a. any change in the Constitution of the Bank or
    b. any arrangement entered into between VMC and the Company with or without our consent;
    c. any forbearance or indulgence shown to the Company,
    d. any variation in the terms, covenants or conditions contained in the said AGREEMENT;
    e. any time given to the Company, OR
    f. any other conditions or circumstances under which in a law a surety would be discharged.

c) Our liability hereunder shall be joint and several with that of the Company as if we were the principal debtors in respect of the said sum of Rs..... (Rupees ……. Only).

d) We shall not revoke this guarantee during its currency except with the previous consent of VMC in .....department in writing;

e) Provided always that notwithstanding anything herein contained our liabilities under this guarantee shall be limited to the sum of Rs.....(Rupees .....only) and shall remain in force until VMC certifies that the terms and conditions of the said AGREEMENT have been fully and properly carried out by the Company.

f) Bank hereby agrees and covenants that if at any stage default is made in payment of any installment or any portion thereof due to VMC under the said AGREEMENT or if the Company fails to perform the said AGREEMENT or default shall be made in fulfilling any of the terms and conditions contained in the said AGREEMENT by the Company, the Bank shall pay to VMC demand without any demur, such sum as may by demanded, not exceeding Rs……. (Rupees.....) and that the Bank will indemnify and keep VMC indemnified against all the losses pursuant to the said AGREEMENT and default on the part of the Company. The decision of VMC that the default has been committed by the Company shall be conclusive and final and shall be binding on the Bank/Guarantor. Similarly, the decision of VMC as regards the Agreement due and payable by the Company shall be final and conclusive and binding on the Bank /Guarantor.

g) VMC shall have the fullest liberty and the Bank hereby gives its consent without any way affecting this guarantee and discharging the Bank/Guarantor from its liability hereunder, to vary or modify the said AGREEMENT or any terms thereof or grant any extension of time or any facility or indulgence to the Company and Guarantee shall not be released by reason of any time facility or indulgence being given to the Company or any forbearance act or omission on the part of VMC or by any other matter or think whatsoever which under the law, relating to sureties so releasing the guarantor and the Guarantor hereby waives all surety ship and other rights which it might otherwise be entitled to enforce.

h) That the absence of powers on the part of the Company or VMC to enter into or execute the said AGREEMENT or any irregularity in the exercise of such power or invalidity of the said AGREEMENT for any reason whatsoever shall not affect the liability of the Guarantor/Bank and binding on the bank notwithstanding any abnormality or irregularity

i) The Guarantor agrees and declares that for enforcing this Guarantee by.....against it, the Courts at Vadodara only shall have exclusive jurisdiction and the Guarantor hereby submits tothe same

1……

2……

Being respectively the Director of the Company, who in token thereof, has here to set his respective hands in the presence of –

1……

2……